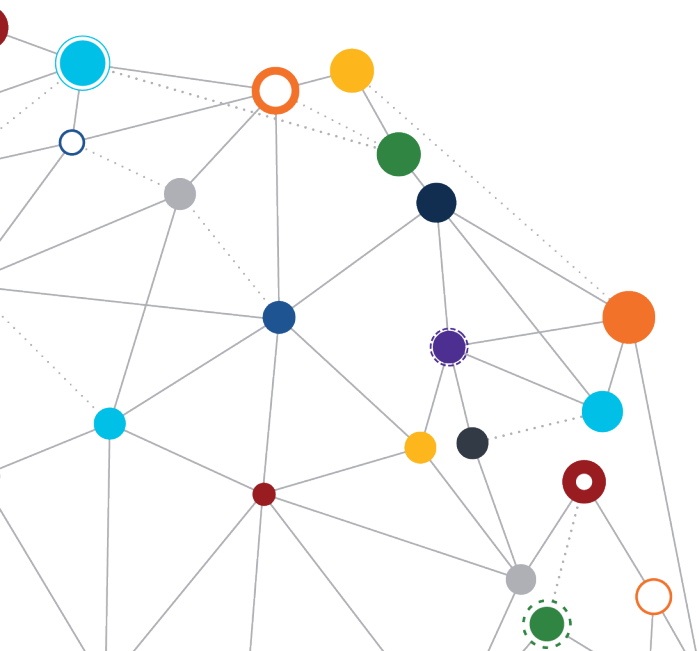




Infrastructure Standard for Telecommunications Spaces

Version 3.1

July 1, 2021 | Solution Delivery-Data Center and
Infrastructure Engineering



VA



U.S. Department of Veterans Affairs
Office of Information and Technology



Version Changes

Table 1: Version Changes

Date	Reason for Changes	Version	Author
2007	Original	0.1	Michael Julian, RCDD
2009	Revision	0.2	Michael Julian, RCDD
2012	Revision	1.0	Michael Julian, RCDD
12/07/2016	Enterprise Data Center Infrastructure Collaboration Team (EDICT) Revision	1.1	EDICT
2/05/2018	-Expanded revision including the addition of Administrative and Operations Standards. -Name changed to include IT Support Infrastructure to reflect broader scope.	1.2	EDICT
6/11/2018	-Minor update to allow for deeper network racks in cooperation with IO's request. -Expanded allowance for fiber cabinets, cassettes, patch panels, and racks. -Changed National Data Center Operations & Logistics (NDCOL) facility ID numbers to VA Record ID numbers. -Clarified reference to OIT responsibility for the enterprise-wide technical framework and IT architecture services.	2.1	EDICT
4/10/2019	-Name updated from "VA Enterprise Facility IT Support Infrastructure Standard" to "Infrastructure Standard for Telecommunications Spaces." -Minor editorial changes resulting from initial review by the Office of Construction and Facilities Management (CFM) Facilities Standards Service (FSS) for publication in Technical Information Library (TIL). -Removed operational information not required for building planning and design for publishing of the Standard on the TIL. Information redacted will be published as part of the Operations and Maintenance Standards for Telecommunications Spaces. In the interim, refer to the June 11th, 2018 version 2.1 of The Standard on the Baseline Configuration Management (BCM) Standards site. -508 compliance updates.	2.2	EDICT CFM
8/21/2020	-Editorial changes resulting from full review by CFM FSS for publication in CFM's TIL. -Updated Telecommunications Room (TR), entrance room, and electrical power standards -Added grounding and bonding, transformers, intra-building/inter-building cabling, lighting, and lightning protection system sections and Request for Variance process instructions.	3.0	EDICT CFM

Date	Reason for Changes	Version	Author
	-Created VA-modified Data Center Facility Environment Conditioning Standards figure to replace ASHRAE's less stringent requirements -Reorganized sequence of chapters for a more logical flow of data center design. Refer to the Table of Contents and the associated links to quickly find any subject of interest. -Relocated Figures, Tables, Definitions, References, and Acronyms sections to front matter for consistency with Telecommunication Industry Association (TIA) formats. -Added appendices to include Request for Variance form and OIT Design Guide Template Design Package		
7/1/2021	-4.2.8 Bonding and Grounding: changed "supplementary" ground path to "dedicated" -4.4.4 Fiber Optic Cable: relaxed authorization for Universal Cassettes to be used when breaking out to LC connectors -Appendix B Sheet 2: corrected split package air conditioner heat rejection requirement to be 17,000 BTU/h (5 kW) per rack -Appendix B Sheet 3: corrected layer issue masking attributes and specifications in the table -Appendix B Sheet 21: corrected layer issue masking a PDU on left side of Plate 1	3.1	EDICT

Table of Contents

Version Changes	i
Table of Contents.....	iii
Definitions	ix
References	xv
Acronyms and Initialisms	xxi
Contributors	xxvi
1 Introduction	1
1.1 Purpose	1
1.2 Scope	1
1.3 Administration.....	1
1.3.1 Request for Variance from Infrastructure Standards.....	2
1.3.2 Request for Change to Infrastructure Standards for Telecommunications Spaces	2
1.3.3 Contact Information.....	2
1.4 Standards Overview	2
1.5 Background	3
1.6 Objectives.....	3
1.7 Authority	4
2 IT Support Space Classification	5
2.1 Core Data Centers	5
2.2 Mission Support Centers	5
2.3 Campus Support Centers.....	6
2.4 Network Support Centers	6
3 Information Technology Support Space Planned Redundancy Levels	8
3.1 Core Data Center and Campus Support Center Planned Redundancy Levels.....	11
3.2 Mission Support Center Planned Redundancy Levels.....	12
3.3 Network Support Center, Telecommunications Room, and Entrance Room Planned Redundancy Levels.....	14
4 Infrastructure Standards	16
4.1 Architectural Standards and Space Design	16
4.1.1 Building Specifications for Common Telecommunications Spaces.....	16
4.1.2 Data Center Layout Standards	17
4.1.3 Telecommunications Room	20
4.1.4 Entrance Room.....	20
4.2 Electrical Power and Grounding Standards.....	21
4.2.1 Topology for Power Distribution Standards.....	22
4.2.2 Uninterruptible Power Supply Specifications	28
4.2.3 Transformers.....	32
4.2.4 Branch Power Circuit to Rack/Cabinet	32
4.2.5 Zone Power Distribution Units	33
4.2.6 Rack Mounted Uninterruptible Power Supplies	35
4.2.7 Vertical Rack Power Distribution Units	36
4.2.8 Bonding and Grounding	37
4.2.9 Lighting.....	44
4.2.10 Metering	45
4.3 Mechanical and Environmental Conditioning Standards and Monitoring.....	50
4.3.1 Introduction	50
4.3.2 Environmental Operating Envelope Conditions.....	51
4.3.3 Data Center Facility Environment Conditioning Standards.....	52
4.3.4 Data Center Facility Environment Conditioning Standards (Amendments and Exceptions) ...	54
4.3.5 Environmental Control Equipment Requirements.....	55
4.3.6 Airflow Control.....	57

4.3.7	Monitoring	64
4.4	Telecommunications Standards	68
4.4.1	Unshielded Twisted Pair	69
4.4.2	Unshielded Twisted Pair Patch Panel Standards	71
4.4.3	Unshielded Twisted Pair Patch Cord	72
4.4.4	Fiber Optic Cable	73
4.4.5	Fiber Distribution Cassettes	76
4.4.6	Fiber Patch Cords	76
4.4.7	Fiber Distribution Panel/Cabinet	78
4.4.8	Cable Support Infrastructure	78
4.4.9	Work Area Outlets	80
4.4.10	Horizontal Distribution	80
4.4.11	Backbone Distribution	80
4.4.12	Server Cabinets	82
4.4.13	Network Equipment Racks	84
4.4.14	Network Equipment Cabinets	85
5	Administration Standards	88
5.1	Data Center Position Identification	88
5.1.1	Cross-Reference to Other Identification Systems	91
5.1.2	Standardized Data Center Facility Type Identification	91
5.1.3	Standardized Data Center Naming Convention	93
5.1.4	Color Coded Identification	95
5.1.5	Information Transport Systems Equipment and Component Labeling	96
5.2	Management Standards	112
5.2.1	Management of Telecommunications Cabling	112
5.2.2	Management of Airflow	117
Appendix A: Request for Variance		A-1
Appendix B: OIT Design Guide Templates		B-1
B.1	Design Narrative	B-1
B.2	Main Computer Rooms	B-1
B.3	Entrance Rooms	B-2
B.4	Telecommunications Rooms	B-2
B.5	IT Design Guide Templates for Critical Infrastructure in Telecommunications Spaces	B-3

Figures

Figure 1: Generic Redundancy Level Standards for Physical Plants, Facilities, and Infrastructure Systems	9
Figure 2: Hot Aisle/Cold Aisle Approach	19
Figure 3: Power Distribution Options Simplified Visualization (CDC, CSC, MSC).....	23
Figure 4: Standard Density Underfloor/Whip Distribution Topology (CDC, CSC, MSC) – 60 A Zone PDU	24
Figure 5: Standard Density Underfloor/Whip Distribution Topology (CDC, CSC, MSC) – 30 A Zone PDU	24
Figure 6: High Density Underfloor/Whip Distribution Topology (CDC, CSC, MSC) – 60 A Zone PDU	25
Figure 7: High Density Underfloor/Whip Distribution Topology (CDC, CSC, MSC) – 30 A Zone PDU	26
Figure 8: Whip Distribution Topology with In-Rack UPS (NSC, TR, Entrance Room) – 60 A Zone PDU .	27
Figure 9: Whip Distribution Topology with In-Rack UPS (NSC, TR, Entrance Room) – 30 A Zone PDU .	27
Figure 10: Single Rack Whip Distribution Topology with In-Rack UPS (NSC, TR, Entrance Room) –20 A Circuits.....	28
Figure 11: International Electrotechnical Commission (IEC) Standard C13/14 cord set	37
Figure 12: Typical Bonding Connection	39
Figure 13: Typical Rack Bonding Layout	40
Figure 14: Typical Bonding Requirements for IT Equipment Enclosures	40
Figure 15: OEM Equipment Ground Lug Connection	41
Figure 16: OEM Equipment Ground Connection	41
Figure 17: Primary Bonding Busbar	42
Figure 18: Power Usage Effectiveness	49
Figure 19: VA-Modified ASHRAE Environmental Classes for Data Center Applications.....	52
Figure 20: Brushed Floor Grommet.....	59
Figure 21: Air Dam Foam	59
Figure 22: Blanking Panel Application	60
Figure 23: Method A Polarity of Horizontal Pre-terminated MPO-based Fiber Optic Cabling	75
Figure 24: Method A Polarity End to End Connectivity (A-B and A-A Patch Cords Required)	75
Figure 25: Polarity of Fiber Optic Patch Cable	77
Figure 26: Typical Position Identification Examples.....	90
Figure 27: Example of Best-Practice Differentiated Power Cord Coloration Implementation	96

<i>Figure 28: Rack Unit Identification Example.....</i>	<i>101</i>
<i>Figure 29: Example of Communications Cable Labeling</i>	<i>109</i>
<i>Figure 30: Detailed Example of Communications Cable Labeling.....</i>	<i>110</i>
<i>Figure 31: Power Whip Label Example.....</i>	<i>111</i>
<i>Figure 32: Equipment-Specific Cable Management Requirements.....</i>	<i>113</i>
<i>Figure 33: Acceptable Cable Management Results</i>	<i>114</i>
<i>Figure 34: Unacceptable Cable Management Results</i>	<i>115</i>
<i>Figure 35: Cable Labels in Bundles.....</i>	<i>117</i>

Tables

<i>Table 1: Version Changes</i>	<i>i</i>
<i>Table 2: Contributors</i>	<i>xxvi</i>
<i>Table 3: Data Center Classification Standards</i>	<i>6</i>
<i>Table 4: Data Center Planned Redundancy Levels</i>	<i>9</i>
<i>Table 5: Core Data Center and Campus Support Center Planned Redundancy Levels</i>	<i>11</i>
<i>Table 6: Mission Support Center Planned Redundancy Levels</i>	<i>12</i>
<i>Table 7: Network Support Center, Telecommunications Room, and Entrance Room Planned Redundancy Levels</i>	<i>14</i>
<i>Table 8: Building Specifications</i>	<i>16</i>
<i>Table 9: Data Center Floor Layout Standards</i>	<i>18</i>
<i>Table 10: Data Center Uninterruptible Power Supply Specifications</i>	<i>30</i>
<i>Table 11: Branch Power Circuit to Rack/Cabinet Standards (Underfloor/Whip Distribution)</i>	<i>32</i>
<i>Table 12: Branch Power Circuit to Rack/Cabinet Standards (Busbar Distribution)</i>	<i>33</i>
<i>Table 13: Zone Power Distribution Unit Standards</i>	<i>33</i>
<i>Table 14: Rack Mounted Uninterruptible Power Supply Standards</i>	<i>35</i>
<i>Table 15: Vertical Rack Power Distribution Unit Standards</i>	<i>36</i>
<i>Table 16: Busbar Standards</i>	<i>42</i>
<i>Table 17: Building Earth Ground System Standards</i>	<i>43</i>
<i>Table 18: Lightning Protection System</i>	<i>43</i>
<i>Table 19: Metered Energy Consumption Data</i>	<i>45</i>
<i>Table 20: Energy Consumption Performance Metrics</i>	<i>48</i>
<i>Table 21: Data Center Facility Environment Conditioning Standards</i>	<i>52</i>
<i>Table 22: Facility Environmental Requirements (CDC Amendments and Exceptions)</i>	<i>54</i>
<i>Table 23: Facility Environmental Requirements (MSC Amendments and Exceptions)</i>	<i>54</i>
<i>Table 24: Facility Environmental Requirements (NSC Amendments and Exceptions)</i>	<i>54</i>
<i>Table 25: Environmental Control Equipment</i>	<i>55</i>
<i>Table 26: Monitored Conditions</i>	<i>64</i>
<i>Table 27: Unshielded Twisted Pair Standards</i>	<i>70</i>
<i>Table 28: Unshielded Twisted Pair Patch Panel Standards</i>	<i>71</i>
<i>Table 29: Unshielded Twisted Pair Patch Cord Standards</i>	<i>72</i>

<i>Table 30: Fiber Optic Cable Standards</i>	<i>73</i>
<i>Table 31: Fiber Distribution Cassettes</i>	<i>76</i>
<i>Table 32: Fiber Patch Cord Standards</i>	<i>76</i>
<i>Table 33: Fiber Distribution Panel/Cabinet Standards</i>	<i>78</i>
<i>Table 34: Cable Support Infrastructure Standards</i>	<i>78</i>
<i>Table 35: Cable Distribution Standards</i>	<i>79</i>
<i>Table 36: Server Cabinet Standards</i>	<i>82</i>
<i>Table 37: Network Equipment Racks</i>	<i>84</i>
<i>Table 38: Network Equipment Cabinet Standards</i>	<i>85</i>
<i>Table 39: Data center Position Identification Conventions</i>	<i>88</i>
<i>Table 40: Standard FACILITYTYPE Naming Conventions</i>	<i>92</i>
<i>Table 41: Color Identification Conventions</i>	<i>95</i>
<i>Table 42: Information Technology Systems Equipment and Component Labeling Conventions</i>	<i>97</i>
<i>Table 43: Support Infrastructure Identification Conventions</i>	<i>99</i>
<i>Table 44: Transport Media and Interface Identification Conventions</i>	<i>102</i>
<i>Table 45: Power Distribution Identification Conventions</i>	<i>106</i>

Definitions

Access Provider: The operator of any facility that is used to convey telecommunications signals to and from a customer premises.

Administration: The method for labeling, identification, documentation and usage needed for installation, moves, additions and changes of the telecommunications infrastructure.

Alarming: Communicating when an absolute limit has been exceeded.

Alerting: Communicating when a condition threshold has been reached or exceeded.

American Society of Heating and Air-Conditioning Engineers (ASHRAE) Recommended Monitoring Locations (Modified):

- Front Top (FT): Centered horizontally, inside the front door, no more than 1 ft from the top of the cabinet
- Front Middle (FM): Centered horizontally, inside the front door, 4 ft +/- 6 in. from the finished floor surface
- Front Bottom (FB): Centered horizontally, inside the front door, no more than 1 ft from the finished floor surface
- Rear Top (RT): Centered horizontally, inside the rear door, no more than 1 ft from the top of the cabinet
- Rear Middle (RM): Centered horizontally, inside the rear door, 4 ft +/- 6 in. from the finished floor surface (not required by VA but recommended)
- Rear Bottom (RB): Centered horizontally, inside the rear door, no more than 1 ft from the finished floor surface (not required by VA but recommended)

Backbone:

- A facility (e.g., pathway, cable, conductors, optical fibers) between any of the following spaces: telecommunications rooms (TRs), telecommunications enclosures (TE), common TRs, floor-serving terminals, entrance facilities (EF), equipment rooms, and common equipment rooms.
- In a data center, a facility (e.g., pathway, cable, conductors, optical fibers) between any of the following spaces: entrance rooms or spaces, Main Distribution Areas (MDA), Horizontal Distribution Areas (HDA), Zone Distribution Areas (ZDA), and TRs.

Backbone Bonding Conductor: A telecommunication bonding connection which interconnects telecommunications bonding.

Busway: A pre-fabricated electrical distribution system used as an effective alternative to cable and conduit consisting of bus bars in a protective enclosure.

Bonding: The permanent joining of metallic parts to form an electrically conductive path that ensures electrical continuity and the capacity to conduct safely any current likely to be imposed.

Cabinet: A container that may enclose connection devices, terminations, apparatus, wiring, and equipment.

Channel: All telecommunications cable elements, permanent cable (link) and patching elements, that connect two pieces of IT equipment.

Channel Rack: A network four-post rack with sides having a cable management pattern of staggered shapes (circles, hexagons, etc.) to allow for connection points for cable management accessories used for housing networking equipment.

Compression Connection: A means of permanently bonding a conductor to a connector by permanently deforming the connector using a compression tool.

Computer Room: An architectural space whose primary function is to accommodate data processing equipment.

Concurrently Maintainable: The ability to have planned maintenance performed at any time on any path, equipment or component of the system without interrupting the operation of the system.

Cross-connect: A facility enabling the termination of cable elements and their interconnection or cross connection.

Conduit: (1) A raceway of circular cross-section. (2) A structure containing one or more ducts.

Data Center: A building or portion of a building whose primary function is to house a computer room and its support areas.

Demarcation Point: A point where the operational control or ownership changes.

Dew Point: The temperature to which air must be cooled (assuming constant air pressure and moisture content) to reach a relative humidity of 100 % (i.e. saturation).

Electromagnetic Interference (EMI): Radiated or conducted electromagnetic energy (noise) of any frequency that has an undesirable effect on electronic equipment or signal transmissions.

Entrance Room: A space in which the joining of inter or intra building telecommunications cabling takes place.

Equipment Distributor (ED): Reservation of the top 5 Rack Units (RU) of space in a server cabinet for copper and fiber patch panel equipment.

Exothermic Weld: A method of permanently bonding two metals together by a controlled heat reaction resulting in a molecular bond.

Facility Total Electrical Consumption (FEC): Total of all energy used from all sourced by the building housing the data center. This metric requires totaling the amount of electrical

energy going into the building through the building's main step-down transformer(s) and the amount of electrical energy generated through running any facility or data center generator equipment.

Ground: A conducting connection, whether intentional or accidental, between an electrical circuit (e.g., telecommunications) or equipment and the earth, or to some conducting body that serves in place of earth.

Grounding: The act of creating a ground.

Grounding Electrode: A conductor, usually a rod, pipe or plate (or group of conductors) in direct contact with the earth for the purpose of providing a low-impedance connection to the earth.

Grounding Electrode System: One or more grounding electrodes that are connected.

Horizontal Cabling (HC): (1) The cabling (media) between and including the telecommunications outlet and connector and the horizontal cross connect, usually in a telecommunications room. (2) Within a data center, the media between the Equipment Distributor (ED) and the Horizontal Distributor (HD).

Horizontal Distribution Area (HDA): Distribution used to serve equipment areas when the horizontal cabling is not located in the main distribution area. When used, the horizontal distribution area may include the horizontal cabling, which is the distribution point for cabling to the equipment distribution areas.

Information Transport Systems (ITS): The physical connectivity elements required to deliver a high-performance, reliable network system infrastructure for all requirements including data, voice, and video.

Infrastructure (telecommunications): A collection of those telecommunications components, excluding equipment, that together provide the basic support for the distribution of information within a building or campus.

Link: A single permanently-installed telecommunications cable element. See also permanent link.

Listed: Equipment included in a list published by an organization, acceptable to the authority having jurisdiction, that maintains periodic inspection of production of listed equipment, and whose listing states either that the equipment or material meets appropriate standards or has been tested and found suitable for use in a specified manner.

Main Distribution Area (MDA): The central point of distribution for the data center structured cabling system which includes the main cross-connect (MC) and may include horizontal cross-connect (HC) when equipment areas are served directly from the main distribution area.

Mechanical Connection: A reversible means of connecting a conductor to a connector through the use of a set screw or other bolt and nut device.

Mechanical Room: An enclosed space serving the needs of mechanical building systems.

Media: Wire, cable, or conductors used for telecommunications (transmission of signal).

Mesh Bonding Network: The set of metallic components that are interconnected to form the principle means for effectively bonding equipment inside a building to the grounding electrode system.

Metering: Measurement of a condition or set of conditions over a period of time. Metering is primarily a data collection function, intended to support analysis of operational characteristics of a data center.

Monitoring: Continual real-time observation of a condition or set of conditions affecting a data center.

Multimode Optical Fiber: An optical fiber that carries many paths of light.

Patch Cord: A cord used to establish connections on a patch panel.

Patch Panel: A connecting hardware system that facilitates cable termination and cabling administration using patch cords.

Pathway: A facility for the placement of telecommunications cable.

Permanent Link: The permanent, or fixed part of the network comprised of components that will permanently stay in place, such as patch panels, horizontal or backbone cable, and an outlet.

Plenum: A compartment or chamber to which one or more air ducts are connected and forms part of the air distribution system.

Pod: A modular subset of the data center.

Power Usage Effectiveness (PUE): A dimensionless metric which describes how efficiently a data center is designed and operated to conserve power. PUE is defined as the ratio: Total Data Center Energy Consumption (TEC)/Total IT Equipment Energy Consumption (TITEC) or TEC/TITEC.

Primary Bonding Busbar (PBB): The dedicated extension of the building grounding electrode system for the telecommunications infrastructure. The PBB also serves as the central attachment point for the TBB(s) and equipment.

Rack: Supporting frame equipped with side mounting rails to which equipment and hardware are mounted.

Rack Bonding Busbar (RBB): A busbar within a cabinet, frame or rack.

Rack Bonding Conductor (RBC): The bonding conductor from the rack or rack bonding busbar to the telecommunications equipment bonding conductor.

Rack Unit (RU): Vertical mounting space of 1.75 in. for cabinets or racks compliant with IEC 60297 or EIA/ECA-310-E.

Radio Frequency Interference (RFI): Radiated or conducted electromagnetic energy (noise) of a specific frequency that has an undesirable effect on electronic equipment or signal transmissions.

Raised Floor aka Access Floor: A platform with removable panels where equipment is installed, with the intervening space between it and the main building floor used to house the interconnecting cables and at times is used as a means for supplying conditioned air to the IT equipment and the room.

Secondary Bonding Busbar (SBB): A common point of connection for telecommunications system and equipment bonding to ground located in the distributor room.

Service Provider: The operator of any service that furnishes telecommunications content (transmissions) delivered over access provider facilities.

Single-Mode Optical Fiber: An optical fiber that carries only one path of light.

Site Infrastructure Energy Efficiency Rating (SIEER): A dimensionless metric which describes how efficiently a data center is designed and operated to conserve power. SIEER is defined as the ratio: Total Data Center Energy Consumption (TEC)/Total Uninterruptible Power Supply (UPS) Load Energy Consumption (TUPEC) or TEC/TUPSC.

Telecommunications: Any transmission, emission, and receptions of information by cable, radio, optical, or other electromagnetic systems including signaling commonly referred to as “data.”

Telecommunications Room (TR): A telecommunications space that differs from equipment rooms and entrance rooms in that this space is a floor-serving space that provides a connection point between backbone and horizontal cabling. The terms “closet,” “telecom closet,” “phone closet,” “Intermediate Distribution Frame (IDF),” etc. are no longer used. Blueprints are to use the abbreviation “TR.”

Telecommunications Space: An area used for housing the installation and termination of telecommunications equipment and cable. Telecommunications spaces include telecommunications rooms, entrance rooms, data centers, computer rooms, horizontal pathways, vertical pathways, etc. The term “telecommunications spaces” is used interoperably with the term “IT support space.”

Total Data Center Energy Consumption (TEC): Total of all energy used from all sources required to power all infrastructure and equipment within the confines of a data center.

Total Data Center Physical Infrastructure Energy Consumption: Summation of all energy used by facilities physical infrastructure equipment that supports the computer space. This is the total amount of electricity necessary to run all mechanical and electrical equipment

providing support to the computer space but does not include the IT Equipment Load Electrical Consumption.

Total IT Equipment Energy Consumption (TITEC): Summation of all energy directly consumed by IT equipment in the data center. IT equipment includes, but is not limited to, computers, storage systems, and networking gear.

Total Renewable Energy Usage by Data Center (RENEW): All electricity consumed by the facility housing the VA computer facility generated from renewable resources (solar, wind, hydro, wave, and similar) locally or through the supplying utility. Hydrocarbon and nuclear utility sources are not considered renewable for the purposes of this value.

Total UPS Load Energy Consumption (TUPEC): Summation of all energy used on the load (not source) side(s) of all UPS system(s) supporting the critical load in the computer space.

Uninterruptible Power Supply (UPS): A buffer between utility power or other power source and a load that requires continuous precise power.

Work Area: A building space where the occupants interact with telecommunications terminal equipment.

Work Area Outlet (WAO): A fixed connector in an equipment outlet assembly consisting of faceplate, body, housing, and one or more telecommunications cabling interface in the building space where the occupants interact with telecommunications terminal equipment. The term “jack” is no longer used.

Zone Distribution Areas (ZDA): A space in a data center where an equipment outlet or a consolidation point is located.

References

References are effective from their date of publication unless otherwise stated and are inclusive of supplements or addenda published by the standards body, amendments adopted by Authority Having Jurisdiction (AHJ), and superseding standards. The list is not comprehensive; compliance with all applicable standards is required. References consist primarily of nationally recognized “model” building codes, and technical standards developed or adopted by voluntary consensus standards bodies, use of which is required by P.L. 100-678, P.L. 104-113, and OMB Circular A119.

ANSI/ASHRAE Addendum15c, Safety Code for Mechanical Refrigeration

ANSI/ASHRAE Addendum15d, Safety Code for Mechanical Refrigeration

ANSI/ASHRAE Standard 90.4, Energy Standard for Data Centers

ANSI/ASHRAE Standard 188, Legionellosis: Risk Management for Building Water Systems

ANSI/BICSI 004-2018, Information Communication Technology Systems Design and Implementation Best Practices for Healthcare Institutions and Facilities

ANSI/TIA -455-C, General requirements for standard test procedures for optical fibers, cables, transducers, sensors, connecting and terminating devices, and other fiber optic components

ANSI/TIA-455-78B, FOTP-78 IEC 60793-1-40 Optical Fibres - Part 1-40: Measurement Methods and Test Procedures Attenuation

ANSI/TIA-470.110-D, Telecommunications Telephone Terminal Equipment - Transmission Requirements for Analog Telephones with Handsets

ANSI/TIA-492AAAA-B, Detail Specification for 62.5-µm.Core Diameter/125-µm Cladding Diameter Class Ia Graded-Index Multimode Optical Fibers

ANSI/TIA-492AAAB-A, Detail Specification for 50-µm Core Diameter/125-µm Cladding Diameter Class Ia Graded-Index Multimode Optical Fibers

ANSI/TIA-492AAAC-B, Detail Specification for 850-nm Laser- Optimized, 50-µm Core Diameter/125-µm Cladding Diameter Class Ia Graded-Index Multimode Optical Fibers

ANSI/TIA-526-7-A, Measurement of Optical Power Loss of Installed Single-Mode Fiber Cable Plant, A doptiono fIE C61280 -4-2e dition 2: Fibre-Optic Communications Subsystem Test Procedures; Part 4-2: Installed Cable Plant Single-Mode Attenuation and Optical Return Loss Measurement

ANSI/TIA-526-14-C, Optical Power Loss Measurement of Installed Multimode Fiber Cable Plant; Modification ofIE C61280-4-1 edition 2, Fiber-Optic Communications Subsystem Test Procedures- Part4 -1: Installed Cable Plant-Multimode Attenuation Measurement

ANSI/TIA-568.0-D, Generic Telecommunications Cabling for Customer Premises

ANSI/TIA-568.1-D, Commercial Building Telecommunications Infrastructure Standard

ANSI/TIA-568.2-D, Balanced Twisted-Pair Telecommunications Cabling and Components Standard

ANSI/TIA-568.3-D, Optical Fiber Cabling and Components Standard

ANSI/TIA-568-C.2, Balanced Twisted-Pair Telecommunications Cabling and Components Standards

ANSI/TIA-568-C.2 ERTA, Balanced Twisted-Pair Telecommunications Cabling and Components Standards

ANSI/TIA-568-C.2-1, Balanced Twisted-Pair Telecommunications Cabling and Components Standard, Addendum 1: Specifications for 100 Ω Category 8 Cabling

ANSI/TIA-568-C.2-2, Balanced Twisted-Pair Telecommunications Cabling and Components Standard, Addendum 2: Alternative Test Methodology for Category 6A Patch Cords - Addendum to TIA-568-C

ANSI/TIA-568-C.4, Broadband Coaxial Cabling and Components Standard

ANSI/TIA-569-D, Telecommunications Pathways and Spaces

ANSI/TIA-569-D-1, Telecommunications Pathways and Spaces: Addendum 1- Revised Temperature and Humidity Requirements for Telecommunications Spaces - Addendum to TIA-569-D

ANSI/TIA-598-D, Optical Fiber Cable Color Coding

ANSI/TIA-606-C, Administration Standard for Telecommunications Infrastructure

ANSI/TIA-607-D, Generic Telecommunications Bonding and Grounding (Earthing) for Customer Premises

ANSI/TIA-607-C-1, Generic Telecommunications Bonding and Grounding (Earthing) for Customer Premises - Addendum to ANSI/TIA-607-C (Superseded by 607-D)

ANSI/TIA-758-B, Customer-Owned Outside Plant Telecommunications Infrastructure Standard

ANSI/TIA-810-B, Telecommunications Telephone Terminal Equipment Transmission Requirements for Narrowband Digital Telephones

ANSI/TIA-862-B, Structured Cabling Infrastructure Standard for Intelligent Building Systems

ANSI/TIA-942-B, Telecommunications Infrastructure Standard for Data Centers

ANSI/TIA-968-B, Telecommunications Telephone Terminal Equipment Technical Requirements for Connection of Terminal Equipment to the Telephone Network

ANSI/TIA-968-B-1, Telecommunications Telephone Terminal Equipment Technical Requirements for Connection of Terminal Equipment to the Telephone Network-Addendum 1 - Addendum to TIA-968-B

ANSI/TIA-968-B-2, Telecommunications Telephone Terminal Equipment Technical Requirements for Connection of Terminal Equipment to the Telephone Network- Addendum 2 - Addendum to TIA-968-B

ANSI/TIA-968-B-3, Telecommunications Telephone Terminal Equipment Technical Requirements for Connection of Terminal Equipment to the Telephone Network- Addendum 3 - Addendum to TIA-968-B

ANSI/TIA-1005-A, Telecommunications Infrastructure Standard for Industrial Premises

ANSI/TIA-1005-A-1, Telecommunications Infrastructure Standard for Industrial Premises

Addendum 1- M12-8 X-Coding Connector (Addendum to TIA-1005-A)

ANSI/TIA-1152-A, Requirements for Field Test Instruments and Measurements for Balanced Twisted-Pair Cabling

ANSI/TIA-1179, Healthcare Facility Telecommunications Infrastructure

ANSI/TIA/EIA-455-1-B, Cable Flexing for Fiber Optic Interconnecting Devices

ANSI/TIA/EIA -455-2C, Impact Test Measurements for Fiber Optic Devices

ANSI/TIA/EIA-455-5C, FOTP-5 Humidity Test Procedure for Fiber Optic Components

ANSI/TIA/EIA-455-8, FOTP-8 Measurement of Splice or Connector Loss and Reflectance Using an OTDR

ANSI/TIA/EIA-455-34A, FOTP-34 Interconnection Device Insertion Loss Test

ANSI/TIA/EIA-455-171A, FOTP-171 Attenuation by Substitution Measurement for Short-Length Multimode Graded-Index and Single-Mode Optical Fiber Cable Assemblies

ANSI/TIA/EIA-455-185, FOTP-185 Strength of Coupling Mechanism for Fiber Optic Interconnecting Devices

ANSI/TIA/EIA-455-188, FOTP-188 Low Temperature Testing of Fiber Optic Components

ANSI/TIA/EIA-492CAAB, Detail Specification for Class IVa Dispersion-Unshifted Single-Mode Optical Fibers with Low Water Peak

ANSI/TIA/EIA-568-Set, Commercial Building Telecommunications Cabling Standard

American Society of Heating and Air-Conditioning Engineers (ASHRAE) Standard 52.2, Method of Testing General Ventilation Air-Cleaning Devices Removal by Particle Size (ANSI approved)

ASHRAE 62.1, Ventilation for Acceptable Indoor Air Quality

ASHRAE 135, BACnet - A Data Communication Protocol for Building Automation and Control Networks

ASHRAE 135 ERTA, BACnet - A Data Communication Protocol for Building Automation and Control Network

ASHRAE 135 INT 1, BACnet - A Data Communication Protocol for Building Automation and Control Network

ASHRAE 135 INT 2, BACnet - A Data Communication Protocol for Building Automation and Control Network

ASTM B258, Standard Specification for Standard Nominal Diameters and Cross-Sectional Areas of AWG Sizes of Solid Round Wires Used as Electrical Conductors

ASTM B539, Standard Test Methods for Measuring Resistance of Electrical Connections (Static Contacts)

ASTM D4565, Standard Test Methods for Physical and Environmental Performance Properties of Insulations and Jackets for Telecommunications Wire and Cable

ASTM D4566, Standard Test Methods for Electrical Performance Properties of Insulations and Jackets for Telecommunications Wire and Cable

ATIS 0600311, DC Power Systems Telecommunications Environment Protection

ATIS 0600336, Design Requirements for Universal Cabinets and Framework

BICSI 002, Data Center Design and Implementation Best Practices

BICSI ITSIMM, Information Technology Systems Installation Methods Manual (ITSIMM) - 7th Edition

BICSI OSPDRM, Outside Plant Design Reference Manual (OSPDRM) - 5th Edition

BICSI TDMM, Telecommunications Distribution Methods Manual - 13th Edition; Volumes I & II

BICSI/NECA 568, Standard for Installing Commercial Building Telecommunications Cabling

BICSI/NECA 607 Standard for Telecommunications Bonding and Grounding Planning and Installation Methods for Commercial Buildings 2011

CSI MASTERFORMAT™ 2004 EDITION NUMBERS & TITLES Construction Specifications Institute (CSI)

ECIA EIA/ECA-310-E, Cabinets, Racks, Panels, and Associated Equipment

Executive Order No. 13693 “Federal Leadership on Climate Change and Environmental Sustainability” 2015

Executive Order No. 13834 “Efficient Federal Operations” 2018

Federal Information Processing Standard (FIPS) 200, Minimum Security Requirements for Federal Information and Information Systems

IEEE STD 81, Guide for Measuring Earth Resistivity, Ground Impedance, and Earth Surface Potentials of a Grounding System.

IEEE STD 446, Emergency and Standby Power Systems for Industrial and Commercial Applications (Orange Book)

MIL-STD-188-124B, Grounding, Shielding, and Bonding

National Energy Policy Act of 2005-SEC. 203. FEDERAL PURCHASE REQUIREMENT

NEMA FG 1, Fiberglass Cable Tray Systems - Revision No.1

NEMA VE 1, METAL CABLE TRAY SYSTEMS

NEMA VE 2, Cable Tray Installation Guidelines - Incorporates Errata 2016

NFPA 70, National Electrical Code®

NFPA 70E, Standard for Electrical Safety in the Workplace

NFPA 72, National Fire Alarm and Signaling Code

NFPA 75, Standard for the Fire Protection of Information Technology Equipment

NFPA 99, Healthcare Facilities Code

NFPA 101, Life Safety Code

NFPA 780, Standard for the Installation of Lightning Protection Systems

NFPA 2001, Standard on Clean Agent Fire Extinguishing Systems

PG 18-12 OIT Design Guide, Office of Construction and Facilities Management Design Guide

Physical Security and Resiliency Design Manual

Presidential Memorandum 131205, “Federal Leadership on Energy Management”

TIA-232-F, Interface Between Data Terminal Equipment and Data Circuit-Terminating Equipment Employing Serial Binary Data Interchange

TIA-455-6-B, FOTP-6 Cable Retention Test Procedure for Fiber Optic Cable Interconnecting Devices

TIA-455-13-A, FOTP-13 Visual and Mechanical Inspection of Fiber Optic Components, Devices, and Assemblies

TIA-455-21-A, FOTP-21 Mating Durability of Fiber Optic Interconnecting Devices

TIA-455-36-A, FOTP-36 Twist Test for Fiber Optic Connecting Devices

TIA-455-107-A, FOTP-107 Determination of Component Reflectance or Link/System Return Loss Using a Loss Test Set

TIA-470.000-E, Telecommunications Telephone Terminal Equipment Overview of Performance Standards for Analog Telephones

TIA-485-A, Electrical Characteristics of Generators and Receivers for Use in Balanced Digital Multipoint Systems

TIA-492CAAA, Detail Specification for Class IVa Dispersion-Unshifted Single-Mode Optical Fibers

TIA-561, Simple 8-Position Non-Synchronous Interface Between Data Terminal Equipment and Data Circuit-Terminating Equipment Employing Serial Binary Data Interchange

TIA-568.2-D, Balanced Twisted-Pair Telecommunications Cabling and Components Standard

TIA-604-3-B, FOCIS 3 Fiber Optic Connector Intermateability Standard- Type SC and SC-APC

TIA-968-B ERTA, Telecommunications Telephone Terminal Equipment Technical Requirements for Connection of Terminal Equipment to the Telephone Network

TIA/EIA-455-4C, FOTP-4 Fiber Optic Component Temperature Life Test

TIA TSB-153, Static Discharge Between LAN and Data Terminal Equipment

TIA TSB-155-A, Guidelines for the Assessment and Mitigation of Installed Category 6 Cabling to Support 10GBASE-T

TIA TSB-184-A, Guidelines for Supporting Power Delivery Over Balanced Twisted-Pair Cabling

TIA TSB-184-A-1 (Addendum to TSB-184-A), Guidelines for Supporting Power Delivery Over Balanced Twisted-Pair Cabling-Addendum 1- use of 28 AWG cords

UL 444, UL Standard for Safety Communications Cables - 5th Edition

UL 444 BULLETIN, UL Standard for Safety Communications Cables

VA DIRECTIVE 6500, INFORMATION SECURITY PROGRAM

VA Electrical Design Manual

VA Fire Protection Design Manual

VA HVAC Design Manual

VA Lighting Design Manual

VA OIT Design Guide

VA Operations and Maintenance Standard for Telecommunications Spaces (Pending publication)

VA Physical Security and Resiliency Design Manual

VA Sustainable Design Manual

VA Telecommunications and Special Telecommunications Systems Design Manual

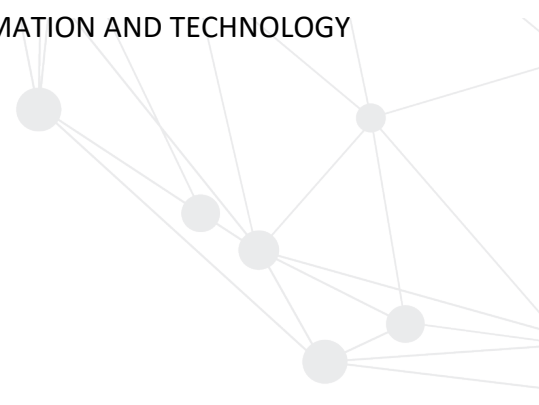
VHA Directive 1061, Prevention of Healthcare-Associated Legionella Disease and Scald Injury from Potable Water Distribution Systems

Acronyms and Initialisms

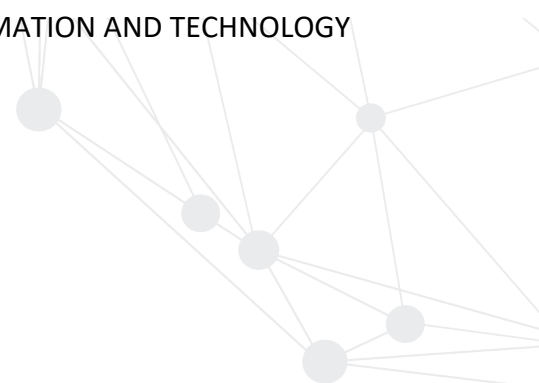
For other acronyms and initialisms, VA employees and contractors with VA intranet access may refer to the full [VA Acronym List](#) or [VA Acronym Lookup](#). All others may inquire with Enterprise Data Center and Infrastructure Engineering (DCIE) via e-mail at VAITESEDatacenterEngineering2@va.gov.

<u>Acronym</u>	<u>Definition</u>
AC	Alternating Current
ACEG	Alternating Current Equipment Ground
AES	Advanced Encryption Standard
AHJ	Authority Having Jurisdiction
ANSI	American National Standards Institute
APC	Angled Physical Contact
ARU	Air Removal Units
ASD	Aspirated Smoke Detection
ASHRAE	American Society of Heating and Air-Conditioning Engineers
ATS	Automatic Transfer Switch
AWG	American Wire Gauge
BACnet	Building Automation and Control Networking Protocol
BAS	Building Automation System
BCM	Branch Circuit Monitoring
BCP	Branch Circuit Panelboard
BICSI	Building Industry Consulting Service International
BMS	Battery Monitoring System
CAB	Change Advisory Board
CANbus	Controller Area Network Bus
CDC	Core Data Center
CE	Clinical Engineering
CFD	Computational Fluid Dynamics
CFM	Cubic Feet Per Minute
CFM	VA Office of Construction and Facilities Management
CM	Communications
CMMS	Computerized Maintenance Management System

<u>Acronym</u>	<u>Definition</u>
CMOP	Consolidated Mail Outpatient Pharmacy
CMP	Communications Plenum
CMR	Communications Riser
CPAC	Consolidated Patient Account Center
CRAC	Computer Room Air Conditioner
CRAH	Computer Room Air Handler
CSC	Campus Support Center
DC	Direct Current
DCIE	Data Center and Infrastructure Engineering
DCIM	Data Center Infrastructure Management
DR	Disaster Recovery
DX	Direct Expansion
EBSR	Enterprise Baseline and Standard Review
EC	Electronically Commutated
ED	Equipment Distributor
EDICT	Enterprise Data Center Infrastructure Collaboration Team
EGS	Earth Ground System
EIA	Electronic Industries Alliance
EMCS	Energy Management Control System
EPO	Emergency Power Off
ESD	Electrostatic Discharge
DX	Direct Expansion
FB	Front Bottom
FCU	Fan Coil Units
FEC	Facility Total Electrical Consumption
FM	Front Middle
FMS	Facilities Management Service
FSC	Financial Services Center
FSS	Facilities Standards Service
FT	Front Top
GbE	Gigabit Ethernet



<u>Acronym</u>	<u>Definition</u>
Gb/s	Gigabit per second
HA	High Availability
HAC	Health Administration Center
HC	Horizontal Cross Connect
HCC	Health Care Centers
HEC	Health Eligibility Center
HD	High Density
HD	Horizontal Distribution
HDA	Horizontal Distribution Area
HVAC	Heating, Ventilation, and Air Conditioning
IAW	In Accordance With
ICC	Intermediate Cross Connect
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronics Engineers
IO	Infrastructure Operations
ISP	Inside Plant
IT	Information Technology
ITOPS	IT Operations and Services
ITS	Information Transport Systems
LAN	Local Area Network
LED	Light Emitting Diode
LFMC	Liquid-tight Flexible Metal Conduit
LP	Limited Power
LPS	Lightning Protection System
MCC	Main Cross Connect
MM	Multimode
MPDU	Master Power Distribution Unit
MPO	Multi-Fiber Push On
MSC	Mission Support Center
NCA	National Cemetery Administration
NDCOL	National Data Center Operations and Logistics



<u>Acronym</u>	<u>Definition</u>
NEC®	National Electrical Code®
NEMA	National Electrical Manufacturers Association
NFPA	National Fire Protection Association
NIC	Network Interface Card
NSC	Network Support Center
O&M	Operations and Maintenance
OIFO	Office of Information Field Office
OIT	Office of Information and Technology
OM	Optical Multimode
OMB	Office of Management & Budget
OPC	Outpatient Clinic
PBB	Primary Bonding Busbar
PDU	Power Distribution Unit
PoE	Power-over-Ethernet
PVC	Polyvinyl Chloride
PS	Power Supply
PUE	Power Usage Effectiveness
R&D	Research & Development
RB	Rear Bottom
RBB	Rack Bonding Busbar
RBC	Rack Bonding Conductor
RCDD	Registered Communications Distribution Designer
RDC	Regional Data Center
RDC	Remote Distribution Cabinet
RH	Relative Humidity
RM	Rear Middle
rPDU	Rack Power Distribution Unit
RT	Rear Top
RT	Refrigeration Ton
RU	Rack Unit
SBB	Secondary Bonding Busbar

<u>Acronym</u>	<u>Definition</u>
SAN	Storage Area Network
SD	Solution Delivery
SD	Standard Density
SIEER	Site Infrastructure Energy Efficiency Rating
SM	Single-mode
SSL	Secure Sockets Layer
SWDM	Short Wavelength Division Multiplexing
TBB	Telecommunication Bonding Backbone
TEBC	Telecommunications Equipment Bonding Conductor
TEC	Total Data Center Energy Consumption
TIA	Telecommunications Industry Association
TIL	Technical Information Library
TITEC	Total IT Equipment Energy Consumption
TPIEC	Total Data Center Physical Infrastructure Energy Consumption
TR	Telecommunications Room
TUPSC	Total UPS Load Energy Consumption
UL	Underwriters Laboratories Inc
UPS	Uninterruptible Power Supply
UTP	Unshielded Twisted Pair
VA	Department of Veterans Affairs
VAMC	Veterans Affairs Medical Center
VACO	VA Central Office
VBA	Veterans Benefits Administration
VED	Vertical Exhaust Duct
VESDA	Very Early Smoke Detection Apparatus
VHA	Veterans Health Administration
VLA	Vented Lead Acid
VRLA	Valve Regulated Lead Acid
VSD	Variable Speed Drive
WC	Water Column
zPDU	Zone Power Distribution Unit

Contributors

This Standard is maintained by the Enterprise Data Center Infrastructure Collaboration Team.



The following subject matter experts have contributed to the development of this document as indicated:

Table 2: Contributors

Name	Solution Delivery	Role
Michael Julian	Office of Information and Technology (OIT), IT Operations and Services (ITOPS), Solution Delivery (SD)-Data Center and Infrastructure Engineering (DCIE) (OIT ITOPS SD DCIE)	SME, Telecommunications Distribution Design, Original Standard Author
Kelly Bates	OIT ITOPS SD DCIE	SME, Electrical and Mechanical Design
Kevin Grzelka	OIT ITOPS SD DCIE	SME, Telecommunications Space Design, Template Author and Custodian
John Wernau	OIT ITOPS SD DCIE	SME, Energy Efficiency and Airflow Design, Standard Custodian
Curtis Derbigny	OIT ITOPS SD DCIE	SME, Systems Integration
Name	Infrastructure Operations	Role
William Pittman	OIT ITOPS Infrastructure Operations (IO) National Data Center Operations & Logistics (NDCOL)	SME, Director, NDCOL
Jack McLain	OIT ITOPS IO NDCOL	SME, Director, Data Center Facilities, Connectivity and Physical Security
Mitch Cobb	OIT ITOPS IO NDCOL	SME
Lawrence Manzanera	OIT ITOPS IO NDCOL	SME
Robert Burke	OIT ITOPS IO NDCOL	SME
Clay Whitney	OIT ITOPS IO NDCOL	SME
Wesley Patton	OIT ITOPS IO NDCOL	SME

Name	Infrastructure Operations (cont.)	Role
Ken Girardi	OIT ITOPS IO NDCOL	SME
Robert Hoag	OIT ITOPS IO NDCOL	SME
Robert Skipper	OIT ITOPS IO NDCOL	SME
Name	Construction and Facilities Management	Role
Bryan Unger	Office of Construction and Facilities Management (CFM) Facilities Standards Service (FSS)	SME, Coordinator – CFM Technical Information Library (TIL)
Dave Tashakkori	CFM FSS	SME
Gary Krupa	CFM FSS	SME
Name	508 Office	Role
Myra Coffield	Section 508 Office	Technical Writing, CommonLook Office

1 Introduction

1.1 Purpose

The Infrastructure Standard for Telecommunications Spaces (The Standard) serves as the master reference document of criteria for Department of Veterans Affairs (VA) facility Information Technology (IT) support infrastructure. It defines the technical requirements necessary to maintain optimum reliability and efficiency within VA facilities and computing centers.

1.2 Scope

The Standard shall be applied to all aspects of IT support infrastructure at all VA owned, operated, and leased spaces. “All aspects” includes, but is not limited to planning, design, construction, sustainment (operations, maintenance, repair), restoration, modernization, and administration. “All VA spaces” includes but is not limited to VA Central Office (VACO) facilities and all field facilities managed by Veterans Health Administration (VHA), Veterans Benefits Administration (VBA), and the National Cemetery Administration (NCA).

IT Support Infrastructure is defined as all passive telecommunications and IT equipment, and supporting physical space and equipment, supporting:

- Copper and fiber cabling Information Transport Systems (ITS)
- IT equipment and systems (servers, storage, network switching and distribution, etc.)
- Telecommunications spaces, Telecommunications Rooms (TR), entrance rooms, data centers, horizontal distribution pathways, vertical distribution pathways, etc.
- Telecommunications equipment and systems (local and wide area network, telephone, cable television, physical security IT system, etc.)
- Physical infrastructure equipment and systems necessary to support active equipment operations at the required availability and sustainability levels (power distribution, environmental conditioning, monitoring and metering equipment, grounding, etc.)

1.3 Administration

The Standard shall be implemented without modification, except when required by an Authority Having Jurisdiction (AHJ), Federal, state, or local laws and codes. Additional requirements of AHJs shall be provided to Enterprise Data Center and Infrastructure Engineering (DCIE) for evaluation of project application and possible amendment of The Standard.

Projects in the planning, design, or implementation phases shall contact DCIE for assistance in telecommunications space design and standards compliance.

Existing IT support infrastructure that does not conform to The Standard shall be brought into compliance during routine tech refresh, lifecycle replacement, upgrades, new installations, or renovations of existing space.

Facility managers should take every opportunity to bring telecommunications spaces into compliance with The Standard through incremental changes between scheduled sustainment activities.

Area Managers shall ensure that all contracting officers, IT personnel, and engineering staff are aware and make use of The Standard.

1.3.1 Request for Variance from Infrastructure Standards

Requests for variance for allowable deviations from specific requirements of any standards published within may be authorized where a proposed alternate method of installation or operation will provide equal or higher safety, reliability, redundancy, or sustainability objectives. Requests for variance shall be submitted to DCIE for consideration using a Request for Variance form (see *Appendix A*) through a [Data Center and Infrastructure Engineering Engagement](#) Services Request on the [Your IT Services portal](#). If a variance is granted, it shall be limited to the particular site and installation covered in the application and will NOT be considered as a precedent for other installations.

NOTE: This is a separate process from VHA's process and form for requesting variances from other TIL standards and design manuals.

1.3.2 Request for Change to Infrastructure Standards for Telecommunications Spaces

Requests for Standard revision shall also be submitted to DCIE for consideration using a Request for Variance form (see *Appendix A*) through a [Data Center and Infrastructure Engineering Engagement](#) Services Request on the [Your IT Services portal](#). If approved, the proposed change to The Standard will be ratified through the Enterprise Baseline and Standard Review (EBSR) process (formerly Systems Engineering Design Review (SEDR)) and submitted to the Enterprise Change Advisory Board (CAB) for final approval (if required) before publication.

1.3.3 Contact Information

Transmit all communications to DCIE via e-mail at VAITSEDatacenterEngineering2@va.gov.

1.4 Standards Overview

The Standard includes standard specifications, decisions supporting the standard specifications, guidelines or recommendations for implementing the standard specifications, and supplemental factors to consider when evaluating subject components.

The Infrastructure and the Administration portions of The Standard define the topologies and specifications for facilities and systems in the VA enterprise.

The Infrastructure Standards also provide guidance on procuring components that meet the standard specifications, guidance on integrating them with existing components, and explanation of how the subject components fit into the VA Architecture.

The Administration Standards define the acceptable and recommended specifications for Inside Plant (ISP) naming conventions, including racks, equipment enclosures, fiber and

Unshielded Twisted Pair (UTP) transport media, power distribution for active and passive elements, cable plant, and data center naming conventions.

At the time of this publishing, a separate standard for Operations and Maintenance (O&M) of telecommunications spaces is being developed as a companion piece to The Standard. The O&M Standard will describe facility computer room operation specifications, management of telecommunications cabling, and management of airflow. It is anticipated that the O&M Standard will be published and socialized prior to the next annual release of v4.0 of this Standard in FY21.

1.5 Background

A standard is a set of rules or requirements that are determined by a consensus opinion of subject matter experts and prescribe criteria for a product, process, test, or procedure. The general benefits of a standard are quality, interchangeability of parts or systems, consistency, and reduction of lifecycle costs.

Telecommunications infrastructure standards are based on business needs provided through or supported by IT services. IT services are designed to support business processes and are constructed from software, hardware and infrastructure components.

Establishing and enforcing standards for the selection and configuration of these supporting components improves the capacity, sustainability, maintainability, reliability, and availability of IT services within projected economic constraints in alignment with business needs.

1.6 Objectives

This Standard provides recommended specifications across all VA facilities to support:

- Requirement development
- Solution design
- Solution evaluation
- Solution procurement and bid evaluation
- Evaluation of architectural specifications
- Standardization of passive ISP and Outside Plant (OSP) infrastructure
- Standardization of design criteria for telecommunications spaces, facilities physical infrastructure systems, and operation and maintenance of systems supporting VA services
- Standardization of naming and identification conventions for VA data centers
- Standardization of ISP and rack power infrastructure naming conventions
- Standardization of labeling and identification conventions for physical plant systems and components
- Standardization of operations
- Operational solution procurement and bid evaluation
- Evaluation of maintenance, repair, and replacement specifications
- Standardization of lifecycle sustainment activities across all VA data centers
- Standardization of the operational, organizational, and aesthetic results of Sustainment, Restoration, and Modernization (SR&M) activities

1.7 Authority

Office of Information and Technology (OIT) is the chartered organization responsible for the enterprise-wide technical framework and IT architecture services for VA systems and projects. Centralized responsibility promotes one technology vision across VA, which supports system optimization, integration, and interoperability throughout the enterprise.

One way in which OIT executes this responsibility is determining technical standards for deployed infrastructure technologies. These standards are also known as configuration baselines. The authority of The Standard derives from the approved enterprise configuration baselines.

2 IT Support Space Classification

VA has four segregated IT Support Space types to support the enterprise distributed computing environment.

- Core Data Centers (CDC) provide enterprise-level support services for VA-wide and cross-component systems and applications.
- Mission Support Centers (MSC) are highly specialized data centers that provide unique, focused services that are best provided from smaller, mission-focused facilities that cannot or should not be supported by CDCs or Campus Support Centers (CSC) due to MSCs' association with mission-specific infrastructure or equipment.
- Campus Support Centers (CSC) operate local data centers to support VA tenants located on a VA campus and within geographically supported areas. CSCs provide services that cannot be provided (technically or economically) from a CDC. Only one CSC is authorized per VA campus.
- Network Support Centers (NSC), provide lower-level application and IT services distribution across the geographic breadth of the VA's computing enterprise, enabling special functions and distributed users to satisfy the specific mission (Note: TRs generally follow NSC rating classification and redundancy requirements).

2.1 Core Data Centers

Core Data Centers (CDC) are facilities that provide shared and distributed enterprise services to supported sites. CDCs host cloud services to provide complete end-to-end IT services (including but not limited to application, backup, and storage) to the supported VA sites.

The CDC architecture is considered mission-essential and is designed, operated, and maintained to provide availability consistent with current VA national data center standards for supported IT applications. CDCs are the target architecture component for IT services that span multiple Department components and user groups (services and applications used by many or all users, such as messaging, productivity, and collaboration).

CDCs may be VA-owned, leased, or government or commercial outsourced (external cloud) data centers. CDCs meet (or are planned to meet) all published CDC physical infrastructure requirements to ensure enterprise reliability, redundancy, High Availability (HA), and Disaster Recovery (DR) requirements.

CDCs may also provide services designated to be provided by an NSC if no local NSC to provide those services is available. CDCs may also provide MSC-type application support.

2.2 Mission Support Centers

Mission Support Centers (MSC) provide element-specific and single-instance special enterprise services that may be inappropriate for consolidation to CDC facilities. MSCs are the target architecture for specialized-function systems and applications, particularly those that support a specific but limited local or distributed user group (services allowing the Department to function but with a limited user base such as internal financial management, or applications requiring centralized processing for a small number of user facilities, such as consolidated pharmacies).

MSC physical requirements supporting the computing environment may be more lenient than those for CDCs. Specific requirements will vary based on the supported functions, criticality of the mission, and similar criteria. For example, MSCs designated as enterprise Test/Development (Test/Dev) environments will not require the same level of high-availability physical and network redundancy as a mission-critical production environment that directly supports patient care.

MSCs may also provide services designated to be provided by an NSC if no local NSC to provide those services is available.

2.3 Campus Support Centers

Campus Support Centers (CSC) provide geographically-specific, operational IT services in support of campus services to Veterans and VA employees that cannot be effectively consolidated or provided over cloud architectures to the campus.

CSCs shall provide NSC services for their supported campuses.

IT services at CSCs are provided by multiple organizations, including Clinical Engineering (CE) aka biomed, health informatics, Veterans Affairs Medical Centers (VAMC), VBA Regional Offices, Office of Information Field Office (OIFO), and Research & Development (R&D), etc.

CSCs are intended to provide for collocation of IT equipment and systems from all provider organizations into the minimum number of operational spaces on a campus; that is, consolidated data centers providing environmental and area network support appropriate to all IT equipment needed by tenant organizations on a particular campus to perform their missions.

2.4 Network Support Centers

Network Support Centers (NSC) provide local IT services that cannot be effectively consolidated or provided over a network-supported distributed architecture to the local operational site. Local means to the immediate facility and to facilities within the local commuting/networking area for the purposes of this architecture; NSCs may provide application support for all VA facilities in a metropolitan area, for example.

Where application architecture and network requirements require a closer data center point of presence to enable end users effective, efficient system access, an NSC is the appropriate target environment.

Table 3: Data Center Classification Standards

ID	Primary Attribute	Secondary Attribute	Specification
1	Core Data Center (CDC)	Description	CDCs are centralized data centers that provide enterprise-level services to geographically-distributed VA organizations to support business functions. CDCs shall maintain complete HA and DR
		Planned ANSI/TIA-942-B Rating	3

ID	Primary Attribute	Secondary Attribute	Specification
		Examples	Information Technology Center (ITC) Regional Data Center (RDC)
2	Mission Support Center (MSC)	Description	MSCs are stand-alone data centers that provide specific enterprise IT functionality to VA organizations and business functions
		Planned ANSI/TIA-942-B Rating	2 or 3
		Examples	• Test/Dev data centers at Rating 2 • Financial Services Center (FSC) • Centralized Mail Outpatient Pharmacy (CMOP) • Consolidated Patient Account Center (CPAC) • Health Administration Center (HAC)
3	Campus Support Center (CSC)	Description	CSCs are stand-alone data centers that provide specific IT functionality to a VA campus supporting all VA tenants on that campus and in the geographically supported area
		Planned ANSI/TIA-942-B Rating	3
		Examples	• VA Medical Center (VAMC) • Large Community-Based Outpatient Clinics (CBOC) • Health Care Center (HCC)
4	Network Support Center (NSC)	Description	NSCs are local data centers that provide local IT application and network support to local operating locations
		Planned ANSI/TIA-942-B Rating	1
		Examples	• Typical size Community-Based Outpatient Clinics (CBOC) • VBA Regional Office • TRs • Entrance rooms • Extra small server rooms (not “shallow rooms”) • Administrative office space server rooms

Evaluation Factors

- Data center facility appropriately and centrally categorized by function and usage
- Capability gaps between existing and planned ANSI/TIA-942-B Rating requirements identified

3 Information Technology Support Space

Planned Redundancy Levels

VA IT support spaces shall be designed and operated in accordance with the ANSI/TIA-942-B Ratings for each data center type, except as detailed specifications are provided in this and other VA data center facilities standards.

These specifications define the minimum system redundancy levels for the physical plants and facilities physical infrastructure systems that support enterprise data centers. This Standard shall be used to quantify and qualify designs of physical infrastructure supporting data center facilities and spaces when those spaces are being designed, built, retrofitted, refreshed, and updated.

The minimum design redundancy levels for new and replacement systems in IT support spaces are detailed below. These designs balance system implementation and operation cost against availability needs for each type of space. In planning physical infrastructure system projects, reduction of potential single points of failure is generally considered more critical than increasing the redundancy level of a system.

From a design perspective on a system+system (2N) design, the smaller capacity of the two individual systems is considered the overall system capacity. For instance, a 2N system with a 500k side and a 750k side has a maximum capacity of 500k.

Redundancy is represented relating to the need (N). If no redundancy is required, the system simply has a redundancy level of N. To gain the simplest form of redundancy, one additional system is added represented by N+1. For example, if two Computer Room Air Conditioners (CRAC) are needed to meet the cooling need of the data center, then to obtain an N+1 redundancy level a third CRAC must be added. To provide further redundancy a 2N system could be put in place. In the CRAC example, four CRACs would create a 2N system. Furthermore, for a 2N+1 system, five CRACs would be required to meet this level of redundancy. Finally, for a 2(N+1) system, six CRACs would be required.

Implementation Guidance

Where N+C redundancy is required for environmental control systems in CDCs, be aware that design considerations including total number and capacity of CRAC units, spacing of units in planned designs, and geometries of computer room spaces may dictate that one, two, three, or more units more than the N requirement may be necessary to ensure appropriate environmental control in all contingency situations. Computational Fluid Dynamics (CFD) modeling must be accomplished to understand the airflow and determine the effect of teaming various combinations of CRAC units and adjustment of settings.

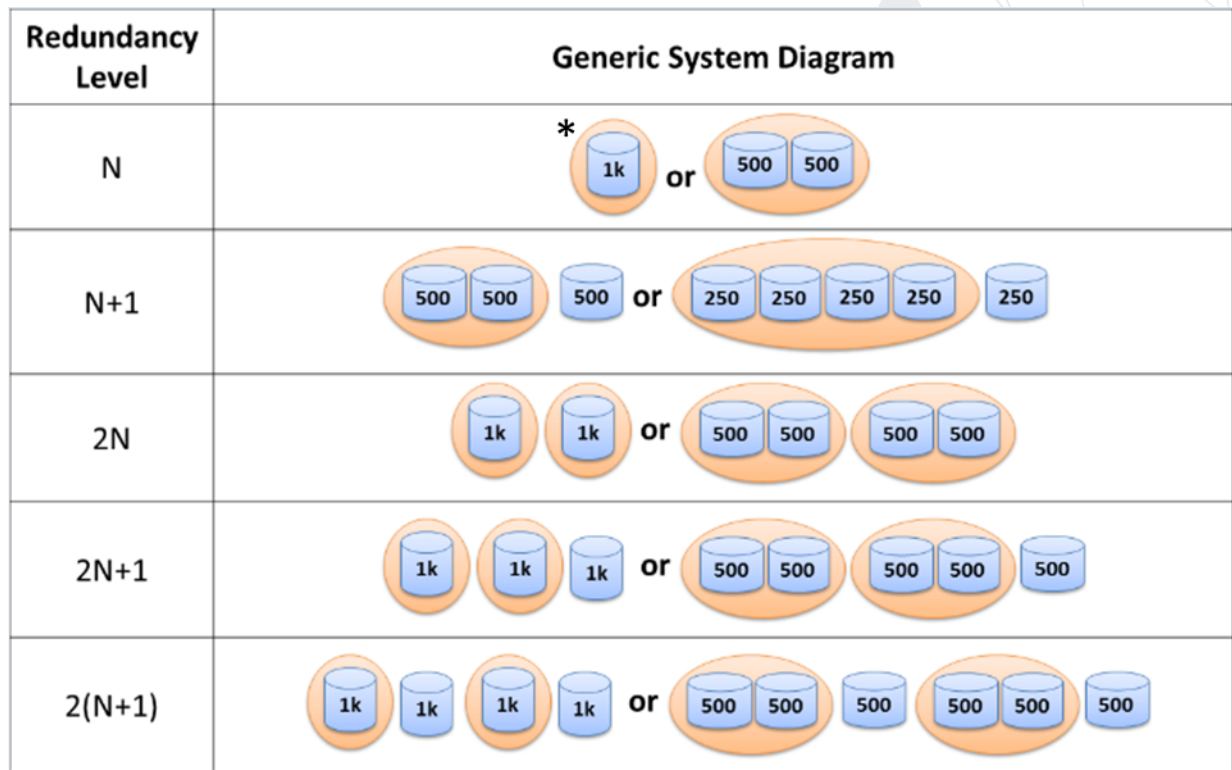


Figure 1: Generic Redundancy Level Standards for Physical Plants, Facilities, and Infrastructure Systems

*Circled loads are active

Table 4: Data Center Planned Redundancy Levels

ID	Primary Attribute	Secondary Attribute	Specification
1	Facility Power Input Sources	Core Data Center (CDC)	N+1 redundant feed: one commercial power source (also known as utility), one generated power (also known as emergency)
		Mission Support Center (MSC)	N+1 redundant feed (one commercial, one generated) for an ANSI/TIA Rating 3 facility or single feed (N) for an ANSI/TIA Rating 2 facility
		Campus Support Center (CSC)	N+1 redundant feed (one commercial, one generated)
		Network Support Center (NSC)	N (one commercial)
2	Emergency Power Generation Source	Core Data Center (CDC)	N+1
		Mission Support Center (MSC)	N for rating 2 facility and N + 1 for rating 3 facility.
		Campus Support Center (CSC)	N+1

ID	Primary Attribute	Secondary Attribute	Specification
		Network Support Center (NSC)	- Provide emergency power to TRs IAW the VA Electrical Design Manual when supporting medical center with a CSC - Not required but recommended supporting other medical and non-medical facilities
3	Uninterruptible Power Supply (UPS) Technical Power Systems	Core Data Center (CDC)	2N
		Mission Support Center (MSC)	N+1 (distributed redundant modules or block redundant system with dedicated battery string for each module)
		Campus Support Centers (CSC)	2N
		Network Support Center (NSC)	N (single UPS system, single battery string)
4	Electrical Distribution (UPS to IT equipment)	Core Data Center (CDC)	2N (mirrored A/B distribution from UPS to rack in support of all IT systems)
		Mission Support Center (MSC)	2N (mirrored A/B distribution from UPS to rack in support of all IT systems)
		Campus Support Center (CSC)	2N (mirrored A/B distribution from UPS to rack in support of all IT systems)
		Network Support Center (NSC)	N
5	Environmental Support (HVAC)	Core Data Center (CDC)	- N+1. If more than five CRACs, then N+C where C=1 for every five to eight CRACs needed - Brief interruptions to electrical power (<10 min) will not cause loss of cooling but may cause temperature to elevate within operational range of critical equipment
		Mission Support Center (MSC)	- N+1. - Brief interruptions to electrical power (<10 min) will not cause loss of cooling but may cause temperature to elevate within operational range of critical equipment

ID	Primary Attribute	Secondary Attribute	Specification
		Campus Support Center (CSC)	- N+1. - Brief interruptions to electrical power (<10 min) will not cause loss of cooling but may cause temperature to elevate within operational range of critical equipment
		Network Support Center (NSC)	N

3.1 Core Data Center and Campus Support Center Planned Redundancy Levels

Table 5: Core Data Center and Campus Support Center Planned Redundancy Levels

ID	Primary Attribute	Secondary Attribute	Specification
1	Facility Power Input Sources	System Redundancy Components	N+1 redundant feed (one commercial, one generated)
		Distribution Paths	One active, one standby (one commercial feed, one standby generation, Automatic Transfer Switch (ATS))
		System (N) Capacity	Total facility design capacity (either commercial feed or generator can provide 100 % of the facility's design load)
2	Emergency Power Generation Source	System Redundancy Components	N+1
		Distribution Paths	Single path ATS
		System (N) Capacity	Total facility design capacity (generator source can provide all critical mechanical and electrical support for the facility)
3	UPS Technical Power Systems	System Redundancy Components	2N
		Distribution Paths	Two simultaneously active (A/B configuration)
		System (N) Capacity	- Each UPS system (A/B) rated for 100 % of data center design load - Modular systems sized for ultimate design load but only populated to current load needs is preferred
		UPS Technology	Two delta conversion or double conversion UPS systems
4	Electrical Distribution (UPS to IT equipment)	System Redundancy Components	2N (mirrored A/B distribution from UPS to rack in support of all IT systems)
		Distribution Paths	Two simultaneously active (A/B distribution from UPSs)

ID	Primary Attribute	Secondary Attribute	Specification
		System (N) Capacity	Maximum design loading of all components with one distribution path offline
5	Environmental Support (HVAC)	System Redundancy Components	- N+1. If more than five CRACs, then N+C where C=1 for every five to eight CRACs needed - Brief interruptions to electrical power (<10 min) will not cause loss of cooling but may cause temperature to elevate within operational range of critical equipment
		Redundancy Configuration	One alternate
		System (N) Capacity	Total facility design capacity

Implementation Guidance

Uninterruptible Power Supply (UPS) systems shall be sized for the ultimate design load of the telecommunications space. For initial cost, maintenance, reliability, and efficiency reasons, modular UPS systems built out to provide sufficient UPS and battery capacity to meet near-term requirements are preferred over full-capacity systems.

3.2 Mission Support Center Planned Redundancy Levels

Table 6: Mission Support Center Planned Redundancy Levels

ID	Primary Attribute	Secondary Attribute	Specification
1	Facility Power Input Sources	System Redundancy Components	N+1 Redundant Feed (one commercial, one generated) for an ANSI/TIA-912 rated 3 facility or Single feed (N) for an ANSI/TIA-942-B rated 2 facility
		Distribution Paths	One active, one standby (one commercial utility feed, one standby generation, ATS)
		System (N) Capacity	Total facility design capacity (either source can provide 100 % of the facility's design load)
2	Emergency Power Generation Source	System Redundancy Components	- N+1 for an ANSI/TIA-942-B rated 3 facility - N for an ANSI/TIA-942-B rated 2 facility
		Distribution Paths	Single path ATS

ID	Primary Attribute	Secondary Attribute	Specification
		System (N) Capacity	Total facility design capacity (generator source can provide all critical mechanical and electrical support for the facility) for an ANSI/TIA-942-B rated 3 facility or sized for UPS and mechanical system without redundancy for a Rated 2 facility
3	UPS Technical Power Systems	System Redundancy Components	N+1 (distributed redundant modules or block redundant system with dedicated battery string for each module)
		Distribution Paths	Two simultaneously active (A/B configuration)
		System (N) Capacity	- Each UPS system (A/B) rated for 100 % of data center design load - Modular systems sized for ultimate design load but only populated to current load needs are preferred
		UPS Technology	Delta conversion or double conversion UPS systems
4	Electrical Distribution (UPS to IT equipment)	System Redundancy Components	2N (mirrored A/B distribution from UPS to rack in support of all IT systems)
		Distribution Paths	- Two simultaneously active (A/B distribution from UPSs to PDUs to the Master Power Distribution Unit (MPDU)) - Redundant, diverse path, and follows rectilinear pathways when not contained in a cable tray
		System (N) Capacity	Maximum design loading of all components with one system offline

ID	Primary Attribute	Secondary Attribute	Specification
5	Environmental Support (HVAC)	System Redundancy Components	• N+1 • Temporary loss of electrical power will not cause loss of cooling, but may cause temperature to elevate within operational range of critical equipment for an ANSI/TIA-942-B rated 3 facility or N+1 redundancy for mechanical equipment) • Loss of electrical power can cause loss of cooling in an ANSI/TIA-942-B rated 2 facility
		Redundancy Configuration	One alternate
		System (N) Capacity	Total facility design capacity

Implementation Guidance

UPS systems shall be sized for the ultimate design load of the telecommunications space. For initial cost, maintenance, reliability, and efficiency reasons modular UPS systems built out to provide sufficient UPS and battery capacity to meet near-term requirements are preferred over full-capacity systems.

3.3 Network Support Center, Telecommunications Room, and Entrance Room Planned Redundancy Levels

Table 7: Network Support Center, Telecommunications Room, and Entrance Room Planned Redundancy Levels

ID	Primary Attribute	Secondary Attribute	Specification
1	Facility Power Input Sources	System Redundancy Components	N (one commercial)
		Distribution Paths	One path
		System (N) Capacity	Total facility design capacity (commercial feed can provide 100 % of the facility's design load)
2	Emergency Power Generation Source	System Redundancy Components	• N+1 when supporting medical center with a CSC • N required at all sites with facility generation capabilities
		Distribution Paths	• Minimum one path when supporting medical center with a CSC • N required at all sites with facility generation capabilities
		System (N) Capacity	When provided, based on planned facility ultimate load

ID	Primary Attribute	Secondary Attribute	Specification
3	UPS Technical Power Systems	System Redundancy Components	- N (single UPS system, single battery string; may be individual rack-mount UPS systems for each rack) for NSCs and TRs - N for Entrance Rooms supporting NSCs and rating 2 MSCs 2N for Entrance Rooms supporting CSCs, CDCs, and rating 3 MSC)
		Distribution Paths	- One path - Two simultaneously active (A/B configuration) for Entrance Rooms supporting CSCs, CDCs, and rating 3 MSC
		System (N) Capacity	- UPS system(s) rated for 100 % of critical IT system design load - Modular systems sized for ultimate design load but only populated to current load needs are preferred
		UPS Technology	- Rack-mounted line interactive UPS systems (mechanical ATS optional) - Facility UPS
4	Electrical Distribution (UPS to IT equipment)	System Redundancy Components	- N 2N for Entrance Rooms supporting CSCs, CDCs, and rating 3 MSC
		Distribution Paths	- One path - Two simultaneously active (A/B configuration) for Entrance Rooms supporting CSCs, CDCs, and rating 3 MSC
		System (N) Capacity	Maximum design loading of all components with one system offline
5	Environmental Support (HVAC)	System Redundancy Components	- N - N+1 for Entrance Rooms supporting CSCs, CDCs, and rating 3 MSC
		Redundancy Configuration	- None - One alternate for Entrance Rooms supporting CSCs, CDCs, and rating 3 MSC
		System (N) Capacity	Total facility design capacity (cooling system designed to ultimate data center load)

Implementation Guidance

UPS systems shall be sized for the ultimate design load of the telecommunications space. For initial cost, maintenance, reliability, and efficiency reasons modular UPS systems built out to provide sufficient UPS and battery capacity to meet near-term requirements are preferred over full-capacity systems. Modular UPS systems may not be available to meet rack-mounted applications.

4 Infrastructure Standards

4.1 Architectural Standards and Space Design

VA telecommunications spaces, including entrance rooms, TRs, and data centers shall be designed and operated in accordance with the ANSI/TIA-942-B ratings for each data center classification, except as detailed specifications are provided in this and other VA data center facilities standards.

4.1.1 Building Specifications for Common Telecommunications Spaces

Table 8: Building Specifications

ID	Primary Attribute	Secondary Attribute	Specification
1	Shell	Security	As per Physical Security and Resiliency Design Manual (PSRDM)
		Access Route	Leads to location outside computer room for access control
		Egress	Observes life safety code
		Door Height (Computer Room)	8 ft minimum
		Door Height (Mechanical Room)	9 ft minimum
		Door Width	- CSC, CDC, and MSCs- A 6 ft double doorway opening is required with or without a mullion. If mullion is used, it must be removable. - NSC- 3 ft
		Slab to Floor Above	16 ft minimum
		Raised Floor to Ceiling	12 ft minimum
		Raised Floor in new data center construction (used only if CFD analysis determines raised floor is the best cooling solution)	24 in. minimum above slab - Height determined by CFD analysis
		Floor Surface	- Electrostatic Discharge (ESD) coating for raised floor systems - Anti-static coating or material for slab floors

ID	Primary Attribute	Secondary Attribute	Specification
		Backboard	4 ft x 8 ft AC grade ¾ in. trade size fire-rated plywood backboard painted high-gloss white with two coats of fire-resistant paint for service provider/security/video/etc. around three perimeter walls of TRs and entrance rooms and as required for computer rooms - Reserve an additional 12 in. dedicated space from backboards to accommodate equipment mounted on them
2	Raised Floor Tile	Dimensions	24 in. x 24 in. nominal
		Standard Tile	Placed by default everywhere perforated tiles and grates are not required by CFD
		Perforated Tile, High-Capacity Grates, and other High Airflow Tiles	- Only placed in the cold aisle in front of equipment inlets as calculated by CFD modeling - Positioned so that tiles are flush to the front of the row with directional airflow aimed towards the air inlets on IT equipment in the racks - Can be removed without interference from racks or cabinets

4.1.2 Data Center Layout Standards

Data centers will be designed in accordance with the specifications below, VA standards, and industry best practices listed in References.

CFD modeling shall be used to determine the most effective and cost-effective layout and containment solution for a given space (including new design/build, capability expansion, and lifecycle replacement). All data center design for new data centers, expansion, or significant modification of existing data centers will be approved through DCIE (VAITSEDatacenterEngineering2@va.gov).

Airflow direction shall be a prime consideration when designing the floor layout and when selecting and procuring IT equipment. Equipment shall be specified, procured, configured, and installed so that the equipment draws cool air from the front face of the rack and flows from the front of the rack to the rear or top of the rack and exiting either the rear door of the rack or the top through a Vertical Exhaust Duct (VED). Do not create conditions for exhaust to be recirculated back into cold air supply spaces.

Table 9: Data Center Floor Layout Standards

ID	Primary Attribute	Secondary Attribute	Specification
1	Aisles	Orientation	- Front of racks will be oriented toward cold aisle as in <i>Figure 2</i> below - Front of racks will be aligned with each other to create a continuous linear aisle - If a raised access floor is used, front of racks will be flush with edge of floor tiles
		Width	4 ft minimum in cold aisles 3 ft minimum in hot aisles (4 ft recommended)
		Cold Aisle	- Power cables if run in an access floor (with data cabling) will be placed in cable trays in cold aisles - Overhead power cables can be run either in the hot or cold aisles
		Hot Aisle	- Data cables if run in an access floor will be placed in cable trays in hot aisles - Power cables if run in an access floor (without data cabling) will be placed in the hot aisle - Overhead power cables can be run either in the hot or cold aisles - Busways will be placed in the hot aisles
		Containment	Hot or cold aisle containment will be determined based on data center layout, HVAC configuration, and CFD analysis
2	Support Equipment	Computer Room Air Conditioners (CRAC)	Placement determined using CFD analysis
		Power Distribution Units (PDU)	- Floor mounted PDUs are not to be placed within cold aisles - A-side and B-side PDU will be aligned in a regular order to indicate power source
		Branch Circuit Panelboards (BCP) also known as Remote Power Panels (RPP)	- Not to be placed within cold aisles - A-side BCP/RPPs aligned in a row - B-side BCP/RPPs aligned in a row
		Uninterruptable Power Supplies (UPS)	Placed in separate dedicated spaces or within the computer room IAW CFD
3	Equipment Rows	Length	No more than 14 racks/cabinets per row
		Orientation	No rows that terminate along a wall

ID	Primary Attribute	Secondary Attribute	Specification
		Cabinet/Rack/Cooling/PD U/RPP Placement	- End of row racks or network cabinets shall be used for the Horizontal Distribution Area (HDA) to accommodate structured cabling - Gaps between adjacent racks and cabinets within each row are closed to eliminate internal paths of bypass and recirculation of air flows - Equipment cabinets will be used between the HDAs
		Clearance	3 ft minimum distance from walls 4 ft minimum (6 ft recommended) distance from air conditioning and power distribution equipment (does not apply to in-row cooling/ PDUs/RPPs) - Per manufacturer requirements - Per AHJ for egress

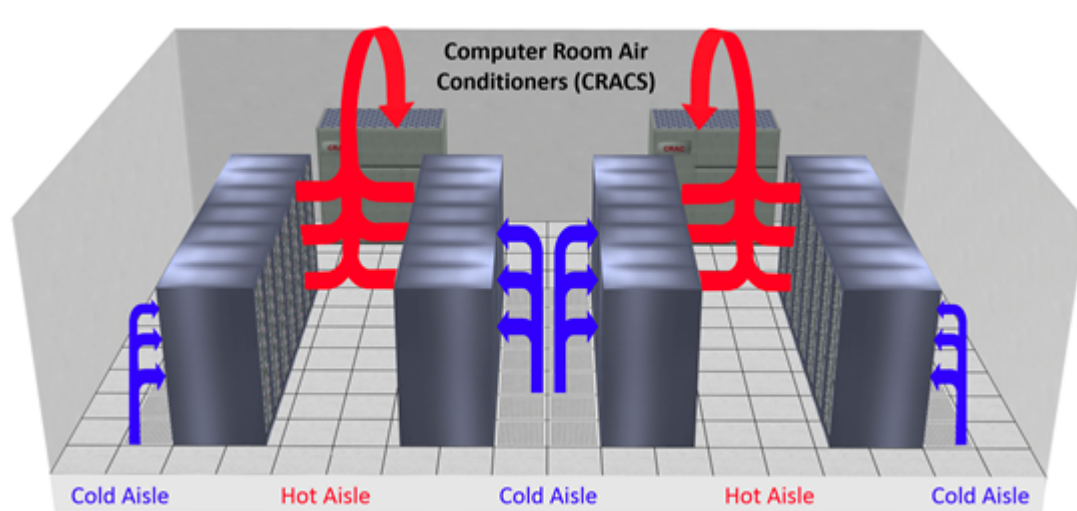


Figure 2: Hot Aisle/Cold Aisle Approach

Implementation Guidance

Determination and maintenance of categorization of enterprise data center facilities shall be conducted by National Data Center Operations & Logistics (NDCOL) in alignment with their mission to report Department data center information to the Office of Management and Budget (OMB).

This Standard applies to new data center and telecommunications space construction as well as expansion, modification, and operation of existing facilities.

VA shall not design or build “shallow rooms” or similar small spaces. Shallow rooms are intended only for access to vertical chases and riser cables installed there. Where a distribution TR is necessary in a location (i.e. the third floor of the west wing of a hospital), VA shall construct and operate out of a standardized TR.

4.1.3 Telecommunications Room

The Telecommunications Room (TR) serves as the transition point between the horizontal and backbone cabling systems. TRs will contain all Local Area Network (LAN), security, video distribution, paging, and any other system that serves the TR zone.

Few TRs are exactly alike. OIT shall design or conduct design reviews of all projects involving new or renovated TRs to assure that the power, cooling, and room layout are consistent with VA and industry standards.

TRs are a subset of the NSC classification archetype with less stringent environmental control requirements.

Evaluation Factors

- Floor area served
- Number of WAO served
- Horizontal distribution distance

Implementation Guidance

For non-medical spaces, provide a minimum of 80 ft² for a one rack TR with an additional 20 ft² per additional rack required.

For health care facility TRs, provide a minimum of 170 ft² for the standard four-rack configuration with an additional 20 ft² per additional rack required. VA shall not design, build, or use “shallow rooms” or similar small spaces.

Where existing shallow rooms are used for telecommunications cabling, they shall only be used for access to vertical chases and riser cabling pass-through installed there. No active or passive network equipment, power requirements, or environmental considerations are required.

There shall be a minimum of one standardized TR per floor. Generally additional standard telecommunications rooms should be provided when:

- The floor area to be served exceeds 10,000 ft²
- The horizontal distribution distance to the work area exceeds 295 ft

Add an additional 12 in. dedicated space from backboards to accommodate service provider, security, video, etc. equipment mounted on them.

See *Appendix B: OIT Design Guide Templates* for additional design considerations.

4.1.4 Entrance Room

Entrance rooms are the entrance to a building where both public and private network service cables and equipment (including wireless) interface with the data center cabling system. Usually this is the demarcation point where operational control or ownership changes, typically between the service provider and the customer.

Entrance rooms will have similar features as TRs including backboard, racks, power, and cable management. DCIE shall design or conduct design reviews of all projects involving new or renovated entrance rooms to assure that the power, cooling, and room layout are consistent with VA and industry standards.

Entrance rooms shall be a minimum of 80 ft² for a one-rack entrance room and an additional 20 ft² for each additional rack required.

There shall be a minimum of one entrance room per building. Two entrance rooms are required for the diverse providers for CDCs and CSCs with a minimum separation of 66 ft between entrance points.

Entrance points must be within 50 ft of the OSP entrance point duct bank to allow for transition from OSP to ISP cabling.

Health care facilities must have two entrance rooms.

Health care facility entrance rooms shall be a minimum of 170 ft². If more than four racks are required than an additional 20 ft² per additional rack must be added to the design.

Evaluation Factors

Number of telecommunications providers

Implementation Guidance

A TR or equipment room (computer room) may serve as an entrance room.

Develop diverse building entrance routes when more than one access provider serves a building so a catastrophic failure along one entrance route will not interrupt the entirety of a building's telecommunications service.

See *Appendix B: OIT Design Guide Templates* for additional design considerations.

4.2 Electrical Power and Grounding Standards

These specifications define a standardized power distribution from branch distribution panel equipment to the active IT equipment enclosure with sufficient power to energize up to 5 kW redundantly in a Standard Density (SD) configuration, or up to 10 kW redundantly in a High Density (HD) configuration. This approach allows the facility engineer to provide a standardized branch circuit (3-phase 60 A) through legacy facilities for the majority of server, LAN, and Storage Area Network (SAN) stacks. Proprietary and equipment-specific power requirements shall be avoided.

This standardized approach creates consistency throughout the data center and reduces overall power complexity. Ultimately, this approach will reduce the capital costs of installing new equipment and the operation costs by reducing or eliminating the need to change branch circuits with each IT equipment refresh. The Minimum Circuit Ampacity (MCA) in IT support spaces, for both critical and convenience distribution, is 20 A. All circuits supporting IT support spaces shall be sized (breaker, conductor, and receptacle equipment) at 20 A or larger.

Electrical distribution in the data center shall be redundant on diverse paths and follow rectilinear pathways when not contained in a cable tray.

Headends for A/B electrical paths to distribution devices (busway headends, PDUs, etc.) follow different paths for physical path redundancy.

Why did the power distribution standards change?

The previous standards for power distribution specified 30 A 3-phase distribution deployed through a 30 A redundant zPDU. This topology was only consistent with legacy facility distribution and did not address sustainable electrical distribution design using busways. In addition, the 30 A circuit can only provide 8.6 kW, which limited the 5 kW standard design's actual consumption to 4.3 kW; while loading design factors typically keep an enclosure with 5 kW (rated) load operating at not more than 70 % (3.5 kW) and within the allotted power budget. Providing the full design 5 kW of power capacity at the enclosure was deemed to be more conservative while remaining within the ability of most existing VA facilities to reject the heat generated.

4.2.1 Topology for Power Distribution Standards

Power distribution topologies in IT support spaces are dependent upon the:

- Classification archetype of the facility or IT support space
- Planned or in-use type of power distribution system, and
- Planned density of the IT equipment cabinets to be supported

Different topologies are used in data center archetype telecommunications spaces (CDCs, CSCs, and MSCs) than in distribution archetype telecommunications spaces (NSCs, entrance rooms, and TRs).

Deviations from the power distribution standards require an approved written variance from DCIE.

4.2.1.1 Topology for Power Distribution (CDC, CSC, MSC)

Topologies employed are dependent upon the type of branch power deployed to IT equipment enclosures. There are two types in use:

- In new construction and in renovations where there is sufficient vertical height, electrical busbar distribution is used
- In legacy facilities without adequate ceiling height, underfloor or cable-tray whips are used

For general purpose equipment enclosure load planning, utilize 50 % of the sum of the IT equipment power supply units on one distribution leg (either the A side or B side) to calculate the total power load for the set of equipment. In aggregate, this provides a sufficiently conservative maximum power requirement for equipment in the enclosure.

Topologies employed are dependent on the designed density of IT equipment to be supported. There are two power density classifications: Standard and High as depicted in *Figure 3* below.

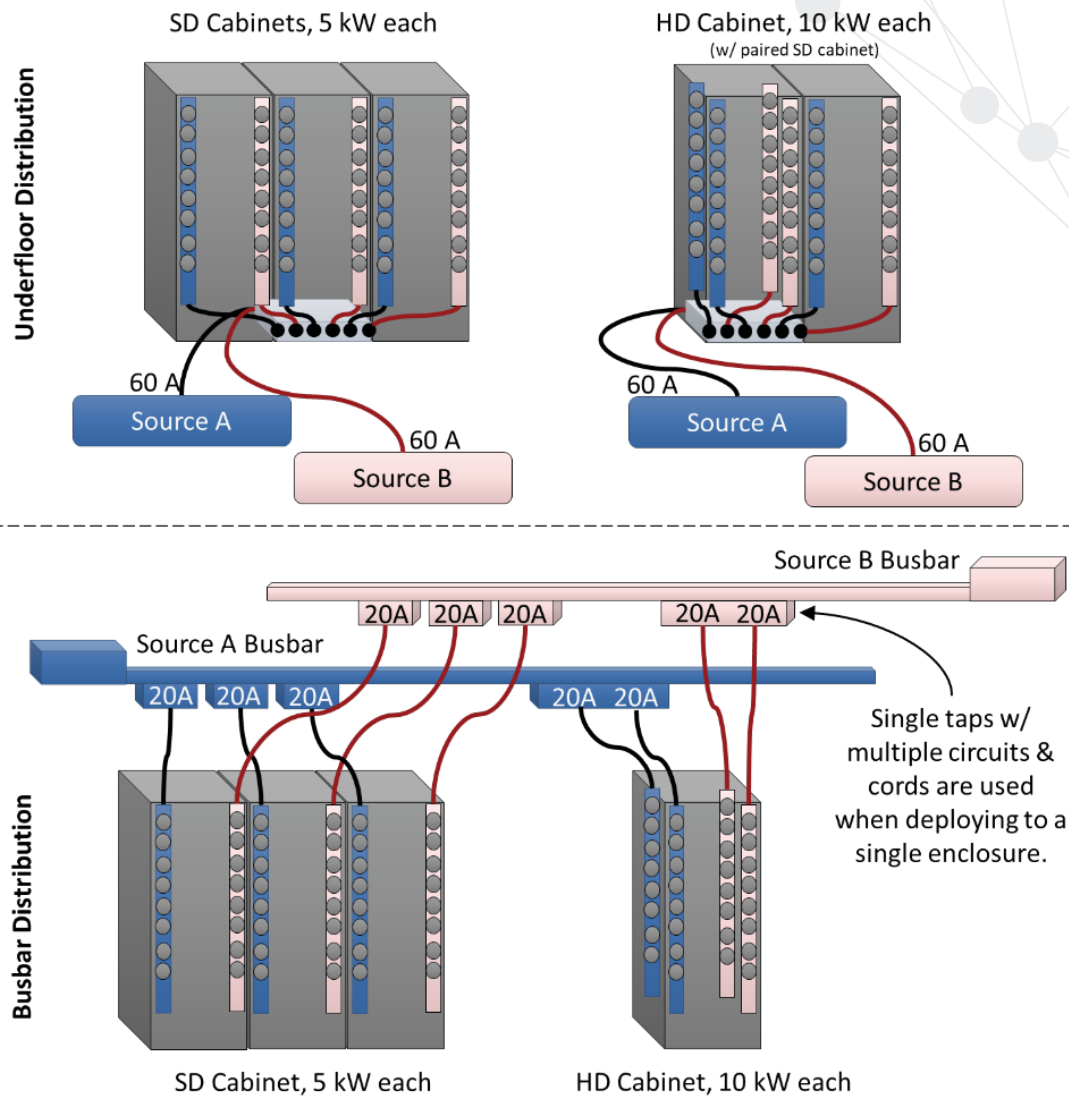


Figure 3: Power Distribution Options Simplified Visualization (CDC, CSC, MSC)

For general IT support space purposes IT equipment cabinets will be Standard Density (SD), utilizing legacy underfloor/whip distribution. Insertion of HD cabinets requires specific design work by DCIE and will not be accomplished ad-hoc without approved, validated designs including CFD models showing acceptable thermal conditions of the overall design at ultimate operating load, in contingency conditions.

SD per-cabinet power is limited to 5 kW. Topology detail for this type of distribution is shown in *Figure 4* and *Figure 5* below. Design and install equipment loads not to exceed this power and heat density without design approval and validation from DCIE.

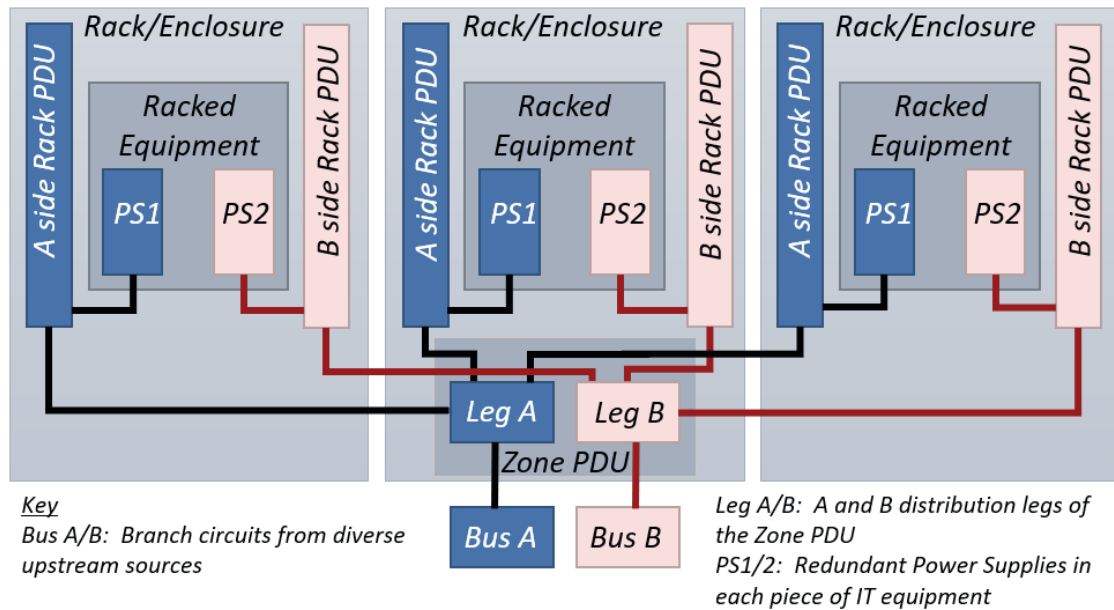


Figure 4: Standard Density Underfloor/Whip Distribution Topology (CDC, CSC, MSC) – 60 A Zone PDU

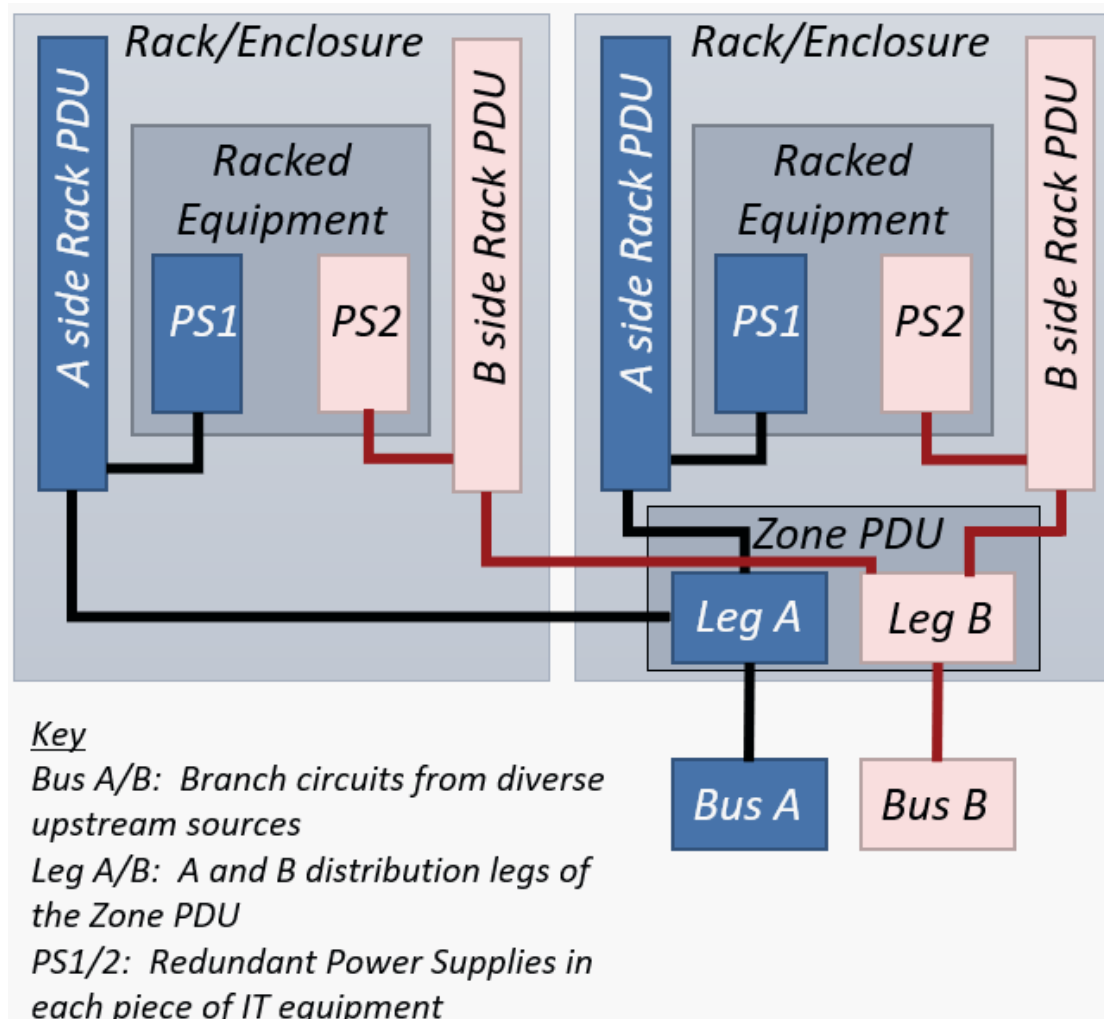


Figure 5: Standard Density Underfloor/Whip Distribution Topology (CDC, CSC, MSC) – 30 A Zone PDU

Only DCIE is authorized to approve installations exceeding the SD level. HD cabinets may be possible in some facilities, if heat rejection systems are properly designed and operated. HD power density is limited to 10 kW. Topology detail for this type of distribution is shown in Figure 6 and Figure 7 below:

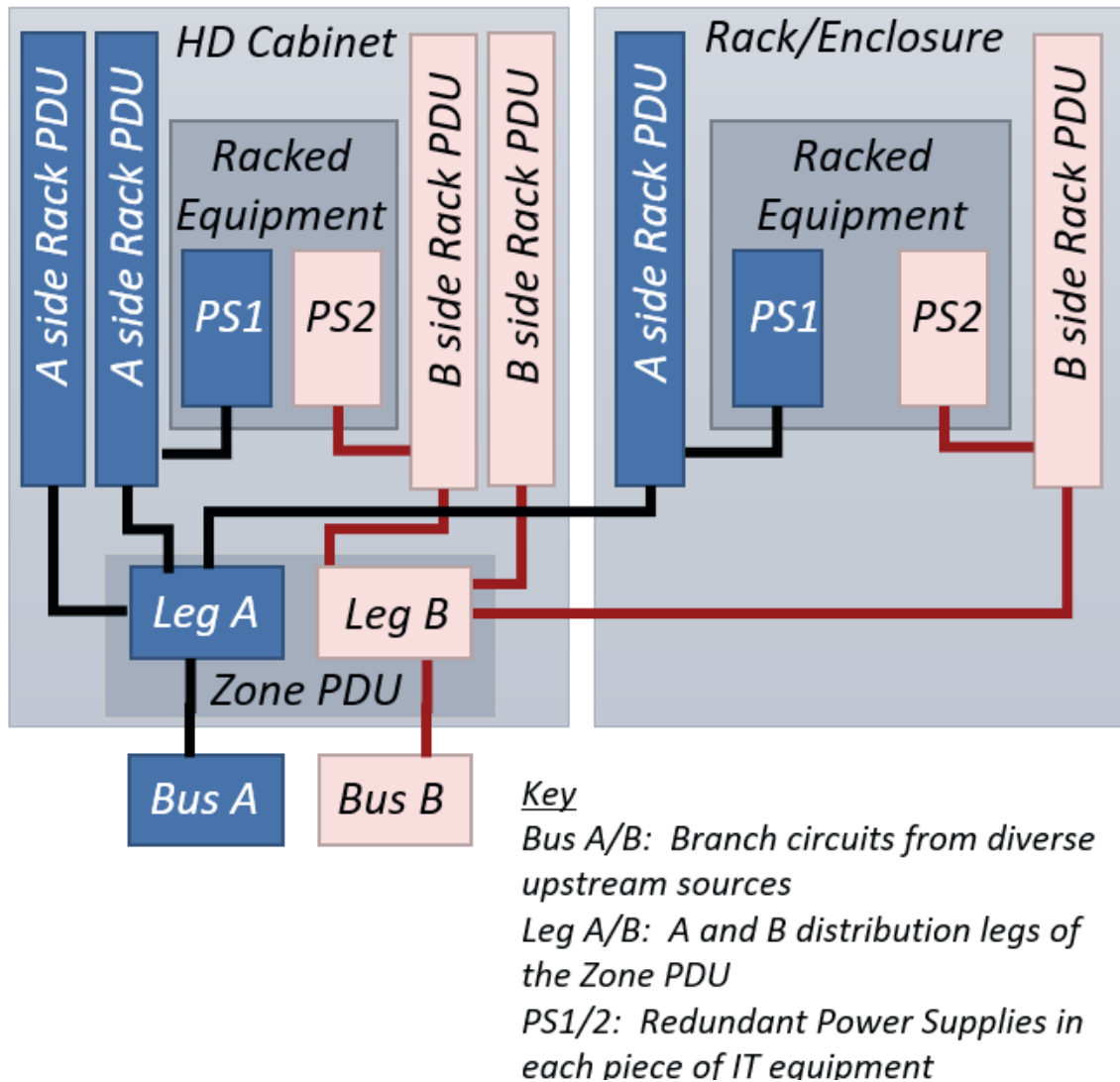


Figure 6: High Density Underfloor/Whip Distribution Topology (CDC, CSC, MSC) – 60 A Zone PDU

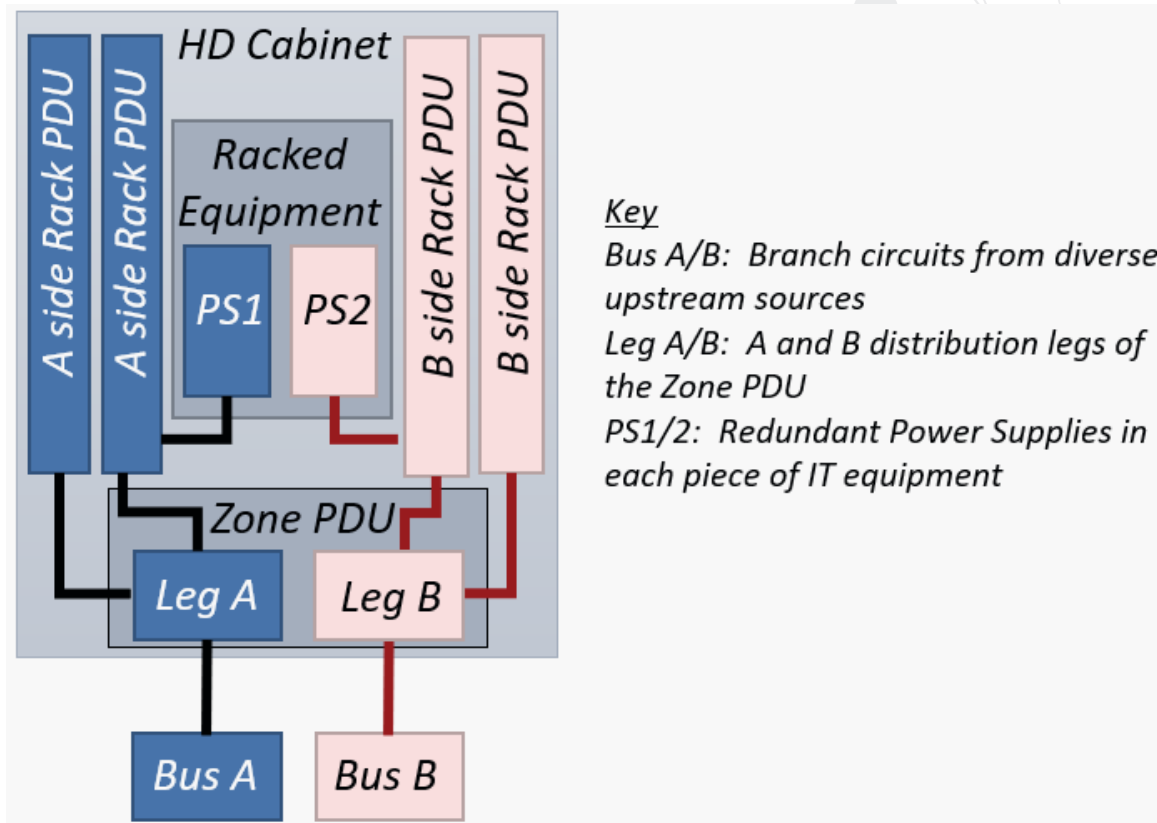


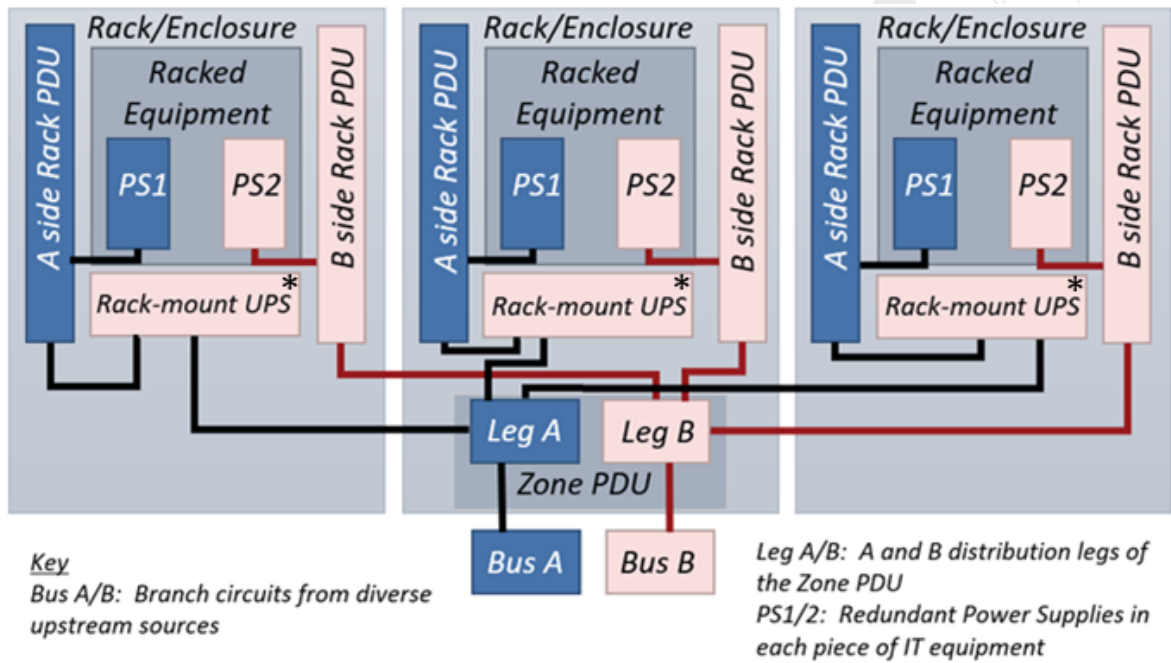
Figure 7: High Density Underfloor/Whip Distribution Topology (CDC, CSC, MSC) – 30 A Zone PDU

4.2.1.2 Topology for Power Distribution (NSC, TR, Entrance Room)

For general purpose equipment enclosure load planning in distribution spaces, utilize 75 % of the sum of the IT equipment power supply units on one distribution leg (either the A or B side) to calculate the total power load for the set of equipment. In aggregate, this provides a sufficiently conservative maximum power requirement for Power-over-Ethernet (PoE) distribution switches in the enclosure.

For distribution telecommunications space purposes IT equipment enclosures will be SD, utilizing legacy whip distribution. HD enclosures are not utilized in these classifications of telecommunications spaces. Power density is limited to 5 kW per enclosure.

Topology detail for this type of distribution is shown in *Figure 8*, *Figure 9*, and *Figure 10* below. Only one distribution leg is required to be supported by UPS in these classification archetypes. If only a single distribution bus is powered by generator (emergency power), the UPS shall be on the same distribution leg to ensure battery recharge during generator operation.



*NOTE: Health care entrance rooms require 2N Rack-mount UPS configuration per rack

Figure 8: Whip Distribution Topology with In-Rack UPS (NSC, TR, Entrance Room) – 60 A Zone PDU

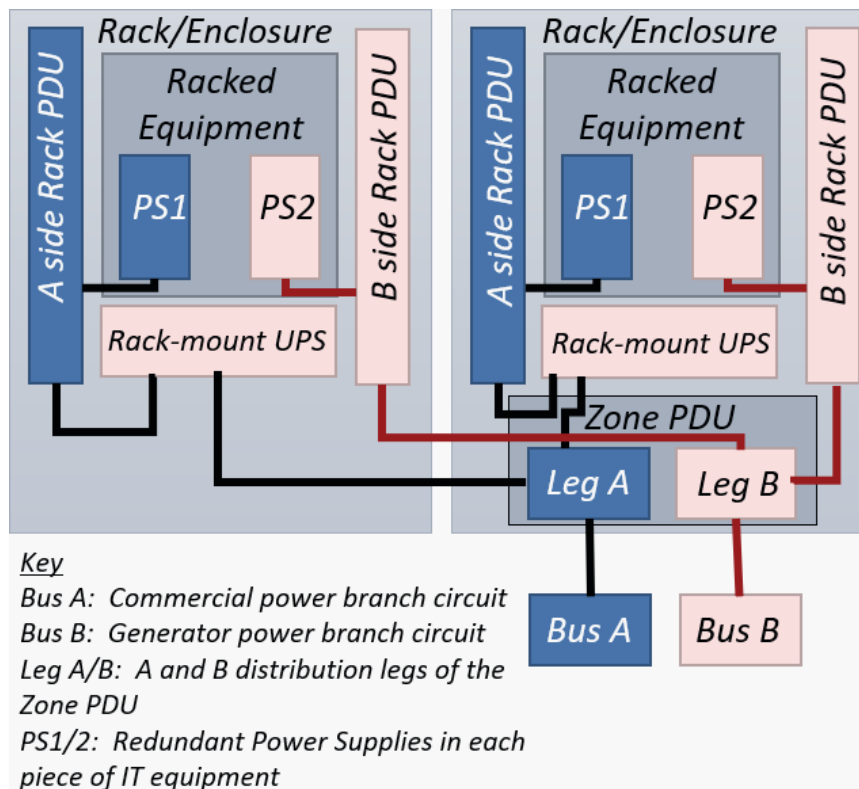


Figure 9: Whip Distribution Topology with In-Rack UPS (NSC, TR, Entrance Room) – 30 A Zone PDU

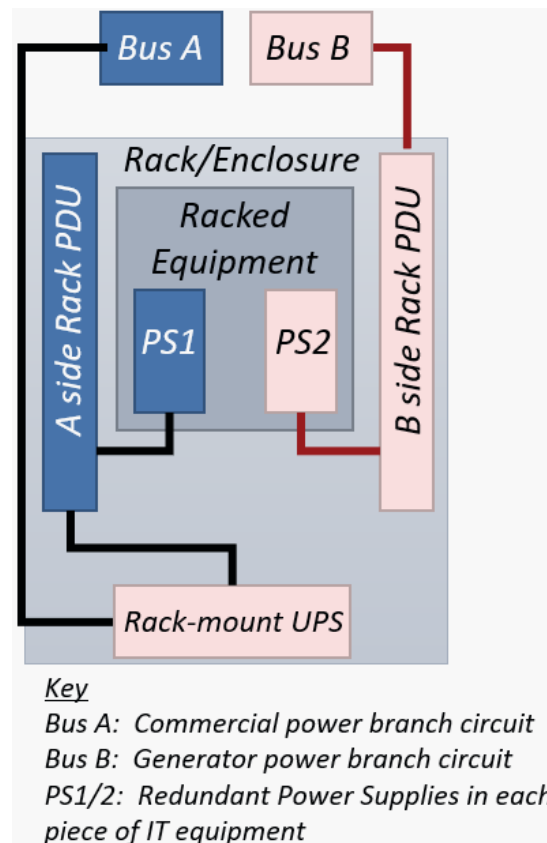


Figure 10: Single Rack Whip Distribution Topology with In-Rack UPS (NSC, TR, Entrance Room) – 20 A Circuits

4.2.2 Uninterruptible Power Supply Specifications

These specifications define the high-level technical requirements for Uninterruptible Power Supply (UPS) systems that support VA enterprise data centers and related IT spaces. This standard shall be used to quantify and qualify designs of UPS systems and the necessary physical infrastructure supporting data center facilities and spaces when those spaces are being designed, built, retrofitted, refreshed, and updated.

Facility UPS systems shall have a planned design load which covers 100 % of the IT equipment electrical loads in the data center sized to provide 10 mins of backup capacity. Because surge and startup loads are not indicated by monitoring steady-state operational UPS usage conditions, actual loading on the UPS system is limited to 85 % of design capacity with one system offline; that is, on a 1000 kVA system, when 850 kVA of equipment is shown on the UPS meter, the system is considered 'full' and no more IT equipment shall be added to the facility.

4.2.2.1 Cascaded UPS Systems

The National Electrical Code® (NEC®) requires the ability to disconnect all electronic equipment and HVAC systems in case of fire or other emergency. In general, these emergency disconnect systems are present in most VA enterprise data centers. If cascaded UPSs are installed, the Emergency Power Off (EPO) will not de-energize the entire data center as the downstream UPS are not turned off by the EPO. This creates a danger to fire and rescue workers in an emergency.

Use of any UPS or additional battery-backed system downstream to the primary (critical bus) UPS is not a Underwriters Laboratories (UL) listed application and is prohibited by VA standards. This may jeopardize the fault clearing capability of engineered breaker coordination and introduces an additional point of failure and unnecessary load to the primary UPS thus degrading overall data center power reliability.

A downstream (cascaded) UPS is assumed to recognize the upstream (main) UPS sine wave. If the downstream UPS does not recognize the upstream UPS sine wave, the downstream UPS (when it switches on) may never synchronize back to the upstream UPS output waveform. If that should occur, the downstream UPS would stay on battery until the battery failed, then shut off.

4.2.2.2 Battery Monitoring

When a power outage occurs, the first system to engage and provide back-up power is the UPS, which draws from batteries until a generator can take the full load. Maintenance of these batteries and knowing how long they can be expected to continue to provide effective service is a critical link in the redundant electrical supply system. At sites equipped with static UPS systems, standardized Battery Monitoring Systems (BMS) shall be installed and maintained.

Replacement of a 'string' of static UPS batteries costs up to \$100k. Each string is designed to last for a 10 to 20-year lifecycle of service if properly maintained and used in an appropriately controlled environment. Battery monitoring systems ensure that VA can maximize the lifecycle of battery systems and preserve scarce resources. These systems shall meet the following minimum requirements:

- System shall meet all current Institute of Electrical and Electronics Engineers (IEEE) recommendations for battery monitoring. System shall not be powered by the UPS batteries being monitored.
- Minimum battery parameters measured shall include cell and inter-cell impedance; system voltage; float, charge and discharge currents; and cell/ambient temperatures. Impedance measurement specification shall be to 0.1 ± 0.01 milli-ohms.
- System shall use a pulsed AC impedance measuring method that does not discharge the batteries during the measurement cycle. AC ripple impedance testing methods are not acceptable. Discharge testing below cell open circuit voltages is not an acceptable method.
- System shall allow individual cell replacement without disconnection and removal of the wiring harness to reduce replacement and maintenance time per cell.
- System shall use measured baseline and follow-on information for individual cells for alarm and trend analysis. Jar-level and multi-cell testing are not acceptable. Discharge data logging as well as charge and float logging shall be enabled.
- System shall be programmable to set custom test and out-of-limit alarms. System shall be integrable with third-party Building Automation Systems (BAS).
- System shall have trending, graphing, and reporting software. The software shall analyze voltage, impedance and temperature measurements, allowing the predictability of battery performance. Software shall acquire and archive measurements and trending

information for off-line presentation in report and graphical format. Software shall archive and display power outage events, their duration, decaying voltage data (if the power outage was long enough), and current out on discharge.

- System software shall be capable of being controlled remotely using Building Automation and Control Networking Protocol (BACnet), Modbus, Controller Area Network Bus (CANbus), or Fieldbus via ethernet or modem to view and modify settings and limits, acquire data and alarms, reset alarms, and perform on-line diagnosis. Software shall be capable of alarming via BAS, SNMP, email, and phone notification.
- Software shall automatically gather measurement units on a user-determined schedule for trending database construction.

Preventative maintenance services on battery monitoring systems shall be outsourced to a qualified service provider and shall include all consumable parts, remedial repairs, labor, and expenses. The service provider shall provide not less than annual maintenance services.

Table 10: Data Center Uninterruptible Power Supply Specifications

ID	Primary Attribute	Secondary Attribute	Specification
1	Technology	Inverter Preferred	UPS systems at CDCs, CSCs, and MSCs shall be inverter preferred systems, such that the power distribution path is through the inverter whether on commercial or stored power
		Double Conversion	Double conversion systems are acceptable for use as inverter preferred technologies in designs above 10 kVA capacity
		Delta Conversion	Delta conversion systems are acceptable for use as inverter preferred technologies in designs above 10 kVA capacity
		Line Interactive	Line interactive UPS systems (typically rack-mounted systems up to 5 kVA capacity) are specified for use in NSCs and other small IT spaces
		Standby/Offline	Standby/offline UPS technologies (commonly used for desktop computer applications) are prohibited in VA data center applications
2	Design Configuration and Capacity	Serial UPS Systems	• Serial (or cascaded) UPS systems are prohibited in VA environments • Under no circumstance shall a UPS system be on a circuit supported by a separate UPS system • For the purposes of this specification, any infrastructure system that uses stored battery energy as its backup is considered a UPS system (this includes server and storage IT equipment configured with external battery backup systems)

ID	Primary Attribute	Secondary Attribute	Specification
		Planned Design Load	Each “N” need on a facility UPS system shall have planned design load covering 100 % of the IT equipment loading planned for the facility
		Design Load Limits	Limited to 85 % of design capacity (because surge and startup loads are not indicated by steady-state usage conditions, actual loading is)
		Appropriate Loads	- Only IT data processing equipment, peripherals, and communications equipment are authorized to be powered by the data center UPS - Do not place administrative loads or mechanical equipment loads on UPS power
3	Energy Storage System	Battery Types	Valve Regulated Lead Acid (VRLA) battery systems for system designs less than 400 kVA
		Battery Sizing	Sized to maintain the UPS system full load rating for 10 min
		Maintenance Determination	VRLA batteries shall only be installed in locations where preventative maintenance is readily accessible and effective
		Kinetic Energy Systems	- Rotary (flywheel) UPS systems require detailed, consistent maintenance that must be taken into consideration if they are to be considered for any particular installation - Do not install kinetic energy systems unless these maintenance requirements can be met without risk or constraints
4	Environmental and Monitoring System	Battery Monitoring System (BMS)	- Dedicated BMS equipment shall be installed on all battery strings on UPS systems with design capacities of 40 kVA and greater - BMS systems shall not be powered by the UPS batteries being monitored
		Monitoring System Technology	- BMS shall use a pulsed AC impedance measuring method that does not discharge the batteries during the measurement cycle - AC ripple impedance testing methods and discharge testing below cell open circuit voltages are not acceptable methods
		Battery Environment	Optimal battery life when maintained at 77 °F
		Spill Control	Required for installations using Vented Lead Acid (VLA) batteries

ID	Primary Attribute	Secondary Attribute	Specification
		Ventilation	Room monitoring and automatic ventilation is required for installations using VRLA and VLA batteries

EVALUATION FACTORS

- UPS Technology
- Energy Storage Technology
- Battery System Environment
- BMS

IMPLEMENTATION GUIDANCE

UPS systems supporting VA data center facilities shall be designed and operated in accordance with the guidance provided in this standard.

Cascaded UPS Systems are expressly forbidden.

4.2.3 Transformers

Transformers supporting power distribution for critical loads in Rating 3 facilities (CDCs, CSCs, and some MSCs) shall be minimum K-13 rated for harmonics reduction. This includes transformers located downstream of UPS equipment supporting all critical load distribution panels, including in PDUs. No harmonics reduction is required for transformers supporting Rating 2 MSCs, NSCs, and other IT support spaces.

4.2.4 Branch Power Circuit to Rack/Cabinet

This section describes the power circuit characteristics from branch distribution panel equipment to the active IT equipment enclosure distribution equipment.

Table 11: Branch Power Circuit to Rack/Cabinet Standards (Underfloor/Whip Distribution)

ID	Primary Attribute	Secondary Attribute	Specification
1	Branch Circuit (underfloor/whip distribution)	Voltage	120/208 V
		Phase	3-Phase WYE
		Amp	30 A or 60 A
		Neutral Conductor	One gauge larger than the phase conductors
		Receptacle	• L21-30R for 30 A zone PDUs (zPDU) • 5-wire cable hardwired junction enclosures (no receptacle) for 60 A zPDU • L21-20R may be used in legacy single-rack TRs to directly feed rack Power Distribution Units (rPDU)
		Whip Color	Differentiated by source bus (jacket or other marking)
		Whip Pathway (underfloor)	Rectilinear with respect to the equipment aisles with no diagonal runs
		Whip Length	Following rectilinear path and no longer than 4 ft of slack

Table 12: Branch Power Circuit to Rack/Cabinet Standards (Busbar Distribution)

ID	Primary Attribute	Secondary Attribute	Specification
1	Branch Circuit (busbar distribution)	Busbar Amperage	Size selected based on full load anticipated for all enclosures
		Busbar Orientation	• Diverse supply pathways to busbar headends • Headends for source A at one end of the row and headend for source B at opposite end
		Tap Circuit Amperage	• 20 A • Multiple circuits on same tap enclosure acceptable
		Tap Whip	• Strain relief at exit from tap enclosure • L21-20 connector
		Tap Whip Neutral Conductor	One gauge larger
		Tap Whip Color	Differentiated by source bus (jacket or other marking)

Implementation Guidance

A test plan is required to ensure that each bus is properly represented in a rack or cabinet upstream through each entire distribution path.

4.2.5 Zone Power Distribution Units

This section describes the characteristics of the zone Power Distribution Unit (zPDU) used in underfloor/whip distribution topologies. The zPDU allows the implementation engineer to specify a standard branch circuit to support a pair of SD, 5 kW redundant and single HD, 10 kW redundant enclosure power configurations.

Table 13: Zone Power Distribution Unit Standards

ID	Primary Attribute	Secondary Attribute	Specification
1	60 A Zone zPDU	Mounting	Standard 19 in. rack mount
		Agency Approval	• UL Listed • UL Listed 60950 • UL Recognized Component Mark
		Form Factor	Two Rack Units (RU)
		AC Voltage Input	120/208V WYE (3-Phase)
		Current Input	60 A
		Current Output	20 A
		Full Load	• 43,200 W (21,600 W redundant) • 60 A (3-phase)
		Hardwired Junction Enclosures (Input)	Two 60 A 5-wire DIN copper terminal blocks, accommodating 4 American Wire Gauge (AWG) input conductors

ID	Primary Attribute	Secondary Attribute	Specification
		National Electrical Manufacturers Association (NEMA) Outlets for Vertical rPDUs (Output)	Six L21-20R
		Cord Length	10 ft
		Breaker Groups	• 60 Amp A Side and B Side (input) • 20 A secondary (branch) A Side and B Side (output)
2	30 A Zone zPDU	Mounting	Standard 19 in. rack mount
		Agency Approval	• UL Listed • UL Listed 60950 • UL Recognized Component Mark
		Form Factor	2 RU
		AC Voltage Input	120/208 V WYE (3-Phase)
		Current Input	30 A
		Current Output	20 A
		Full Load	21,600 W (10,800 W redundant)
		Input	Two L21-30P
		Output NEMA Outlets for Vertical rPDUs	Four L21-20R
		Output NEMA Convenience Outlets	Six 5-20R
		Cord Length	10 ft
		Breaker Groups	• 30 A both A Side and B Side (input) • 20 A secondary (branch) A Side and B Side (output)

Evaluation Factors

- Power efficiency to reduce electricity losses for lowest overall lifecycle cost
- Number and type of receptacles provided

Implementation Guidance

One 60 A zPDU for each set of three SD racks or cabinets. Alternatively, one 30 A zPDU for each set of two SD racks or cabinets.

One 60 A zPDU for each set of one HD cabinet and one SD rack or cabinet. Alternatively, one 30 A zPDU for each HD cabinet.

Zone PDUs shall be mounted starting in RU1 at the bottom of the rack/cabinet due to excess weight to ensure stability, or immediately above the rack mounted UPS (if provided).

The correct specification for the zPDU is to feed it with two power sources. Power inputs should originate from two independent power sources. Each input will use identical specs: WYE (5-wire) configured, 208 V, 60 A, three-phase, terminating in vendor-provided junction boxes with terminal blocks for hardwiring conductors. The neutral conductor should be

upsized one gauge to match the upsized neutral conductors in the PDU units. The neutral "upsizing" should ideally be continued in the power distribution system back to the UPS or transformer winding pole to increase the efficiency of the power distribution system and suppress harmonics in the system.

4.2.6 Rack Mounted Uninterruptible Power Supplies

This section describes the UPSs that are mounted in racks located in NSCs, TRs, and entrance rooms where no facility UPS is available. The rack mounted UPS will provide 5 kW to 6 kW to the vertical PDU on the A-side distribution leg, supporting the SD, 5 kW design. Install the rack mounted UPS between the zPDU and the rPDU on the A-side distribution, or between the 20 A branch circuit and the rPDU (in legacy single-rack TRs).

Rack mounted UPS systems are not utilized in the CDC, CSC, or MSC classification archetypes.

Table 14: Rack Mounted Uninterruptible Power Supply Standards

ID	Primary Attribute	Secondary Attribute	Specification
1	UPS (in rack/ enclosure)	Mounting	19 in. rack mount
		AC Voltage Input	208 V 3-phase
		Current Input	20 A
		Current Output	20 A
		Output Receptacle	L21-20R
		Input Plug	L21-20P
		Phase Type	Three-phase
		Battery Capacity	10 min at full load
		kW Rating	5 kW to 6 kW

Evaluation Factors

- Battery Capacity
- kW Rating
- Not cascaded (no upstream UPS feeding another UPS)

Implementation Guidance

Rack mounted UPS shall be mounted starting in RU1 at the bottom of the rack due to excess weight to ensure stability.

Health care center entrance rooms have two UPS systems, one on the A side and one on the B side.

4.2.7 Vertical Rack Power Distribution Units

This section describes the vertical Rack PDUs (rPDU) (power strips) that will energize IT equipment in racks and cabinets.

Each SD rack or cabinet will receive two vertical rPDUs, one each for the A-side and B-side distribution legs. Each HD rack or cabinet will receive four vertical rPDUs, two each for the A-side and B-side distribution legs.

Mount A-side rPDU on the left side of the enclosure, and B-side rPDU on the right side of the enclosure, as viewed from the back of the enclosure.

Table 15: Vertical Rack Power Distribution Unit Standards

ID	Primary Attribute	Secondary Attribute	Specification
1	rPDU (in rack/ enclosure)	Mounting	Zero RU, toolless
		Receptacle Type(s)	• C-19 • C-13
		Receptacle Quantity	• C-19 (6 Minimum) • C-13 (30 Minimum) • NEMA 5-20 (4 Maximum) – TRs and Entrance Rooms only
		Power Cord Plug	L21-20P
		kW Rating	5kW to 6 kW
		Input Voltage Rating	208
		Phase Type	Three-phase
		Output Voltage	110/208 V
		Monitoring	Power utilization per phase. Light Emitting Diode (LED) display and SNMP v3 with Secure Sockets Layer (SSL) Encryption
2	IT Equipment Power Cords	Color Code	• A Side- black • B Side- a distinctly different color (gray preferred) • Differentiated by source bus (jacket or other marking)
		Securement	Locking configuration to match rPDU
		International Electrotechnical Commission (IEC) Standard	• C13/14 for 10 A • C19/20 for 16 A

Evaluation Factors

- Type of receptacles provided
- Voltage delivered (208 V)

Implementation Guidance

All dual-corded IT equipment should connect to both A and B bus rPDU for power redundancy.

Order IEC C13/14 (10 A) and IEC C19/20 (16 A) cord sets with all equipment requiring 16 A or less. Color coding of power cords is required to distinguish A and B sides for installation and troubleshooting. Match power cord color scheme with upstream distribution color scheme where feasible. Use of colors reserved for other purposes (e.g. emergency bus distribution (red), grounding (green)) for alternate color is discouraged as these already have distinct meanings in electrical distribution.

Rack PDU models with a limited quantity of additional NEMA 5-20 outlets may be required in TRs and entrance rooms and are authorized in lieu of the standard computer room model. They must still meet the salient characteristics of the standard rPDU, particularly the input type (L21-20P) and outlet type and quantity (C13/C19).

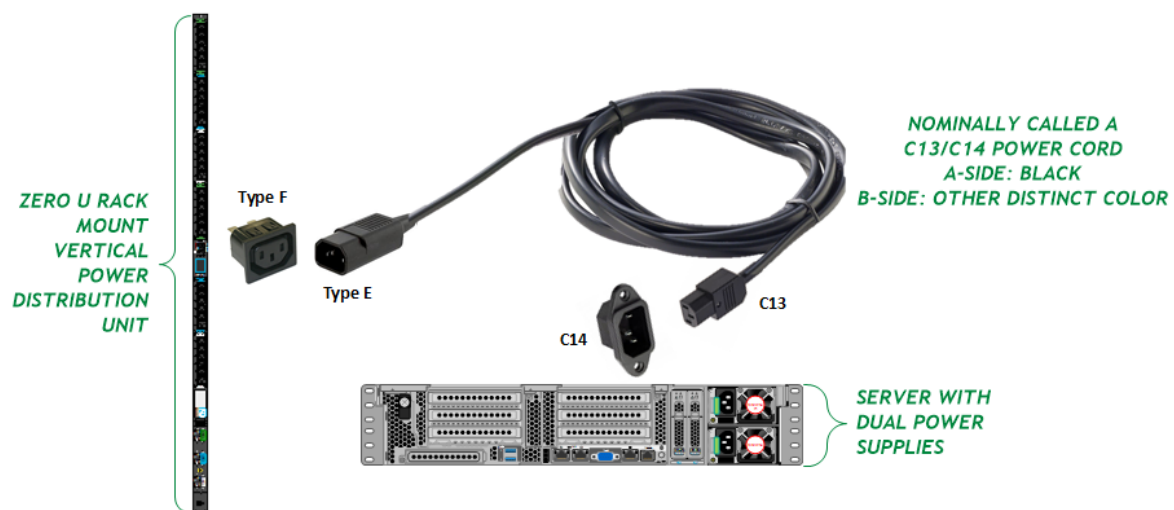


Figure 11: International Electrotechnical Commission (IEC) Standard C13/14 cord set

A test plan is required to ensure that each bus is properly represented in a rack or cabinet upstream through each entire distribution path.

4.2.8 Bonding and Grounding

4.2.8.1 Bonding Practices in Telecommunications Spaces

All telecommunications spaces are to be bonded according to ANSI/TIA-607-D *Generic Telecommunications Bonding and Grounding (Earthing) for Customer Premises*, local codes, the [VA Electrical Design Manual](#), National Electric Code, and National Electrical Safety Code. Typical VA telecommunications spaces utilize Alternating Current (AC) only. Variations to the guidance below may be necessary to accommodate Direct Current (DC) installations.

Grounding through the equipment power cord(s) does not meet the intent of ANSI/TIA-607-D. The guidance below is intended to ensure that the information technology equipment be provided a dedicated and specific ground path over and above the required AC power cord ground path. While the equipment typically has a power cord that contains a

grounding/bonding wire, the integrity of this path to ground cannot be easily verified. Rather than relying on the power cord(s) grounding/bonding wire, equipment should be grounded in a verifiable dedicated manner as described in this overview.

Many types of equipment do not require individual bonding conductors and as such do not have an attachment point for bonding conductors. Equipment that does not have attachment points for bonding conductors may be bonded either through the equipment rail or the power cord. Refer to the manufacturer's documentation for guidelines.

LAN elements, such as switches and routers, have bonding to chassis requirements. These requirements should not be ignored.

Rack busbars are specified below to facilitate bonding for LAN equipment. Metallic pathways under 3 ft in length (e.g., wall and floor sleeves, J-hooks) are not required to be bonded.

All wall mounted and floor mounted racks, cabinets, and pathways exceeding 3 ft in length must be bonded to the busbar within the same telecommunications space.

Other equipment, such as ventilation systems and desktop equipment outside of the equipment room, are out-of-scope for this document but must be addressed per the manufacturer's requirements.

All exterior-mounted PoE or LAN-attached equipment, such as IP cameras, must be bonded to the telecommunications bonding infrastructure described below.

4.2.8.2 Topology Overview

A Primary Bonding Busbar (PBB) (specifications below), will be located in a telecommunications entrance room and will be bonded to the Alternating Current Equipment Ground (ACEG) bus (when equipped) or the panelboard enclosure located in the electrical entrance facility via the telecommunications bonding conductor.

A Secondary Bonding Busbar (SBB) will be installed in each telecommunications space and will connect to the PBB via a Telecommunication Bonding Backbone (TBB). The SBB should be bonded to structural steel and the electrical panelboard ACEG bus (when equipped) located within the telecommunications space. When a panelboard for telecommunications equipment is not in the same room or space as the SBB, that SBB should be bonded to the panelboard that feeds the distributor. When SBBs are located at the top of distinct vertical risers, they will be bonded together via a bonding backbone conductor.

Typically, a single PBB will be specified per building and one SBB will be specified for each telecommunications space.

Multiple SBBs are permitted to maintain optimal conductor routing.

All bonding conductors, including the TBB, will follow sizing requirements specified in ANSI/TIA-607-D.

Implementation Guidance

Bonding conductors for telecommunications should not be placed in ferrous metallic conduit. If it is necessary to place bonding conductors in ferrous metallic conduit the

conductors should be bonded to each end of the conduit using a grounding bushing or using a minimum sized conductor of 6 AWG at both ends of the conduit.

Busbars will be mounted with 2 in. clearance between the rear of the busbar to the wall. The PBB and SBB will be mounted on listed insulators. Busbar location should be determined by the location of bonding conductors to maintain the shortest and straightest path. Typically, the SBB will be mounted directly in-line with the TBB.

Computer rooms will contain an SBB and, where an access floor is installed, a Supplementary Bonding Grid (SBG) mesh-bonding network for bonding of cabinets, racks, frames, and pedestals. The connection of the TBB to the SBB will utilize exothermic welding, listed compression two-hole lugs, or listed exothermic two-hole lugs. The connection of conductors for bonding telecommunications equipment and telecommunications pathways to the SBB shall utilize exothermic welding, listed compression two-hole lugs, or listed exothermic two-hole lugs.

A Telecommunications Equipment Bonding Conductor (TEBC) will connect the racks and cabinets to the PBB/SBB. The TEBC will be a minimum of AWG 6. Metallic objects will not serve as replacements for the TEBC; a continuous TEBC is required. The TEBC will be secured at no greater than 3 ft intervals. The TEBC should be minimum of 2 in. from other cable groups. Bonding to the TEBC will utilize permanent compression-type fittings to create a permanent connection. Where no access flooring is used, the TEBC will be routed through cable tray. Cable trays will be installed with cable tray bonding conductors.

The TEBC will be connected to the cabinets/racks, to a Rack Bonding Conductor (RBC) or to a vertical/horizontal Rack Bonding Busbar (RBB). An RBC is a bonding conductor from the rack or RBB to the TEBC. Each cabinet or equipment rack will have a suitable connection point to which the bonding conductor can be terminated. Properly sized listed two-hole compression lugs or listed terminal blocks with two internal hex screw or equivalent torque characteristics shall be used at this connection point.

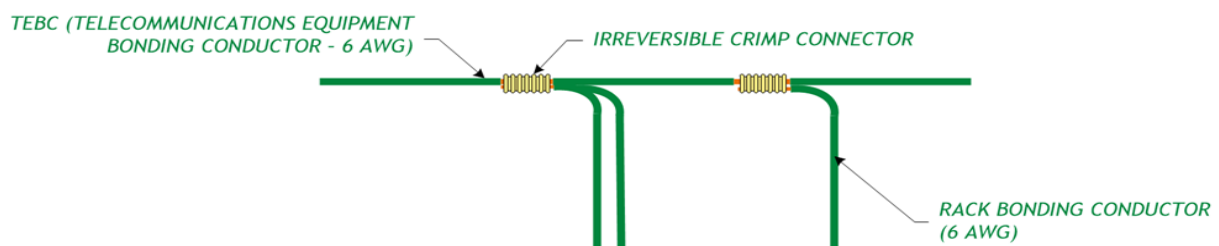


Figure 12: Typical Bonding Connection

Bends of bonding conductors terminating at the PBB or SBB will have a minimum inside bend radius of 8 in. or 90°. At other locations, bends in bonding conductors should be made with the greatest practical inside bend radius. A minimum bend radius of ten times the bonding conductor diameter is recommended. In all cases, a minimum included angle of 90° shall be used.

To limit the potential difference between telecommunications pathways or between telecommunications pathways and power pathways, all metallic telecommunications

pathways will be bonded to the RBB, PBB, or SBB. Cable tray sections will be bonded together via cable tray bonding conductors and will be bonded to the PBB or SBB.

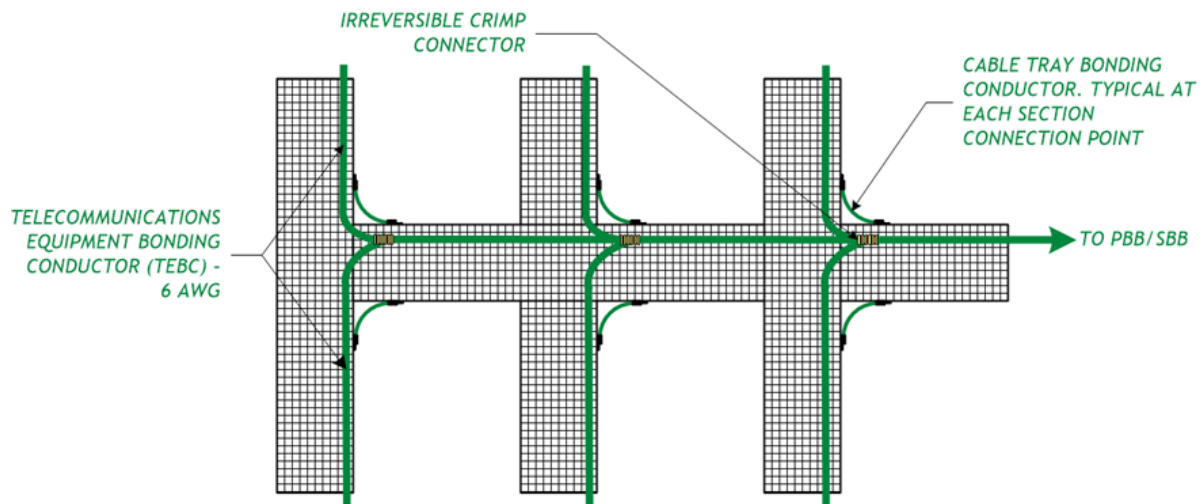


Figure 13: Typical Rack Bonding Layout

An RBB will be installed in the uppermost Rack Unit (RU) in telecommunications racks where LAN elements will be mounted. This document does not specify RBBs to be installed in server cabinets. The connection of the RBB or the TEBC to the rack will utilize exothermic welding, listed compression two-hole lugs, or listed exothermic two-hole lugs. The unit bonding conductor should be connected to the RBB using a listed compression connector and to the grounding post of the telecommunications equipment if provided. Example B in the graphic below is the preferred style and mounting approach of the RBB. Example A will be used in server cabinets.

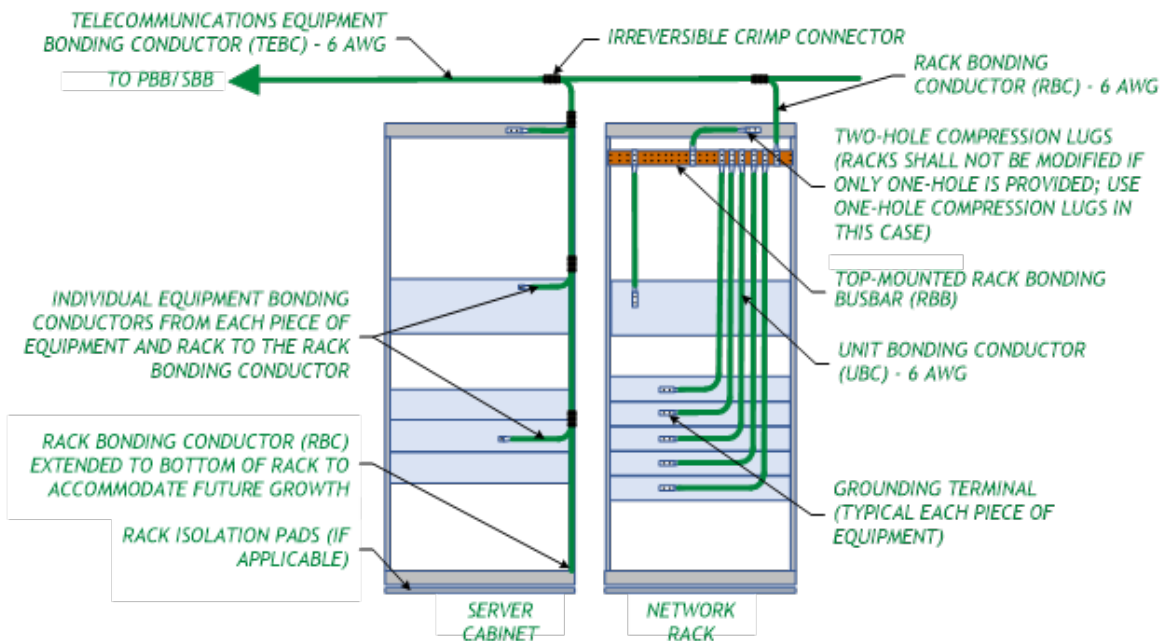


Figure 14: Typical Bonding Requirements for IT Equipment Enclosures

This equipment must be grounded. Never defeat the ground conductor or operate the equipment in the absence of a suitably installed ground conductor. These devices will be bonded to the RBB in the same rack. Each device will have a dedicated Unit Bonding Conductor (UBC). The UBC will not be shared.

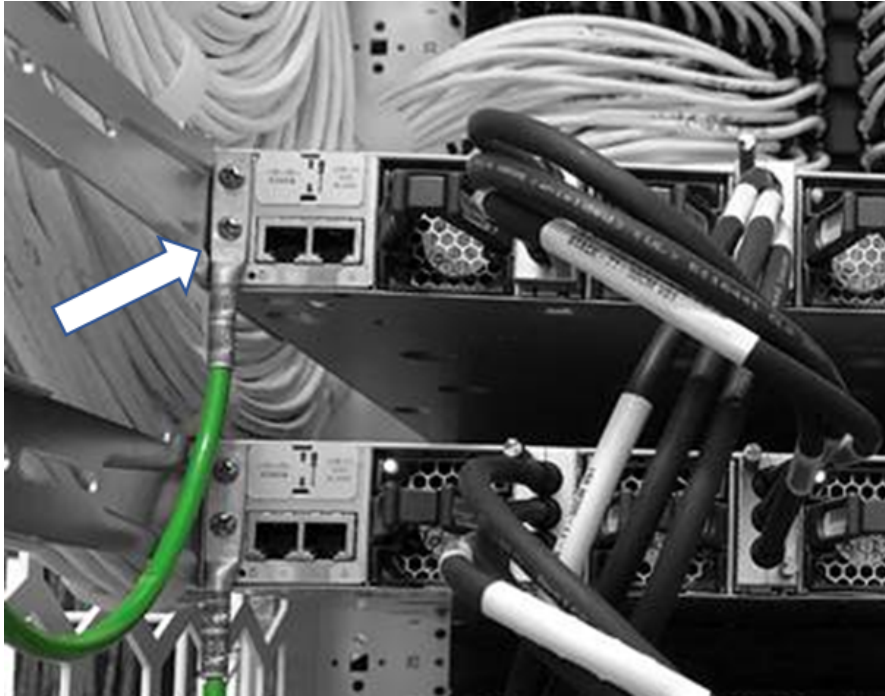


Figure 15: OEM Equipment Ground Lug Connection



Figure 16: OEM Equipment Ground Connection



Figure 17: Primary Bonding Busbar

Table 16: Busbar Standards

ID	Primary Attribute	Secondary Attribute	Specification
1	Busbars	RBB, PBB, and SBB	• Provided with holes for use with correctly matched listed lugs and hardware • Copper, or copper alloys having a minimum of 95 % conductivity when annealed • As specified by the International Annealed Copper Standard (IACS) • Minimum dimensions of 0.25 in. thick x 4 in. wide and variable in length • Maximum 4.0 Ω to ground system resistance (from any point in the system, including IT equipment chassis) • Listed

Evaluation Factors

- Grounding system resistance
- All ancillary equipment, rack systems, cabinets/racks, backboards, etc. bonded to the bonding conductor
- Bonding conductor materials
- Bonding conductors installed using appropriate techniques
- Bonding connections provide low impedance path to ground (tight bolts, removal of non-conductive materials between metal-to-metal contacts)

Implementation Guidance

Connect PBBs to SBBs in other rooms as required.

Provide, inspect, and maintain in accordance with the [VA Electrical Design Manual](#).

Remove powder-coated paint from racks, cabinets, ladder racks, etc. prior to installation to maximize the quality of grounding bonds.

Star-prong washers that bite into the metal may be used with caution that they must fully penetrate the paint for effective contact.

4.2.8.3 Building Earth Ground System

Note: The following sections are provided for reference only. Consult with [VA Electrical Design Manual](#) for design requirements.

A Building Earth Ground System (EGS) is required on all VA buildings regardless of the type, classification, or ANSI/TIA-942-B Rating of IT support spaces operating in those facilities.

It is included here for reference only.

Table 17: Building Earth Ground System Standards

ID	Primary Attribute	Secondary Attribute	Specification
1	Building Earth Ground System	NFPA 70 Article 250-compliant system	Required in all facilities
		Resistance Between Bonded Objects and SBB	4.0 Ω maximum

Evaluation Factors

Visual inspection of each telecom space for grounding system compliance required annually at a minimum.

Implementation Guidance

Provide, inspect, and maintain a facility EGS that meets the guidelines of NFPA 70 (National Electric Code) Article 250, Grounding and Bonding, and the [VA Electrical Design Manual](#).

The building EGS should obtain a resistance-to-earth of 5 Ω or less as measured by the four-point fall-of-potential method according to IEEE 81.

4.2.8.4 Lightning Protection System

Note: The following sections are provided for reference only. Consult with [VA Electrical Design Manual](#) for design requirements.

A facility Lightning Protection System (LPS) is required on all VA buildings containing a Rating 3 computer room and recommended for all VA buildings regardless of the IT support spaces operating in those facilities. Included for reference only.

Table 18: Lightning Protection System

ID	Primary Attribute	Secondary Attribute	Specification
1	Lightning Protection System	CDC	NFPA 780-compliant LPS required
		CSC	NFPA 780-compliant LPS required
		MSC	NFPA 780-compliant LPS required for Rating 3 MSCs
		NSCs and Other IT Support Spaces	LPS recommended

Evaluation Factors

- Facilities containing Rating 3 computer rooms have NFPA 780-compliant LPS protecting the entire facility

- LPS terminals and roof-mounted equipment properly bonded
- Down conductors properly connected to earth ground system and tested for ground resistance triennially

Implementation Guidance

Where required and in accordance with the [VA Electrical Design Manual](#), provide, inspect, and maintain a facility LPS that meets the guidelines of NFPA 780 (Standard for the Installation of Lightning Protection Systems).

4.2.9 Lighting

Lighting requirements in IT support spaces enable safe and reliable operation and maintenance of IT equipment and supporting infrastructure equipment. IT support spaces are intended to be lights-out operating environments except when access by operations and maintenance personnel.

ANSI/TIA-942-B recommends using LED lighting in IT support spaces to improve lifecycle cost. Managed Intelligent LED lighting in a data center can reduce server inlet temperatures by 2 °F to 3 °F. This can reduce cooling costs by up to 10 %.

Zoned and automated lighting systems meeting these requirements are acceptable.

Utilize energy-efficient motion sensors to activate lights in the immediate area soon after an employee enters an IT support space. All aisles and passageways shall be illuminated to allow for proper identification of visitors by security equipment and to allow for effective operation and maintenance of systems in the space.

Utilize a three-level lighting protocol in IT support spaces depending on human occupancy:

- Level 1: IT support space unoccupied - lighting should be sufficient to allow effective use of video surveillance equipment. Level 1 shall be provided in all IT support spaces.
- Level 2: Initial entry into the IT support space - motion sensors should be used to activate lights in the immediate area of entry and be programmed to illuminate aisles and passageways. Sufficient lighting should be provided to allow safe passage through the space and to permit identification via security cameras. Level 2 shall be provided in all IT support spaces.
- Level 3: Occupied space - when the IT support space is occupied for purposes of maintenance or interaction with equipment, lighting shall be 500 lux in the horizontal plane and 200 lux in the vertical plane, measured 3 ft above the finished floor in the middle of all aisles between cabinets. In IT support spaces larger than 2,500 ft², zone lighting shall provide Level 3 in the immediate area of work and Level 2 in all other zones. Level 3 is a requirement in all IT support spaces except for entrance rooms and TRs.

Lighting plan in IT support spaces should match the IT equipment layout. Attempt to place light lighting fixtures in both cold and hot aisles so that technicians have sufficient

illumination to work by. Lighting fixtures should be located above aisles between cabinets rather than directly above cabinets or overhead cable pathway systems.

Emergency lighting and signs shall be properly placed per Code and AHJ requirements such that an absence of primary lighting will not hamper emergency exit.

4.2.10 Metering

Metering is used to collect energy consumption data that can be used to track long-term trends, benchmark progress and performance, perform diagnostics, and alert data center operators to impending or ongoing system failures. The level of metering described is necessary for VA to:

- Baseline utility consumption
- Determine priorities for utility intensity reduction
- Measure the effectiveness of energy efficiency initiatives, and
- Show the use of power in telecommunications over time.

4.2.10.1 Metered Energy Consumption Data

All VA data center facilities shall install and operate an Automated Energy Metering system to meet the metered energy consumption data collection requirements of the metering and monitoring standards. This system may combine elements of BAS, Data Center Infrastructure Management (DCIM) systems, Data Center Operations Systems, and others as needed to achieve the required outcomes.

All VA data center facilities shall report the following energy consumption data within the period specified:

Table 19: Metered Energy Consumption Data

Data	Units	Measured Location(s)	Period
Facility Total Electrical Consumption (FEC)	Kilowatt-hours (kWh)	Main building step-down transformer(s) and Generator plant	Monthly
Total Data Center Energy Consumption (TEC)	Kilowatt-hours (kWh)	• Utility feed(s): Output to data center or • Automatic Transfer Switch (ATS): Output to data center or • Trunk circuit(s): Output to data center	Monthly
Total Data Center Physical Infrastructure Energy Consumption (TPIEC)	Kilowatt-hours (kWh)	• Branch circuit(s) supporting physical infrastructure equipment: Output or • Physical infrastructure equipment: Input	Monthly
Total IT Equipment Energy Consumption (TITEC)	Kilowatt-hours (kWh)	• Rack-mounted PDU: Output or • Power consumption data reported by IT system firmware: Input or • Branch Circuit Monitoring (BCM) of circuit(s) supporting IT equipment: Output	Monthly
Total UPS Load Energy Consumption (TUPSC)	Kilowatt-hours (kWh)	UPS System(s): Output	Monthly

Data	Units	Measured Location(s)	Period
Total Renewable Energy Usage by Data Center (RENEW)	Kilowatt-hours (kWh)	Utility billing detail and/or Facility Engineering staff	Monthly

4.2.10.1.1 Facility Total Electrical Consumption

Facility Total Electrical Consumption (FEC) energy from the local utility is metered by the local utility provider. This portion of the metric can be obtained from the utility electrical bill or can be calculated if local resources have capabilities to read the electrical meter(s) directly.

Generator energy may be estimated by the number of hours that the generator plant(s) run times the power capacity of those generators. The time that the generators run includes building (live) load tests where the building (or some portion thereof) is supported by the generator plant but does not include offline generator tests and maintenance.

4.2.10.1.2 Total Data Center Energy Consumption

Total Data Center Energy Consumption (TEC) is the sum of Total Data Center Physical Infrastructure Energy Consumption (TPIEC) and Total IT Equipment Energy Consumption (TITEC).

4.2.10.1.3 Total Data Center Physical Infrastructure Energy Consumption

Total Data Center Physical Infrastructure Energy Consumption (TPIEC) includes the amount of energy needed to operate power supply/distribution equipment (transformers, switchgear, meters, UPSs and PDUs, wiring to the IT devices), heating/cooling equipment (chillers, pumps, cooling towers, CRAC units), lighting in the data center, physical security systems in the datacenter (access control, intrusion detection, surveillance), fire protection and similar systems, and system electrical losses (through transmission, conversion, etc.).

VA facilities generally have shared/integrated physical plants (mechanical and electrical rooms) that provide support to both data center and non-data center (administrative, patient care, etc.) spaces. Separate metering of the portion(s) of physical plants that solely provide support to the data center should not be expected to be available because of the mixed use and legacy nature of VA facilities. There are no simple, standardized methods of obtaining this information in the shared/integrated physical infrastructure environments typical in VA facilities.

VA uses a measure/estimate methodology to obtain this information. Estimate the fraction of the load of the device supporting the data center and multiply this fraction by the monthly energy usage of the device. This fraction is an engineering estimate of what percentage of the device's load is used to support the data center (as opposed to supporting non-data center loads). For example, based on how much chilled water flow goes to administrative office spaces versus to the datacenter (perhaps based on water pipe size), the site estimates that 65 % of the chilled water goes to the data center. The total energy consumed by the chillers is measured at 1,460 kWh. The energy consumed to support the data center for that chiller for the month is $1,460 \text{ kWh} * 0.65 = 949 \text{ kWh}$.

Individual sites are responsible for documenting the assumptions that they use to calculate and summate physical infrastructure electrical consumption. There are no standard templates for collecting, aggregating, or estimating this information because of the unique configuration of physical infrastructure systems at each VA facility. Consult with local and enterprise facility and engineering staff as necessary and document the assumptions and process that will be used at the facility.

4.2.10.1.4 Total IT Equipment Energy Consumption

There are multiple ways to obtain Total IT Equipment Energy Consumption (TITEC) data, depending on the equipment that is installed in the data center. Use the most accurate method (methods are described from most to least accurate) to gather the metric information.

- If power distribution equipment (PDUs) in the data center have branch level metering, or if rPDU in the racks have rack-level power monitoring, whether this is connected to a facility Building Automation System (BAS)/Energy Management Control System (EMCS) (BAS/EMCS) or not, collect actual energy use data at the level closest to the IT equipment. Use of a BAS/EMCS to automatically collect and calculate this information provides the most accurate measurement.
- If PDUs in the facility have the capability to record kWh that they supply (and most modern units do), perform a manual (or automated, if BAS/EMCS capabilities are available) recording of the kWh reading on the PDU at a specified time for each reporting period (e.g., calendar month). For example, collect kWh numbers for all PDUs supporting the data center at 9 a.m. on the first calendar day of the month, and summate the change in kWh numbers from the previous period reading.
- Data centers generally have dedicated UPS systems to ensure IT equipment power is continuously available. At a specified time for each reporting period (e.g., calendar month), collect kWh numbers for all UPS systems supporting the data center and summate the change in kWh numbers from the previous period reading.
- For non-dedicated data centers and facilities where UPS systems also support non-data center administrative functions, follow the guidance for UPS kWh reporting above, and use a documented, realistic engineering estimate of the percentage of the UPS system power that is supplied to the data center IT equipment. This engineering estimate could involve counting the approximate number of amp-hours used by IT equipment in the data center and the approximate number of amp-hours that are used in a UPS-backed administrative space, and determining the ratio (3:1, or 75 %, for example). Use this estimated ratio (75 %, in our example) times the change in kWh numbers from the previous period reading to report the metric.
- Where the data center does not have any of the previous metering equipment to assist in developing the metric, provide a realistic engineering estimate of IT equipment power use by another method. Consult with local and enterprise facility and engineering staff as necessary and document the assumptions and process that will be used at the facility. An example of how to estimate a monthly total IT electrical consumption is to use a

clamp-on electrical meter on the input power cables to a dedicated electrical distribution circuit panel, determine the amperage being provided during an assumed representative time period, multiply the amperage by the panel voltage, divide by 1.73 for three-phase distribution, and convert from power basis to energy basis by multiplying by 730 (average hours per month). Other methods may be more appropriate given the equipment in and configuration of a particular data center facility.

4.2.10.1.5 Total Uninterruptible Power Supply Load Energy Consumption

Data centers generally have dedicated UPS systems to ensure IT equipment power is continuously available. At a specified time for each reporting period (e.g., calendar month), collect Total UPS Load Energy Consumption (TUPSC) kWh numbers for all UPS systems and summate the change in kWh numbers from the previous period reading. For example, last month's kWh reading on UPS 1 was 220,000 and this month's reading is 235,000. Last month's kWh reading on UPS 2 was 350,000 and this month's reading is 375,000. The Total UPS Load Energy Consumption (TUPSC) for the month is $(235,000 - 220,000 + 375,000 - 350,000 =) 40,000$ kWh.

Where UPS information is not available as kWh on the UPS, consult with local and enterprise facility and engineering staff as necessary and document the assumptions and process that will be used at the facility.

4.2.10.1.6 Total Renewable Energy Usage by Data Center

Total Renewable Energy Usage by Data Center (RENEW) is the ratio of consumed energy to be produced by renewable sources and the total electrical consumption. For example, Facility X's total electrical consumption provided by electric companies for August is 4,500,000 kWh. Facility X also pays for 250,000 kWh of electricity each month to be provided from wind farms. Facility X will report 250,000 kWh (5.56 percent) renewable energy usage for August.

4.2.10.2 Rating 3 Power Metering Aggregation System

A data center level power metering aggregation system is a requirement for data center facilities intended to be Rating 3 (CDCs, CSCs, and some MSCs) to collect Energy Consumption Performance Metrics. All VA data center facilities shall report the following data within the period specified:

Table 20: Energy Consumption Performance Metrics

Performance Metrics	Units	Performance Goal	Period
Power Usage Effectiveness (PUE)	N/A	- New data centers < 1.4 with a goal of 1.2 - Existing data centers < 1.5	Monthly
Site Infrastructure Energy Efficiency Rating (SIEER)	N/A	N/A	Monthly

Evaluation Factors

- Data collection capabilities and intervals
- Data reporting capabilities and intervals
- Site-specific collection processes and engineering assumptions documented
- Performance metrics meet performance goals

Implementation Guidance

To meet aggregation system requirements for CDC, CSC, and some MSC data center facilities intended to be Rating 3, connect the data center aggregation system to enterprise DCIM system to enable Automated Energy Metering. Options for power consumption metering capability at the vertical rPDU (or PDU/BCP/RPP) level to satisfy this requirement include:

- Operate Branch Circuit Power Monitoring (BCPM) on all PDUs/BCPs/RPPs in the data center, with a single aggregation system.
- Operate single-vendor rPDU and aggregation system throughout the data center.
- Operate a single-vendor aggregation system. All rPDU must be compatible with the aggregation system. Replace noncompatible rPDU as necessary.

4.2.10.2.1 Power Usage Effectiveness

Power Usage Effectiveness (PUE) is a dimensionless metric which describes how efficiently a data center is designed and operated to conserve power. PUE is defined as the ratio: (Total Data Center Energy Consumption)/(Total IT Equipment Energy Consumption) or (TEC/TITEC)

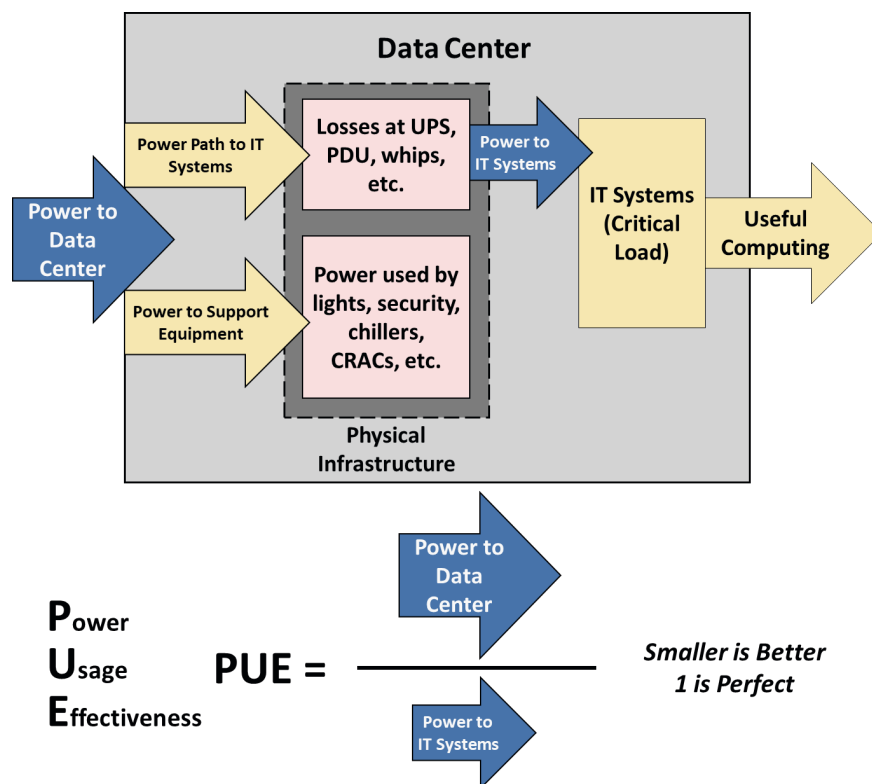


Figure 18: Power Usage Effectiveness

4.2.10.2.2 Site Infrastructure Energy Efficiency Rating

Site Infrastructure Energy Efficiency Rating (SIEER) is a dimensionless metric which describes how efficiently a data center is designed and operated to conserve power. SIEER is defined as the ratio: (Total Data Center Energy Consumption)/(Total UPS Load Energy Consumption) or (TEC/TUPSC).

SIEER measures facility and data center efficiency differently than PUE, requires significantly less metering and monitoring equipment investment, allows VA flexibility in determining what is measured in shared-use (data center and administrative or hospital) facilities, and

provides information that shows efficiency improvements over time between the data center and its host facility.

SIEER is defined as the total power coming into the facility divided by the total power being supported by the UPS system (technical power load, including the power necessary to operate the UPS and downstream power distribution equipment). While SIEER and PUE are similar efficiency metrics, they are not directly comparable.

4.3 Mechanical and Environmental Conditioning Standards and Monitoring

4.3.1 *Introduction*

4.3.1.1 *Efficient Federal Operations*

In accordance with Executive Order No. 13834 “Efficient Federal Operations” 2018, agencies are mandated to:

- Achieve and maintain annual reductions in building energy use and implement energy efficiency measures that reduce costs;
- Meet statutory requirements relating to the consumption of renewable energy and electricity;
- Utilize performance contracting to achieve energy, water, building modernization, and infrastructure goals;
- Ensure that new construction and major renovations conform to applicable building energy efficiency requirements and sustainable design principles; consider building efficiency when renewing or entering into leases; implement space utilization and optimization practices; and annually assess and report on building conformance to sustainability metrics;
- Implement waste prevention and recycling measures and comply with all Federal requirements with regard to solid, hazardous, and toxic waste management and disposal;
- Acquire, use, and dispose of products and services, including electronics, in accordance with statutory mandates for purchasing preference, Federal Acquisition Regulation requirements, and other applicable Federal procurement policies; and
- Track and, as required by section 7(b) of this order, report on energy management activities, performance improvements, cost reductions, greenhouse gas emissions, energy and water savings, and other appropriate performance measures.

4.3.1.2 *Federal Leadership on Climate Change and Environmental Sustainability*

In accordance with Executive Order No. 13693 “Federal Leadership on Climate Change and Environmental Sustainability” 2015, federal agencies were required to promote building energy conservation, efficiency, and management to reduce agency building energy intensity by 2.5 percent annually through the end of fiscal year 2025, relative to the baseline of the agency's building energy use in fiscal year 2015.

Although the specific requirements of Executive Order No. 13693 are now generalized under Executive Order No. 13834, VA's metering and monitoring practices that were implemented under the original guidance still support the new guidance to "achieve and maintain annual reductions in building energy," "ensure that new construction and major renovations conform to applicable building energy efficiency requirements," and to "track and...report on energy management activities, performance improvements, cost reductions, [and] energy...savings."

These practices are maturing in many data centers across VA's enterprise and VA remains committed to the continued improvement and expansion of the progress made over the last decade.

4.3.2 Environmental Operating Envelope Conditions

These specifications define the environmental operating envelope conditions and provide design and operations guidance for how to achieve and maintain those conditions in enterprise data centers. This Standard shall be used to specify and design computer room spaces and the environmental control systems and equipment to be operated therein.

VA data center environmental requirements are classified as Environmental Class A1 per American Society of Heating and Air-Conditioning Engineers (ASHRAE) Technical Committee TC9.9 thermal guidelines for data processing environments, requiring "tightly controlled environmental parameters (dew point, temperature, and RH)" to support mission-critical operations.

IT equipment load shall not exceed the overall cooling capacity of the data center. In general, 1 W of power consumed requires 1 W of cooling.

Excess demand requires re-engineering or self-contained high-density cooling solutions.

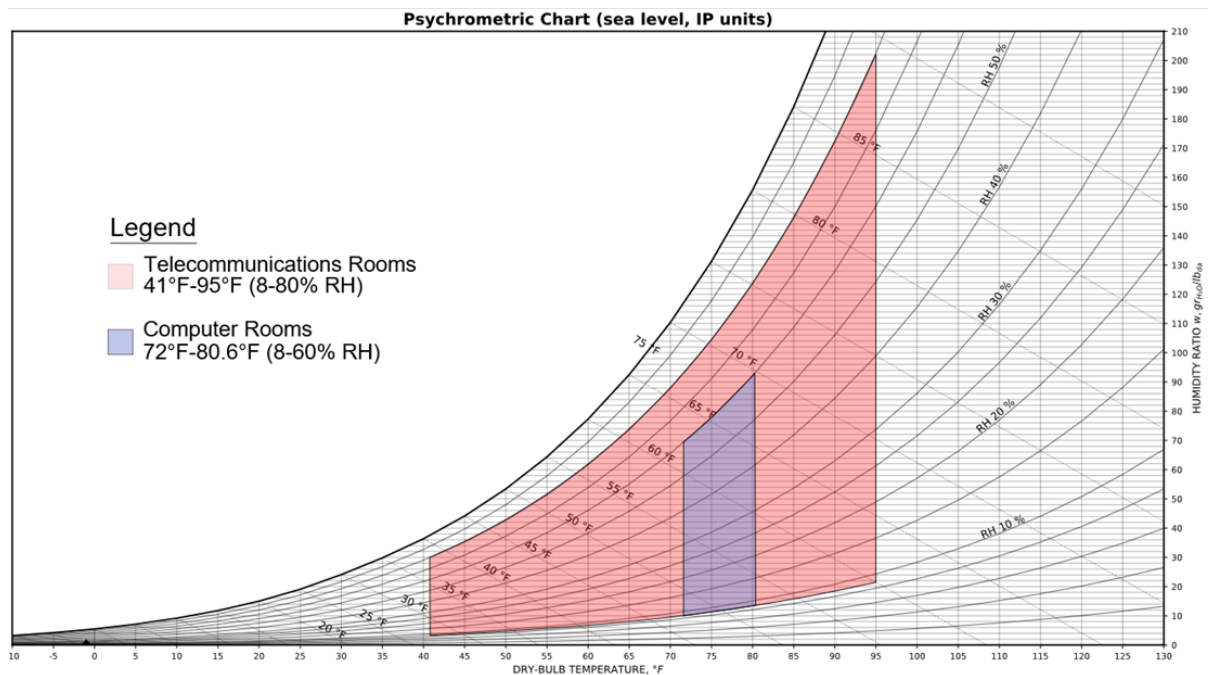
VA data center facilities (with the exception of TRs) shall maintain an average rack-face inlet temperature range of 72 °F to 80.6 °F. In summary, the average rack-face inlet temperature will be determined as the average of a standard three-sensor temperature measurement at the bottom, mid, and top height levels as measured 2 in. distance from the IT equipment inlets. For implementations that do not yet have the standard three-sensor installation shall be measured at 48 in. height above the floor at 2 in. distance in the interim. Data center temperature and humidity conditions shall be measured at the air intake to the IT equipment as described, rather than measuring the ambient or average condition(s) throughout the data center.

Cooling equipment set points will vary depending on equipment, set point thermocouple location, equipment location with respect to the IT equipment, and other factors. Establish and modulate the cooling equipment setpoint(s) to achieve the 72 °F to 80.6 °F equipment inlet temperature range while staying within the other (humidity, dew point) specifications. IT equipment exhaust temperatures shall not be used for determination of cooling equipment set points.

VA computing facilities shall utilize engineered verified layout designs to optimize energy usage in the environmental control systems supporting the facilities. Proposed deviation from standard layout design requires verification and validation by DCIE and CFD analysis.

In general, HD racks shall be distributed across the entire data center floor to keep cooling systems from becoming ineffective.

Alternatively, dedicated HD areas may be employed with cold aisle containment to provide additional capability to power and cool to an average value below the peak enclosure value for HD equipment. Typically, this includes shortening the airflow path between the cooling system and the rack.



Note: Envelopes Represent Conditions at IT Equipment Inlet

Figure 19: VA-Modified ASHRAE Environmental Classes for Data Center Applications

4.3.3 Data Center Facility Environment Conditioning Standards

Table 21: Data Center Facility Environment Conditioning Standards

ID	Primary Attribute	Secondary Attribute	Specification
1	Environmental Envelope	Temperature (at IT equipment inlet)	72 °F to 80.6 °F dry bulb (VA Modified ASHRAE Recommendation)
		Humidity (at IT equipment inlet)	8 % RH to 60 % RH
		Dew Point (at IT equipment inlet)	≤ 59 °F
		Rate of Temperature Change	≤ 9 °F per hour
		Room Height (vertical dimension)	16 ft height (12 ft legacy) from finished floor to floor above, with larger heights preferable
		Monitoring	Varies by classification (see Amendments and Exceptions Tables in 4.3.4 below)

ID	Primary Attribute	Secondary Attribute	Specification
2	Environmental Control Equipment	Primary Air Conditioning Type	CRAC
		Humidity Control Equipment	Dedicated, low-pressure dry fog humidification systems (only as necessary)
		Airflow Plenum	Non-ducted architectural plenums (underfloor, above ceiling, etc.) should not be used in new construction or renovation of existing spaces
		Static Pressure (raised floor)	0.15 in. of Water Column (WC), Design distribution shall be not less than 5 kW (1.422 ton) of cooling per each 25 % perforated tile
		Airflow Tile (raised floor)	Placement per CFD analysis; typically: • 25 % open for SD racks • 50 % open for HD racks • Air Removal Units (ARU)/active tiles for unique hot spots
		Airflow Control Equipment (raised floor)	Fill in all floor penetrations to eliminate leakage to minimize cooling plenum static pressure loss
		High-Capacity Grate Tiles (raised floor)	May be used in the floor near HD areas (over 5 kW per rack) to eliminate hot-spots and air shortages
		Overhead Supply Air Ducts	• Vents directly placed over cold aisle • Placed as closely as possible to equipment intake • No lateral diffusing vents
		Ducted Systems Hot Air Return Vents	When used, placed as closely as possible to equipment exhausts or directly connected via flexible duct connections
		Blanking Panels	Required in every RU where no equipment resides on the front (cold aisle) of each cabinet
		Computational Fluid Dynamics (CFD)	Required for all new or significantly modified computer rooms (except NSCs)
		Containment Systems	• Must be considered for all new or significantly modified data centers • Required where close-coupled air conditioning is employed
		Rules-based Borrowed Cooling (existing data centers)	• Utilized for high-density deployment to borrow adjacent underutilized cooling capacity • Racks separated to prevent exceeding peak enclosure power and cooling values
		Close-Coupled Air Conditioning Systems	Used only with a containment system and when shown to be effective in CFD modeling

Evaluation Factors

- Temperature and humidity controlled within designated parameters
- Temperature and humidity measured at appropriate locations
- Bypass and recirculation air measures implemented
- CFD modeling used for data center design

Implementation Guidance

Data center environment shall be $\geq 8\%$ RH to preclude static buildup and $\leq 60\%$ RH to preclude water condensation, hardware corrosion, and early system and component failure.

Identify an environmental control window that is compatible with all the different thermal specifications, types of equipment, and equipment vintages in the data center for optimal energy saving.

4.3.4 Data Center Facility Environment Conditioning Standards (Amendments and Exceptions)

Amendments and exceptions to the requirements in *Section 4.3.3* are listed in tables below for each class of data center to include NSC standards which are not covered above.

Table 22: Facility Environmental Requirements (CDC Amendments and Exceptions)

ID	Primary Attribute	Secondary Attribute	Specification
1	Environmental Envelope	Alarming	Automatic alarming when parameters are exceeded is recommended where automated systems are available
		Monitoring	Monitoring integrated into BAS with automatic alarming when conditions exceed prescribed limits

Table 23: Facility Environmental Requirements (MSC Amendments and Exceptions)

ID	Primary Attribute	Secondary Attribute	Specification
1	Environmental Envelope	Monitoring	Monitored by automatic system(s) in multiple locations, with indicators allowing operators to determine the general condition of the computer room

Table 24: Facility Environmental Requirements (NSC Amendments and Exceptions)

ID	Primary Attribute	Secondary Attribute	Specification
1	Environmental Envelope	Temperature (at IT equipment inlet)	For TRs only: 41 °F to 95 °F dry bulb
		Humidity	For TRs only: 8 % RH to 80 % RH
		Dew Point	For TRs only: ≤ 82.4 °F

ID	Primary Attribute	Secondary Attribute	Specification
		Room Height (vertical dimension)	No raised floor or suspended ceiling, maximize internal room height
		Monitoring	Monitored with indicators allowing operators to determine the general condition of the computer room
2	Environmental Control Equipment	Primary Air Conditioning Type	Air Conditioner Units in any NSC with active IT server and storage components
		Humidity Control Equipment	No dedicated humidity control equipment required
		Airflow Plenum	Not applicable
		Airflow Control Equipment	Not applicable
		Computational Fluid Dynamics (CFD) Modeling	Not required
		Containment Systems	Not specified

4.3.5 Environmental Control Equipment Requirements

Note: The classification of CRAC units, for the purposes of this document, includes Chilled Water (CW) Fan Coil Units (FCU), Computer Room Air Handler (CRAH) units, split system Direct Expansion (DX) CRAC units, and similar types of dedicated floor-mounted systems.

Table 25: Environmental Control Equipment

ID	Primary Attribute	Secondary Attribute	Specification
1	Computer Room Air Conditioner (CRAC) Equipment	Type	Only downflow CRACs shall be used except where modeled as most acceptable in a CFD model
		Total Cooling Capacity	Not less than 10 Refrigeration Ton (RT) per unit (Note: NSCs and their subclass telecommunications spaces are covered separately)
		Placement	CRAC units employed with raised access floors should be placed perpendicular to hot isles for effective air distribution supply to the cold isles. Applies to all configurations that use CRACs
		Fan Type	- New and replacement units shall utilize Electronically Commutated (EC) fan technology - Upgrade centrifugal fan units with EC plug fans where feasible

ID	Primary Attribute	Secondary Attribute	Specification
		Variable Speed Drive (VSD)	• Shall be used in all possible applications • Fan(s) should be controlled by an automated VSD system provided internally to the CRAC or as supplementary external equipment powering the fans • Older CRAC units shall be retrofitted with VSDs
		Intake Chimneys	If not fully ducted return or connected to a return plenum: • Utilize CRAC intake hoods, extended returns, or chimneys on downdraft models to draw in heated air from the highest, warmest part of the room when possible to maximize ΔT across the cooling coils • Do not extend hoods into a void ceiling space • Leave at least minimum 18 in. between top of chimney and the ceiling to allow return air to flow naturally into the chimney without constriction • Retrofit existing CRAC units with hoods where feasible
		Floor Stand	Where CRAC units are used in computer spaces with a raised access floor, they shall be mounted level with the finished floor level on an adjustable, seismic-rated floor stand designed for the equipment
		Service Clearance	Provide clear space around units as recommended by the manufacturer (typically 36 in. on the front and each side)
		Control System	• Install a management control system for all computer rooms with two or more CRAC units • The control system shall operate the system(s) in the most energy-efficient manner possible maintaining planned system redundancy levels, specified IT equipment inlet conditions, and maximizing the ΔT between the CRAC supply and return

ID	Primary Attribute	Secondary Attribute	Specification
		Local Monitoring	Each CRAC shall have a local monitoring panel affixed to the unit accessing the following information: • System automatic restart with programmable delay • Sequential load activation • Sensor calibration • Current temperature set point and location of that set point • Current supply and return temperature and humidity • Unit diagnostics (fan, valve, alarm) • Alarm log history • Runtime log • Audible and visual alarm
2	Split System Air Conditioner Equipment	Operation	Automatic operation to maintain set point temperature in the telecommunications space for outside environmental conditions between the ASHRAE Fundamentals Weather Data 99.6 % column for Heating (Winter) and 0.4 % column for Cooling (Summer)
3	Portable Air Conditioner Equipment	General Usage	Shall not be used as part of a planned cooling system in VA data centers
		Emergency Usage	May be used for supplemental emergency cooling for short durations when primary equipment is unable to support the design load (e.g. until repaired)
4	Portable Fire Suppression	Type	Use only AHJ-approved clean agent portable fire suppression devices in telecommunications spaces

4.3.6 Airflow Control

4.3.6.1 Room Height

Legacy facilities are encouraged to maximize room height (vertical dimension) through the removal of aesthetic suspended ceilings to maximize cooling efficiency.

4.3.6.2 Air Distribution Ceiling Plenums

In new designs, ceiling void spaces (above a suspended ceiling) shall not be used as plenum spaces for air distribution (return) except when fully ducted above a suspended ceiling. Legacy facilities may continue to use down flow CRAC units with in-room and above-ceiling return designs. When used as a plenum (without ducting), the horizontal surfaces of the plenum must be periodically cleaned (vacuumed). Taking the friability of typical suspended ceiling acoustic tiles, their location above IT equipment, and the interface of other infrastructure equipment (lighting, fire suppression, structured cabling systems, etc.) into account, the maintenance requirement is not offset by the additional risks and costs incurred.

In existing spaces, unless directly ducted to the IT equipment rack discharges, do not extend CRAC intakes through a closed suspended ceiling, as this makes the space a plenum.

A recommended best practice is to remove suspended ceiling systems in data center facilities altogether to allow additional room for exhausted hot air to rise. Because cooling equipment efficiency is directly proportional to the change in temperature, extending CRAC intake ducts to the hottest exhaust air near the ceiling presents the greatest opportunity for efficiency.

Where the ceiling void space is used as a plenum return in existing (legacy) facilities, future physical infrastructure technical refresh projects shall change the air path to eliminate this usage.

4.3.6.3 Air Distribution Floor Plenums

The goal of using air distribution floor plenums is to provide uniform tile air velocity cooling effect across all perforated tiles or grilles.

Overhead cabling is used whenever possible to eliminate possible airflow restriction. Under-floor obstructions such as network and power cables are minimized to prevent obstruction of airflow and cooling supply to the racks. Excess unused cables and equipment cables shall be reclaimed to minimize blockage of air flow. Floor vents are not placed too close to CRAC units to avoid producing negative pressure which causes room air to be drawn back under the floor.

CFD analysis is required before selecting raised-floor solutions.

4.3.6.4 Perforated Tiles and High-Capacity Grates

Perforated tiles or grates are used to introduce cold air directly into the equipment intake in the cold aisle. An appropriate number of tiles or grates are used to match the load that is calculated in the cold aisle. Improperly placed tiles increase non-functional airflow and reduce the pressure of cold air flow where it is required. Placing too few in the cold aisle will cause recirculation and too many will increase the amount of bypass. If airflow efficiency cannot be optimized, bypass is preferred over recirculation.

High-capacity grate tiles may be used in the floor near HD areas (over 5 kW per rack) to eliminate hot-spots and air shortages. These should not be automatically addressed by installing high-capacity grates without considering the complications that may be introduced:

- High-capacity grates pass three times more air than normal capacity tiles and may alter the fixed pressure that is required to feed cold air to other servers
- Additional forced air may bypass the intended target equipment and flow right past the faces to the top of the aisle providing no benefit
- Creates large airflow variations
- Dramatically alters under-floor pressure gradients, making cooling non-uniform and unpredictable
- Impacts airflow in neighboring areas. Some locations will NOT receive sufficient cooling
- Air Removal Units (ARU) or active floor tiles are used to improve targeted air flow for higher density racks

A recalculation of CFD modeling is required when high-capacity grates or active floor tiles are added or relocated.

4.3.6.5 Air Containment Devices

Where an underfloor plenum is used for cooling air distribution, all access floor penetrations shall be sealed with authorized materials such as brushed floor grommets and air dam foam sheets to maintain the rating of any fire and smoke barrier and to minimize cooling plenum static pressure loss.

4.3.6.5.1 Brushed Floor Grommets



Figure 20: Brushed Floor Grommet

4.3.6.5.2 Air Dam Foam Sheets

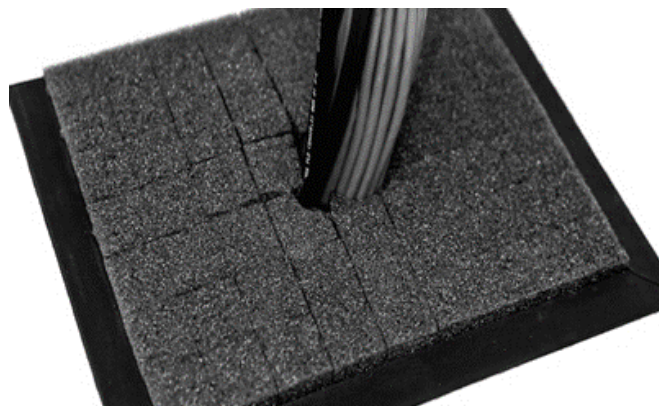


Figure 21: Air Dam Foam

4.3.6.5.3 Blocking Recirculation Paths

Recirculation air from gaps between and under racks and cabinets has been shown in CFD analysis and thermal survey measurements to be a significant contributor to cooling inefficiency in VA data centers.

Ensure that IT equipment racks and cabinets are installed flush to one another without air gaps between the racks. Use appropriate materials to fill gaps between racks to prevent recirculation of exhaust air to the cold aisle.

Install baying kits to keep cabinets aligned.

Install blanking materials under IT equipment racks and cabinets in the same plane as blanking panels.

4.3.6.5.4 Blanking Panels

Blanking panels reduce hot spots and recirculation of air by preventing equipment exhaust air or hot-aisle air from migrating to the conditioned air-intake stream at the front of IT equipment racks or cabinets.

All IT equipment racks and cabinets shall have blanking panels and air dams installed in all positions within the enclosure that do not contain equipment to prevent recirculation of cooling airflow.

Failure to do so causes hot exhaust air to mix with the cool supply air (see *Figure 22: Blanking Panel Application A*) and does not allow for an adequate amount of cool air to reach the intake of all the equipment present in the rack. With blanking panels installed (see *Figure 22: Blanking Panel Application Example B*) the hot air does not have a path back to recirculate so only cold air is provided to the equipment intake.

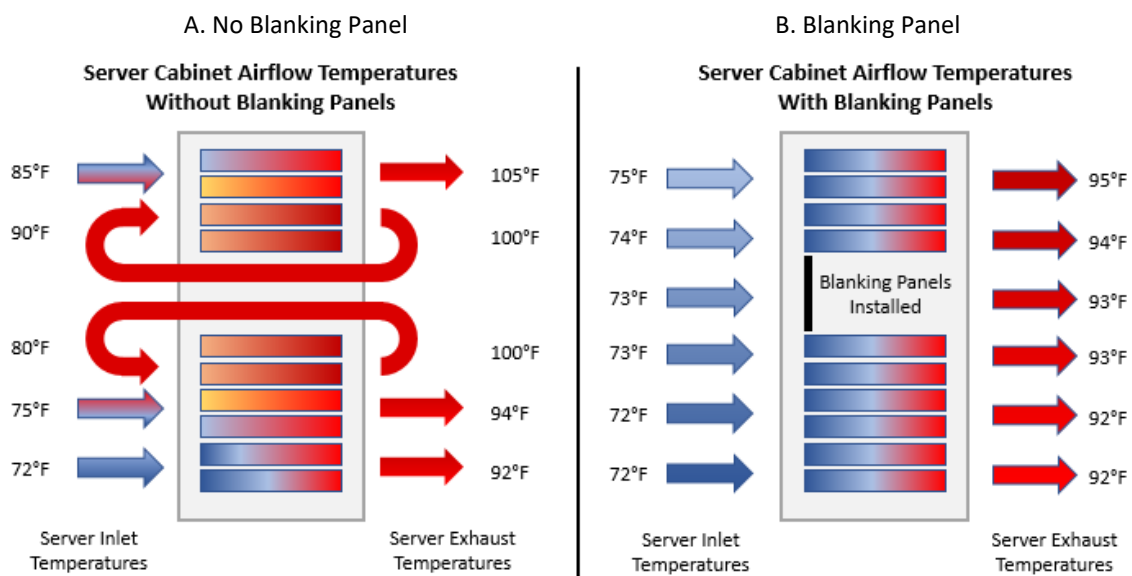


Figure 22: Blanking Panel Application

When angled patch panels are used, angled panel covers are a requirement to block airflow over and under the patch panel. Angled blanking panels can be used between angled patch panels; however, angled panel covers must be used at the top and bottom of an angled stack.

Seal other cabinet holes that traditional blanking panels and panel covers do not fit with manufacturer sealing kits.

Other air containment solutions including expanding foam pillows, fire-stop caulking, temporary metal or fiberboard cover plates, and commercial expanding foam are not acceptable for use in VA data centers; they are either temporary, are not flexible enough to allow for re-cabling, require replacement during any changes, or may present fire and safety hazards.

4.3.6.5.5 Air Distribution Containment Systems

Legacy VA data centers typically use the entire room volume as a heat sink, the least efficient and least effective manner of controlling temperatures and humidity.

Air containment systems may increase the efficiency of data center HVAC systems by separating cold supply and hot return air streams, allowing hotter and drier air to be introduced to the CRAC unit or close-coupled cooling system. The efficiency of these systems is directly proportional to the ΔT between the air streams.

Containment systems may be installed in any VA data center to improve the efficiency and/or effectiveness of the HVAC system. Containment systems are required to be installed in all VA data centers where close-coupled cooling systems are employed. Both hot and cold containment system designs are acceptable and may be appropriate for use in VA data center facilities. Use CFD modeling to determine which design solution(s) will be most effective and cost-effective prior to implementation.

Local AHJ fire and safety standards may supersede this Standard or make containment solutions cost prohibitive.

4.3.6.5.6 Containment Equipment

Use the most cost-effective containment solution available that will satisfy the efficiency design criteria for a given implementation. Hard walls, sliding doors, and hard roofing containment which may be difficult to retrofit and cost prohibitive, while often the most effective containment, are not required. Polyvinyl chloride (PVC) strip curtain walls (vinyl sheeting) solutions, which are inexpensive and the least complicated solutions to install, are acceptable.

When containment is used, consideration must be given to fire alarm and suppression systems to ensure containment systems do not interfere.

4.3.6.5.7 Cold Containment

Cold containment solutions are intended to collect and contain cold (supply) air from cooling units (CRACs) at the inlets of the IT equipment. Design and operate cold containment solutions such that appropriate volumes of air are presented (generally a 10 % overpressure) to the cold containment zone to provide enough pressure to inhibit reintroduction of hot air into the containment but not overly pressurized to avoid inefficiencies with bypass airflow (either excess air being forced through IT equipment or leakage due to over-pressurization). Cold containment retrofits will generally require the least amount of equipment to implement.

Cold containment may be used in conjunction with heat containment solutions to achieve air stream separation goals. Partial cold containment solutions, when used with proper airflow management equipment, can provide significant efficiency returns with minimal

amounts of investment. For example, the addition of end-of-row containment doors, with no other containment, may improve airflow efficiency up to 15 % in some data centers.

4.3.6.5.8 Heat Containment

Heat containment solutions are intended to direct hot (exhaust) air from IT equipment racks away from areas that it can mix with cold (supply) air entering the IT equipment rack face. Heat containment may be used in conjunction with cold containment solutions to achieve air stream separation goals.

4.3.6.5.9 Chimney Return Systems

An effective subset of heat containment, chimney VED return systems direct the exhaust from IT equipment racks vertically to a ducting system that returns the hot air stream directly to the CRAC unit intakes without mixing. This variant on heat containment requires extensive installation work, limiting its ability to be implemented in operational data centers.

Do not employ chimney systems without ducted returns unless the finished ceiling is at least 12 ft and chimney has 18 in. clearance below the ceiling to allow for natural flow of air without restriction.

4.3.6.6 Heating, Ventilation, and Air Conditioning Environmental Control Equipment

In general, legacy VA data centers use a pressurized subfloor air distribution plenum for cooling air distribution to IT equipment. It is not necessary for future VA data centers to use this model if other, more efficient and effective environmental control models can be used.

Alternative air conditioning types are allowable provided they increase the efficiency of the HVAC system and do not provide more cooling than the ultimate design load of the data center (for example, do not install a 5-ton CRAC unit in a NSC designed to contain one ton of IT equipment).

4.3.6.7 Computer Room Air Conditioner Units

Floor-mounted Computer Room Air Conditioner Units (CRAC) are generally used in conjunction with raised access floor distribution plenums in legacy VA data center facilities. Where used with a raised access floor, CRACs shall be supported by manufacturer-specified, seismic-rated floor stands independent of the access floor.

Utilize high-efficiency Electronically Commutated (EC) plug fans in CRAC units. Retrofit legacy CRAC units with EC plug fans for increased operational efficiency where feasible.

4.3.6.8 Split-System Air Conditioning Units

Split-system air conditioning units are air-cooled Direct Expansion (DX) or water-cooled units which are systems typically with a ductless connection between a compact heat exchanger component in the interior space and a condenser (heat rejection) component external to the controlled environment that is either wall mounted or is mounted within a suspended ceiling system. These units are only appropriate for use in small, non-critical, distributed VA data centers and other small IT-related spaces. The typical configuration of these rooms is a local server room or TR. Size these systems to match the ultimate design load of the space.

Due to the non-critical nature of IT equipment operating in these facilities, redundant equipment is not required.

4.3.6.9 Portable Air Conditioning Units

Portable air conditioning units (air-cooled or water-cooled) are considered inappropriate for long-term use in VA data center spaces and shall only be used for temporary spot cooling purposes in the event a primary system is inoperative. Portable units shall not be used to increase the cooling capacity of a VA data center to increase the amount of IT equipment that can be installed.

4.3.6.10 Close-Coupled Air-Conditioners

The use of close-coupled air conditioning systems is permitted in VA data centers when the systems are specifically designed into the support model. In general, use close-coupled systems in portions of the data center designed for HD (> 5 kW per IT equipment rack). When close-coupled systems are used, containment must be implemented in the zone(s) where the systems are installed.

Close-coupled systems shall not be used to increase the amount of cooling available within a data center unless the power being supplied to the data center is simultaneously being increased (increased design capacity in both power and cooling must match).

4.3.6.11 Humidity Control Equipment

Humidity control equipment shall be used in VA data center facilities to allow operators to maintain the computing environment within the allowable operating parameters and to minimize the energy required to provide a stable and efficient environment for IT equipment.

The introduction of additional humidification in VA data centers should be conducted using dedicated, low-pressure dry fog humidification systems. Acceptable systems to support VA IT equipment produce uniform, non-wetting dry fog with a mean droplet diameter not to exceed 10.0 µm. Provide compressed air, humidity sensors, an automatic control system, and appropriate water strainer, filter, purifier, and deionizer equipment for the humidification system.

System design shall comply with ANSI/ASHRAE Standard 188 (current version) and VHA Directive 1061 for Legionella control.

Steam humidification systems introduce additional heat into the data center that must be removed, using additional energy; ultrasonic humidification systems are effective but require significant maintenance. Both types are designated as inappropriate for use in VA data centers due to identified resource capabilities.

Evaluation Factors

- CFD Analysis
- IT equipment inlet environmental conditions
- CRAC set points
- CRAC fan speed control
- CRAC intake chimneys

- Floor to ceiling height
- Design redundancy levels maintained

Implementation Guidance

Environmental control of VA data center spaces shall be based on the IT equipment requirements and not cooled for human comfort. Significant energy intensity reductions can be made by modifying these environmental control parameters and ensuring appropriate airflow paths and systems are designed, operated, and maintained. Retrofit all VA data center spaces for maximal energy savings as rapidly as feasible. Operate CRAC units only as required to maintain prescribed operational parameters.

4.3.7 Monitoring

Table 26: Monitored Conditions

Primary Attribute	Secondary Attribute	Specification
Monitored Conditions	Fire and Smoke Detection	• All VA data centers shall have a fire and smoke detection system in all data and infrastructure spaces as prescribed by National Fire Protection Association (NFPA) standards, Local Fire Codes, the AHJ, and/or CFM • The fire and smoke detection system may be incorporated into an automatic fire suppression system • Where allowed by the AHJ, fire and smoke detection shall use an early-warning Aspirated Smoke Detection (ASD) system such as Very Early Smoke Detection Apparatus (VESDA)
	Water Detection	All CDCs, CSCs, and MSC shall install a liquid water detection system in all data and infrastructure spaces (not required for NSCs). Required in: • Raised floor spaces • IT spaces where water sources exist in or above electronic equipment • HVAC chiller rooms • Fire suppression system areas (pumps and controls)
	Power	All VA data centers shall monitor continuity and quality of: • Utility power delivery • UPS power output • Auxiliary generator power output • Branch circuit power output
	Temperature - Equipment Cabinets	• All VA data centers shall monitor air temperatures at multiple points in IT equipment cabinets. Exact quantity and placement of sensors shall be determined according to the design and implementation of airflow management measures to ensure an adequate supply of cooling air

Primary Attribute	Secondary Attribute	Specification
		All equipment cabinets in facilities lacking best-practice airflow management shall have at least four sensors per cabinet as per Sensor Location specification below
	Temperature - CRACs	All VA data centers shall monitor supply air and return air temperatures: - Supply air temperature shall be monitored at the output of each physical infrastructure device supplying cold air, whether the device is active or not - Return air temperature shall be monitored at the intake of each physical infrastructure device supplying cold air, whether the device is active or not
	Humidity - CRACs	All VA data centers shall monitor supply air and return air Relative Humidity (RH): - Supply air RH shall be monitored at the output of each physical infrastructure device supplying cold air, whether the device is active or not - Return air RH shall be monitored at the intake of each physical infrastructure device supplying cold air, whether the device is active or not
IT Equipment Cabinets	Temperature and Humidity Sensors	To ensure an adequate supply of cooling air, all IT cabinets in facilities lacking best-practice airflow management shall have at least four temperature sensors per cabinet and one humidity sensor as per Sensor Location specification below
		IT Cabinets in facilities <i>with</i> best-practice airflow management may use fewer sensors, meeting the following minimum criteria: - Sensors in cabinets at the ends of a contiguous row/string - Sensors in every third cabinet in the middle of a contiguous row/string - At least two sensors per monitored cabinet - Temperature sensors in the FM and RT locations - Humidity sensor in the FM location

Primary Attribute	Secondary Attribute	Specification
	Sensor Locations	IT Equipment Cabinet temperature sensor locations shall correspond to VA modified ASHRAE-recommended monitoring locations: • Front Top (FT): Centered horizontally, inside the front door, no more than 1 ft from the top of the cabinet • Front Middle (FM): Centered horizontally, inside the front door, 4 ft +/- 6 in. from the finished floor surface • Front Bottom (FB): Centered horizontally, inside the front door, no more than 1 ft from the finished floor surface • Rear Top (RT): Centered horizontally, inside the rear door, no more than 1 ft from the top of the cabinet One humidity sensor in the FM location
Communication Protocols	IT Devices	Monitored IT Devices shall use the Simple Network Communication Protocol (SNMP), configured in a secure mode as required by the TRM. Implementations shall adhere to the VA LAN Security Standard: • SNMPv3 secured using SSL • SSL using Secure Hash Algorithm (SHA-1) authentication • Advanced Encryption Standard (AES) Encryption, at 128-bit or the highest level supported by the attached hardware
	Monitoring Systems	To support data stream integration, monitoring systems shall support Secured SNMP v3.0 for upstream communication and at least one of the following industrial communication protocols to receive sensor data: • BACnet or BACnetIP • Modbus or Modbus IP • CANbus or CANbus IP • Fieldbus or Fieldbus-IP
	Monitoring Devices	Monitoring devices and systems shall be capable of communication using at least one of the following protocols: • Secured SNMP v3.0 • BACnet or BACnetIP • Modbus or Modbus IP • CANbus or CANbus IP • Fieldbus or Fieldbus-IP

Primary Attribute	Secondary Attribute	Specification
	Physical Infrastructure Devices	Infrastructure device communication modules shall support at least one of the following protocols: • Secured SNMP v3.0 • BACnet or BACnet IP • Modbus or Modbus IP • CANbus or CANbus IP • Fieldbus or Fieldbus-IP

Evaluation Factors

- Facility actively monitors all required monitoring points
- IT equipment rack temperature/humidity sensors installed as described in Standard
- Water detection system design
 - Point leak detection units
 - Zone leak detection panels
 - Linear detection systems
 - Wire-free sensors
 - DCIM integration
- Monitoring systems and equipment comply with communications protocol requirements

Implementation Guidance

Monitoring shall be defined as continual real-time observation of a condition or set of conditions affecting computing spaces.

Connect all vertical rPDU to the rPDU monitoring system. Maximize connectivity of facility equipment within the data center to the rPDU monitoring system.

Provide output feeds from the system to the Facilities Management Service (FMS) building monitoring system for remote monitoring capabilities.

Functional definitions of monitoring and metering shall not be interpreted to require separate systems; a single sensor, device, or system may support multiple monitoring and metering functions.

Detection, alerting, and alarming functions, as well as automatic triggering of interventions, shall be considered part of a monitoring system.

Monitoring systems collect forensic data during a period when a detection threshold has been exceeded.

Alerts are communicated when a condition threshold has been reached or exceeded. Alert triggers shall be set to provide sufficient reaction time to respond to, investigate, and correct the underlying cause of the condition.

Alarming systems shall communicate when an absolute limit has been exceeded.

4.4 Telecommunications Standards

This Standard supports structured cabling as a vendor-neutral, extensible approach to information transport system design in VA facilities, *not limited to data center applications*. The specifications for horizontal cabling for LAN applications and vertical cabling for intra- and inter-building backbone are laser-enhanced fiber optics and high-bandwidth UTP. The minimum bandwidth requirement for fiber is 40 Gigabit Ethernet (GbE) and 10 GbE on UTP. All local and campus area networks shall be installed in a structured topology with patch panels serving transition points. Individual “home run” cables are not permitted for LAN applications, including data centers.

Both HDAs, which serve as the distribution point for horizontal cabling and houses Horizontal Cross-connects (HC) and equipment for connecting to the equipment distribution area, and the Main Distribution Areas (MDA) where the main cross-connect is located as defined by ANSI/TIA-942-B specifications should be located adjacent to switching components within the rack to minimize patch cord length and cable management requirements. This approach is consistent with ITS industry standards, and it supports the SD LAN and data center network design baselines.

Specification of structured cabling (both copper and fiber) requires calculation of total length of the link pathway in all dimensions (orthogonally) from dimensionally accurate plans. Procure structured cabling lengths to not exceed 6 ft at each end from the calculated length to avoid coiling of excess structured cabling.

All cabling system components shall be of the same performance category across each entire channel. Components typically may not be mismatched with respect to performance category. For example, where copper telecommunications cabling is used, all components – channel cabling, patch panels, jacks, receptacles, and patch cabling – shall have the same category designation. However, mixed environments are acceptable when supporting a planned upgrade path. For instance, Category 6A (Cat 6A) U/UTP patch cords can be used with Cat 6 patch panels knowing that the performance of the entire channel is at the capability level of the lowest performing component until the upgrade plan is complete.

Communications cabling must not be painted. Paint, plaster, cleaners, abrasives, corrosive residues, or other contaminants may result in an undetermined alteration of the cable properties.

Over the past decade, power-over-LAN cable technologies such as PoE have become a viable powering option for a wide range of applications. Anticipating future standards, device manufacturers are pushing the envelope and designing more sophisticated equipment that demands increased power.

As the power is increased, the heat generated within the cable increases as well. This is especially true when the cables are bundled. The additional heat generated by the increased current could push the cables beyond their rated temperatures.

To address this concern, UL has introduced a Limited Power (LP) certification to simplify the cable choice and installation considerations. The “-LP” cable designation indicates that the cable has been evaluated to carry the marked current under reasonable worst-case installation scenarios without exceeding the temperature rating of the cable. The

certification takes into account large bundle sizes, high ambient temperatures, and other issues related to environmental effects, such as enclosed spaces or conduits. If LP certified cable is not used, the number of cables that can be bundled or enclosed in a sealed raceway will be limited per the specifications in the associated NEC ampacity chart.

Where practicable, this Standard recommends Cat 6A UTP with a minimum LP rating 0.5 to support PoE technology up to 100 W. This specification is shown as “UL Listed as x-LP (0.5 A).” If running Type 4 PoE, cables without LP certification are subject to inspection to ensure bundle size and current limitations per the NEC ampacity table are not exceeded.

Work-Area Outlets (WAO), media, media connectors, and patch panels shall support the minimum performance and capacity requirements of this Standard.

This Standard has been harmonized with OIT LAN Baselines.

4.4.1 *Unshielded Twisted Pair*

This Standard specifies pre-terminated Category 6A (Cat 6A) U/UTP for horizontal distribution. This performance category is applicable to field-terminated or factory pre-terminated cabling for UTP with each cable terminating in an individual connector. While pre-terminated cables require that distances between termination points are known this approach results in the rapid installation of horizontal cabling without risks associated with field termination including wire fragments, impaired cabling performance due to poor installation techniques, faulty terminations, etc. Spiral wrapping pre-terminated bundles is an effective way to maintain the bundle of six cables without impacting ease of installation.

Factory pre-termination is the preferred termination method, but field-termination is acceptable for work-area distribution applications and where pre-termination is impractical such as in horizontal distribution from TRs.

Cat 6A cabling is specified in standard ANSI/TIA-1179 Healthcare Facility Telecommunications Infrastructure for new installations.

Copper Clad Aluminum (CCA) cable is not permitted.

Table 27: Unshielded Twisted Pair Standards

ID	Primary Attribute	Secondary Attribute	Specification
1	Performance Characteristics (Horizontal and First Level Backbone)	Performance Category	- Cat 6A (10 GbE) - Cat 6 (exempted and authorized variance only)
		Performance Specifications	Meets or exceeds TIA-EIA-568-C.2-10 and TSB-155
		Conductor Size	22 AWG to 24 AWG
		Limited Power (LP) Certification	UL Listed as x-LP (0.5 A)
		Jacket Color	Blue (to match 606C HC patching field designation)
		Termination Method	- Pre-terminated, 8P8C with TIA 568B termination method preferred OR - Field-terminated with TIA 568B termination method to WAOs only where distances cannot be precisely calculated
		Media Connector	Pre-terminated with split 8P8C preferred
		Bundling	Multiple cable harness

Evaluation Factors

- Performance category tuned to match all channel components
- Performance characteristics
- Conductor size
- Termination (factory or field)
- PoE
- Compatibility with specified patch panels
- Combustion rating

Implementation Guidance

Horizontal cabling, including fiber and UTP, shall be installed in a structured topology with patch panels serving as at all vertical and horizontal transition points including Equipment Distributors (ED), Horizontal Distributors (HD), and throughout the MDA for all backbone levels. The patch panels at the HC and MDA shall be located adjacent to switching components to minimize patch cord length and cable management requirements.

There should be two simultaneous A/B redundant, diverse distribution paths.

Horizontal channel distance for UTP is limited to 295 ft with 32 ft reserved for patch cords.

Category 6A UTP is required for all new installations for clinical spaces to meet ANSI/TIA-1179 and 10 GbE performance objective.

Cat 6 UTP may be specified for installation for strictly administrative areas with an approved variance from DCIE (see *Appendix A*).

Retrofit/replacement decision points to meet minimum 1 GbE objective on legacy cable plant:

- Cat 5e replacement with Cat 6A strongly recommended
- Interim full testing of Cat 5e channels to verify 1 GbE performance.
- Cat 5 or below – replacement with Cat 6A required
- Failed Cat 5e permanent link test – Interim repair, replacement with Cat 6A required

Pre-termination requires that distances between termination points are known; however, it assures maximum performance and rapid installation. Only use field-terminations to WAOs as required where distances cannot be precisely calculated.

See Sections 4.4.8 and 5.2.1 on cable routing and implementation.

4.4.2 Unshielded Twisted Pair Patch Panel Standards

This Standard anticipates the bandwidth demands of virtualization by specifying Cat 6A UTP in production and test facilities. This Standard can be applied to field-terminated or factory pre-terminated applications. The preferred termination method is factory pre-terminated cabling for UTP. Cat 6A cabling is specified in standard ANSI/TIA-1179 Healthcare Facility Telecommunications Infrastructure for new installations.

Modularity of structured cabling components in the TRs, entrance rooms, and data centers improves capacity planning and move/add/change outcomes. This Standard specifies a one RU high, 24 or 48 position patch panel that is performance tuned to the UTP. Angled panels eliminate the need for horizontal cable management in the rack and are the typical configuration. No horizontal cable management is required when specifying angled panels.

Table 28: Unshielded Twisted Pair Patch Panel Standards

ID	Primary Attribute	Secondary Attribute	Specification
1	Performance Characteristics	Performance Category	• Cat 6A (10 GbE) • Cat 6 (exempted and authorized variance only)
		Position Count	24/48
		Form Factor	• One RU • Angled
		Color Coding	Black
		Components	• Angled panel cover • Rear cable manager

Evaluation Factors

- Performance category tuned to match all channel components
- Form factor
- Compatibility with pre-terminated cable interfaces

Implementation Guidance

Horizontal cabling, including fiber and UTP, shall be installed in a structured topology with patch panels serving as all vertical and horizontal transition points including EDs, HDs, and throughout the MDA for all backbone levels. The HC shall be located adjacent to switching components to minimize patch cord length and cable management requirements.

Where installed, the ED patch panels will be installed in the top five rear RUs of each enclosure or within small (five RUs or less) racks mounted on cable tray above the associated equipment cabinet.

4.4.3 Unshielded Twisted Pair Patch Cord

This Standard specifies center-tuned Cat 6A (or Cat 6 exempted and authorized variances only) patch cords. This Standard does not specify color-coding for patch cable jackets. Based on the observations of hundreds of VA facilities, color-coded patch cords provide little or no utility and create inventory management issues. Effective color coding has been observed for in-rack, dual-homed server/appliance applications using a simple white/black scheme. Patch cords shall be factory pre-terminated and shall match or exceed the category of the associated patch panel and connected UTP.

Table 29: Unshielded Twisted Pair Patch Cord Standards

ID	Primary Attribute	Secondary Attribute	Specification
1	Performance Characteristics	Performance Category	- Cat 6A (Cat 6 exempted and authorized variance only) - Stranded
		Jacket Color	Not specified
		Termination Method	Factory pre-terminated
		Conductor Size	22 AWG to 28 AWG

Evaluation Factors

- Performance category tuned to match all channel components
- Performance characteristics
- Center-tuned to installed patch panels
- Patch cords of shortest appropriate length used

Implementation Guidance

See the O&M Standard for Telecommunications Infrastructure (publication pending) on cable routing and implementation.

Use most appropriate length to meet cable management standards.

28 AWG patch cords are allowable but reduce overall horizontal channel distance.

Maximum individual patch cable length is limited to 10 ft to eliminate use of patch cords for horizontal or backbone connectivity.

ISP design best practices dictate that component-level compliance to a performance standard must be maintained across the entire horizontal link. The quality of patch cords and adherence to performance standards is critical to the reliable operation of a high-speed LAN.

4.4.4 Fiber Optic Cable

This Standard specifies laser-optimized Optical Multimode (OM) OM4- and OM5-rated multimode (MM) fiberoptic cable primarily for ISP use.

OM4 is primarily used to support single-pair 10 GbE, four-pair 40 GbE or ten-pair 100 GbE, access-layer to aggregation or core layers. Typical intra-building backbone between TRs and MDAs will consist of a minimum of two diversely routed 24-strand cable assemblies from both A and B-side MDAs and two 12-strand Single-mode (SM) cables to both A and B as required by the LAN baseline.

OM5 is designed to support Short Wavelength Division Multiplexing (SWDM) where switch transceivers can support 40 Gigabit per second (Gb/s) and 100 Gb/s on two fibers. OM5 is included in this Standard for specialized data center applications only.

In backbone distribution, SM OS1 12/24-strand fiber optic cable assemblies using Multi-fiber Push On (MPO) connectors are specified. Typical intra-building backbone between TRs and MDAs will consist of a minimum of two diversely routed 12-strand cable assemblies for path diversity from both A and B MDAs as required by the LAN baseline. However, if intra-building distances exceed the OM4 limit and only SM is used, then two 24-strand cable assemblies are required.

OS2 12/24-strand indoor/outdoor rated assemblies SM fiber optic cable is primarily used for OSP and long distance transmission for requirements exceeding the capabilities of MM fiber. Additionally, it is necessary for some special systems and for intra-building as well as inter-building connectivity. Typical inter-building backbone will consist of a minimum of two diversely routed 24-strand cable assemblies from both A and B MDAs as required by the LAN baseline. Where separate buildings are serviced with SM fiber, each TR in that building requires the minimum redundant SM cable distribution.

Pre-termination requires that distances between termination points are known; however, it assures maximum performance and rapid installation.

If fiber media is required for horizontal distribution to WAO, a structured approach is required. A stated need for fiber to work WAOs should be scrutinized by the planner and justified by sound technical requirements.

Table 30: Fiber Optic Cable Standards

ID	Primary Attribute	Secondary Attribute	Specification
1	Multimode	Performance Category	- OM4 (horizontal and backbone) - OM5 (specialized data center applications)
		Performance Specifications	Laser Optimized 50/125 μ m fibers with minimal effective modal bandwidth of 4,700 MHz·km at 850 nm
		Combustion Rating	- Communications Riser (CMR) rated for vertical runs through floors - Communications Plenum (CMP) rated for plenum spaces

ID	Primary Attribute	Secondary Attribute	Specification
			- Communications (CM) minimum jacket rating suitable when riser or plenum are not required - Or as per AHJ requirement
		Jacket Color	- Aqua (OM4) - Lime (OM5)
		Termination Method	Factory pre-terminated
		Media Connector	Pre-terminated with MPO
		Strand Count	12 or 24 per assembly
		Bundling	Dielectric, Tight Buffered
		Polarity	- Straight (or Type A-Key up one end & key down on the other) - Type B with “universal” cassettes
2	Single Mode	Performance Category	- OS1 (vertical backbone) - OS2 (OSP backbone)
		Performance Specifications	Laser Optimized 9/125 μm with effective modal bandwidth of at least 850 MHz·km at 1310 nm
		Combustion Rating	- Riser cable for vertical runs through floors - Plenum rated for plenum spaces - Tight-buffered OS1 for ISP - Loose-tube gel-filled OS2 riser rated for OSP indoor/outdoor transition - Or as per AHJ requirement
		Jacket Color	Yellow
		Termination Method	Factory pre-terminated
		Media Connector	Pre-terminated with MPO
		Strand Count	12 or 24 per assembly
		Bundling	Dielectric - OS1- Tight Buffered - OS2- Loose-tube gel-filled
		Polarity	- Straight (or Type A-Key up one end & key down on the other) - Type B with “universal” cassettes

Evaluation Factors

- Warranty
- Performance category tuned to match all channel components
- OM4 performance characteristics for up to 10 GbE intra-building backbone
- OM5 performance characteristics for data center distribution for specialized application
- OS1 performance characteristics for vertical backbone
- OS2 performance characteristics for inter-building backbone and OSP

- Combustion rating requirements
- Technical requirements for inclusion in horizontal distribution to the WAO

Implementation Guidance

Dual 24-strand OM4 50/125 MM and dual 12-strand OS1 per TR with path diversity will support most small/medium capacity distribution needs. Some larger TRs may require additional fiber counts.

Pre-termination requires that distances between termination points are known; however, it assures maximum performance and rapid installation. This Standard specifies the polarity of horizontal pre-terminated MPO-based fiber optic cabling as Type A or “straight-through” for LC cassette breakout. This will ensure that position one will correspond to position one on both ends of the cable. If a polarity flip for the channel is necessary, use a Type A/A patch cable on one end and a Type A/B patch cable on the other end. In some cases, an A/B cord can be converted to an A/A by switching the connector position on one end of the patch cord. Alternatively, Type B polarity can be used with “universal cassettes.”

There should be two simultaneous A/B redundant, diverse distribution paths minimally to all telecommunications rooms in clinical spaces. Diverse path routing is also required for the service provider circuits to diverse entrance rooms and within the data center between the equipment distributors and HDAs, and between the HDAs and the MDAs. Path diversity to administrative spaces should have diverse distribution paths wherever possible. Where not possible, media should not be in the same conduit.

MPO-8 cassettes for SR4 should be planned directly with the distribution designer or DCIE.

See Sections 4.4.8 and 5.2.1 on cable routing and implementation.

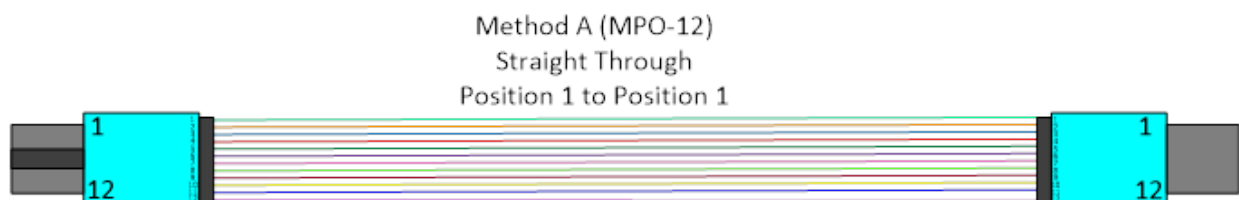


Figure 23: Method A Polarity of Horizontal Pre-terminated MPO-based Fiber Optic Cabling

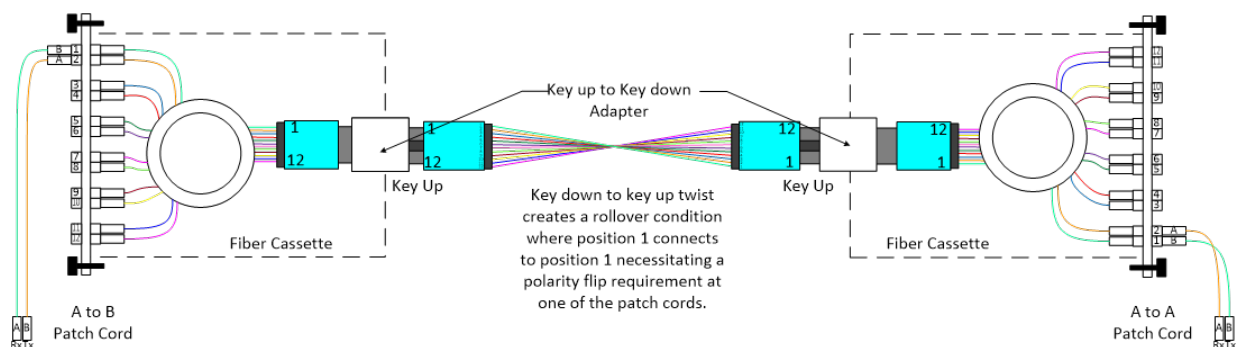


Figure 24: Method A Polarity End to End Connectivity (A-B and A-A Patch Cords Required)

4.4.5 Fiber Distribution Cassettes

This Standard specifies the transition component of the structured cabling system for fiber optic cabling in telecommunication spaces. It is the fiber equivalent to UTP patch panels. This Standard specifies 12/24-strand, OM4- or OM5-rated MM cassettes for ISP and use SM cassettes for both OS1 and OS2 applications. The cassette will be equipped with dual, 12-strand or single 24-strand MPO interfaces as the backbone or horizontal interface.

Table 31: Fiber Distribution Cassettes

ID	Primary Attribute	Secondary Attribute	Specification
1	Performance Characteristics	Form Factor	One RU
		Capacity	3-12 cassettes
		User Interfaces	LC connectors (6/12 duplex per cassette) MPO connectors (8 fibers)
		Backbone Interfaces	12-strand or 24-strand MPO (50 micron)
		Type	OM4 Laser Enhanced 10 GbE or OM5 Laser Enhanced 50/125 Multimode to match media

Evaluation Factors

- Performance category tuned to match all channel components
- Performance characteristics
- Form factor
- Interfaces

Implementation Guidance

This Standard specifies Optical Multimode-4 and -5 (OM4 and OM5 rated 850 nm laser-enhanced multi-cable fiber cable assemblies pre-terminated with MPO connectors. One cassette will be used to provide LC interfaces at both ends of the horizontal or backbone link and will support one or two 12-strand cable assemblies or one 24-strand cable assembly for a total capacity of 6 or 12 duplex LC interfaces. Cassettes can also be used to breakout 24-strand MPO backbone trunks into three MPO connections of eight fiber strands.

4.4.6 Fiber Patch Cords

This Standard specifies high-bandwidth, factory pre-terminated fiber patch cords. The standard media interface for fiber patch cords is duplex LC. Patch cord polarity requirements can be accommodated by specifying a polarity-reversible duplex interface or by using universal cassettes. When polarity requirements are not known, the polarity-reversible duplex LC interface should be specified.

Table 32: Fiber Patch Cord Standards

ID	Primary Attribute	Secondary Attribute	Specification
1	Performance Characteristics	Performance Category	• OM4 • OM5 (used only for specific data center applications (see <i>Section 4.4.4</i>))
		Performance Specifications	Laser enhanced, 50/125 MM fiber, Duplex LC

ID	Primary Attribute	Secondary Attribute	Specification
		Jacket Color	- Aqua (OM4) - Lime (OM5)
		Termination Method	Factory pre-terminated
		Polarity	- Type A/A (Straight) or - Type A/B (Crossover) as required

Evaluation Factors

- Performance category tuned to match all channel components
- Polarity reversibility
- Performance specifications

Implementation Guidance

ISP design best practices dictate that component-level compliance to a performance standard must be maintained across the entire horizontal link. The quality of patch cords and adherence to performance standards is critical to the reliable operation of a high-speed LAN or SAN.

The polarity of the fiber optic channel may need to be flipped depending on the application. This will ensure that the equipment sending or transmitting will be connected to the receiving port of the equipment at the other end. If typical A/B fiber patch cords are used on both sides of the channel, continuity may be lost. As shown in *Figure 25*, this is corrected by either switching the connector position of both strands at one end of the patch cord or by using a Type A/A patch cable on only one end of the channel.

Pre-termination requires that distances between termination points are known; however, it assures maximum performance and rapid installation.

See Sections 4.4.8 and 5.2.1 on cable routing and implementation.

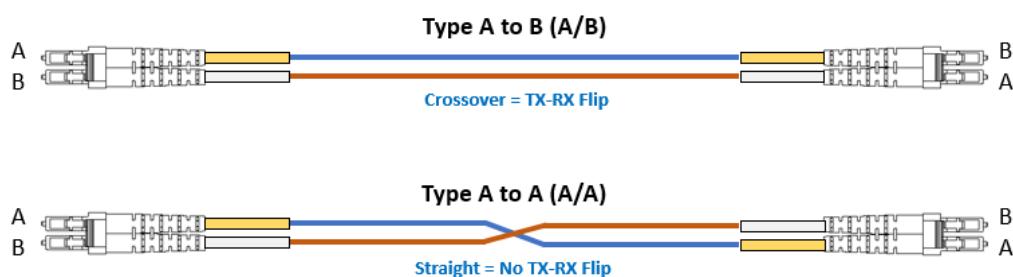


Figure 25: Polarity of Fiber Optic Patch Cable

4.4.7 Fiber Distribution Panel/Cabinet

This Standard specifies a one RU panel for mounting fiber cassettes. The smallest form-factor has been selected for application flexibility.

Table 33: Fiber Distribution Panel/Cabinet Standards

ID	Primary Attribute	Secondary Attribute	Specification
1	Performance Characteristics	Form Factor	1 RU
		Cassette Capacity	3-12 cassettes
		Type	- Cabinet - Panel

Evaluation Factors

- Form factor
- Cassette capacity
- Locking requirements

Implementation Guidance

ED Fiber Distribution Panels will be installed in RU45 of each VED cabinet where required.

4.4.8 Cable Support Infrastructure

This Standard creates uniform high-capacity horizontal and vertical cable management for Cat 6A patch cords.

Table 34: Cable Support Infrastructure Standards

ID	Primary Attribute	Secondary Attribute	Specification
1	Horizontal Cable Management Panels	Form Factor	- One RU (up to 50 Cat 6A patch cords) - Two RU (up to 100 Cat 6A patch cords)
		Cover	Double-hinged
		Finger Spacing	To meet requirements; typically, 6-port spacing (five fingers)
2	Vertical Cable Management Panels	Door	Double-hinged
		Dimensions	6 in. to 16 in.

Evaluation Factors

- RU Finger spacing
- Double-hinged cover capacity

Implementation Guidance

Vertical cable management will be utilized for the HDA and the MDA. Horizontal cable management within the rack will be utilized as necessary to encourage defined and traceable routing of individual patch cords from switches to the vertical cable managers.

4.4.8.1 Cable Distribution

This Standard offers consistency in the construction characteristics of overhead cable tray to achieve common mounting, hanging, support, bonding, and interconnecting features for the benefit of data center and ITS designers and end-users.

Network cable distribution should be distributed in overhead cable trays in all new installations when overhead space allows. Existing underfloor cable trays will be phased out during major upgrades to existing data centers and ITS.

Table 35: Cable Distribution Standards

ID	Primary Attribute	Secondary Attribute	Specification
1	Cable Tray	Design	Wire Basket
		Capacity	- Maximum fill ratio not to exceed 50 % - Depth of telecommunications cables not to exceed 6 in.
		Bonding Kit	Mechanically bond cable pathway sections and trays to one another and to the Supplemental Bonding Network in accordance with ANSI/TIA-607-D Generic Telecommunications Bonding and Grounding (Earthing) for Customer Premises
		Clearance	2 in. from top of cable tray to bottom of access floor in legacy underfloor installation 6 in. clearance between cable trays in stacked overhead installations - Media types will be segregated
		Installation	Per manufacturer's recommendations

Evaluation Factors

- Wire-basket construction
- Toolless joining kits
- ASTM coating

Implementation Guidance

Cable pathways shall meet the clearance requirements of fire detection, suppression, and prevention systems, and these systems must be coordinated with other systems (e.g., electrical, mechanical, telecommunications) and meet the requirements of the manufacturer and the AHJ.

If both power and telecommunications cabling are distributed from below the access floor, then:

- The power cabling should be routed either adjacent to or within the cold aisle
- The telecommunications cabling should be routed adjacent to or within the hot aisle
- Power and communication pathways should be positioned at different heights off the floor so that they can cross each other without interference. Alternatively, at every point

where the power and copper cabling cross the path of each other, the crossing should be at a right (90 degree) angle

4.4.9 Work Area Outlets

Work Area Outlets (WAO) are user interfaces to the cabling system within a workspace, floor, or building. This Standard specifies the *typical* quantity and type of faceplate, wall box, conduit, and outlets for *typical* administrative work areas. However, this guidance does not preclude variations based on atypical capacity requirements nor does it supersede guidance or requirements set forth by CFM concerning WAO specifications for specialized spaces.

Typical in-wall workboxes shall be a single-gang telecommunications-rated work box with rigid conduit extending to the horizontal pathway.

Floor boxes shall be single-gang or “poke-thru” type. Cable that extends through a slab penetration shall terminate in the nearest practicable TR on the same serving floor.

The typical standard density WAO will consist of two RJ45 interfaces. This provides connectivity for one IP telephone and one workstation. Some IP phone base units have a data port. This allows the workstation to interface with the network via the phone base unit to leave one RJ45 at the faceplate available for future use.

WAO media interfaces shall be tuned to the horizontal media, specified to the same performance rating, and installed according to best practices.

No “phone outlets” are specified in this Standard. A second gang with outlets for analog telephony purposes are acceptable.

Single-outlet posted faceplates shall be configured with one RJ45 interface. The high-density WAO configuration consists of four RJ45 interfaces. This configuration is typically used for desktop support and other technical staff or in cases where multiple devices need network access from the same WAO.

Evaluation Factors

- Serving zone capacity requirements
- Special safety or medical device requirements

4.4.10 Horizontal Distribution

Horizontal distribution is defined as structured cabling that connects the WAO or the ED to a HD. The HD will be located in the end of row HDA in a computer room or in a TR depending on the application.

Horizontal media shall be specified per *Section 4.4.1*. If the application, such as a data center, requires fiber, specify the media per *Section 4.4.4*.

4.4.11 Backbone Distribution

Backbone distribution is defined as structured cabling that connects the HD to the Main Distributor (MD). The MDA will be located in a centralized location within the building, data center, or LAN serving area. Typically, this will be in the computer room.

The A-side and B-side MDAs shall be a separate and distinct telecommunications spaces within the LAN serving area. It is preferred that the MDAs are built in separate locations, but they can be combined into one space. The MDAs shall not be located in a TR.

This Standard only recognizes 50/125 Laser-Enhanced Multimode (OM4 and OM5) and 9/125 Single Mode (OS1 and OS2) as acceptable backbone media (see *Section 4.4.4*).

UTP shall be utilized only if fiber media cannot serve specialized applications such as legacy phones, fax machines, or security panels. Where copper backbone is required, this Standard specifies quantity two riser-rated Cat 5e, 25-pair cables to be installed between the TR HDA and the MDA. This approach will support 12 patch panel positions. One 12-position Cat 6 patch panel shall be installed in the HDA and the MDA. Individual 4-pair Cat 6 riser-rated cables may be used, but this approach is not recommended and will require technical review by DCIE.

4.4.11.1 Intra-building

4.4.11.1.1 Multimode

- Laser Optimized 50/125 OM4 required for new installations to meet single-pair 10 GbE objective
- Retrofit/replacement decision points:
 - Laser Optimized 50/125 (OM3 is exempted; OM4 for all tech refresh projects required)
 - 62.5 core (OM1) – replace
 - Non-optimized (OM2) – replace

4.4.11.1.2 Single Mode

- 9/125 OS1 MPO
- SM may be required for backbone distribution in support of specialized applications, such as DAS.
- Cassette LC, LC/Angled Physical Contact (APC) or as required by application.

4.4.11.2 Inter-building

Inter-building backbone shall not transition at an intermediate cross-connect. Therefore, the MDA will connect directly to each TR in all campus buildings outside of the computer room building.

For all inter-building connectivity, 9/125 indoor/outdoor rated OS2 SM fiber will be utilized. when paths are outside of MM range and not connected by environmentally protected conditioned spaces.

4.4.12 Server Cabinets

These specifications define a standardized server enclosure with sufficient depth to accommodate active hardware and cable management.

Table 36: Server Cabinet Standards

ID	Primary Attribute	Secondary Attribute	Specification
1	Dimensions	Height	7.5 ft maximum
		Width	24 in. 30 in. permitted in groups of four to maintain alignment with floor tiles in raised floor environments only
		Depth	48 in. nominal with all doors and accessories installed
		RU	45
2	Design	Rails	- Square-punched (two pair = front + rear) - Toolless adjustable
		Rack Marking	Present on front and rear rails graduated starting with RU1 from the bottom
		Static Capacity	2,500 lbs minimum
		Color	White or existing match
		Bonding Connection	Doors, frame, and side, top, and bottom panels
		Green Technology	Heat containment per data center cooling architecture
3	Panels	Front Door	Single perforated (minimum 63 % open)
		Rear Door	- Single or split solid when VEDs are used, or - Split, perforated where VEDs cannot be implemented
		Latches	Keyed lock upgradable to keyless system compression latch
		Top	- VEDs (heat containment) required in facilities equipped with overhead return air ducting systems - High-capacity cable access w/brushes
		Side	- Solid - Locking

ID	Primary Attribute	Secondary Attribute	Specification
		Bottom	• Solid • With high-capacity cable access w/ brush grommets or air dam foam in raised access floor locations
4	Required Accessories	Air Dam and Sealing Kit	Required under each cabinet and between rails and cabinet sides
		Castors	Required for safe cabinet movement
		Leveling Legs	Required
		Baying Kit	Required to keep cabinets aligned
		Grounding	• RBC • Bonded in accordance with ANSI/TIA-607-D Generic Telecommunications Bonding and Grounding (Earthing) for Customer Premises • Tagged with “Do Not Disconnect” • See <i>Section 4.2.8</i> for specific requirements and guidance
		rPDU brackets	• Zero U vertical single mount brackets for SD cabinets • Double mount brackets for HD cabinets

Evaluation Factors

- Dimensions
- VEDs/heat containment
- Available RUs
- Grounding equipment and bonding

Implementation Guidance

This Standard specifies green technology in the form of heat containment. This approach to thermal management does have corresponding data center design requirements and, therefore, should be selected based on the cooling architecture of the target data center.

Server cabinets with VEDs shall be specified and provided for deployment to computer rooms with overhead return air ducting systems installed.

Enclosures shall be equipped with seismic bracing when deployed in areas prone to earthquakes or as required by Federal regulation or other AHJs and best practices.

No networking equipment can be installed in server cabinets. This includes fabric interconnects, fabric extenders, as well as other traditional networking equipment.

White server cabinets are specified for reduced energy absorption and increased reflectivity for improved visibility. Black servers, switches, patch panels, blanking panels, and other equipment makes it difficult for technicians to work within a dark cabinet due to the low light levels.

Cabinet height of 7.5 ft maximum is required to leave room for structured cabling, power busbars, containment systems, fires suppression, etc. A larger cabinet does not provide an advantage because the limiting load factor is the power and heat rejection capability of the cooling systems, not rack space. A taller cabinet also makes the equipment difficult to access without a ladder.

4.4.13 Network Equipment Racks

Table 37: Network Equipment Racks

ID	Primary Attribute	Secondary Attribute	Specification
1	Dimensions	Height	7.5 ft maximum
		Width	24 in. nominal minimum
		Depth	30 in. minimum
		Rail Style (Front/Rear)	- EIA threaded or - Square punched
		RU	45
2	Design	Style	Channel rack
		Static Capacity	2,000 lbs minimum
		Cable Management	Built-in overhead waterfall and cable management strap attachment points
		Rail Marking	Present on front and rear rails graduated starting RU number one at the bottom-most position
		Color	White or existing match
3	Required Accessories	Grounding	- Rack Bonding Busbar (RBB) - Bonded in accordance with ANSI/TIA-607-D Generic Telecommunications Bonding and Grounding (Earthing) for Customer Premises - Tagged with "Do Not Disconnect" See <i>Section 4.2.8</i> for specific requirements and guidance
		rPDU	rPDU brackets
		Side Air Intake Scoops	Required for racks with side to side airflow equipment

Evaluation Factors

- Dimensions
- Cable management capacities

- Available accessories for horizontal and vertical cable management
- Grounding equipment and bonding
- Static capacity
- Locking front or rear doors
- Side air intake scoops

Implementation Guidance

Racks shall have proper seismic rating when deployed in areas prone to earthquakes or as required by Federal regulation or other AHJs and best practices.

Racks and network equipment cabinets shall be used for all networking gear to include the MDA and the HDA.

Racks are mandatory for use in TRs.

When racks are used, this Standard specifies a deep channel open rack to accommodate the depth and weight of most data networking equipment while providing high capacities for horizontal cable and large diameter Cat 6A patch cords.

White racks are specified for reduced energy absorption and increased reflectivity for improved visibility. Black networking equipment, patch panels, blanking panels, and other equipment makes it difficult for technicians to work within a dark rack due to the low light levels.

Server cabinets shall not be used in place of network racks due to inherent cable management limitations. Specifically designed network cabinets greater than 30 in. wide may be used in place of network racks if the proper cable management accessories are installed.

4.4.14 Network Equipment Cabinets

Table 38: Network Equipment Cabinet Standards

ID	Primary Attribute	Secondary Attribute	Specification
1	Dimensions	Height	7.5 ft maximum
		Width	• 30 in. minimum • 40 in. maximum and preferred
		Depth	48 in nominal with all doors and accessories installed
		Rack Unit	45
2	Design	Rails	• Square-punched (2 pair = front + rear) • Toolless adjustable
		Rack Marking	Present on front and rear rails graduated starting one RU from the bottom
		Static Capacity	2,500 lbs minimum
		Color	White or existing match

ID	Primary Attribute	Secondary Attribute	Specification
		Bonding Connection	Doors and frame
		Green Technology	Heat containment per data center cooling architecture
3	Panels	Front Door	Single or double perforated (minimum 63 % open)
		Rear Door	- Split solid when VEDs are used, or - Split, perforated where VEDs cannot be implemented
		Latches	Keyed lock upgradable to keyless system compression latch
		Top	- VEDs (heat containment) required in facilities equipped with overhead return air ducting systems - High-capacity cable access w/brushes
		Side	- Solid with brush sealed cable openings to allow for horizontal distribution - Vertical cable management - Locking
		Bottom	- Solid - With high-capacity cable access w/ brush grommets or air dam foam in raised floor environments
4	Required Accessories	Air Dam and Sealing Kit	Required under each cabinet and between rails and cabinet sides
		Castors	Required for safe cabinet movement
		Leveling Legs	Required
		rPDU brackets	Zero U vertical single mount brackets
		Grounding	- RBC bonded in accordance with ANSI/TIA-607-D Generic Telecommunications Bonding and Grounding (Earthing) for Customer Premises - Tagged with “Do Not Disconnect” - See <i>Section 4.2.8</i> for specific requirements and guidance
		Baying Kits	Required to keep cabinets aligned

Evaluation Factors

- Dimensions
- VEDs/heat containment
- Available RUs

- Grounding equipment and bonding

Implementation Guidance

This Standard specifies green technology in the form of heat containment. This approach to thermal management does have corresponding data center design requirements and, therefore, should be selected based on the cooling architecture of the target data center.

Network equipment cabinets with VEDs shall be specified and provided for deployment to computer rooms with overhead return air ducting systems installed.

Enclosures shall be equipped with seismic bracing when deployed in areas prone to earthquakes or as required by Federal regulation or other AHJs and best practices.

White network cabinets are specified for reduced energy absorption and increased reflectivity for improved visibility. Black networking equipment, patch panels, blanking panels, and other equipment makes it difficult for technicians to work within a dark network cabinet due to the low light levels.

Cabinet height of 7.5 ft maximum is required to leave room for structured cabling, power busbars, containment systems, fires suppression, etc. A taller cabinet does not provide an advantage because the limiting load factor is the power and heat rejection capability of the cooling systems, not rack space. A taller cabinet also makes the equipment difficult to access without a ladder.

5 Administration Standards

These specifications are intended to allow complete and consistent identification and physical location information for VA data center and computing facilities, physical plant and distribution equipment supporting VA data center facilities, equipment located within the VA data center environment, and the component types and elements described for use in physical and logical information management systems describing the connectivity of physical elements present in the data center environment.

Identification specifications are based on the current version of ANSI/TIA-606-C, with modifications for the VA operating environment.

5.1 Data Center Position Identification

The purpose of this specification is to establish guidance for identifying physical locations for equipment and components in the VA data center environment. This specification contains instruction for uniform identification of equipment and component locations used at telecommunications spaces across the enterprise.

Specifically, because of the legacy nature of VA facilities, no single facility location system is implemented enterprise-wide. Individual VA locations (i.e., VAMCs) have developed and implemented individual location identification methodologies that are beyond the scope of this document. Telecommunications spaces within VA facilities shall follow the location identification guidance herein.

Table 39: Data center Position Identification Conventions

ID	Primary Attribute	Secondary Attribute	Specification
1	Data Center Floor Grid Location Identification	Alignment	Aligned with plan North in building drawings
		Location Origin	Northwest corner
		Location Increments	- Two-foot increments with alphanumeric identification - Begin at the appropriate designator such as A1 for first full floor tile, depending on number of tiles in the data center - Increase letters from left to right - Increase numbers from top to bottom
		Variance/Waivers	Required but granted automatically if existing (legacy) system is compatible

Evaluation Factors

Each data center uses a compliant grid identification scheme allowing detailed location identification of physical equipment.

Implementation Guidance

Data center layouts shall be master-planned. Equipment layouts can be adequately defined by location in the data center master floor plan. This location identification scheme shall be

applicable to all enterprise data centers, existing and new. Individual facilities shall apply for a variance if there is an existing location schema differing from the one stated. Variance for compatible legacy systems will be approved automatically in the DCIM system of record for facilities with an approved variance. A characteristic watermark or similar shall be assigned to the plan view to indicate a non-standard layout.

While some buildings are built to align with true north, many are not – but the architectural plans for all facilities will have a “Plan North,” aligned with the top of the drawing. This specification assumes that data center access floor grids will align with Plan North.

The ANSI/TIA-606-C standard initiates the computer room location system at the top left (plan NW) corner of the computing floor space. The access floor grid (where used) shall be alphanumeric with tile location A1 (*depending on data center size; see below*) denoting the (plan) northwestern-most full tile on the floor. For grid location identification purposes, beginning with alphanumeric tile location A01, letters shall increase across the X-axis (left to right) from A to Z, or from AA to AZ, then BA to BZ, and so on. Numbers shall increase down the Y-axis (top to bottom) from 1, or 01, onward (spreadsheet orientation). Representative examples of grid identification are shown below for reference.

If a facility has more than 26 tiles on the x-axis, use AA as the starting location (rather than A) to allow sorting in database tools. For example, AA through AZ, then BA through BZ – do not start with A through Z in these larger facilities. If the data center has:

- Fewer than 10 tiles on the Y-axis, then use “A1-A9.”
- 10 or more tiles on the Y-axis, then use “A01-A99.”
- 100 or more tiles on the Y-axis, then use “A001-A999.”
- The same follows for the X-axis where “A1-Z1” are used for 26 tiles or fewer in the Y direction, and “AA01-ZZ01” are used for data centers with over 26 tiles in the Y direction.

Where an irregular arrangement of the facility means that there is no tile in what would typically be the A1 position, label floor tiles from this location using a two ft increment. This system is of use in designating equipment locations for non-standard raised floor items, including CRAC units and larger storage or mainframe units. Note that there may be partial tiles around the first full tile(s) in any row or column that are not numbered in this system.

In data centers operated without an access floor grid, apply stenciled grid cross-hairs at 24 in. centers to the floor (where the cross-hair locations are accessible). The cross-hairs should extend in each cardinal direction at least 4 in. and 4 in. on each axis). Note that the grid need not begin in the physical corner of the room and may be moved a simulated ‘partial tile’ in to better align with installed equipment.

	A	B	C	D	E	F	G	H	I	J
01										
02										
03			RACK B04	RACK D04	RACK E04	RACK F04	RACK G04	RACK H04		
04										
05										
06										
07			RACK C07	RACK E07	RACK F07	RACK G07	RACK H07			
08										
09										
10										

Grid Identification– Typical Small Computer Room

					E	F	G	H	I	J
01										
02										
03							RACK G02			
04							RACK G04			
05							RACK G05			
06							RACK G06			
07							RACK G07			
08			RACK D08				RACK G08			
09										
10										

Grid Identification– Offset Start Small Computer Room

Figure 26: Typical Position Identification Examples

Identification of individual computer spaces within a single building depends upon the configuration of those spaces. Where access floor spaces are adjacent and use the same grid (i.e., rooms separated by partition walls for security or other isolation purposes), both spaces shall use the same grid beginning with the A01 tile in the northwestern-most space. Where access floor spaces are separated and/or are based on separate grids (i.e., one room in one portion of a building and the other in a different portion), each space shall use individual grids, each starting with its own A01 tile.

Grid identification within each data center space shall be posted to allow occupants to understand their current location. Post location identifiers on walls, columns, and/or from hanging tags as necessary to allow occupants to reasonably determine all grid locations. Identifiers shall be a minimum of 4 in. high characters, semi-gloss black lettering, with top of character not more than 6 in. below ceiling, concrete slab, truss line or typically accepted physical divider between workspace and ceiling line.

Items on the floor shall be noted as being located at locations in accordance with ANSI/TIA-606-C, meaning the grid location of the tile containing the left front corner of the item (when facing the front of the item). If the item has no identifiable front, the first full tile covered closest to the grid origin shall be the location identifier.

5.1.1 Cross-Reference to Other Identification Systems

Nothing in the DCIM specification is designed or intended to conflict with any other mandatory standard(s) affecting the VA data center environment.

VA information management tools such as DCIM, BAS, and Computerized Maintenance Management Systems (CMMS) shall include database fields (where applicable) to identify the standardized location identifier (and other location identifiers) as specified herein. These fields are not intended to supplant or replace other identification numbers already in use by other systems.

5.1.2 Standardized Data Center Facility Type Identification

Data centers across VA generally support the local mission and are tailored to the mission of the local office. This leads to multiple types of (and names for) the facilities, which must be standardized to ensure that the intent of using a facility's name is clearly understood.

The list of FACILITYTYPE attributes is intended to align with (but will not be replaced by) the data center classification schema described in *Section 2* of The Standard as VA undergoes further transformation and data center consolidation activities.

Evaluation Factors

- Each data center is classified as a single FACILITYTYPE.
- All data centers are classified as a FACILITYTYPE on the current FACILITYTYPE list.
- Use of FACILITYTYPE is consistent in OIT-standardized data structures.

Implementation Guidance

The FACILITYTYPE attribute describes at a high-level what type of facility the data center supports, the specific mission of the facility, or what type of information is processed by the data center.

The FACILITYTYPE list is maintained by NDCOL and is anticipated to be simplified as the enterprise completes transformative data center consolidation activities.

Classify all data center facilities as exactly one of the facility types in the list below. Computer room facilities in VAMCs, whether OIT, biomedical, facility management service, R&D, or otherwise, are classified as VAMCs in the FACILITYTYPE list.

FACILITYTYPE names shall only consist of three or four alpha characters and not contain ampersands, numbers, or other special characters to prevent issues with DCIM systems or other database queries.

Table 40: Standard FACILITYTYPE Naming Conventions

FACILITYTYPE	Description	Classification Archetype
VAMC	VA Medical Center	Campus Support Center (CSC)*
OPC	Large Outpatient Clinics	
CBOC	Large Community-Based Outpatient Clinics	
HCC	Health Care Center	
CMOP	Consolidated Mail Outpatient Pharmacy	Mission Support Center (MSC)
CPAC	Consolidated Patient Account Center	
HAC	Health Administration Center	
OIFO	Office of Information Field Office	
HEC	Health Eligibility Center	
FSC	Financial Services Center	
ARC	Allocation Resource Center	
NCPS	National Center for Patient Safety	
PBM	Pharmacy Benefits Management Office	
ALC	Acquisition Logistics Center	
CSP	Cooperative Studies Program	
HSRD	Health Services Research & Development Center	
NSC	Network Support Center (generic, if no other designation is appropriate)	Network Support Center (NSC)
CBOC	Typical size Community-Based Outpatient Clinics (CBOC)	
VARO	VBA Regional Office	
VRE	Vocational Rehabilitation & Employment Center	
RMC	Records Management Center	

** There are some specific instances of overlap between CDCs and VAMC facilities (e.g. Cleveland, Temple) as the data centers in these locations operate both as CSCs and also provide or are designed and built to meet the intent of operation as a CDCs.*

5.1.3 Standardized Data Center Naming Convention

Facility legacy and common names are not standardized and do not yield sufficient clarity on where an OIT-operated or OIT-supported computer or telecom space is located, what its function is, or what the intended mission supported by the facility or space is, in the context of VA's data center strategy.

Evaluation Factors

- Each data center is assigned a single standardized data center name.
- Use of CITYNAME is consistent in OIT-standardized data structures for all data center facilities.
- Use of FACILITYTYPE is consistent in OIT-standardized data structures for all data center facilities.

Implementation Guidance

The basic Standardized Data Center Name is formatted as [CITYNAME FACILITYTYPE]. The master naming list for Standardized Data Center Names shall be maintained by NDCOL.

The CITYNAME attribute is intended to be the physical municipality in which the site or facility is primarily located which is unique from other municipality names in use and is the most discrete and descriptive. This field is qualitative depending on the mission and function of the type of VA facility; for example, the new VAMC supporting the Denver, CO metropolitan area is actually located in Aurora, CO, but the CITYNAME will be Denver.

Although the VAMC supporting Augusta, ME, is located in Togus, ME; because Augusta, GA is the more commonly known metropolitan area, the Georgia facility will be assigned CITYNAME Augusta, while the Maine facility will be assigned CITYNAME Togus.

The FACILITYTYPE attribute describes at a high-level what type of facility the data center supports, the specific mission of the facility, or what type of information is processed by the data center. The available types align with the data center classification archetypes. As classes of facilities are simplified and collapsed through data center consolidation efforts, this list may be revised.

A site or facility will be referred to by its "standard name" where possible. Where there is more than one reportable computer room at a single site or facility, both computer rooms share the same "standard name" and are differentiated using its OMB assigned DCOI facility ID number. This naming convention includes the ability to identify all OIT computer support spaces (computer rooms, TRs, non-OIT computer rooms, and similar) uniquely, drilling down from the site or facility 'standard name' with additional record fields as described. Due to the wide variety of legacy configurations at individual facilities, additional detailed guidance on how to linguistically identify these individual spaces is not provided.

Example: The fictional VAMC in Timbuktu is classified as a Campus Support Center for the hospital campus; its standardized data center name is Timbuktu CSC. There are other IT support spaces on the campus supported by this CSC, including a biomed server room that is determined to not be able to be consolidated to the main data center, two main TRs servicing remote buildings on campus, auxiliary TRs servicing other floors in the remote buildings, and auxiliary TRs servicing different wings and floors of the main hospital building. All these spaces are considered hierarchically under (attached to) the Timbuktu CSC. They are specifically identified by building number and room number; for example, Timbuktu CSC, Bldg 3, Rm 3-102.

Not all computing support spaces will be assigned a Standardized Data Center Name. All computing support spaces, from computer rooms to TRs to non-reportable rooms containing IT equipment that requires IT or physical environment management, shall be assigned a VA Record ID number, separate from the DCOI-assigned facility ID number. The VA Record ID numbers shall be of the format “VA-XXXXX,” where XXXXX is a unique five-digit number. VA Record ID numbers are not intended to be sequential, either at time of assignment or in the future. ID numbers are assigned at the enterprise data warehouse level by NDCOL and distributed by the NDCOL DCIM team.

Example: Timbuktu CSC is assigned DCOI facility ID number DCOI-99999. The VA Record ID number for the reportable data center is VA-12345, and the TRs supporting two-story remote building 3 are VA-34201 and VA-58456. The TRs, -34201 and -58456, are supported by the reportable data center but are not assigned a DCOI facility ID number.

Example: A fictional National Cemetery has a computing support space, operating a router connecting the facility with the VA wide area network. NCA spaces do not operate data centers. This facility will not be assigned a Standardized Data Center Name or DCOI facility ID number but will be assigned a VA Record ID number, VA-56789.

Where a computing support space is directly supported by a data center with a DCOI facility ID number, include record field information that the space is supported by that data center, indicating the hierarchical arrangement. Not all computing support spaces are located in facilities with a data center, or directly supported by a facility subject to the standardized naming convention. Assign these spaces a VA Record ID number; they will be further identified by their physical address and having blank record field information where the standard name and DCOI facility ID number of the supporting facility would otherwise be recorded.

The legacy (or official) site or facility name shall be retained and record fields available in all databases and systems to cross-reference the standardized name with the legacy name. An example of a legacy name that may not be sufficiently descriptive is “John J Pershing VAMC.” Without additional information, the location of this facility could be in doubt.

5.1.4 Color Coded Identification

Table 41: Color Identification Conventions

ID	Primary Attribute	Secondary Attribute	Specification
1	Color Identification	Backbone Telecommunications Cabling	See <i>Table 27</i> : Unshielded Twisted Pair Standards
		Telecommunications Patch Cabling	Not specified, follow local site schema
		Power Distribution	Colors shall be black on A Side and other distinguishable color on B Side to differentiate electrical bus power sources

5.1.4.1 Data Communication Cabling Color Coding

The Standard does not specify an enterprise color scheme for UTP patch cords. Color coding scheme for patch cords should be avoided, however; individual site schemas are acceptable if consistent and documented. Horizontal and backbone cabling is color coded per *Table 27* and *Table 29*.

5.1.4.2 Power Distribution Color Coding

All redundant power distribution within the computing spaces shall incorporate specific color identification to differentiate between different electrical bus power types. Use of specific color identification for each electrical bus shall stay consistent throughout building distribution. No enterprise color specification is prescribed. A suggested strategy is to use blue Liquid-tight Flexible Metal Conduit (LFMC) conduit for A-bus circuits and grey LFMC conduit for B-bus circuits. Use of different label colors to differentiate electrical bus power is an acceptable solution.

Within IT equipment racks/cabinets, separate A-side and B-side power and segregate equipment power cords (between vertical rack power distribution units and IT equipment) by color (black on A Side and other distinguishable color on B Side) for identification of A/B power to each piece of IT equipment (see example below). Follow best practices for managing the power cords similarly to other cabling.

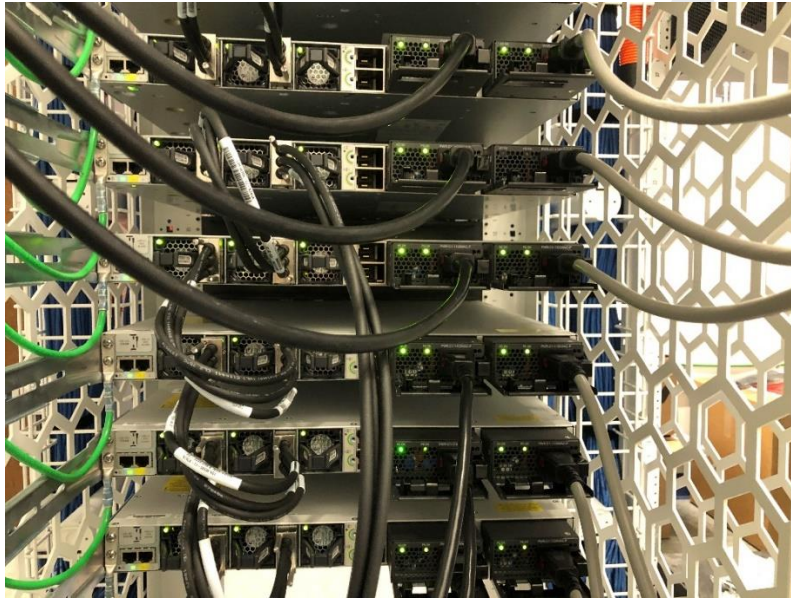


Figure 27: Example of Best-Practice Differentiated Power Cord Coloration Implementation

5.1.5 Information Transport Systems Equipment and Component Labeling

The purpose of this specification is to establish guidance for labeling physical locations, equipment, and components in the VA data center environment congruent with the identification specifications in this Standard.

This specification contains instruction for uniform and appropriate labeling of data center and ITS equipment at VA data center facilities. Applications include:

- Identifying and labeling new systems when installed
- Re-labeling existing systems as field verification documentation is generated
- Uniform labeling of equipment and components used at data processing facilities across the enterprise

5.1.5.1 Label Materials

Materials for labeling purposes are not specifically defined in this Standard. Use materials appropriate to the purpose, intent, environment, and equipment being labeled.

5.1.5.1.1 Approved Materials

Labels should be resistant to the environmental conditions at the point of installation (such as moisture, heat, or ultraviolet light), and should have a design life equal to or greater than that of the labeled component.

5.1.5.1.2 Unapproved Materials

Materials which can be lost, damaged, rendered unreadable, easily removed, soiled or degraded by operating environment, or which are of a non-permanent nature shall not be used for labeling or identification purposes. This includes (but is not limited to) paper, paper

in clear plastic protectors, and direct labeling of equipment and components with markers or paint.

Table 42: Information Technology Systems Equipment and Component Labeling Conventions

ID	Primary Attribute	Secondary Attribute	Specification
1	Equipment Labeling	Attachment	Permanent
		Position	• Readily visible • Horizontal orientation where feasible • Top right corner of equipment faceplate where feasible • Shall not interfere with operation of labeled equipment
		Materials	• Appropriate for the installation environment • Durable and permanent • Heat resistant in high-temperature areas
		Coloration	Not specified
2	IT Equipment Rack/Cabinet Labeling	Governing Industry Standard	ANSI/TIA-606-C, NFPA 70
		Labeling Requirements	• Floor Grid Location • Power Source(s)
3	Data Communications Cabling Labeling	Governing Industry Standard	ANSI/TIA-606-C
		Label Location	• Both ends of all installed cables
		Coloration	Following local site schema
4	Power Distribution Labeling	Governing Industry Standard	ANSI/TIA-606-C, NFPA 70
		Label Location	• Both ends of all installed cables • Within 36 in. of leaving distribution panel enclosure • Within 12 in. of the point-of-use end of conduit
		Coloration	• No enterprise specification is prescribed • Colors shall be used to differentiate electrical bus power sources

Evaluation Factors

- All equipment and components are labeled as specified
- Labeling is done in a legible, professional manner
- Facility location systems are in place in the data center environment
- Power distribution is labeled appropriately from end-to-end and uses bus color scheme allowing rapid visual identification

Implementation Guidance

Data center equipment and components shall be labeled per operational requirements before being designated operable or in service.

Labels shall be permanently attached and designed to withstand wear, erosion, and corrosion.

Position labels so they are readily visible and, where feasible, oriented in a horizontal position. Where feasible, labels should be placed near the top right corner of permanently installed equipment faceplates. Labels shall be legible without manipulation by site personnel. Although not recommended, vertical orientation may be used only where space is limited. Vertical orientation shall be read from top to bottom.

Labels or tags installed under this specification shall not interfere or alter existing manufacturer's nameplate or code information. Labels described in this specification are intended for identification within the local facility and are not expected to support or substitute for asset management labeling.

Labels shall be located to eliminate any possible confusion as to the item being identified and shall not obscure other items nearby.

Labels shall not interfere with an operator's ability to read the instrument, display, or gauge, or to operate the equipment.

Labels shall be mounted in such a way as to preclude accidental removal.

Labels shall be mounted on a flat surface, if possible.

Use ANSI/TIA-607-D identifier scheme for telecommunications bonding and grounding system elements to use compatible terms in ISO/IEC 30129 for all new construction or renovation.

For existing infrastructure, labels using legacy terms from earlier revisions of ANSI/TIA-607-D are exempted until routine tech refresh, lifecycle replacement, upgrades, new installations, or renovations of existing space.

5.1.5.2 Support Infrastructure Identification

This section identifies equipment support elements covered under this Standard. Equipment support elements are passive components such as racks and cabinets that serve to house or mount both passive and active components such as patch panels or servers.

Table 43: Support Infrastructure Identification Conventions

ID	Primary Attribute	Secondary Attribute	Specification
1	Equipment Support	Row	ROW-xxx ROW designates row xxx is a sequential number (pad number of digits in designator (x, xx, xxx) to match the total number) - Example: Row 01
		Rack/Cabinet	RK-aann RK designates rack (or cabinet) labeled left to right when looking at the front of the rack aa is the alphabetic portion of the floor grid location (pad number of digits in designator (a, aa) to align with the number of floor location tiles in the plan E-W axis) nn is the numeric portion of the floor grid location (pad number of digits in designator (n, nn, nnn) to match the total number) - Example: RK-AM45 - For DESIGN PURPOSES ONLY, racks may be identified by row designation and sequential numbers. These will be converted to alphanumeric locations upon physical installation for sustainment n is the numeric row designator (pad number of digits in designator (n, nn) to match the total number of rows) xxx is a sequential number (pad number of digits in designator (x, xx, xxx) to match the total number) - Example: RK-101
		Pathway Rack	PRK-aann PRK designates pathway rack, which is a small enclosure (typically five RU) that is generally attached to an overhead cable tray assembly above an IT equipment rack or cabinet for the purposes of terminating structured cabling to a patch panel in the pathway rack aa is the alphabetic portion of the floor grid location (pad number of digits in designator (a, aa) to align with the number of floor location tiles in the plan E-W axis) nn is the numeric portion of the floor grid location (pad number of digits in

ID	Primary Attribute	Secondary Attribute	Specification
			designator (n, nn, nnn) to match the total number) • Example: PRK-AM45 • For DESIGN PURPOSES ONLY, racks may be identified by row designation and sequential numbers. These will be converted to alphanumeric locations upon physical installation for sustainment • n is the numeric row designator (pad number of digits in designator (n, nn) to match the total number of rows) • xxx is a sequential number (pad number of digits in designator (x, xx, xxx) to match the total number) • Example: PRK-101
		RU	ru • ru is the RU number present (stamped or painted) on front and rear rails corresponding to the bottom left corner of the item in the rack or server • Example: SERVER AD02-35 with 35 indicating the RU
2	Cable Tray	Not Defined	Not Defined
3	Cable Management Panels	Horizontal	CMH-aann-ru • CMH designates cable management panel • aann designates installed location by rack or cabinet identifier • ru is the RU in which the panel is installed Example: CMH-AM45-30 (cable management panel at position RU30 in rack or cabinet AM45)
		Vertical	CMV-aann-aann; CMV-aann-L; CMV-aann-R • CMV designates cable management panel • aann-aann designates rack or cabinet identifiers, left-to-right, viewed from front, between which panel is installed • L or R designates on which side of rack panel is installed, as viewed from front of rack, if not installed between racks Example: CMV-AM45-AM46 (cable management panel is between racks or cabinets AM45 and AM46)

5.1.5.3 IT Equipment Cabinet/Rack Location Identification

Use the modified ANSI/TIA-606-C compatible labeling format with cabinet/rack grid location and cabinet/rack RU position (format xy-ru) to identify IT equipment, switch, patch panel, and similar equipment locations. Equipment location in IT equipment racks shall be identified by the bottom RU that the equipment occupies (e.g., a four RU height server position occupying positions 36-39 is located at RU position 36). VA-standard IT equipment racks have 45 RU of spaces for racked equipment. RU numbering shall begin at the bottom-most mounting space and increase upwards (one RU is 1.75" in height, and the numbering is typically printed on the mounting rails).

In the example shown, the server in IT equipment rack AD02 that occupies spaces 27 and 28 RU from the bottom of the rack would be identified with location AD02-27.

5.1.5.4 Transport Media and Interface Identification

This section identifies transport media and associated elements covered under this Standard. Transport media includes, but may not be limited to, all horizontal and vertical (backbone) cabling and patch cords. Associated elements include, but may not be limited to, patch panels and fiber termination elements.

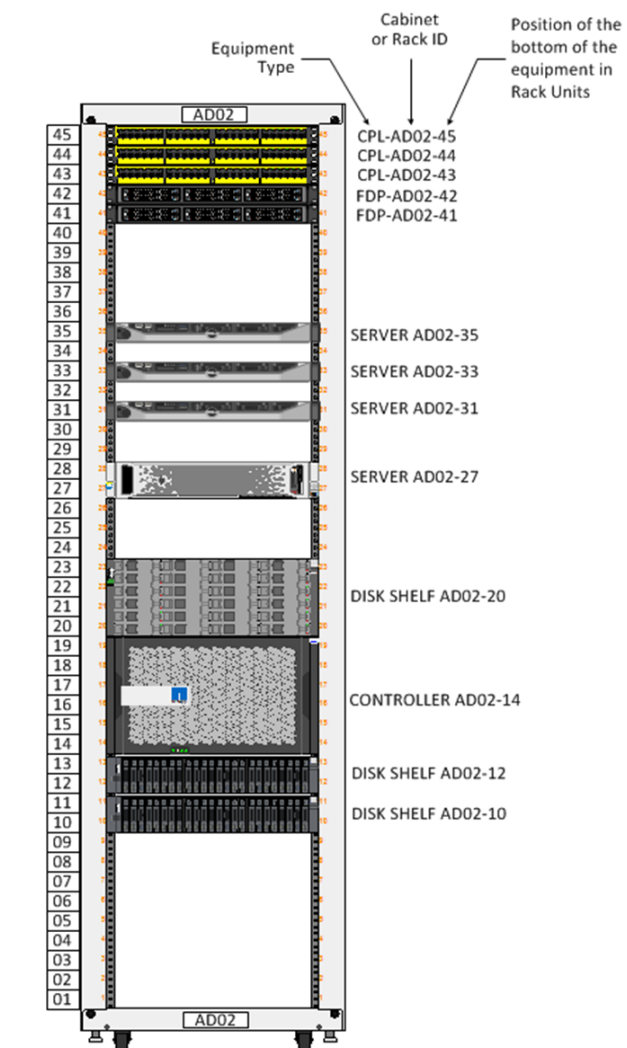


Figure 28: Rack Unit Identification Example

Table 44: Transport Media and Interface Identification Conventions

ID	Primary Attribute	Secondary Attribute	Specification
1	Patch Panel	UTP	CPL-aann-ru • CPL designates UTP patch panel • aann designates installed location by rack or cabinet identifier • ru is the RU in which installed • Example: CPL-AM45-40 (patch panel installed in RU40 in rack, cabinet, or pathway rack AM45) • Note: aann-ru may be utilized as the sole label designator (without CPL). The fully qualified name is intended to support reporting activities through a physical layer asset management system
		UTP Patch Panel Position	CPL-aann-ru.xx • CPL designates UTP patch panel • aann designates installed location by rack or cabinet identifier • ru is the RU the panel is installed • xx is the position number as identified by the patch panel stamped label • Example: CPL-AM45-AA.24 (position 24 in the patch panel designated as AA in rack, cabinet, or pathway rack AM45) • Note: aann-ru.xx may be utilized as the sole label designator. The fully qualified name is intended to support reporting activities through a physical layer asset management system

ID	Primary Attribute	Secondary Attribute	Specification
2	Fiber Distribution Termination Hardware	Fiber Distribution Panel/Cabinets	FPL-aann-ru FPL designates fiber distribution panel aann designates installed location by rack or cabinet identifier ru is the RU the distribution panel is installed - Example: FPL-AM45-30 (fiber cabinet at position RU30 in rack or cabinet AM45)
		Fiber Distribution Cassette	FCS-aann-ru.L FCS designates fiber distribution panel cassette aann designates installed location by rack or cabinet identifier ru is the RU in which the FCS is installed L designates the fiber distribution cabinet/panel slot location ID numbered from left to right starting at 1 - Example: FCS-AM45-40.2 (the cassette in slot location two of FCS in RU40 in rack or cabinet AM45)
		Fiber Distribution Cassette Interface Position	FCS-aann-ru.L.xx FCS designates fiber distribution cassette interface position aann designates installed location by rack or cabinet identifier ru is the RU in which the FDC is installed L designates the fiber distribution cabinet/panel slot ID xx is the position identifier as labeled on the cassette. If a label is not available, then use (L) for the left MPO and (R) for the right MPO - Example: FCS-AM45-40.2.05 (position five in the cassette in slot two of FCS installed in RU40 in rack or cabinet AM45), alternately FCS-101-40.2.L
3	Cable (Data Communications)	Bulk Unshielded Twisted Pair (UTP)	UTP[aann-ru.yy]/[aann-ru.yy] UTP designates unshielded twisted pair aann designates terminal locations by rack or cabinet identifier ru is the patch panel RU designator yy is a two-digit number corresponding to the patch panel position - Example: UTP[AM45-42.24]/[AAR15-03.24] UTP connecting position 24 in patch panel located at RU42 in rack AM45 to position 24 in patch panel located at RU03 in pathway rack AR15

ID	Primary Attribute	Secondary Attribute	Specification
		Pre-terminated Cable Assemblies (UTP)	UTP[aann-ru]/[aann-ru].nn-mm • UTP designates unshielded twisted pair • aann designates terminal locations by rack or cabinet identifier • ru designates RU in which installed • nn-mm are two numbers that identify the start and end ports to be connected by the bundle • Example: UTP[AM45-AA]/[AR15-AB].01-06 UTP bundle connecting patch panel AA located in rack AM45 to patch panel AB in rack AR15 port 01 to port 06 (inclusive)
4	Fiber Cable (Data Communications)	Pre-terminated Cable Assemblies (Multi-mode)	FMM[aann-yy.L]/[aann-yy.L] • FMM designates fiber optic cabling, multimode • aann designates terminal locations by rack or cabinet identifier • yy is a numeric fiber distribution cabinet/panel identifier per rack or cabinet • L designates the fiber distribution cabinet/panel cassette or slot ID and the left or right MPO connector • Example: FMM[AM45-03.2L]/[AR15-03.2L] Multi-mode cable assembly connecting cassette two, left MPO in FCS 03 located in rack AM45 to cassette two, left MPO in FCS 03 located in rack AR15
		Pre-terminated Cable Assemblies (Single-mode)	FSM[aann-yy.L]/[aann-yy.L] • FSM designates fiber optic cabling, single mode • aann designates terminal locations by rack or cabinet identifier • yy is a numeric fiber distribution cabinet/panel identifier per rack or cabinet • L designates the fiber distribution cabinet/panel cassette or slot ID and the left or right MPO connector • Example: FMM[AM45-03.2R]/[AR15-03.2R] Multi-mode cable assembly one connecting cassette two, right MPO in FCS 03 located in rack AM45 to cassette two, right MPO in FCS 03 located in rack AR15

ID	Primary Attribute	Secondary Attribute	Specification
5	Patch Cords	UTP	CCA[aann-ru.xx]/[aann-ru.xx] • CCA designates UTP patch cable • aann designates terminal locations by rack or cabinet identifier • ru is the RU location of the patch panel or device • xx is the port or visually identifiable Network Interface Card (NIC) ID to which the end of the patch cord is connected • Example: CCA[AM45-44.12]/[AR15-30.2] UTP patch cable connecting patch panel 44 position 12, located in rack AM45 to IT equipment NIC port 2, located in RU position 30, rack AR15
		Fiber Optic	FCA[aann-ru.xx]/[aann-ru.xx] • FCA designates fiber patch cable • aann designates terminal locations by rack or cabinet identifier • ru is the RU location of the patch panel or device • xx is the port or visually identifiable NIC ID to which the end of the patch cord is connected • Example: FCA[AM45-44.06]/[AR15-24.2] Fiber patch cable connecting patch panel in RU44, position 06, located in rack AM45 to IT equipment NIC port 2, located in RU position 24, rack AR15

5.1.5.5 Power Distribution Identification

This section identifies power distribution components and associated elements covered under this Standard. Power distribution includes, but may not be limited to, active and passive facilities equipment that transfers electricity from source equipment and panels to the terminal receptacles on rack-mounted power distribution assemblies. Associated elements include, but may not be limited to, PDUs, distribution panels, power whips, and in-rack power distribution systems.

Identification of power distribution is limited to physical location of the equipment and not to power chain relationships. Power chain relationships will be elaborated upon in a future standard revision.

Table 45: Power Distribution Identification Conventions

ID	Primary Attribute	Secondary Attribute	Specification
1	Power Distribution Source (data center level)	Power Distribution Unit (PDU)	PDU-xxxx PDU designates Power Distribution Unit xxxx is the host name matching the VA naming convention for networked PDUs within the facility or local naming convention for non-networked PDUs - In facilities with multiple (A/B) bus distribution to the IT equipment, local naming conventions should indicate which bus the power is supplied from - Example: PDU-3A
		Remote Distribution Cabinet (RDC)	RDC-xxxx RDC designates a Remote Distribution Cabinet, which is a stand-alone expansion distribution panel unit fed from a sub-feed breaker in a PDU xxxx is the local naming convention for the RDC within the facility. The typical RDC designator will be a sub-designation of the parent PDU, i.e., RDC3A-1 indicates an RDC fed from sub-feed breaker 1 on PDU3A - Example: RDC-3A-1
		Distribution Panel (DP)/ Branch Circuit Panelboards (BCP)/ Remote Power Panels (RPP)	DP-xxxx DP designates a wall-mounted distribution panel fed from an upstream DP, sub-feed breaker in a PDU, or other similar source xxxx is the local naming convention for the DP within the facility - Naming conventions for DPs at the facility or data center level are not specified - Example: DP-CR3A
		Busway	BUS-xxxx BUS designates an overhead power distribution busway, which is a ceiling-mounted conductor enclosure fed from an upstream distribution panel, sub-feed breaker in a PDU, or other similar source, to which individual 'taps' with integral power whips are attached xxxx is the local naming convention for the busway within the facility - Busways are typically associated with rows of IT equipment - Example: BUS-ROW3

ID	Primary Attribute	Secondary Attribute	Specification
		Power Tap	TP[ELEMENT1]/[ELEMENT2] • Example: TP[BUS-xxxx]/[RPDU-aann.gl.zn] Note: there are two busways per row in a small data center so a busway element example would be [BUS-03A] or [BUS-03B] to identify the 03 A or B busway. However, in larger data centers there are multiple pods in rows along each line. In that case BUS-03A1, BUS-03A2, etc. will identify which pod along the 03 line is specified • TP designates a power tap off a power busway system • The ELEMENT components select the source (ELEMENT1) and powered item (ELEMENT2) using nomenclature described elsewhere in this specification. Many combinations of taps are possible using this designation depending upon the specific power distribution schema used in individual data center facilities
2	Power Distribution Source (rack/enclosure level)	Zone Power Distribution Unit (zPDU)	• ZPDU-aann.ru • ZPDU designates a Zone Power Distribution Unit, which is a rack-mounted unit allowing A/B distribution to standard and HD power distribution topologies • aann designates terminal locations by rack or cabinet identifier • ru is the RU where a horizontal ZPDU is installed in the rack or cabinets viewed from the rear of the rack Example: ZPDU-AM45. 03 is a horizontal ZPDU installed in rack AM45 at RU position 03
		Subzone Vertical Rack Power Distribution Unit (rPDU)	RPDU-aann.gl.zn or RPDU-aann.aann.gl.zn • RPDU designates Rack Power Distribution Unit, which is a distribution power strip that power cords from terminal IT equipment plug into • aann designates terminal locations by rack or cabinet identifier • gl is the group and leg designation as labeled on the zPDU outlet connected to the rack-mounted PDU • z designates left or right (L or R) mounting of the PDU • n is the number of the PDU starting at the rear of the rack

ID	Primary Attribute	Secondary Attribute	Specification
			• Example: RPDU-AR16.A2.L2 is the second power strip mounted on the left of the rack connecting to the zPDU's A bus in rack • AR16.Alternative: If no zPDU is present (direct connection from the RPDU to a power whip from a DP, RDC, or PDU) • Where there is a single bus supporting a space, the bus is denoted as A. The first RPDU on each bus is number 1, the second 2, and so on Example: RPDU-AR16.A2 is the second power strip on the A bus in rack AR16. RPDU-AR18.B1 is the first power strip in the B bus in rack AR18
3	Power Distribution (conductors)	Power Whips	WHIP[ELEMENT1]/[ELEMENT2] • Examples: WHIP[PDU-xxx.aa,bb,cc]/[zPDU-aann.ru] • WHIP designates a set of electrical conductors in a conduit or other flexible power cord assembly intended to move electricity from a distribution point to a point-of-use component • The ELEMENT components select the source (ELEMENT1) and powered item (ELEMENT2) using nomenclature described elsewhere in this specification. Many combinations of whips are possible using this designation depending upon the specific power distribution schema used in individual data center facilities • aa,bb,cc is the set of circuit breaker positions used at the source end corresponding to the positions in the breaker panel, if applicable. For whips beginning at a busway tap, do not use the .aa,bb,cc portion of the designator Example: WHIP[PDU-3A.13,15,17]/[zPDU-AR14-03] Power whip connecting three-pole circuit breaker in positions 13, 15, and 17 in PDU-3A to zPDU in rack AR14 located at position RU03
		Power Cords (from rPDU to IT Equipment)	Not Defined

5.1.5.6 Data Communications Cabling Labeling

The Administration Standards specify use of ANSI/TIA-606-C compatible formats for identifying data communications cabling (more complicated installations may require reference to that document). Data communications cabling (UTP, fiber, and other) shall be labeled on both ends with information about the network port that it connects to on each end. Machine-printed paper tags secured with clear plastic or acetate adhesive tape are the minimum standard for these tags as in *Figure 29* below. Hand written labels of any type are not permitted.

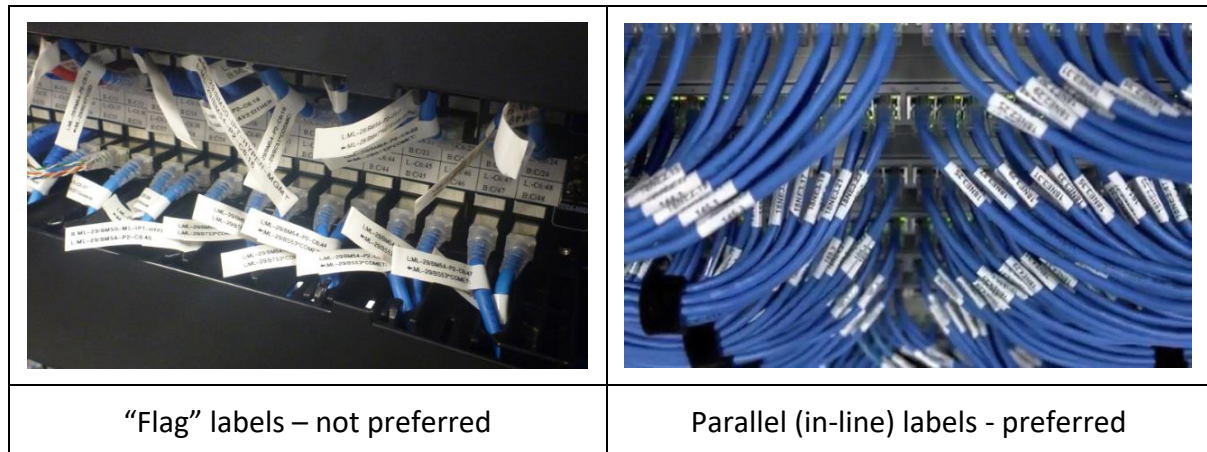


Figure 29: Example of Communications Cable Labeling

IT equipment (including switch and patch panel) network ports shall be individually identified from manufacturer information, typically provided in equipment drawings available for integration with rack elevation drawings and a DCIM tool. Where not otherwise provided, cardinaly number ports along the top row from left-to-right, then along the next row down from left-to-right, and so forth.

Equipment ports and the cables in each (if any) are denoted by the equipment position with the additional identifier xx. Thus, a cable end will be labeled in VA-modified ANSI/TIA-606-C compliant format [aann-ru.xx].

In order to identify the connected path of each data communications cable, the physical position identifiers at both ends are needed. The label for each cable shall identify the near side location (NS) and the far side (FS) location in VA-modified ANSI/TIA-606-C compatible format (e.g. “NS/FS”). Therefore, a UTP patch cable going from AD02-35 port 10 to patch panel BD02-19 port 04 would be identified (on the near end) as CCA[AD02-35.10]/[BD02-19.04], and labeled (on the far end) as CCA[BD02-19.04]/[AD02-35.10]. Automated systems adopted within VA shall understand the reciprocal relationship in these identifiers as representing a single connection.

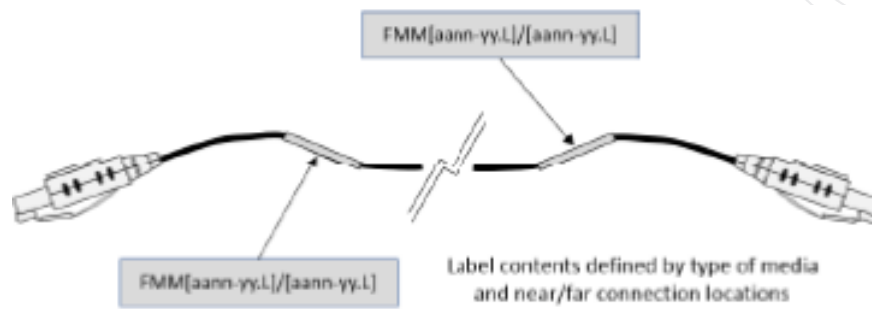


Figure 30: Detailed Example of Communications Cable Labeling

5.1.5.7 IT Equipment Rack/Cabinet Labeling

In order to cross-identify network and IT equipment enclosures (hereafter, generically referred to as ‘racks’) located on the floor with critical safety information (including power sources), each rack shall be labeled with its physical location (grid identification) and power distribution sources. Rack identifying physical location shall be posted on both the front and rear of each rack, on the frame above or at the top of the door (as applicable).

5.1.5.8 IT Equipment In-Rack/In-Cabinet Power Distribution Labeling

Rack power information shall be labeled on each individual vertical rPDU, on the frame behind the door, front and back, as well as on the doors, front and back. Machine-printed paper tags secured with clear plastic or acetate adhesive tape are the minimum standard for these tags. Hand written labels of any type are not permitted.

Alternative IT equipment (non-rack based) shall be labeled with physical location (grid identification) and power distribution sources and with any additional identifying information needed to identify that specific piece of equipment. Install power designator labels near or on the power cord leaving the equipment. Place equipment location tags on the front top, or at location associated with the first full tile closest to the origin.

5.1.5.9 Power Distribution to IT Labeling

Data center power distribution components such as breakers, switches, receptacle junction boxes, and IT equipment power circuits (“whips”) shall be labeled with approved machine generated adhesive tape, cable tags or other approved method. Hand written labels of any type are not permitted.

All non-IT power distribution in the data center space should be labeled in the same manner to ensure a safe environment. All electrical equipment which is served by or which contains multiple sources of power shall comply with NFPA 70 standards to identify both disconnecting means.

The source end of each whip shall be labeled within 3 ft of the conduit leaving the distribution panel enclosure. The point-of-use end of each whip shall be labeled on the junction box, on the connector, or within 1 ft of the end of the conduit.

Labels installed internal to electrical or electronic components shall not provide a ground or short circuit path or interfere with the operation of components.

5.1.5.10 Physical Label Format

Labels on power whips shall contain the following information:

- PDU/Panel Source
- Circuit Number
- Whip Length
- Equipment Location
- Receptacle Type

Figure 31 below shows an example of a self-adhesive label for use on a whip. There is no prescribed format for these labels.

PDU/Panel Source:	PDU A-3 Panel 2	
Circuit Number:	2,4,6	Length: 50'
Equipment Location:	Rack E8	
Receptacle Type:	L21-30R	

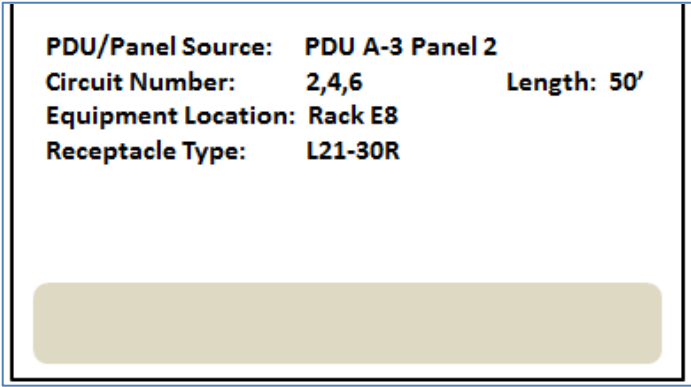


Figure 31: Power Whip Label Example

5.2 Management Standards

5.2.1 *Management of Telecommunications Cabling*

Cable management is largely qualitative in nature, requiring the exercise of judgment by IT staff. Proper cable management is necessary to enable optimal airflow through IT equipment (minimizing energy usage), to allow identification and management of cabling throughout the IT equipment lifecycle, and to minimize risks of damage to the networking system by minimizing opportunities for damage to cables, connectors, and conveyances. Aesthetically, proper cable management reflects the professional approach expected and necessary to support the mission of providing the best possible IT support to the Veteran.

Evaluation Factors

- Cable management allows appropriate airflow
- Cables properly labeled to allow easy identification
- Cable management is organized, logical, systematic, and aesthetic

Implementation Guidance

Cabling plants shall be planned to enable the installation of IT equipment. Install and utilize cable conveyances (cable trays, cable managers, and similar) in all IT installations. Free-run telecommunications cabling is not maintainable or sustainable and has a higher lifecycle cost than properly designed and installed cabling plant systems.

- Cabling shall be run in horizontal, vertical, or overhead cable management systems where available. Procure and install cable management equipment when cabling installations need such to enable aesthetic, managed cabling outcomes.
- Always plan the cabling path and manage the cabling installation, down to the level of the patch cord to the individual piece of IT equipment. Individual cables shall not be free-run.
- Individual cable lengths shall be selected appropriately for their purpose. Cable slack shall be managed with attention to installation criteria such as bend radius. Excessive slack indicates a poor choice of cable length.
- Patch cords shall not be run from one IT equipment rack to another.
- Power cords shall not be run from one IT equipment rack to another. Power distribution detailed in the Infrastructure section specifically calls out zPDUs that supply power to multiple racks to obtain maximum power densities. These devices and their power supplies are not affected by this guidance.
- Power cords shall be matched to the manufacturer specifications for the hardware receptacles intended without the use of adapters.
- Patch and power cords shall not be left unmanaged where they may interfere with exhaust airflow or have the potential to catch or snag on people or equipment.
- Document all cabling components and their linkage between components and make sure that this information is updated on a regular basis. The installation, labeling, and documentation should always match. Maintain cabling documentation, labeling, and logical/physical cabling diagrams.
- No enterprise color scheme is specified for UTP patch cords. Where a local or organizational color scheme for cabling exists, it shall be followed. Color schemes for patch cords should be avoided.

- Consider vendor-specific IT equipment requirements for cabling so as not to impede intended operation of that equipment, such as blocking exhaust ports. For example, cabling management in the picture below is appropriate except for the aqua-colored cables entering from the left, which cover the fan module on the left of the switch. This keeps the fan module from being replaced without disconnecting the network.

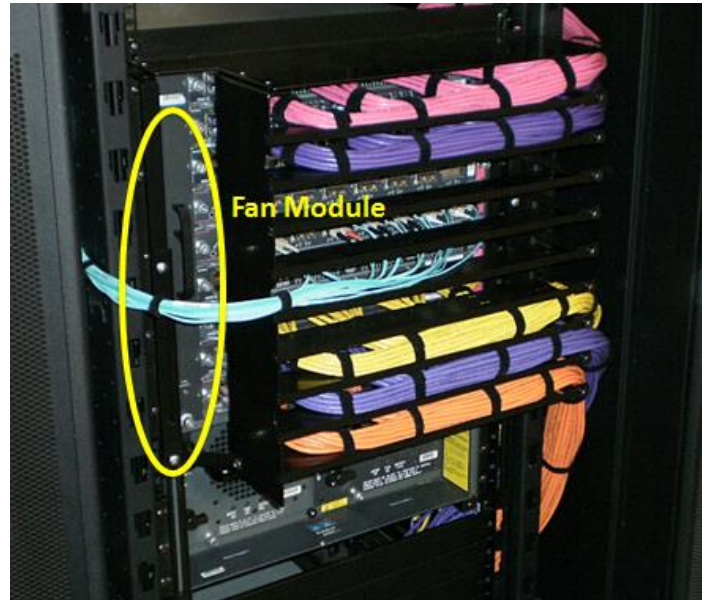


Figure 32: Equipment-Specific Cable Management Requirements

Below are examples of *acceptable* cabling management implementations. No installation is ever optimal for all considerations, but these represent the level of workmanship expected in the VA computer room environment.



Figure 33: Acceptable Cable Management Results

Aesthetic requirements shall be implemented to the ‘reasonable person’ standard. Below are examples of *unacceptable* cabling management implementations that will require remediation.

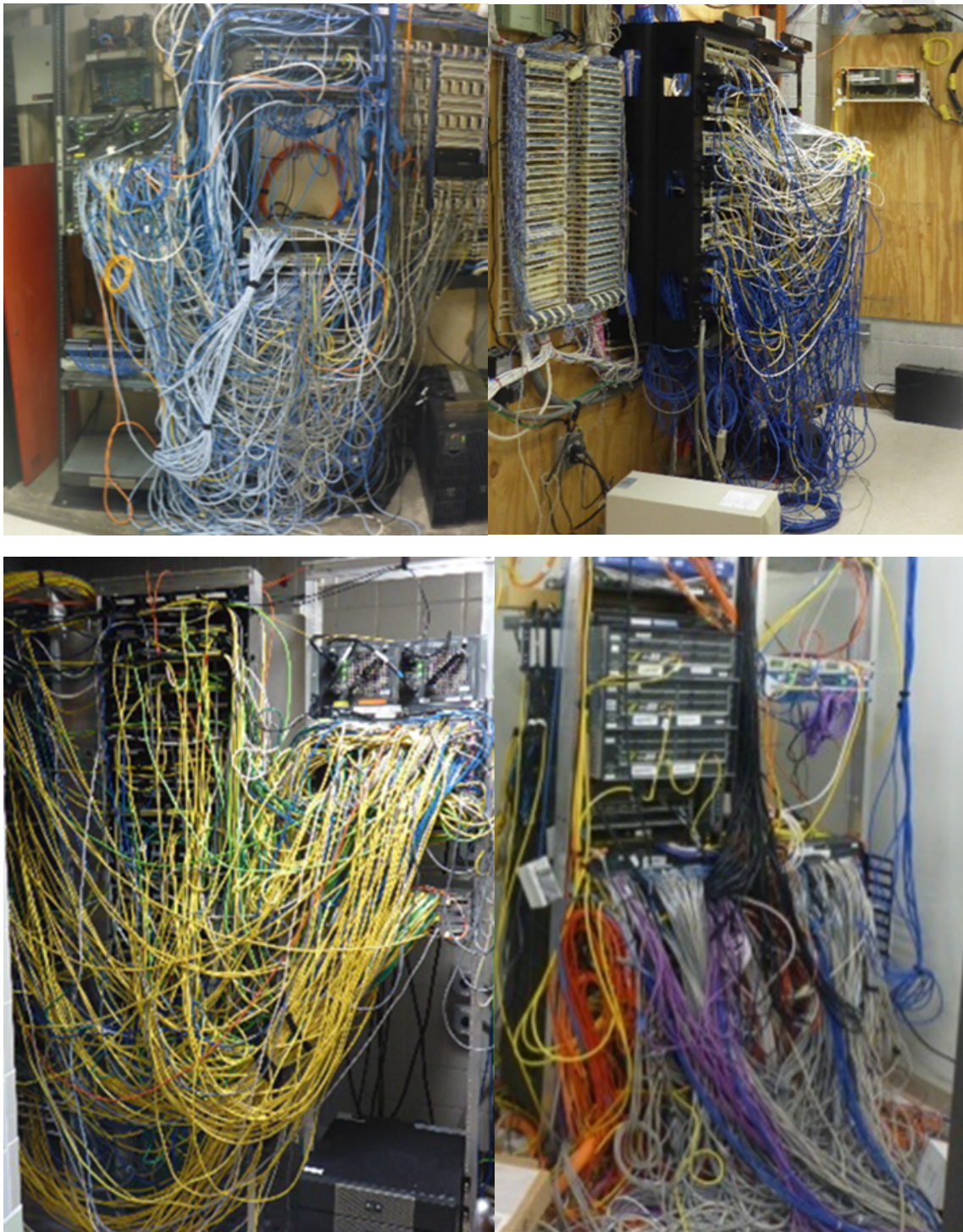


Figure 34: Unacceptable Cable Management Results

5.2.1.1 Installation Guidance

- Cross-connects are only authorized at the MDAs, HDAs, TRs, and entrance rooms.
- No splicing or intermediate cross-connecting are authorized for intra-building or inter-building connectivity.

- All TRs shall be connected directly to the MDAs.
- All cabling shall be installed in a neat, workmanlike manner without twists, kinks, and unnecessary crossovers.
- Cables should not be in contact with the ground. Use cable management components and techniques to maintain a clean, clear, and safe work environment.
- Do not mount cabling in locations that block access to other equipment (e.g. power strip or fans) inside and outside the racks.
- Patch cords should follow the side of the IT equipment rack closest to the assigned NIC.
- Cables should not be looped around themselves or other objects.
- Route cables with gentle loops to avoid damage due to exceeding bend radius limitations. Glass fiber can be easily broken with rough handling or overly tight bends.
- Cables should be tight enough to maintain position but not tight enough to deform the jacket or put tension on the connectors.
- Based on heat exhaust (airflow), serviceability, and excess cable lengths, do not install folding/retractable cable management arms for IT equipment in VA computer spaces. Arms currently installed on existing equipment may be used until the equipment is refreshed and removed.
- All cable slack should be concealed within the rack either vertically or within cable managers. Slack should not be looped. With use of the correct length cables, there should not be sufficient slack to enable looping.
- Use the correct length patch cords.
- Cables should not be twisted or wrapped around other cables, including when bundled.

5.2.1.2 Cable Bundling Guidance

- Bundle cables together in groups of relevance (for example, cables to a single equipment distributor or uplinks to core devices), as this will ease management and troubleshooting.
- Bundles of cables, when necessary, should be built from sub-bundles of no more than 6-8 individual cables. The larger bundles should not exceed four sub-bundles of like quantity.
- When bundling or securing cables, use Velcro-based ties not more than every 3 ft to 6 ft, and more frequently as needed to maintain a clean installation. Do not use zip ties or twist ties for cable bundling or securing, as these apply pressure on the cables.
- Only bundle like cable types. Do not bundle fiber, power, and UTP together.
- Cable labels shall be visually accessible to local personnel following installation, for future identification and troubleshooting purposes. Rather than bundling groups of cabling in a manner that prevents identification of individual cables, bundle in a different manner and/or relocate the cabling to locations where they will not be obscured. Consider bundling so all cables are on the exterior of the bundle so they are visible, as shown in *Figure 35* below. The exception to this is when factory pre-bundled cabling is used.

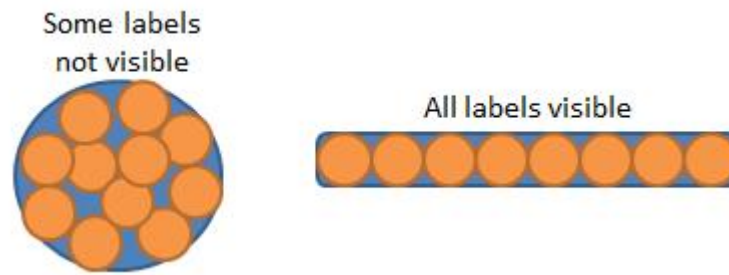


Figure 35: Cable Labels in Bundles

5.2.1.3 Unshielded Twisted Pair, Fiber, and Power Cabling Guidance

- Segregate power and telecommunications cabling in separate cable tray systems; use different conveyance systems located not less than 1 ft apart when these two systems are run in parallel.
- Where possible, run power and telecommunications in separate paths to reduce risks of Electromagnetic Interference (EMI)/Radio Frequency Interference (RFI) and data transmission losses.
- Do not install UTP cabling on top of fiber cabling to prevent damage to the fiber transmission medium.
- Segregate UTP and fiber telecommunications cabling, using different conveyances where possible. Where not possible or provided, ensure fiber cable is protected from damage. Use fiber innerduct where necessary within the same cabling conveyance.
- Use properly-sized equipment power cords; use of 6 ft length power cords where 2 ft cords are needed is not considered a best practice.
- Separate A-side and B-side power and segregate equipment power cords (between vertical rack power distribution units and IT equipment) by color for identification of A/B power to each piece of IT equipment. Follow best practices for managing the power cords similarly to other cabling.

5.2.2 Management of Airflow

VA computing facilities shall utilize engineered, verified layout designs to optimize energy usage in the environmental control systems supporting the facilities. Proposed deviation from standard layout design requires verification and validation by DCIE and CFD analysis.

Evaluation Factors

- Designated hot and cold aisles are utilized in facilities with more than eight IT equipment rack/cabinet equivalents.
- Supply cooling air is not directly mixed with return air through placement of perforated floor tiles and similar vents.
- Blanking panels and air dams are deployed in all IT equipment cabinets to minimize bypass air across the surface area of the cabinets.

Implementation Guidance

Unless otherwise designated through validated engineering design, utilize a hot/cold aisle arrangement for equipment located in VA data centers. This applies to both facilities that use a raised access floor plenum for cooling distribution and those that do not.



Request for Variance

Infrastructure Standard for Telecommunications Spaces

Applicant Name: _____

Physical Facility / VA Organization Represented: _____

Email Address: _____

Contact Phone: _____

Location Where Variance is Requested: _____

Variance
Requested:

Standard Section Number/ Page Number: _____

Standard Section
Text:

Problem Statement:

Proposed
Alternate Method:

Plan to Bring
Project into
Compliance:

Timeline to Bring
Project into
Compliance:

Request Type:

Signature:

Date:

Send variance applications to: OIT ITOPS SD Data Center and Infrastructure Engineering (DCIE)
<VAITSEDatacenterEngineering2@va.gov>

Please attach: Diagrams, one-line drawings, photos, manufacturer's brochures, and/or other supplemental information to support your application. Allow a minimum of two weeks for processing.

What is a variance?

A variance is an allowable deviation from specific requirements of a published standard where a proposed alternate method of installation or operation will provide equal or higher safety, reliability, redundancy, and sustainability objectives. Alternately, *under emergency time-sensitive circumstances* a temporary variance may be granted without meeting the criteria above with an actionable plan to bring the project into compliance within a reasonable time frame.

What will a variance not do?

A variance will not avoid compliance with the intention of a standard section or rule. A variance is not a method to recover from an installation or design error. A variance will not be granted if the alternate method would reduce the safety or effectiveness of the installation or equipment.

What does a variance cover?

If the variance is granted, it shall be limited to the particular site and installation covered in the application and will NOT be considered as a precedent for other installations.

Who may apply for a variance?

Those responsible for compliance with the standards may apply for a variance.

Who grants a variance?

A variance is granted only by Data Center and Infrastructure Engineering, Solution Delivery, upon written request.

Appendix B: OIT Design Guide Templates

B.1 Design Narrative

The Data Center and Infrastructure Engineering (DCIE) team has developed the [OIT Design Guide Templates for Critical Infrastructure in Telecommunications Spaces](#), also located at this link in the Technical Information Library, for Architect/Engineer (A/E) partners to use in their data center design drawing development for new construction, restoration, and modernization projects across VA. These generic Main Computer Room (MCR), entrance room, and Telecommunications Rooms (TR) designs can be applied without significant modifications to future construction projects.

The existence of a standardized, standards-based design will prevent non-compliant, incorrectly sized, unbalanced, and unsustainable data center designs from being produced by VA mission partners, will reduce design project timelines and costs, and prevent unnecessary rework and change orders.

These designs utilize a standardized ANSI/TIA-942-B Telecommunications-Electrical-Architectural-Mechanical (TEAM) approach that will allow A/E designs to separate requirements by discipline. Disciplines not directly related to the telecommunications infrastructure (e.g. structural, plumbing, fire protection) are not addressed.

These templates provide a ninety percent solution for most designs; however, projects in the planning, design, or implementation phases shall contact DCIE for additional assistance in telecommunications space design and standards compliance.

Transmit all communications to DCIE via e-mail at VAITESEDatacenterEngineering2@va.gov.

B.2 Main Computer Rooms

Main Computer Rooms (MCR) provide IT services to Campus Support Centers (CSC) for most typical VA Medical Center (VAMC) campuses.

Legacy MCRs typically support between 25k W and 75 kW of critical load, with significant additional loads anticipated as VA modernizes electronic health record systems and deploys additional medical systems increasingly dependent upon IT systems.

Analysis of projected IT trends and needs has led VA to standardize on a single CSC design appropriate for an estimated ninety percent of VAMCs needs for a twenty-year data center life cycle. Some smaller medical centers will only require a 'small' design to support their IT requirements (see sheet seven of the OIT Design Guide Templates). Only a small number will

require larger design sizes or capabilities. Unless otherwise indicated by DCIE, utilize a 'medium' design (see sheet ten) for computer rooms to support VAMCs.

For Network Support Centers (NSC), Health Care Centers (HCC), and Community Based Outpatient Centers (CBOC) that provide lower-level application and IT services distribution across the geographic breadth of the VA's computing enterprise, an 'extra small' design (see sheet six) or 'small' design (see sheet seven) MCR is typically appropriate but may require larger depending on the number of subscribers they service.

For Core Data Centers (CDC) and Mission Support Centers (MSC), consult DCIE for the appropriate size and design during project scoping for these classifications.

B.3 Entrance Rooms

Carrier circuits shall be terminated in entrance rooms located in the same building as the MCR.

There shall be a minimum of one entrance room per building. Entrance rooms shall be a minimum of 80 ft² for a one-rack entrance room and an additional 20 ft² for each additional rack required (see sheet four).

Two entrance rooms are required for the diverse providers for CDCs and CSCs with a minimum separation of 66 ft between entrance points and entrance rooms.

Health care facility entrance rooms shall be a minimum of 170 ft². See sheet four of the following OIT Design Guide Templates Design Package for construction and outfitting guidance for these redundant entrance rooms.

Follow guidance for the NSC telecommunications space classification (and entrance room specific exceptions) in The Standard for electrical distribution and environmental control requirements for these spaces.

B.4 Telecommunications Rooms

Fiber backbone from the MCR terminates in Telecommunications Rooms (TR) distributed around the buildings on campus, which provide horizontal network distribution to end-user Work Area Outlets (WAO).

There shall be a minimum of one TR per floor. Telecommunications rooms shall be a minimum of 80 ft² for a one-rack TR and an additional 20 ft² for each additional rack required (see sheet five).

Health care facility TRs shall be a minimum of 170 ft².

Telecommunications Room construction and outfitting requirements are detailed in sheet five of the following OIT Design Guide Templates Design Package. Follow guidance for the NSC telecommunications space classification in The Standard for electrical distribution and environmental control requirements for these spaces.

VA



U.S. Department of Veterans Affairs

Office of Information and Technology
IT Operations and Services
| *Solution Delivery*

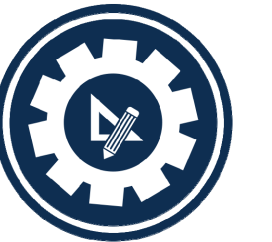
OIT DESIGN GUIDE TEMPLATES FOR CRITICAL INFRASTRUCTURE IN TELECOMMUNICATIONS SPACES (CLINICAL AND NON-CLINICAL ENVIRONMENTS)

DEVELOPED BY:
DATA CENTER AND INFRASTRUCTURE ENGINEERING

DEPARTMENT OF VETERANS
AFFAIRS



IT OPERATIONS AND SERVICES
SOLUTION DELIVERY



EDCT

ENTERPRISE DATA CENTER AND
INFRASTRUCTURE ENGINEERING

PROJECT:

OIT DESIGN GUIDE TEMPLATES

PROJECT No:

N/A



DRAWING No:

FILE: OIT_DESIGN_GUIDE_TEMPLATES-04
162021-V3.1.VSDX

DESIGN BY: KEVIN GRZELKA, CTDC

VERIFIED BY: MICHAEL JULIAN, RCDD
KELLY BATES, DCIE

DOC VERSION No: 3.1

ISSUE DATE: Apr 16, 2021

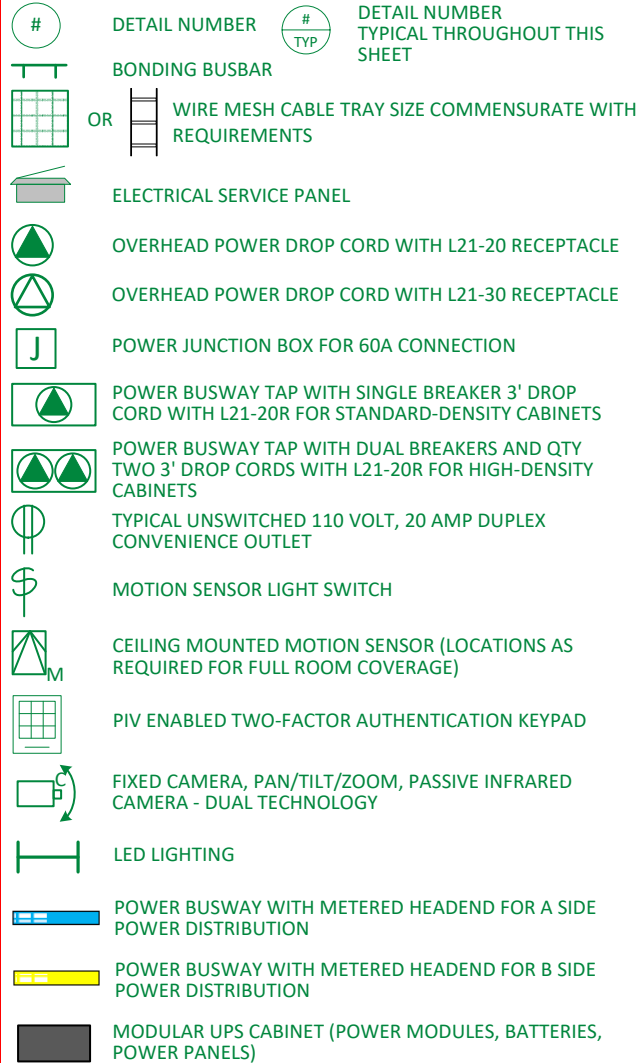
SHEET TITLE

COVER

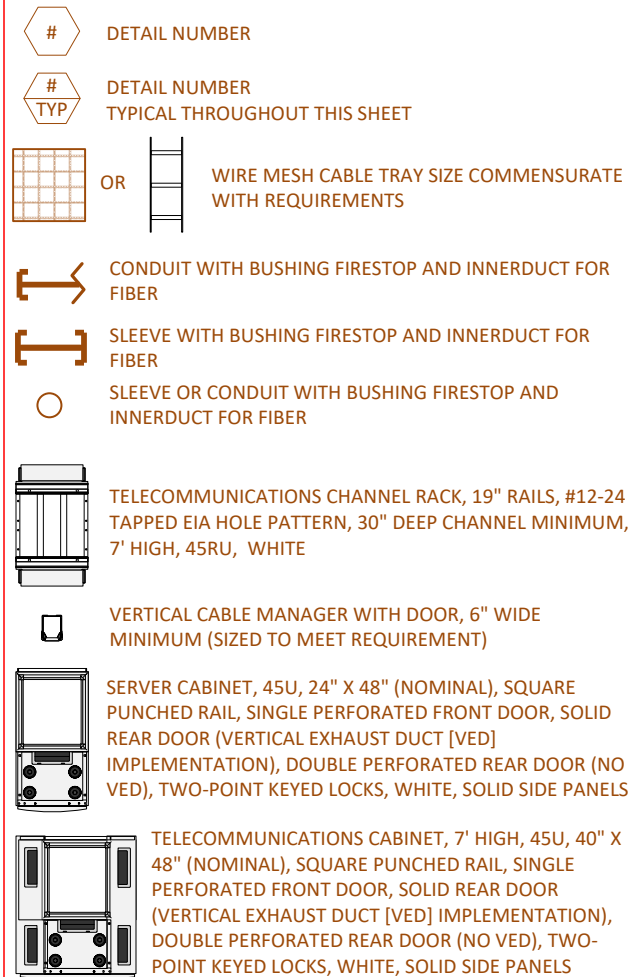
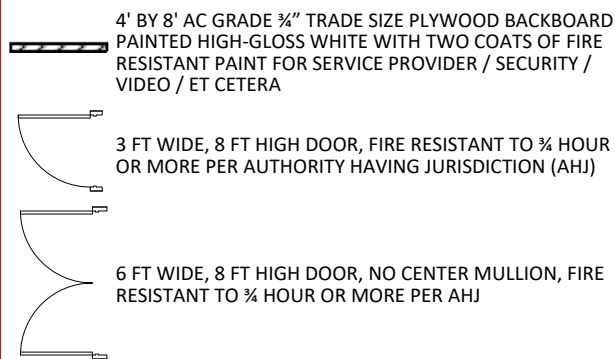
SHEET: 1 OF 37

LEGEND

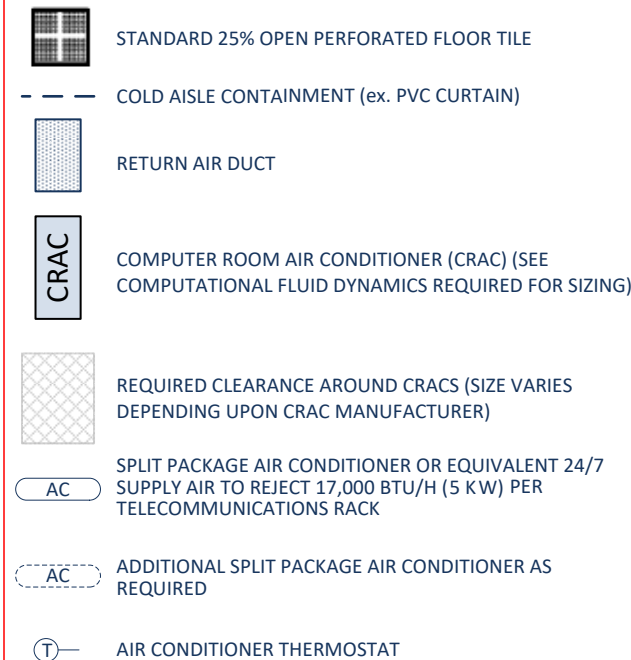
ELECTRICAL



TELECOMMUNICATIONS

ARCHITECTURAL

MECHANICAL



NOTES

ELECTRICAL

- 1 MINIMUM QTY TWO 110 V 20 A CONVENIENCE POWER OUTLETS PER WALL; PLACEMENT DETERMINED BY OTHERS
 - 2 PRIMARY BONDING BUSBAR PLACEMENT DETERMINED BY OTHERS
 - 3 SECONDARY BONDING BUSBAR PLACEMENT DETERMINED BY OTHERS
 - 4 SERVICE PANEL FOR A-SIDE POWER, GENERATOR BACKED IF THE FACILITY HAS GENERATION CAPABILITIES. SHALL BE EQUIPPED WITH POWER METER; PLACEMENT PER OTHERS
 - 5 SERVICE PANEL FOR B-SIDE POWER, GENERATOR BACKED IF POSSIBLE. SHALL BE EQUIPPED WITH POWER METER; PLACEMENT PER OTHERS
 - 6 20 AMP 208 VOLT THREE-PHASE TWIST LOCK RECEPTACLE (L21-20R); 8.5 FT AFF {WYE (5-WIRE)} COORDINATE INSTALLATION LOCATION WITH TELECOMMUNICATIONS CONTRACTOR
 - 7 30 AMP 208 VOLT THREE-PHASE TWIST LOCK RECEPTACLE (L21-30R); 8.5 FT AFF {WYE (5-WIRE)} COORDINATE INSTALLATION LOCATION WITH TELECOMMUNICATIONS CONTRACTOR
 - 8 POWER BUS BAR (VOLTAGE AND AMPERAGE TO SUPPORT ADEQUATE SINGLE & DOUBLE BREAKER 208V, 20 AMP, L21-20R BUSWAY TAPS PER SERVER ROW) FED FROM A-SIDE SOURCE
 - 9 POWER BUS BAR (VOLTAGE AND AMPERAGE TO SUPPORT ADEQUATE SINGLE & DOUBLE BREAKER 208V, 20 AMP, L21-20R BUSWAY TAPS PER SERVER ROW) FED FROM B-SIDE SOURCE
 - 10 POWER BUS BAR (VOLTAGE AND AMPERAGE SUFFICIENT TO SUPPORT ADEQUATE SINGLE BREAKER 208V, 20 AMP, L21-20R BUSWAY TAPS PER NETWORK ROW) FED FROM A-SIDE SOURCE
 - 11 POWER BUS BAR (VOLTAGE AND AMPERAGE SUFFICIENT TO SUPPORT ADEQUATE SINGLE BREAKER 208V, 20 AMP, L21-20R BUSWAY TAPS PER NETWORK ROW) FED FROM B-SIDE SOURCE
 - 12 BLUE CABLE PATH FOR A-SIDE UPS
 - 13 YELLOW CABLE PATH FOR B-SIDE UPS
- LIGHTING:
1. LED LIGHTING PLACED IN AISLES DIRECTLY IN FRONT OF AND BEHIND CABINET ROWS
 2. LIGHTING OPERATED BY MOTION SENSOR PER CABINET ROW OR SECTION
 3. 500 LUMENS IN THE HORIZONTAL PLANE AND 200 LUMENS IN THE VERTICAL PLANE REQUIRED MEASURED AT 3 FT AFF IN FRONT OF AND BEHIND EQUIPMENT CABINETS

TELECOMMUNICATIONS

- 1 SERVICE PROVIDER ENTRANCE POINT
- 2 SERVICE BACKBONE - ENTRANCE ROOM-MAIN COMPUTER ROOM (MCR) WITH DUAL REDUNDANT PATHWAYS
- 3 MAIN CROSS CONNECT (CAMPUS/BUILDING DISTRIBUTOR) – BACKBONE CABLING (MCR-TELECOMMUNICATIONS ROOMS)
- 4 ENTRANCE ROOM INTERCONNECT
- 5 HORIZONTAL DISTRIBUTION TO WORK AREA OUTLETS (WAO)
- 6 ADDITIONAL CABINETS OR TELECOMMUNICATIONS RACKS TO MEET REQUIREMENTS.
- 7 HORIZONTAL DISTRIBUTION AREA (HDA) – SIDE A
- 8 HORIZONTAL DISTRIBUTION AREA – SIDE B
- 9 MAIN DISTRIBUTION AREA (MDA) – SIDE A
- 10 MAIN DISTRIBUTION AREA – SIDE B
- 11 STANDARD-DENSITY (SD) CABINET SUPPORTING 5KW REDUNDANT
- 12 HIGH-DENSITY (HD) CABINET SUPPORTING 10KW REDUNDANT
- 13 40" NETWORK CABINET WITH HORIZONTAL CABLE CUTOUTS AND APPROPRIATE CABLE MANAGEMENT ACCESSORIES, SUPPORTING 5KW REDUNDANT
- 14 WIRE MESH CABLE TRAY MINIMUM OF 4" X 12" WITH SECOND LEVEL OF FIBER CHANNEL PATHWAY (4" X 4")
- 15 BLUE CABLE TRAY PATH FOR CABLES TO THE A-SIDE MDA/HDA
- 16 YELLOW CABLE TRAY PATH FOR CABLES TO THE B-SIDE MDA/HDA
- 17 NETWORK CHANNEL RACK

ARCHITECTURAL

FLOOR TO CEILING HEIGHT:

1. TELECOMMUNICATION ROOMS INCLUDING NETWORK SUPPORT CENTERS: MAXIMUM EXTENT POSSIBLE
2. DATA CENTERS EXCLUDING EXTRA SMALL NETWORK SUPPORT CENTERS: MINIMUM 16 FT SLAB TO DECK ABOVE

FLOOR COMPOSITION:

1. STATIC DISSIPATING FLOOR TILE OR COATING
2. CONCRETE SLAB 5" THICKNESS AT GROUND
3. STEEL DECK AND FILL FOR FLOORS ABOVE GROUND
4. AVOID ACCESS FLOOR PLENUMS FOR AIR DISTRIBUTION PURPOSES

WALLS: FULL HEIGHT TO DECK ABOVE FOR ALL TELECOMMUNICATION SPACES

CEILINGS: NO SUSPENDED CEILINGS ALLOWED

DATA CENTER SIZE ALLOCATION PER CHART BELOW:

DATA CENTER SIZE	SQUARE FOOTAGE	FORM FACTOR
EXTRA SMALL	780	24' X 32'
SMALL	1152	24' X 48'
MEDIUM	1760	40' X 44'
LARGE	2464	44' X 56'

CONTACT DATA CENTER AND INFRASTRUCTURE ENGINEERING (VAITESDATABCENTERENGINEERING2@VA.GOV) TO DETERMINE WHAT SIZE DATA CENTER IS REQUIRED FOR YOUR FACILITY.

FLOORPLANS ARE NOT TO SCALE
DOCUMENTS ARE NOT TO BE USED FOR CONSTRUCTION

MECHANICAL

CHANGES TO DESIGN WILL REQUIRE A NEW COMPUTATIONAL FLUID DYNAMICS (CFD) ANALYSIS TO BE PERFORMED



**IT OPERATIONS AND SERVICES
SOLUTION DELIVERY**



EDUCT
ENTERPRISE DATA CENTER AND
INFRASTRUCTURE ENGINEERING

PROJECT:

OIT DESIGN GUIDE TEMPLATES

PROJECT No:

N/A

MARK	DATE	DESCRIPTION

ISSUE:

DRAWING No:

FILE: OIT_DESIGN GUIDE TEMPLATE
S-04162021-V3.1-VSDX

DESIGN BY: KEVIN GRZELKA, CTDC

VERIFIED BY: MICHAEL JULIAN, RCDD
KELLY BATES, DCIF

DOC VERSION No: 3.1

ISSUE DATE: Apr 16, 2021

SHEET TITLE

LEGEND

SHEET: 2 OF 37

GENERIC TELECOMMUNICATIONS MEDIA AND INTERFACE SPECIFICATIONS FOR TELECOMMUNICATION SPACES			
ID	PRIMARY ATTRIBUTE	SECONDARY ATTRIBUTE	SPECIFICATION
1	COPPER PATCH PANELS	PERFORMANCE CATEGORY	CATEGORY 6A (10 GBE)
		POSITION COUNT	48
		FORM FACTOR	ANGLED
		SIZE	ONE RACK UNIT
		JACK COLOR CODING	A-SIDE BLUE / B-SIDE YELLOW
2	FIBER DISTRIBUTION PANELS	CASSETTE CAPACITY	12 CASSETTES OR 6 DOUBLE CASSETTES
		CASSETTE USER INTERFACES	LC DUPLEX CONNECTORS
		CASSETTE BACKBONE INTERFACES	MPO-12
		PERFORMANCE CHARACTERISTICS	OM4 LASER ENHANCED 40 GBE 50/125 MULTIMODE / OS1 9/125 SINGLE MODE
		FORM FACTOR	ONE (1) RU
3	UTP (HORIZONTAL AND FIRST LEVEL BACKBONE)	PERFORMANCE CATEGORY	CATEGORY 6A (10 GBE)
		PERFORMANCE SPECIFICATIONS	MEETS OR EXCEEDS TIA-EIA-568-C.2-10, T5B-155.
		JACKET COLOR	BLUE (HORIZONTAL), WHITE (1 ST LEVEL BACKBONE)
4	FIBER (HORIZONTAL AND FIRST LEVEL BACKBONE)	PERFORMANCE CATEGORY	OM4 LASER ENHANCED TO 40 GIGABIT ETHERNET (GBE) / OS1 / OS2
		PERFORMANCE SPECIFICATIONS	LASER OPTIMIZED 50/125 MM FIBERS WITH AT LEAST 4,700 MHZ-KM AT 850 NM / OS1 9/125 5M FIBERS (INTRA-BUILDING) / OS2 9/125 5M (INTER-BUILDING)
		MODE	MULTIMODE/SINGLE MODE (DISTRIBUTION TO TRs ONLY)
		JACKET COLOR	AQUA (OM4) / YELLOW (OS1/2)
		MEDIA CONNECTOR	PRE-TERMINATED WITH MPO, TYPE A
		STRAND COUNT	12 OR 24
		BUNDLING	LOOSE TUBE
5	UTP PATCH CORDS	PERFORMANCE CATEGORY	CATEGORY 6A, 26-GAUGE, STRANDED
		PERFORMANCE SPECIFICATIONS	CENTER TUNED TO HORIZONTAL MEDIA
		JACKET COLOR	BLUE
		TERMINATION METHOD	FACTORY PRE-TERMINATED
6	FIBER PATCH CORDS	PERFORMANCE CATEGORY	OM4 / OS1 / OS2
		PERFORMANCE SPECIFICATIONS	LASER OPTIMIZED 50/125 MM FIBERS WITH AT LEAST 4,700 MHZ-KM AT 850 NM / OS1 9/125 5M FIBERS (INTRA-BUILDING) / OS2 9/125 5M (INTER-BUILDING)
		MODE	MULTIMODE/SINGLE MODE (PATCH CORD TO TRs ONLY)
		JACKET COLOR	AQUA (OM4) / YELLOW (OS1/2)
		MEDIA CONNECTOR	PRE-TERMINATED WITH DUPLEX LC



IT OPERATIONS AND SERVICES
SOLUTION DELIVERY



EDCT
ENTERPRISE DATA CENTER AND
INFRASTRUCTURE ENGINEERING

PROJECT:

OIT DESIGN GUIDE TEMPLATES

PROJECT No:
N/A

MARK	DATE	DESCRIPTION

ISSUE:

DRAWING No:

FILE: OIT_DESIGN_GUIDE_TEMPLATE
S-04162021-V3.1.VSDX

DESIGN BY: KEVIN GRZELKA, CTDC

VERIFIED BY: MICHAEL JULIAN, RCDD
KELLY BATES, DCIE

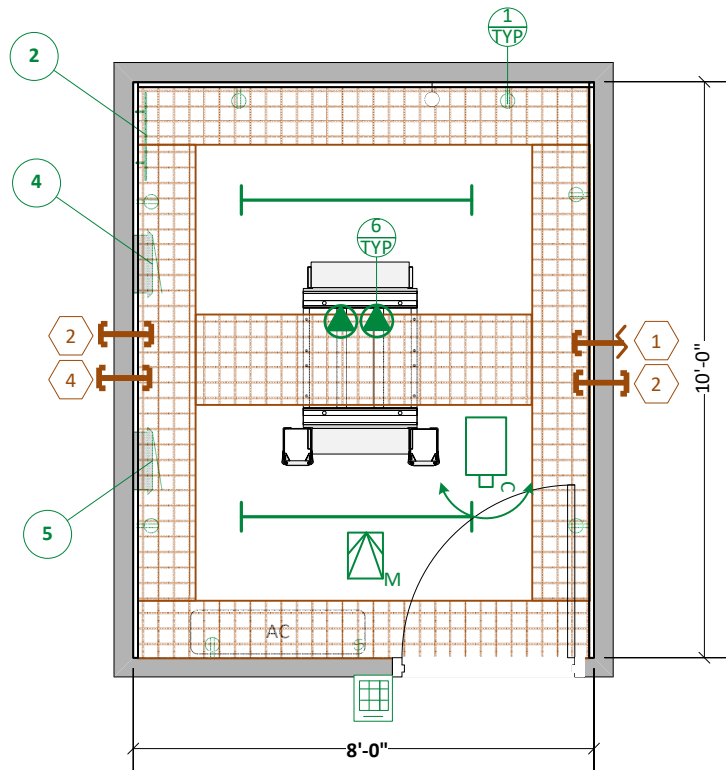
DOC VERSION No: 3.1

ISSUE DATE: Apr 16, 2021

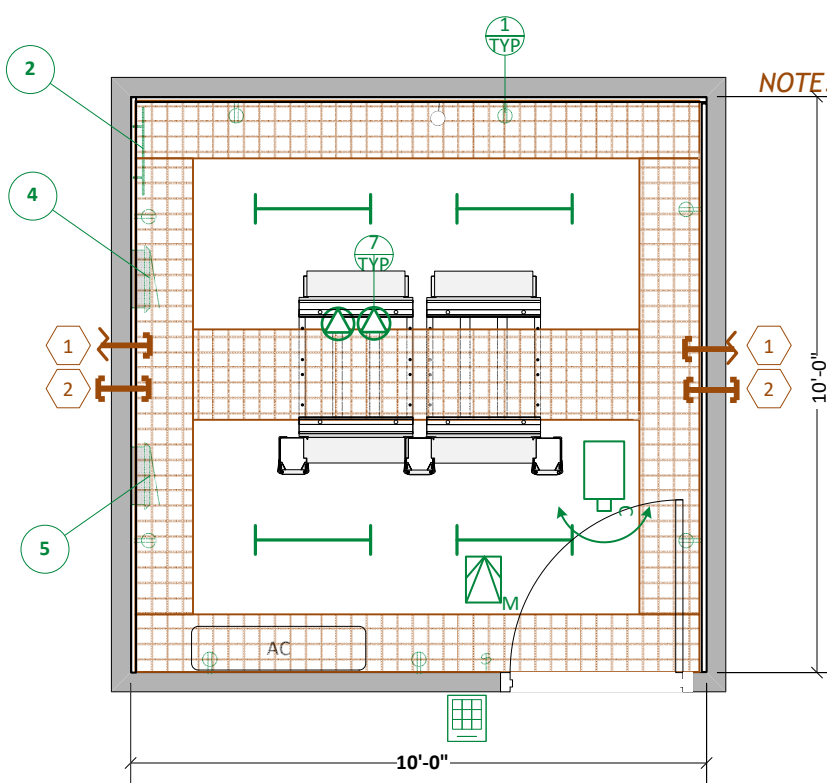
SHEET TITLE

SPECIFICATIONS

SHEET: 3 OF 37

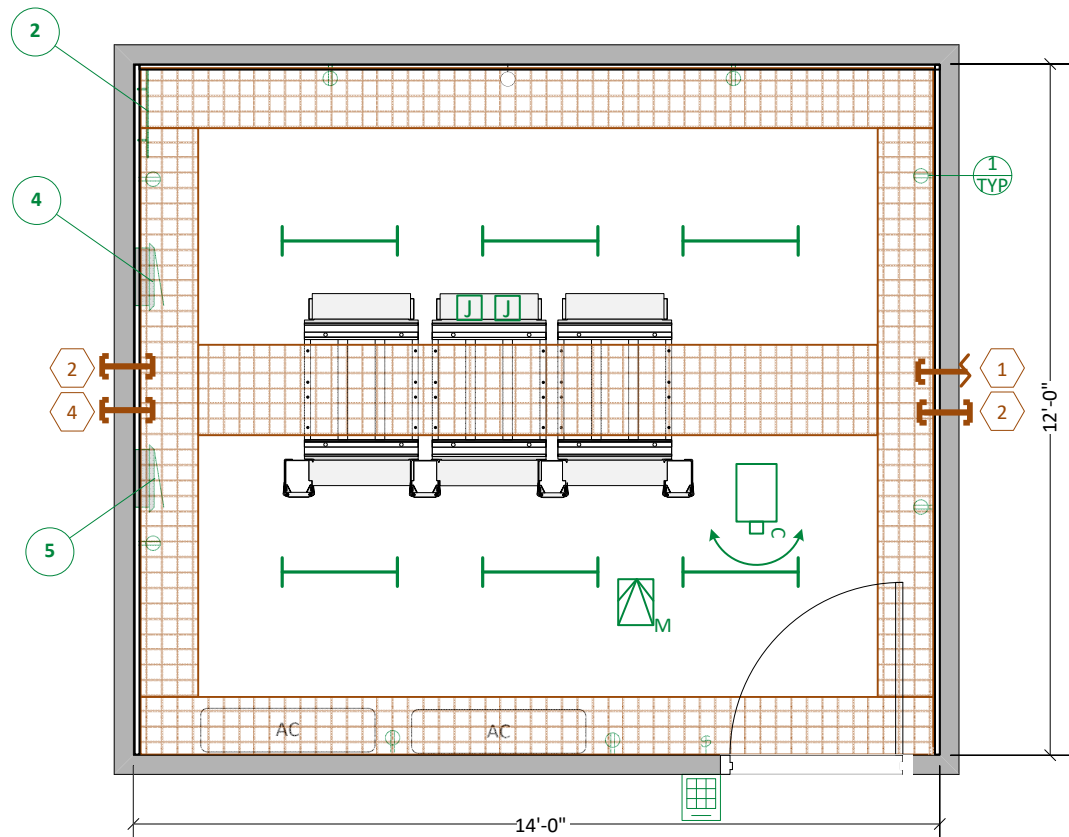


3 GENERIC FLOOR PLAN FOR ENTRANCE ROOM (ONE SERVICE PROVIDER)



NOTE: HEALTH CARE FACILITIES REQUIRE TWO DISTINCT ENTRANCE ROOMS SEPARATED BY A MINIMUM OF 66 FT.

4 GENERIC FLOOR PLAN FOR ENTRANCE ROOM (TWO SERVICE PROVIDERS)



2 GENERIC FLOOR PLAN FOR HEALTHCARE ENTRANCE ROOM

NOTE:
THERE SHALL BE A MINIMUM OF ONE ENTRANCE ROOM PER BUILDING.

WALL SPACE REQUIREMENT

REQUIRED TERMINATION WALL SPACE FOR AN ENTRANCE ROOM IN A TR OR ENTRANCE ROOM PER THE CHART BELOW:

GROSS FLOOR SPACE SERVED (FT ²)	WALL LENGTH FOR ENTRANCE FACILITY TERMINATION
<10,000 FT ²	39"
<20,000 FT ²	42"
<40,000 FT ²	68"
<50,000 FT ²	90"
<60,000 FT ²	96"
<80,000 FT ²	120"
<100,000 FT ²	144"

FLOOR SPACE REQUIREMENT

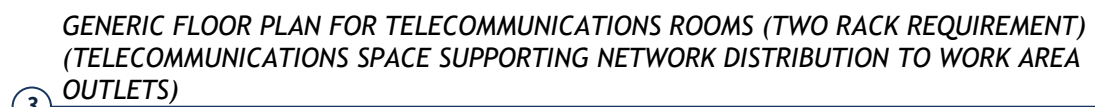
FLOOR SPACE SHOULD BE PROVIDED FOR FACILITIES WITH GROSS SQUARE FOOTAGE PER THE CHART BELOW FOR SHARED ENTRANCE ROOMS:

GROSS FLOOR SPACE SERVED (FT ²)	ENTRANCE FACILITY SIZE ALLOCATION
<200,000 FT ²	10'X10'
<400,000 FT ²	12'X13'
<500,000 FT ²	12'X15.5'
<600,000 FT ²	12'X18.5'
<800,000 FT ²	12'X22.5'
<1,000,000 FT ²	12'X27.5'

HEALTH CARE FACILITY ENTRANCE ROOMS SHALL BE 170 SQ FT MINIMUM

1 TELECOMMUNICATION ENTRANCE ROOM NOTES

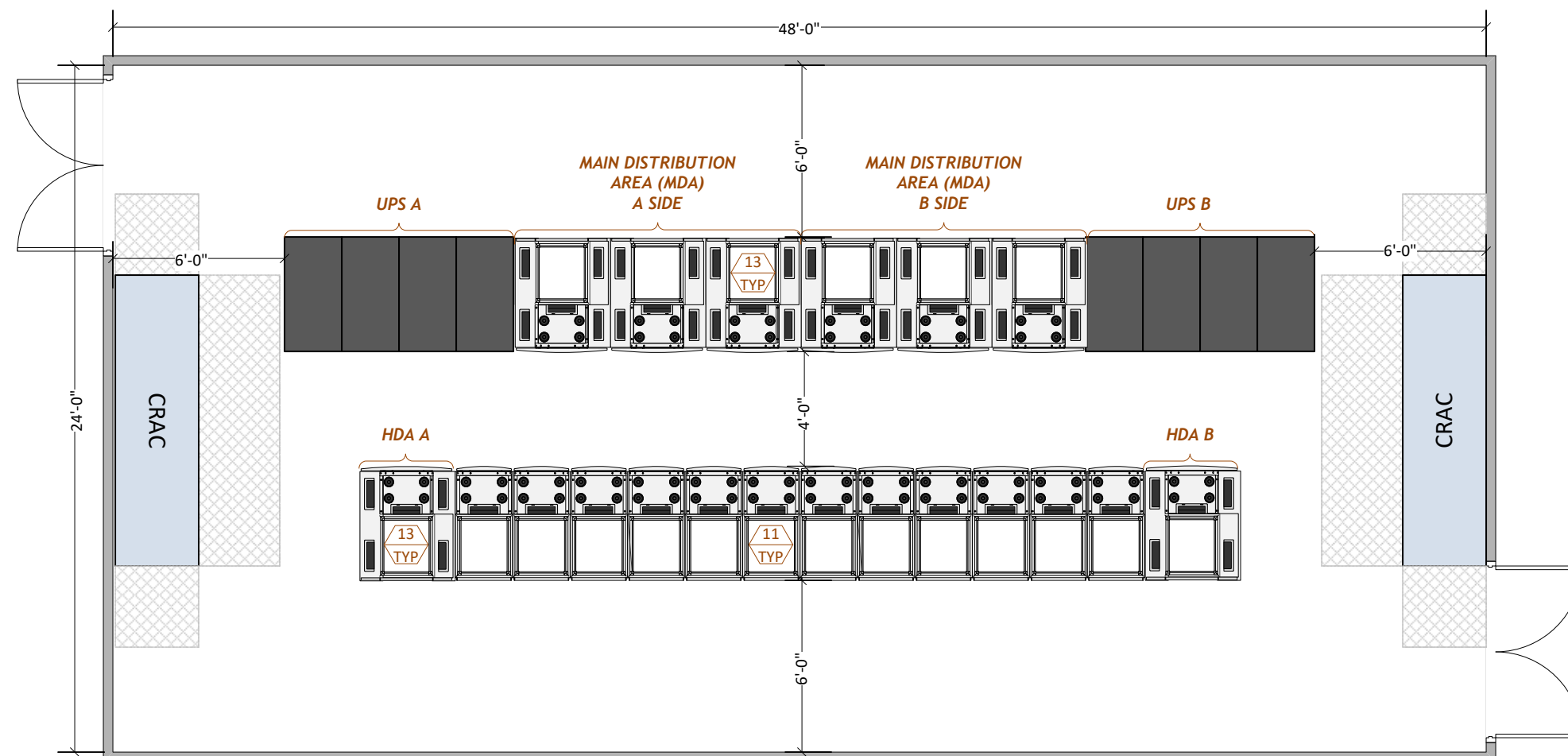




1 TELECOMMUNICATION ROOM NOTES



SHEET: 5 OF 37



1 GENERIC FLOOR PLAN FOR SMALL DATA CENTERS (1,152 FT²)



PROJECT:

OIT DESIGN GUIDE TEMPLATES

PROJECT No:

N/A

MARK	DATE	DESCRIPTION

ISSUE:

DRAWING No:

FILE: OIT_DESIGN_GUIDE_TEMPLATE
S-0416202I-V3.1.VSDX

DESIGN BY: KEVIN GRZELKA, CTDC

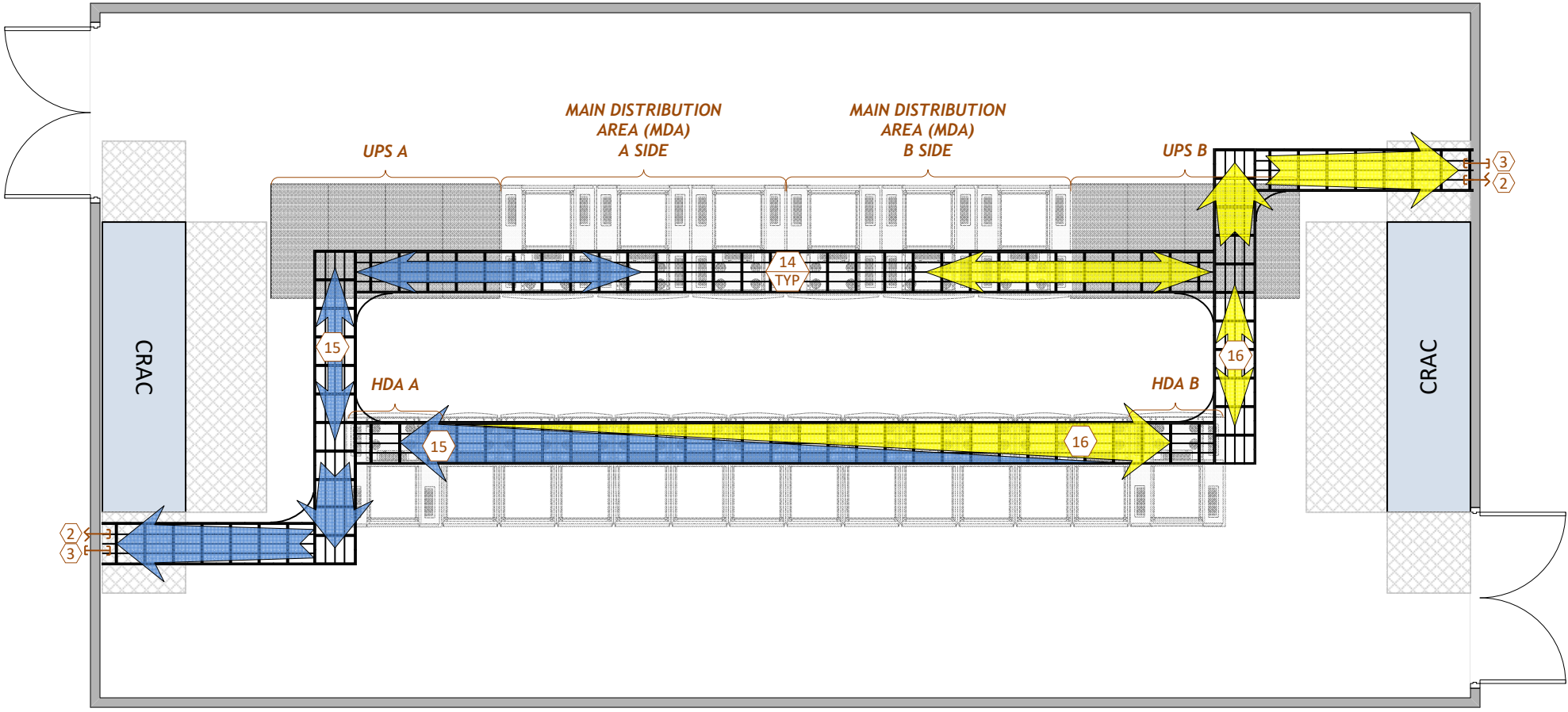
VERIFIED BY: MICHAEL JULIAN, RCDD
KELLY BATES, DCIE

DOC VERSION No:	3.1
-----------------	-----

ISSUE DATE:	Apr 16, 2021
-------------	--------------

SHEET TITLE

SMALL DATA CENTER



NOTE: SEE THE GENERIC SMALL CAMPUS
SUPPORT CENTER (DATA CENTER)
DESIGN PORTFOLIO PACKAGE FOR
DETAILED IMPLEMENTATION
INSTRUCTIONS

1 SMALL DATA CENTER - DATA CABLE TRAY LAYOUT



MARK	DATE	DESCRIPTION

ISSUE:

DRAWING No:

FILE: OIT DESIGN GUIDE TEMPLATE
S-04162021-V3.1.VSDX

DESIGN BY: KEVIN GRZELKA, CTDC

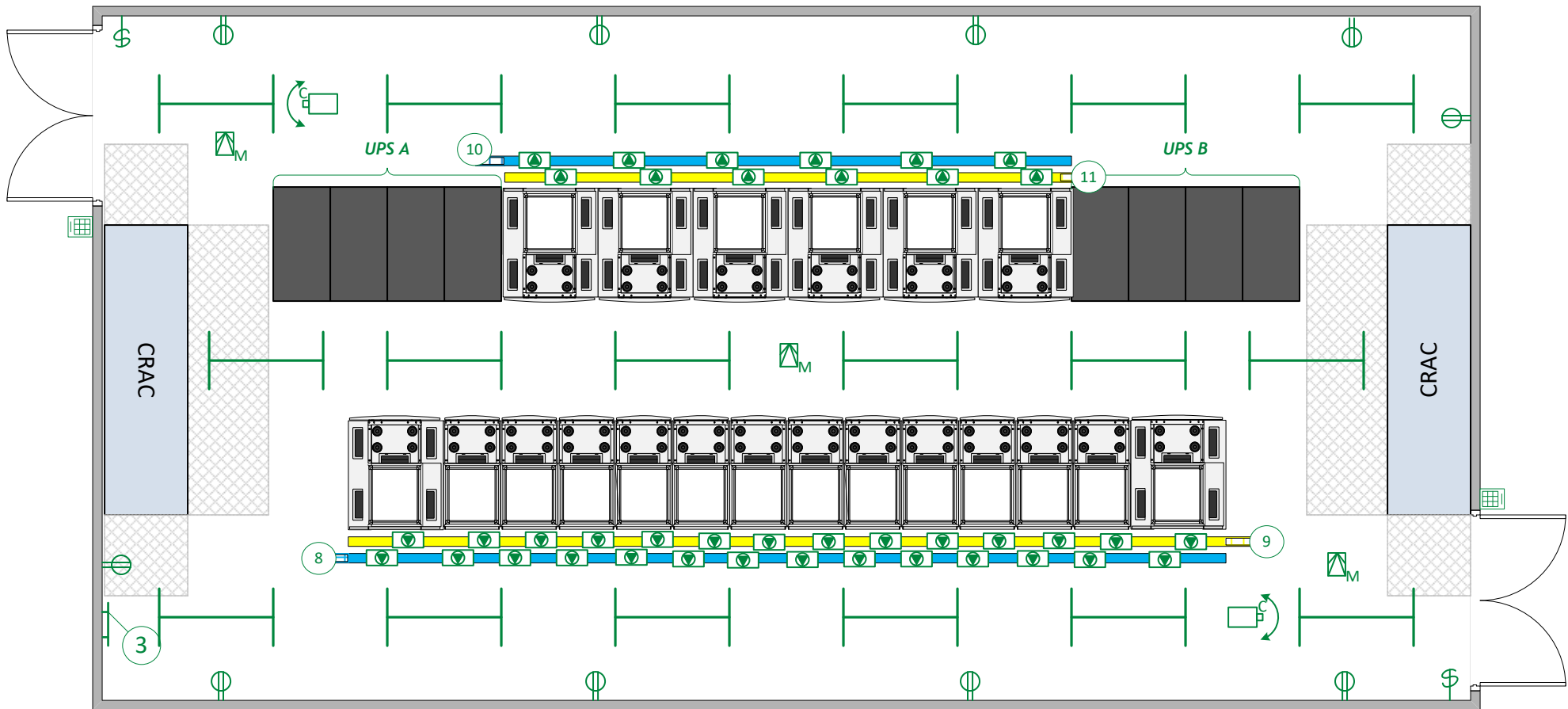
VERIFIED BY: MICHAEL JULIAN, RCDD
KELLY BATES, DCIE

DOC VERSION No: 3.1

ISSUE DATE: Apr 16, 2021

SHEET TITLE

SMALL DATA CENTER - PATHWAY
DETAILS



1 GENERIC FLOOR PLAN FOR SMALL DATA CENTERS - POWER PLAN (72.9 W/FT²)

DEPARTMENT OF VETERANS
AFFAIRS



IT OPERATIONS AND SERVICES
SOLUTION DELIVERY



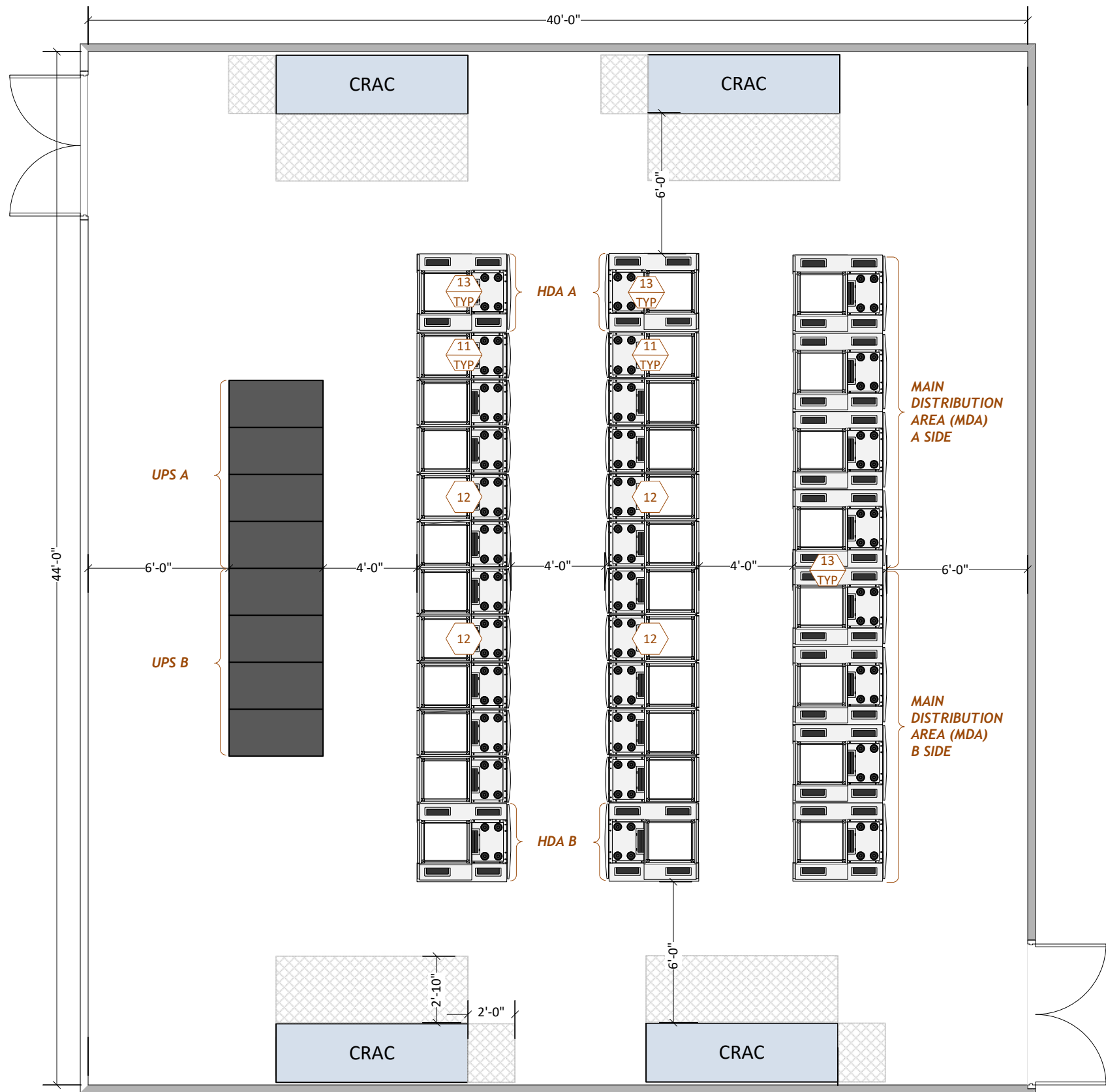
EDCT
ENTERPRISE DATA CENTER AND
INFRASTRUCTURE ENGINEERING

PROJECT:
OIT DESIGN GUIDE TEMPLATES
PROJECT No:
N/A

MARK	DATE	DESCRIPTION

ISSUE:
DRAWING No:
FILE: OIT DESIGN GUIDE TEMPLATE
S-04162021-V3.1.VSDX
DESIGN BY: KEVIN GRZELKA, CTDC
VERIFIED BY: MICHAEL JULIAN, RCDD
KELLY BATES, DCIE
DOC VERSION No: 3.1
ISSUE DATE: Apr 16, 2021

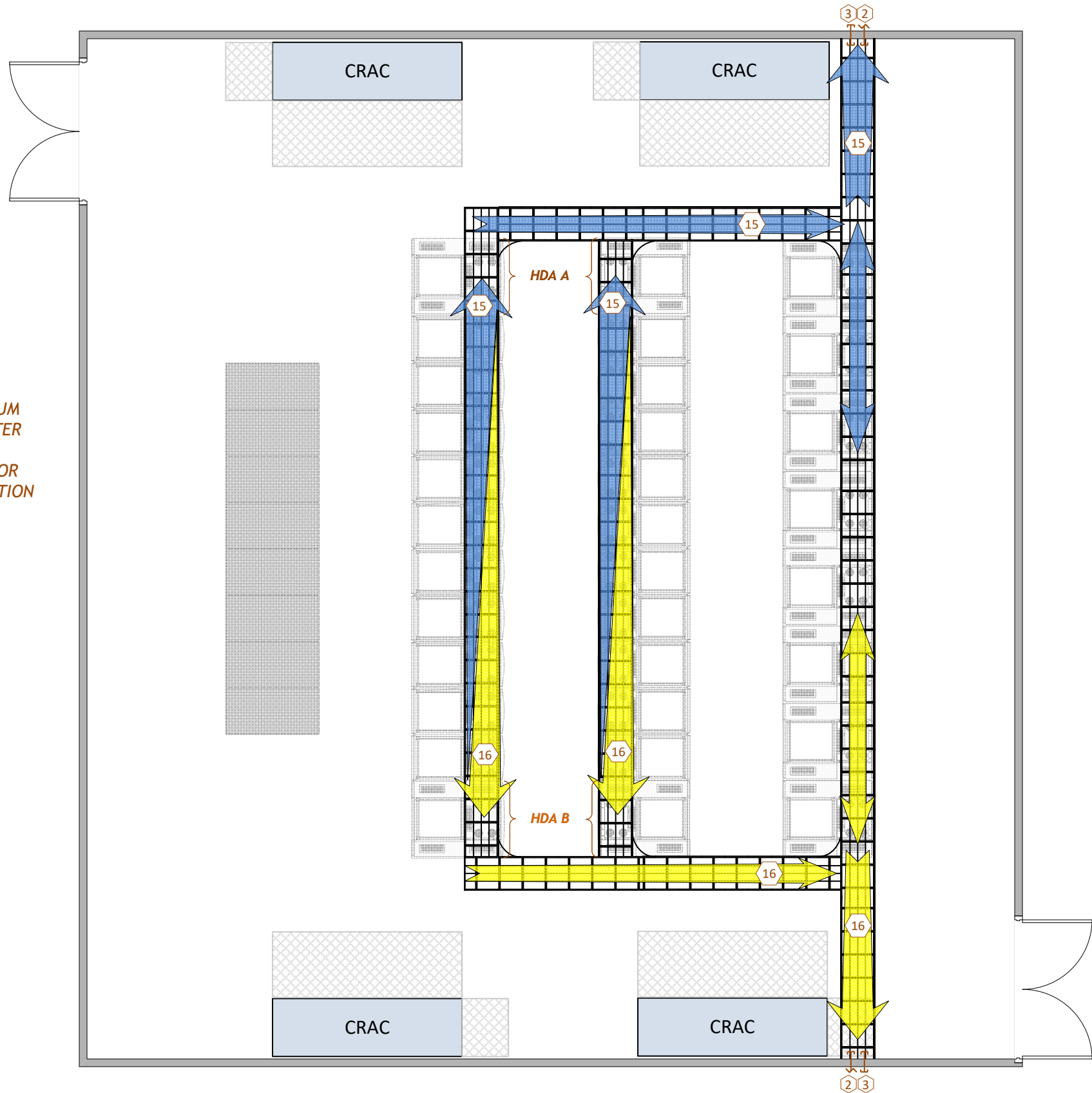
SHEET TITLE
SMALL DATA CENTER - POWER



1 GENERIC FLOOR PLAN FOR MEDIUM DATA CENTERS (1,760 FT²)



NOTE: SEE THE GENERIC MEDIUM CAMPUS SUPPORT CENTER (DATA CENTER) DESIGN PORTFOLIO PACKAGE FOR DETAILED IMPLEMENTATION INSTRUCTIONS



1 MEDIUM DATA CENTER - DATA CABLE TRAY LAYOUT

DEPARTMENT OF VETERANS
AFFAIRS



IT OPERATIONS AND SERVICES
SOLUTION DELIVERY



EDCT

ENTERPRISE DATA CENTER AND
INFRASTRUCTURE ENGINEERING

PROJECT:

OIT DESIGN GUIDE TEMPLATES

PROJECT No:

N/A

MARK	DATE	DESCRIPTION
------	------	-------------

ISSUE:

DRAWING No:

FILE: OIT_DESIGN_GUIDE_TEMPLATE
S-04162021-V3.1.VSDX

DESIGN BY: KEVIN GRZELKA, CTDC

VERIFIED BY: MICHAEL JULIAN, RCDD
KELLY BATES, DCIE

DOC VERSION No: 3.1

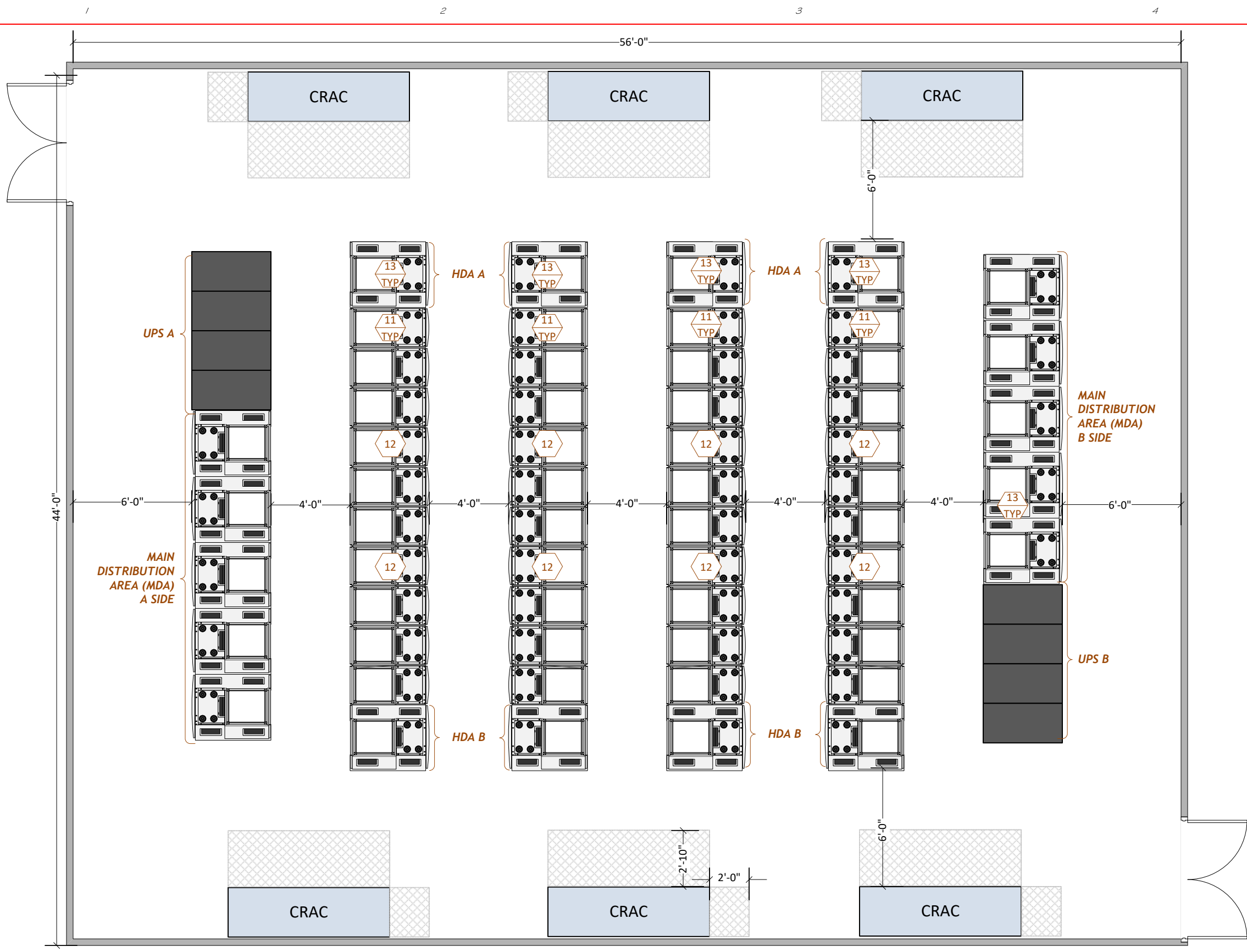
ISSUE DATE: Apr 16, 2021

SHEET TITLE

MEDIUM DATA CENTER -
PATHWAY DETAILS

SHEET: 11 OF 37





1 GENERIC FLOOR PLAN FOR LARGE DATA CENTERS (2,464 FT²)



PROJECT:

OIT DESIGN GUIDE TEMPLATES

PROJECT No:

N/A

MARK	DATE	DESCRIPTION

ISSUE:

DRAWING No:

FILE: OIT_DESIGN_GUIDE_TEMPLATE
S-04162021-V3.1.VSDX

DESIGN BY: KEVIN GRZELKA, CTDC

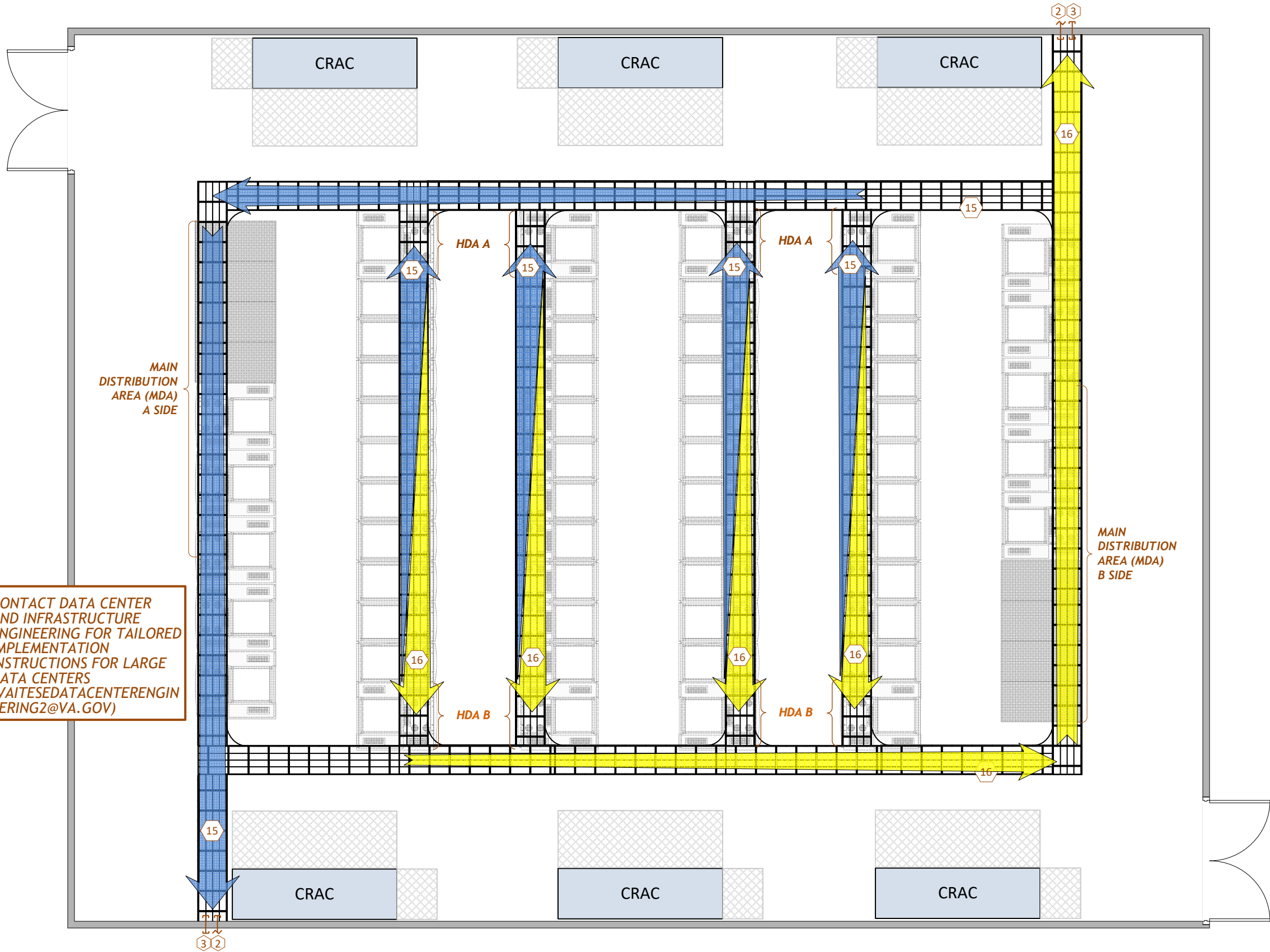
VERIFIED BY: MICHAEL JULIAN, RCDD
KELLY BATES, DCIE

DOC VERSION No: 3.1

ISSUE DATE: Apr 16, 2021

SHEET TITLE

LARGE DATA CENTER



NOTE: CONTACT DATA CENTER AND INFRASTRUCTURE ENGINEERING FOR TAILORED IMPLEMENTATION INSTRUCTIONS FOR LARGE DATA CENTERS (VAITSEEDATACENTERENGINEERING2@VA.GOV)

1 LARGE DATA CENTER - DATA CABLE TRAY LAYOUT



PROJECT:

OIT DESIGN GUIDE TEMPLATES

PROJECT No:

N/A

MARK	DATE	DESCRIPTION
------	------	-------------

ISSUE:

DRAWING No:

FILE: OIT_DESIGN_GUIDE_TEMPLATE
S-04162021-V3.1.VSDX

DESIGN BY: KEVIN GRZELKA, CTDC

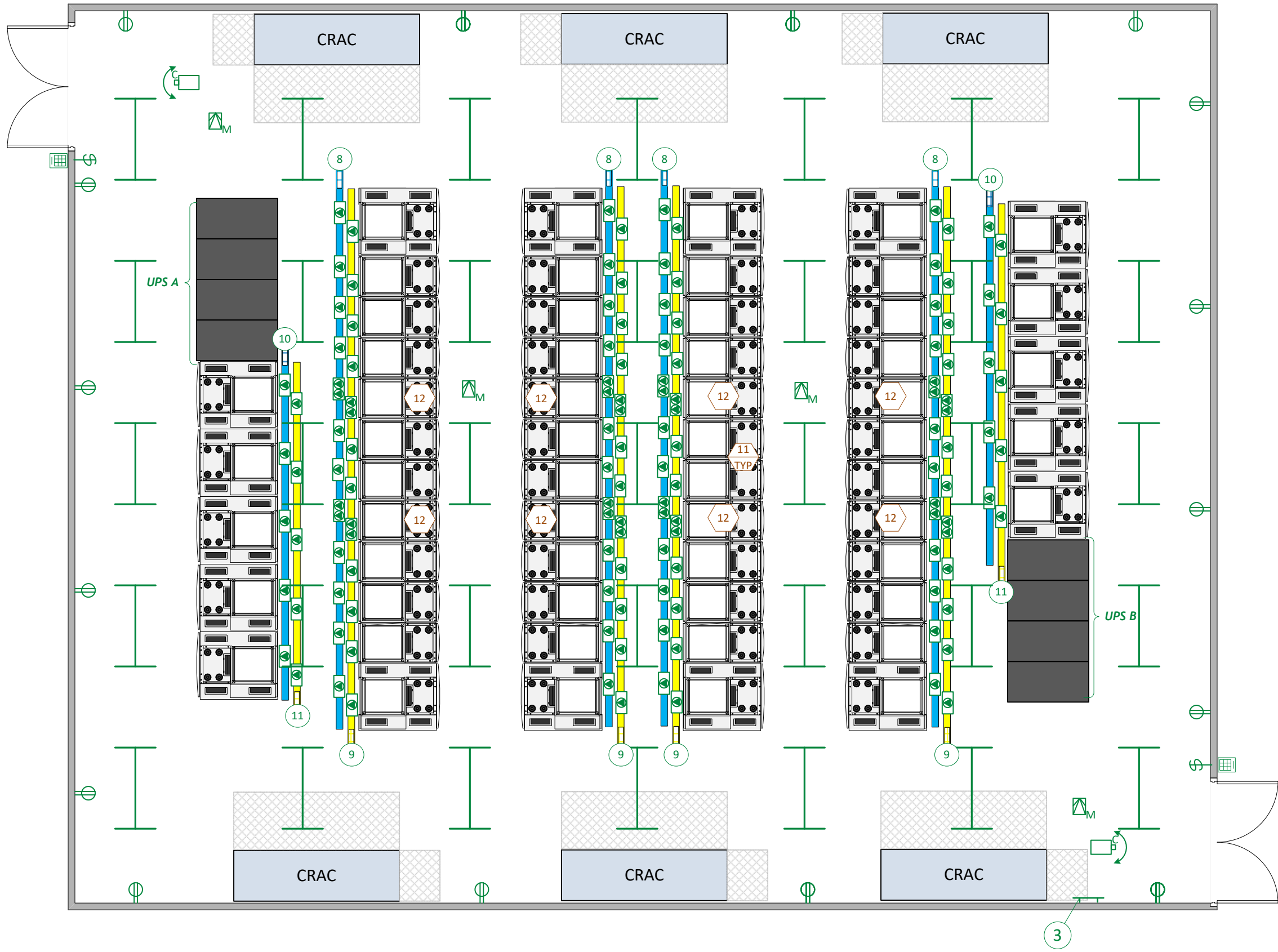
VERIFIED BY: MICHAEL JULIAN, RCDD
KELLY BATES, DCIE

DOC VERSION No: 3.1

ISSUE DATE: Apr 16, 2021

SHEET TITLE

LARGE DATA CENTER - PATHWAY
DETAILS



1 GENERIC FLOOR PLAN FOR LARGE DATA CENTERS - POWER PLAN



PROJECT:

OIT DESIGN GUIDE TEMPLATES

PROJECT No:

N/A

MARK	DATE	DESCRIPTION

ISSUE:

DRAWING No:

FILE: OIT DESIGN GUIDE TEMPLATE
S-04162021-V3.1.VSDX

DESIGN BY: KEVIN GRZELKA, CTDC

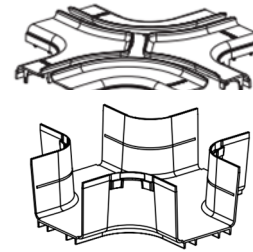
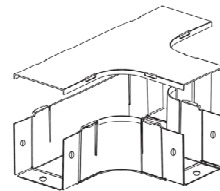
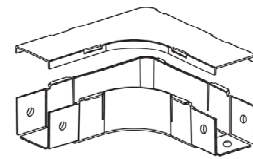
VERIFIED BY: MICHAEL JULIAN, RCDD
KELLY BATES, DCIE

DOC VERSION No: 3.1

ISSUE DATE: Apr 16, 2021

SHEET TITLE

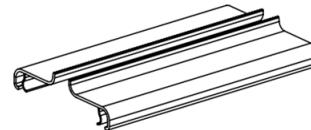
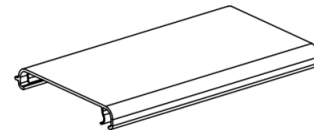
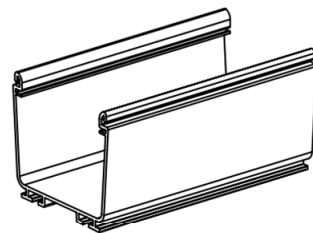
LARGE DATA CENTER - POWER



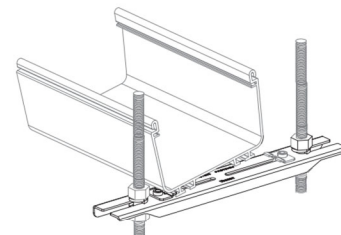
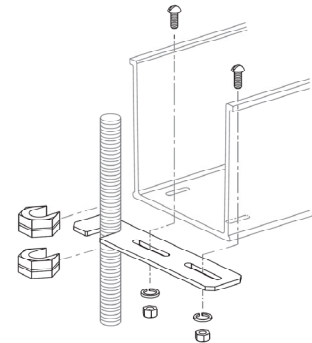
FIBER PATHWAY CHANNEL CORNERS (PANDUIT FR4X4/FRHC4/FRSHC4 OR EQUIVALENT):
 RIGHT ANGLE CORNER
 TEE CONER
 FOUR WAY CROSSING
 SPLIT OR SOLID COVERS



FIBER PATHWAY BEND RADIUS SPILLWAYS (PANDUIT FRVT4X4/FRTR4X4 OR EQUIVALENT):
VERTICAL TEE
TRUMPET SPILLWAY
BEND RADIUS SPILLWAYS REQUIRED AT EVERY RACK LOCATION

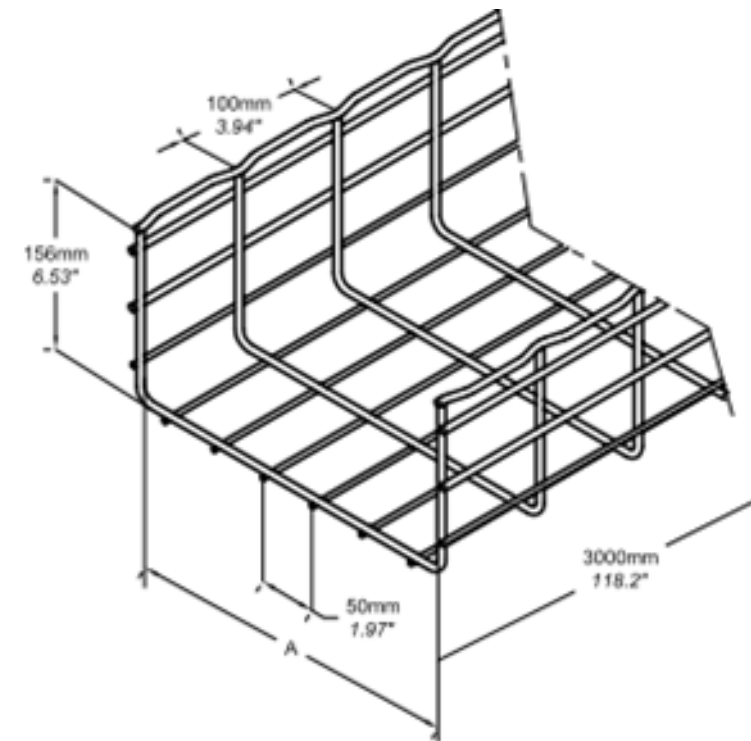


FIBER PATHWAY CHANNEL AND COVERS (PANDUIT FR4X4/FRHC4/FRSHC4 OR EQUIVALENT):
4" X 4" CHANNEL RUNNER
SOLID COVER
SPLIT HINGED COVER



FIBER PATHWAY MOUNTING BRACKET (PANDUIT FR6TRBE/FR6TRBN OR EQUIVALENT):
BRACKET TO MOUNT TO EXISTING THREADED ROD OR
BRACKET TO MOUNT TO NEW THREADED ROD

2 PROPOSED FIBER PATHWAY AND DETAILS



PROPOSED CABLE TRAY:
CABLOFIL
4" DEEP X 12" WIDE MINIMUM (SIZE COMMENSURATE WITH
DISTRIBUTION REQUIREMENT)
WHITE POWERCOAT
TRAY INSERT

CAT. NO	DEPTH (IN)	WIDTH (IN)	CAT 6a 4-pr PLENUM (0.35")
CF105/300	4	12	248
CF105/450	4	18	373
CF105/500	4	20	414
CF105/600	4	24	497
CF150/150	6	6	186
CF150/200	6	8	238
CF150/300	6	12	357
CF150/450	6	18	536
CF150/500	6	20	596
CF150/600	6	24	715
CF150/900	6	36	1078

1 CABLE TRAY CAPACITY PLANNING



**IT OPERATIONS AND SERVICES
SOLUTION DELIVERY**



EDCT
ENTERPRISE DATA CENTER AND
INFRASTRUCTURE ENGINEERING

PROJECT:

OIT DESIGN GUIDE TEMPLATES

PROJECT No:

N/A

MARK	DATE	DESCRIPTION

ISSUE:

DRAWING No:

FILE: OIT_DESIGN GUIDE TEMPLATE
S-0416202T-V3.1.VSDX

DESIGN BY: KEVIN GRZELKA, CTDC

VERIFIED BY: MICHAEL JULIAN, RCDD
KELLY BATES, DCIE

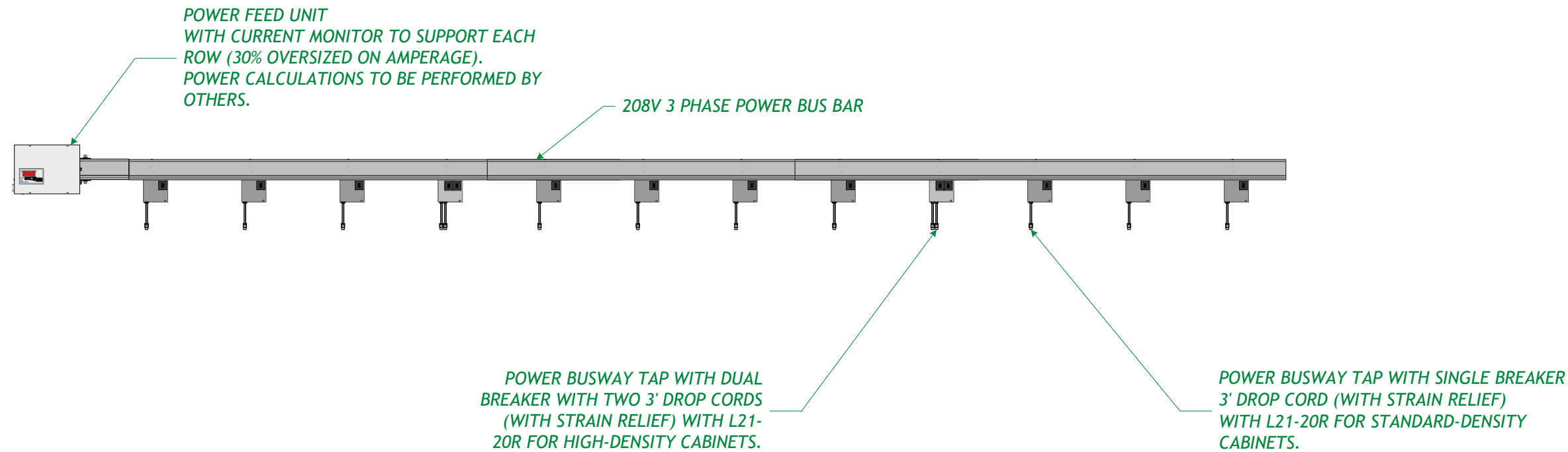
DOC VERSION No:	3.1
-----------------	-----

ISSUE DATE: Apr 16, 2021

SHEET TITLE

PATHWAY DETAILS

SHEET: 16 OF 37



3 TYPICAL POWER BUSWAY COMPONENTS FOR SERVER ROWS



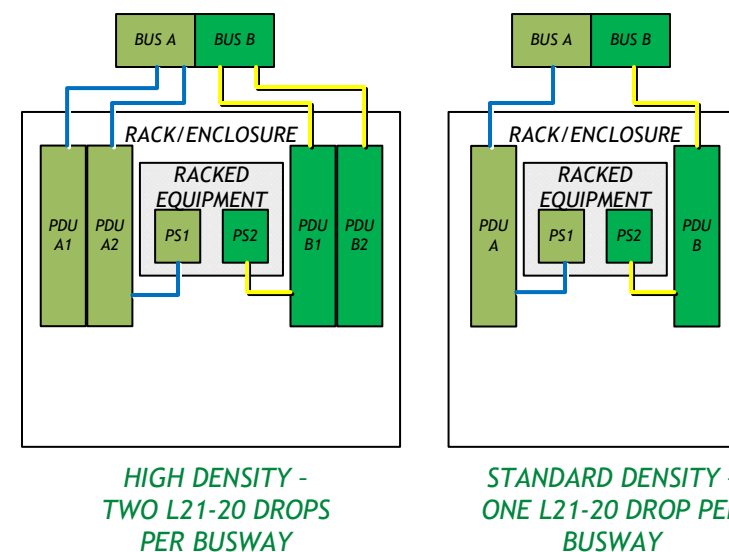
CHATSWORTH ECONNECT PDU; VERTICAL; MONITORED; L21-20 PLUG; THREE PHASE; 120/208V INPUT; 20A; (30) C13 (6) C19 OUTLETS; 208V OUTPUT; 3 X 2P 20A HYDRAULIC MAGNETIC BREAKERS; LCD; ETHERNET, USB, AND ENVIRONMENTAL SENSOR PORTS; IP AND SERIAL MONITORING; IP CONSOLIDATION (PDU LINKING); TOOL-LESS MOUNTING; 70.5"H (1791 MM) X 2.2"W (56 MM) X 2.2"D (56 MM)



REQUIRES ONE L21-20R

2 208 VOLT RACK PDU FOR EQUIPMENT POWER - TO BE ENERGIZED BY BUSWAY TAP

THIS DESIGN PROVIDES DIVERSE POWER INPUTS FOR ACTIVE EQUIPMENT BY SPLITTING THE SOURCE POWER ACROSS TWO BUSWAYS. EACH SERVER CABINET WILL CONTAIN A MINIMUM OF TWO EQUIPMENT-FACING ZERO U VERTICAL RACK PDUS - EACH WILL BE ENERGIZED BY SEPARATE BUSWAY TAP INPUTS ON DISPARATE BUSWAYS.

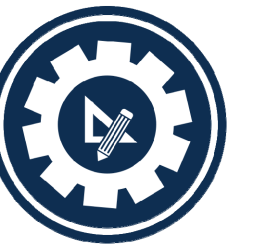


1 POWER SCHEMATIC FOR POWER RACK- LEVEL REDUNDANCY

DEPARTMENT OF VETERANS AFFAIRS



IT OPERATIONS AND SERVICES SOLUTION DELIVERY



EDCT
ENTERPRISE DATA CENTER AND
INFRASTRUCTURE ENGINEERING

PROJECT:

OIT DESIGN GUIDE TEMPLATES

PROJECT No:

N/A

MARK	DATE	DESCRIPTION
------	------	-------------

ISSUE:

DRAWING No:

FILE: OIT DESIGN GUIDE TEMPLATE
S-04162021-V3.1.VSDX

DESIGN BY: KEVIN GRZELKA, CTDC

VERIFIED BY: MICHAEL JULIAN, RCDD
KELLY BATES, DCIE

DOC VERSION No: 3.1

ISSUE DATE: Apr 16, 2021

SHEET TITLE

BUSWAY POWER DISTRIBUTION

SHEET: 17 OF 37



30 AMP 3-PHASE ZONE PDU BASE UNIT - REAR
CIRCUITS WITH FOUR L21-20R AND SIX NEMA 5-20 OUTLETS AND TWO 10' POWER CORDS TERMINATING IN L21-30P
ZONIT ZPDS-208V30A-L21-4L21-20R-CR OR EQUIVALENT



NOTE: THE CORRECT SPECIFICATION FOR THE ZONE PDU IS TO ENERGIZE IT FROM TWO INDEPENDENT POWER SOURCES. EACH INPUT WILL USE IDENTICAL SPECS: WYE (5-WIRE), 208V, 30A, THREE-PHASE, TERMINATING IN A NEMA L21-30R LOCKING RECEPTACLE. THE NEUTRAL CONDUCTOR SHOULD BE UPSIZED ONE GAUGE TO MATCH THE UPSIZED NEUTRAL CONDUCTORS IN THE ZONE PDU UNITS. THE NEUTRAL "UPSIZING" SHOULD IDEALLY BE CONTINUED IN THE POWER DISTRIBUTION SYSTEM BACK TO THE UPS OR TRANSFORMER WINDING POLE. THIS INCREASES THE EFFICIENCY OF THE POWER DISTRIBUTION SYSTEM AND SUPPRESSES HARMONICS IN THE SYSTEM.

2 TYPICAL ZONE PDU UNIT WITH REAR DETAIL

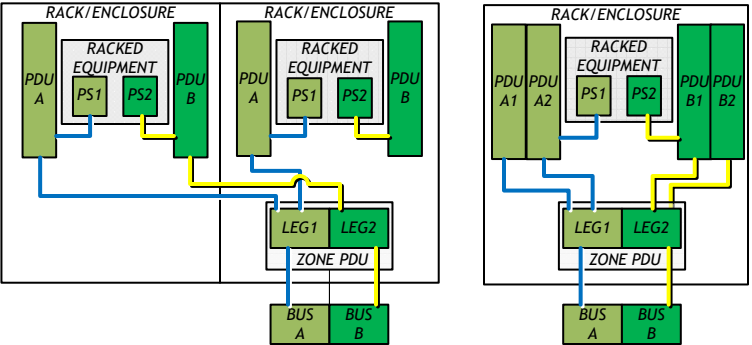


CHATSORTH ECONNECT PDU; VERTICAL; MONITORED; L21-20 PLUG; THREE PHASE; 120/208V INPUT; 20A; (30) C13 (6) C19 OUTLETS; 208V OUTPUT; 3 X 2P 20A HYDRAULIC MAGNETIC BREAKERS; LCD; ETHERNET, USB, AND ENVIRONMENTAL SENSOR PORTS; IP AND SERIAL MONITORING; IP CONSOLIDATION (PDU LINKING); TOOL-LESS MOUNTING; 70.5"H (1791 MM) X 2.2"W (56 MM) X 2.2"D (56 MM)

208 VOLT RACK PDU FOR EQUIPMENT POWER - TO BE ENERGIZED BY ZONE PDU

3

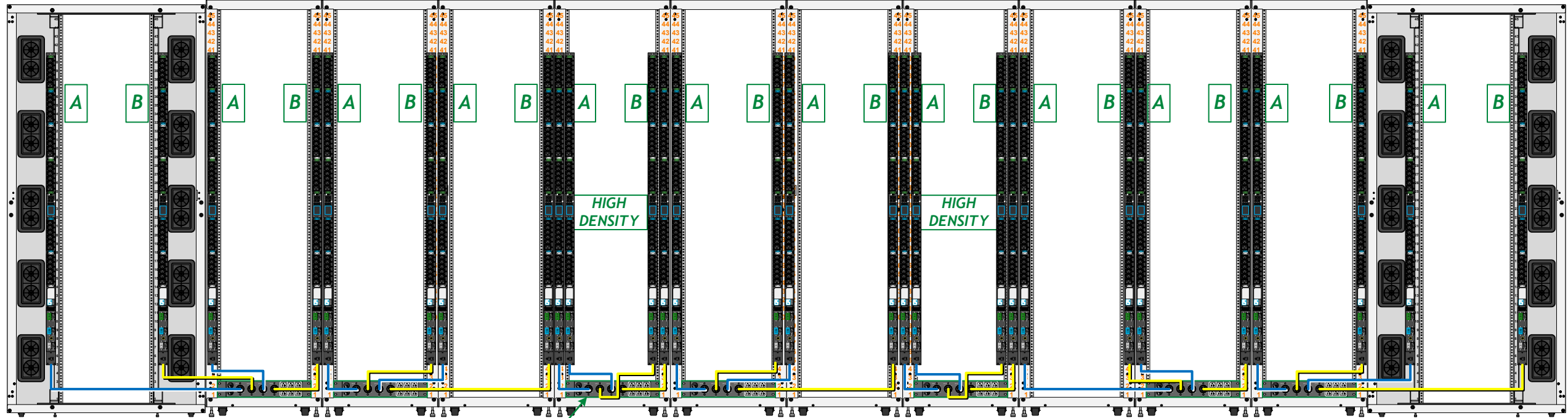
THIS DESIGN PROVIDES DIVERSE POWER INPUTS FOR ACTIVE EQUIPMENT BY SPLITTING THE SOURCE POWER ACROSS TWO BUSWAYS. EACH SERVER CABINET WILL CONTAIN A MINIMUM OF TWO EQUIPMENT-FACING ZERO U VERTICAL RACK PDUS - EACH WILL BE ENERGIZED BY SEPARATE BUSWAY TAP INPUTS ON DISPARATE BUSWAYS.



STANDARD DENSITY -
ONE ZONE PDU PER TWO
RACKS

HIGH DENSITY -
ONE ZONE PDU PER
RACK

4 POWER SCHEMATIC FOR POWER RACK- LEVEL REDUNDANCY



SEVEN ZONE PDUs PER SERVER ROW
(MEDIUM AND LARGE)

NOTE: SMALL SERVER ROWS AND ALL MDA ROWS FOLLOW THE SAME SCHEMA WITH ONE ZONE PDU PER TWO STANDARD RACKS/CABINETS AND ONE FOR EACH HIGH DENSITY CABINET

1 SUBZONE TO 30A ZONE PDU CONNECTION MAP FOR TYPICAL (12 RACK) SERVER ROWS (MEDIUM AND LARGE)



PROJECT:

OIT DESIGN GUIDE TEMPLATES

PROJECT No:

N/A

MARK	DATE	DESCRIPTION
------	------	-------------

ISSUE:

DRAWING No:

FILE: OIT_DESIGN_GUIDE_TEMPLATE
S-04162021-V3.1.VSDX

DESIGN BY: KEVIN GRZELKA, CTDC

VERIFIED BY: MICHAEL JULIAN, RCDD
KELLY BATES, DCIE

DOC VERSION No: 3.1

ISSUE DATE: Apr 16, 2021

SHEET TITLE

30A ZONE PDU POWER
DISTRIBUTION



60A 3-PHASE ZONE PDU BASE UNIT - REAR SIX L21-20R OUTPUT RECEPTACLES WITH 10' POWER CORDS TERMINATING IN HARDWIRE BOX
ZONIT ZPDS-208V60A-HW-6L21-20R OR EQUIVALENT

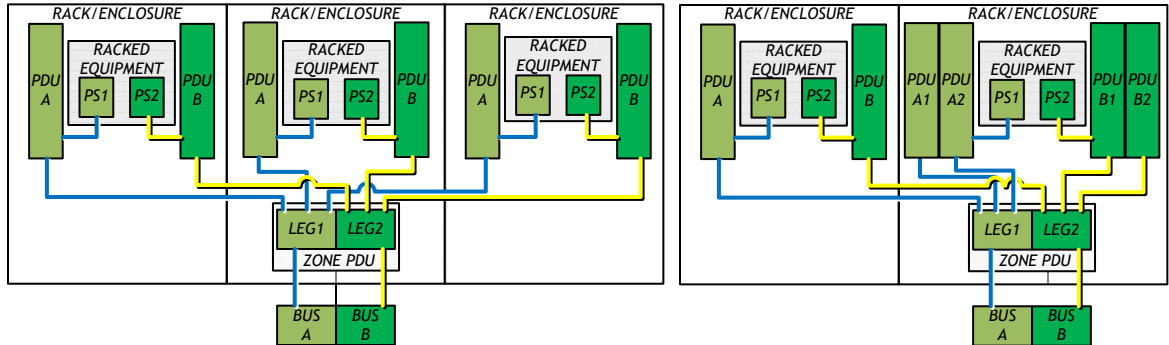
SPECIFICATIONS: RATED INPUT VOLTAGE: 208V 3 Φ ; AMPERAGE: 60A
INPUT CORD TERMINATION: HARDWIRE BOX
OUTPUT RECEPTACLES: SIX L21-20R
OUTPUT BRANCH CIRCUIT PROTECTION:
INDIVIDUAL 20A CIRCUIT BREAKERS PROTECT EACH
OUTPUT PHASE OF EACH OUTPUT RECEPTACLE

NOTE: THE CORRECT SPECIFICATION FOR THE ZONE PDU IS TO ENERGIZE IT FROM TWO INDEPENDENT POWER SOURCES. EACH INPUT WILL USE IDENTICAL SPECS: WYE (5-WIRE), 208V, 60A, THREE-PHASE, TERMINATING IN A JUNCTION BOX TO BE HARDWIRED INTO THE ZONE PDU. THE NEUTRAL CONDUCTOR SHOULD BE UPSIZED ONE GAUGE TO MATCH THE UPSIZED NEUTRAL CONDUCTORS IN THE ZONE PDU UNITS. THE NEUTRAL "UPSIZING" SHOULD IDEALLY BE CONTINUED IN THE POWER DISTRIBUTION SYSTEM BACK TO THE UPS OR TRANSFORMER WINDING POLE. THIS INCREASES THE EFFICIENCY OF THE POWER DISTRIBUTION SYSTEM AND SUPPRESSES HARMONICS IN THE SYSTEM.
TYPICAL ZONE PDU UNIT WITH REAR DETAIL AND ADAPTORS REQUIRED FOR VERTICAL PDU



REQUIRES ONE (1)
L21-20R

208 VOLT RACK PDU FOR EQUIPMENT POWER - TO BE ENERGIZED BY ZONE PDU

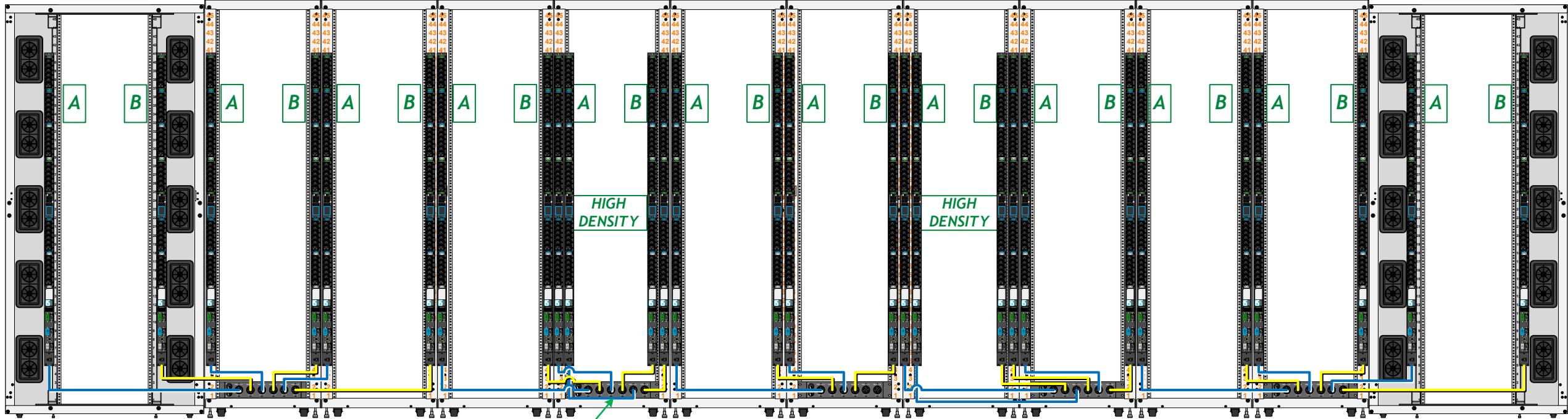


ONE ZONE PDU FOR THREE STANDARD DENSITY
SERVER CABINETS/NETWORK RACKS

ONE ZONE PDU FOR ONE STANDARD
DENSITY SERVER CABINET AND ONE HIGH
DENSITY SERVER CABINET

POWER SCHEMATIC FOR POWER RACK- LEVEL REDUNDANCY

NOTE:
THIS DESIGN PROVIDES DIVERSE
POWER INPUTS FOR ACTIVE
EQUIPMENT BY SPLITTING THE
SOURCE POWER ACROSS TWO
INPUTS ON THE ZONE PDU.
EACH INPUT WILL SUPPORT
THREE EQUIPMENT-FACING
RACK PDUS. EACH SERVER
CABINET WILL CONTAIN A
MINIMUM OF TWO EQUIPMENT-
FACING PDUS - EACH WILL BE
ENERGIZED BY SEPARATE ZONE
PDU INPUTS.



FIVE ZONE PDUs PER ROW (MEDIUM AND LARGE)

NOTE: SMALL SERVER ROWS AND ALL MDA ROWS FOLLOW THE SAME
SCHEMA WITH ONE ZONE PDU PER THREE STANDARD RACKS/
CABINETS AND ONE FOR EACH SET OF ONE HIGH DENSITY
CABINET AND ONE STANDARD DENSITY CABINET

SUBZONE TO 60A ZONE PDU CONNECTION MAP FOR TYPICAL (12 RACK) SERVER ROWS (MEDIUM AND LARGE)

DEPARTMENT OF VETERANS
AFFAIRS



IT OPERATIONS AND SERVICES
SOLUTION DELIVERY



EDCT
ENTERPRISE DATA CENTER AND
INFRASTRUCTURE ENGINEERING

PROJECT:

OIT DESIGN GUIDE TEMPLATES

PROJECT No:

N/A

MARK DATE DESCRIPTION

ISSUE:

DRAWING No:

FILE: OIT DESIGN GUIDE TEMPLATE
S-04162021-V3.1.VSDX

DESIGN BY: KEVIN GRZELKA, CTDC

VERIFIED BY: MICHAEL JULIAN, RCDD
KELLY BATES, DCIE

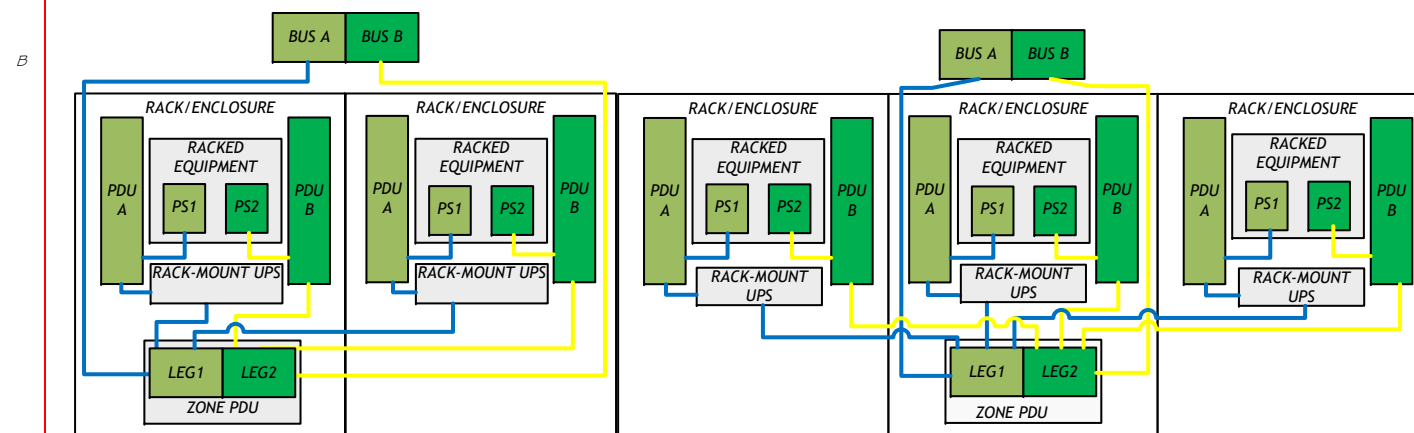
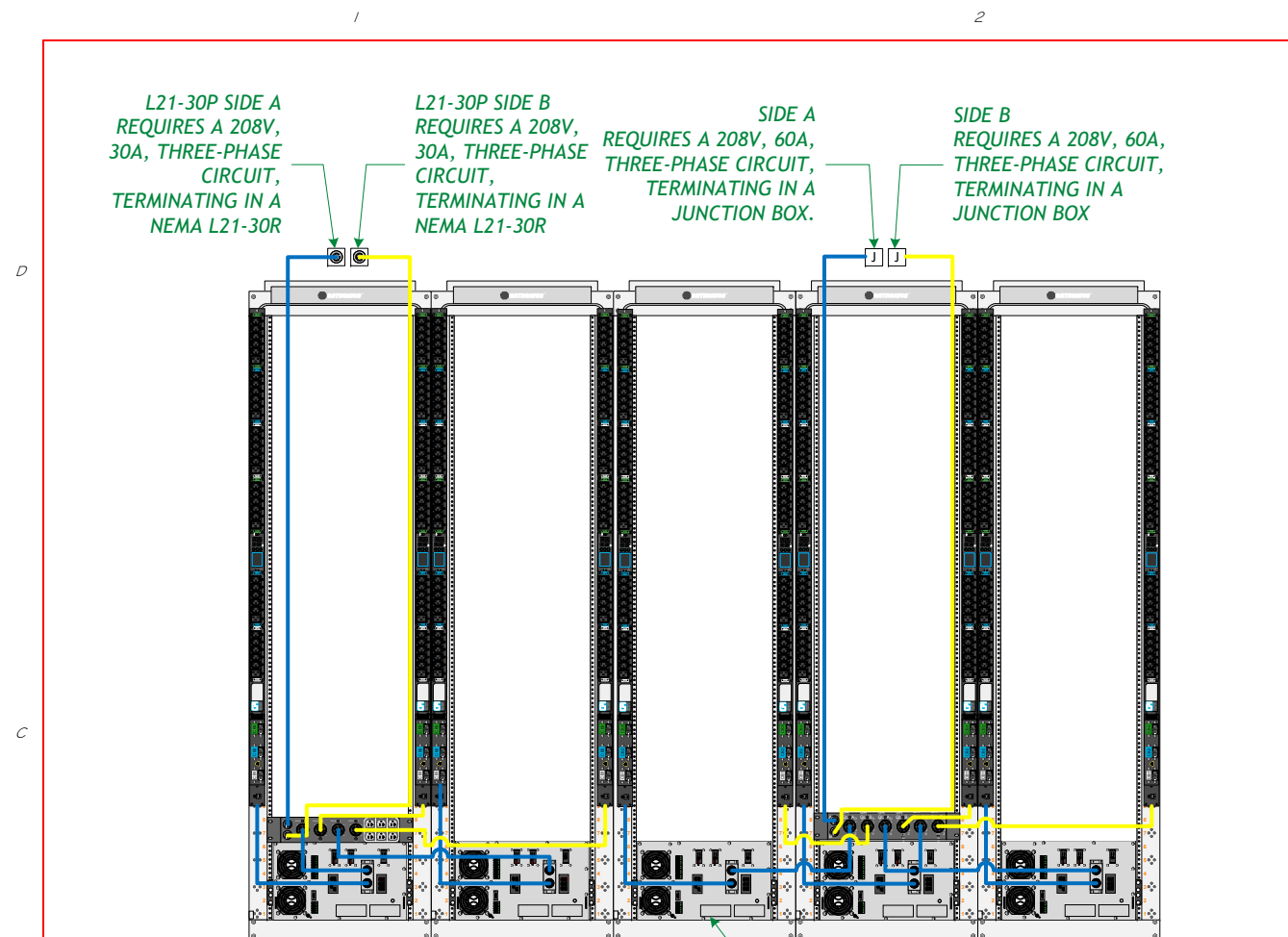
DOC VERSION No: 3.1

ISSUE DATE: Apr 16, 2021

SHEET TITLE

60A ZONE PDU POWER
DISTRIBUTION

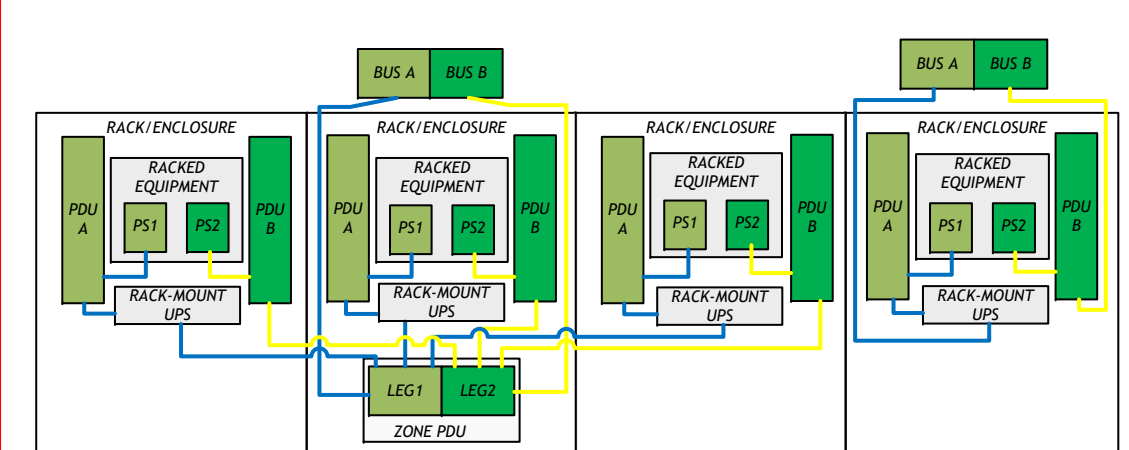
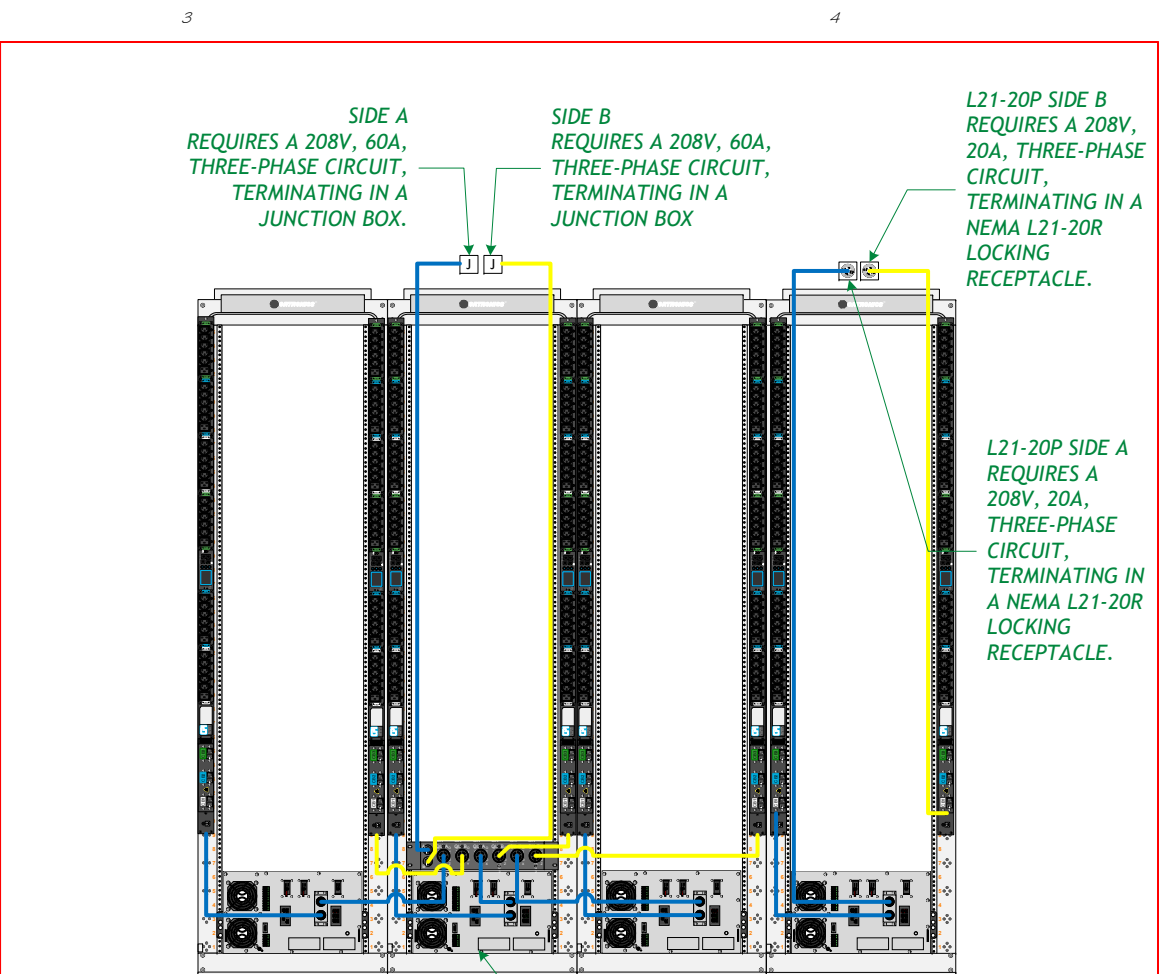
SHEET: 19 OF 37



ONE ZONE PDU
30A 3-PHASE ZONE PDU BASE UNIT -
REQUIRES TWO 30A 3-PHASE (WYE)
CIRCUITS WITH L21-30R RECEPTACLES
TO SUPPORT TWO RACKS

ONE ZONE PDU
60A 3-PHASE ZONE PDU BASE UNIT -
REQUIRES TWO 60A 3-PHASE (WYE)
CIRCUITS HARD-WIRED IN JUNCTION
BOX TO SUPPORT THREE RACKS

2 POWER ELEVATION AND SCHEMATIC FOR A FIVE RACK TR



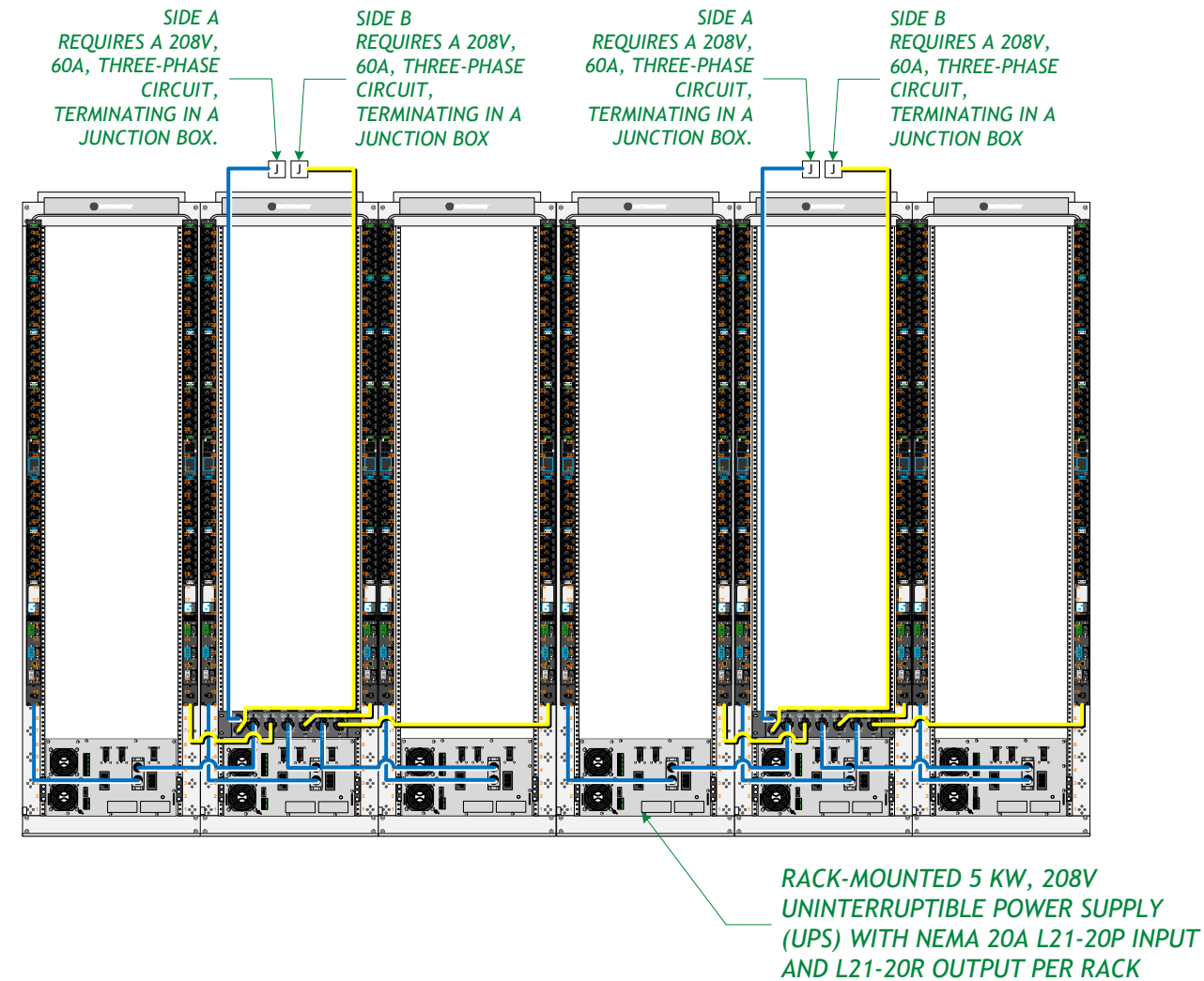
ONE ZONE PDU
60A 3-PHASE ZONE PDU BASE UNIT -
REQUIRES TWO 60A 3-PHASE (WYE)
CIRCUITS HARD-WIRED IN JUNCTION
BOX TO SUPPORT THREE RACKS

REQUIRES TWO 20A 3-
PHASE (WYE) CIRCUITS
TERMINATING IN L21-20Rs

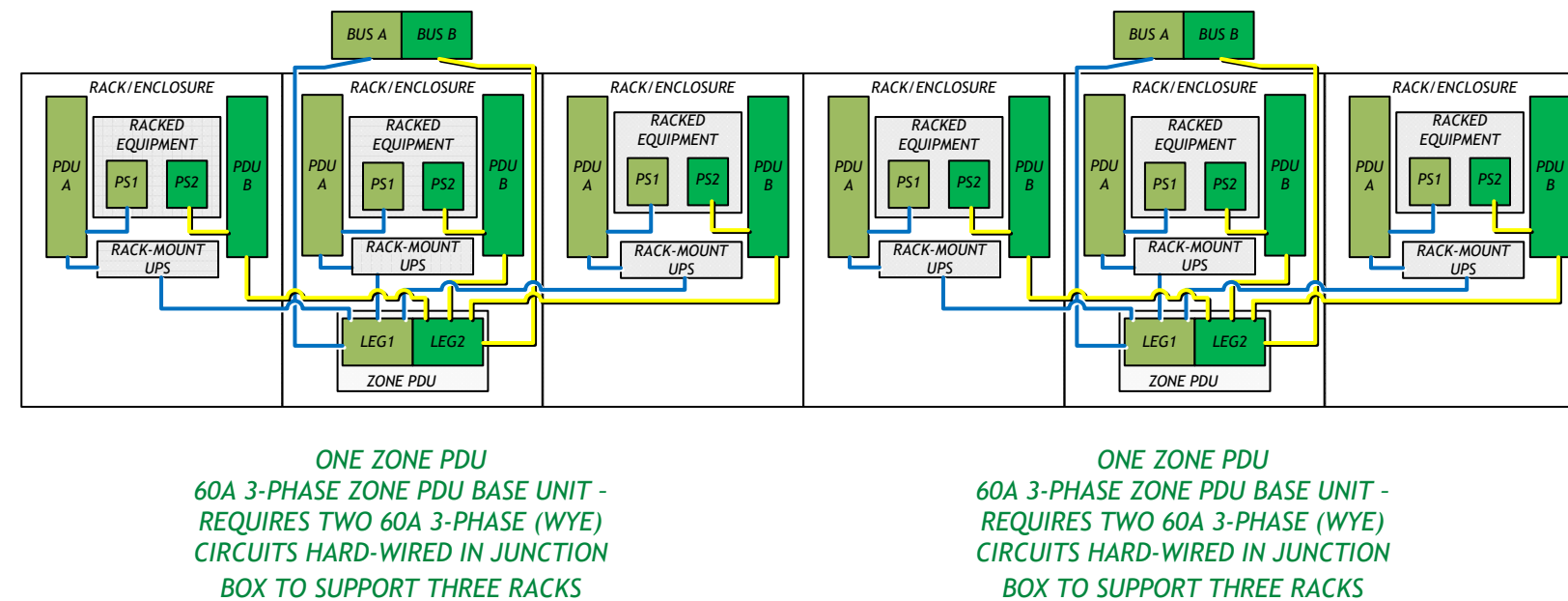
1 POWER ELEVATION AND SCHEMATIC FOR A FOUR RACK TR



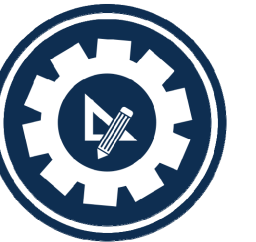
MARK	DATE	DESCRIPTION



2 POWER ELEVATION FOR A SIX RACK TR



1 POWER SCHEMATIC FOR A SIX RACK TR



PROJECT:

OIT DESIGN GUIDE TEMPLATES

PROJECT No:

N/A

MARK	DATE	DESCRIPTION
------	------	-------------

ISSUE:

DRAWING No:

FILE: OIT DESIGN GUIDE TEMPLATE
S-04162021-V3.1.VSDX

DESIGN BY: KEVIN GRZELKA, CTDC

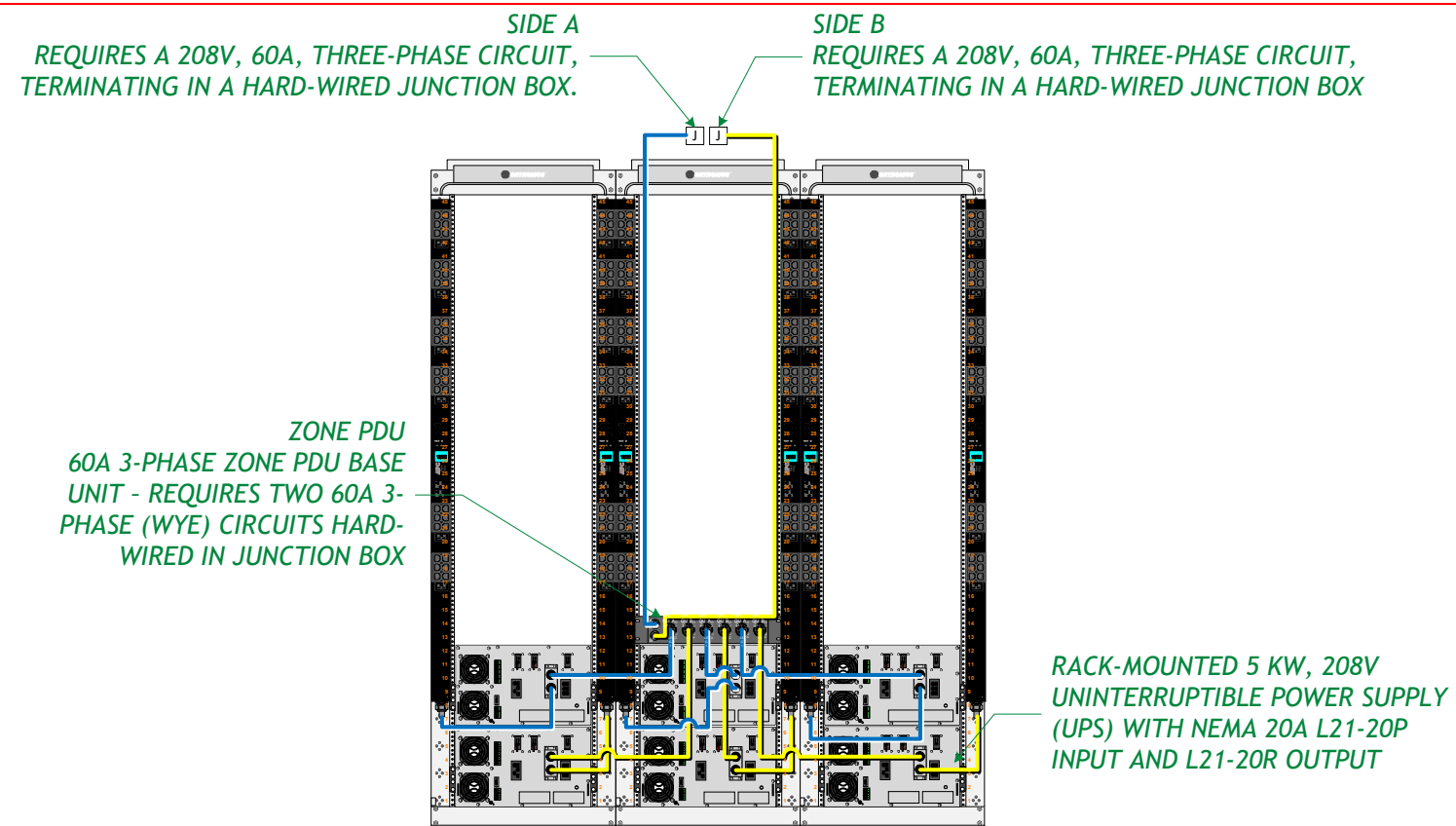
VERIFIED BY: MICHAEL JULIAN, RCDD
KELLY BATES, DCIE

DOC VERSION No: 3.1

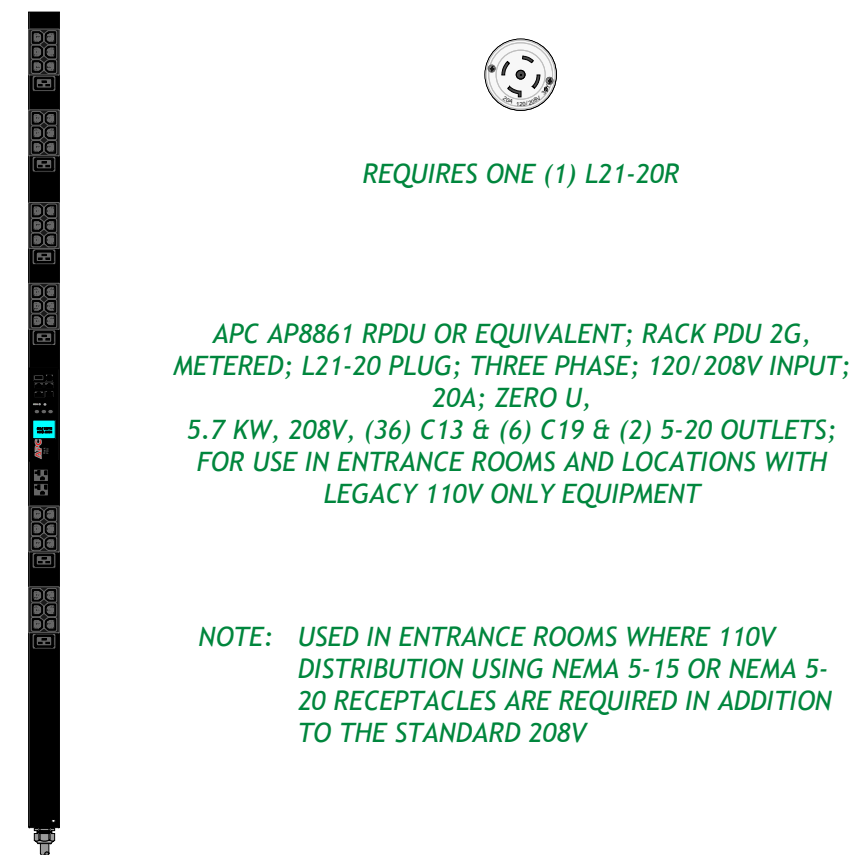
ISSUE DATE: Apr 16, 2021

SHEET TITLE

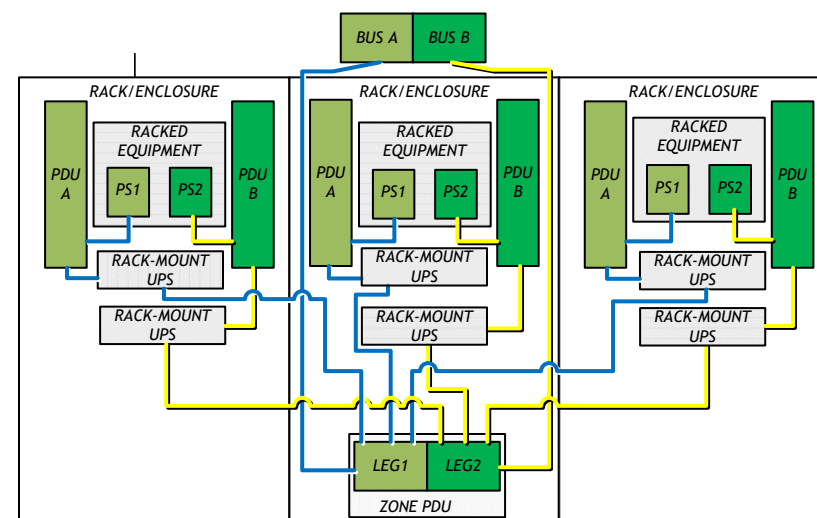
ZONE PDU POWER DISTRIBUTION
IN A SIX RACK TR



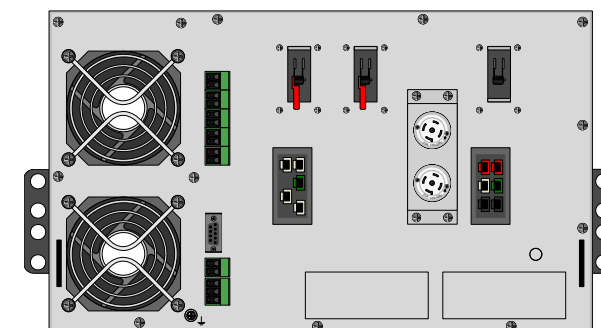
3 POWER ELEVATION FOR A HEALTH CARE FACILITY ENTRANCE ROOM



4 208V RACK PDU FOR ENTRANCE ROOMS



2 POWER SCHEMATIC FOR A HEALTH CARE FACILITY ENTRANCE ROOM



EATON BLADEUPS
CAPACITY: 5KW (EXTERNAL BATTERY MODULES AVAILABLE)
INPUT VOLTAGE: 208V
INPUT CONNECTION: L21-20P, THREE-PHASE, FOUR-WIRE + GROUND
OUTPUT CONNECTION: L21-20R
SIZE: 6RU, 19" RACK-MOUNTABLE
RUNTIME: 16.5 MINUTES AT 50% LOAD

1 RACK-MOUNT 5KW UNINTERRUPTABLE POWER SUPPLY (UPS)



**IT OPERATIONS AND SERVICES
SOLUTION DELIVERY**



EDICT
ENTERPRISE DATA CENTER AND
INFRASTRUCTURE ENGINEERING

PROJECT:

OIT DESIGN GUIDE TEMPLATES

PROJECT No:

N/A

MARK	DATE	DESCRIPTION

ISSUE:

DRAWING No:

FILE: OIT_DESIGN_GUIDE_TEMPLATE
S-04162021-V3.1.VSDX

DESIGN BY: KEVIN GRZELKA, CTDC

VERIFIED BY: MICHAEL JULIAN, RCDD
KELLY BATES, DCIE

DOC VERSION No:	3.1
-----------------	-----

ISSUE DATE:	Apr 16, 2021
-------------	--------------

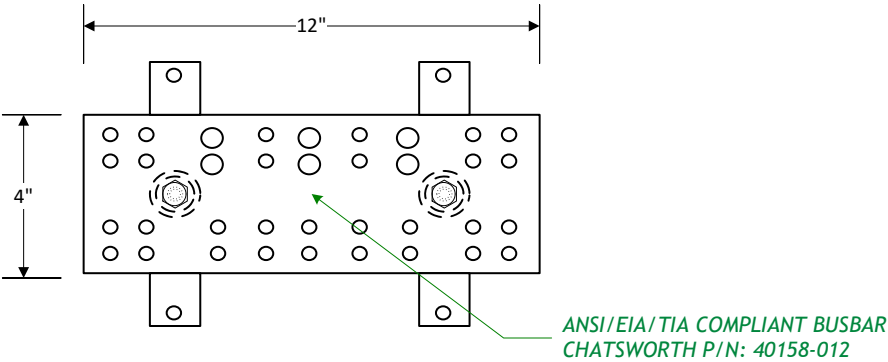
SHEET TITLE

ZONE PDU POWER DISTRIBUTION IN A HEALTH CARE FACILITY ENTRANCE ROOM

SHEET: 23 OF 37

NOTES: ALL GROUNDING SHALL BE DONE IN ACCORDANCE WITH ANSI/TIA-607-D "COMMERCIAL BUILDING GROUNDING (EARTHING) AND BONDING REQUIREMENTS FOR TELECOMMUNICATIONS", NFPA 70, AND IN COMPLIANCE WITH LOCAL CODES.

ALL EQUIPMENT SHALL BE GROUNDED PER MANUFACTURER'S INSTRUCTIONS.

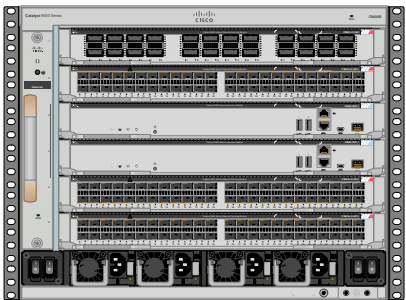
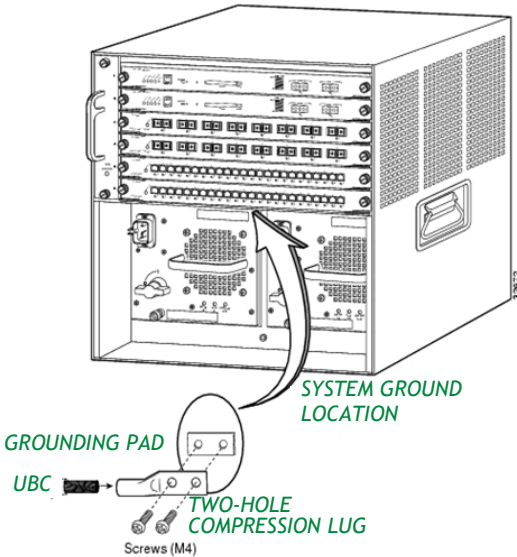


TELECOMMUNICATIONS SECONDARY BONDING BUSBAR (SBB)

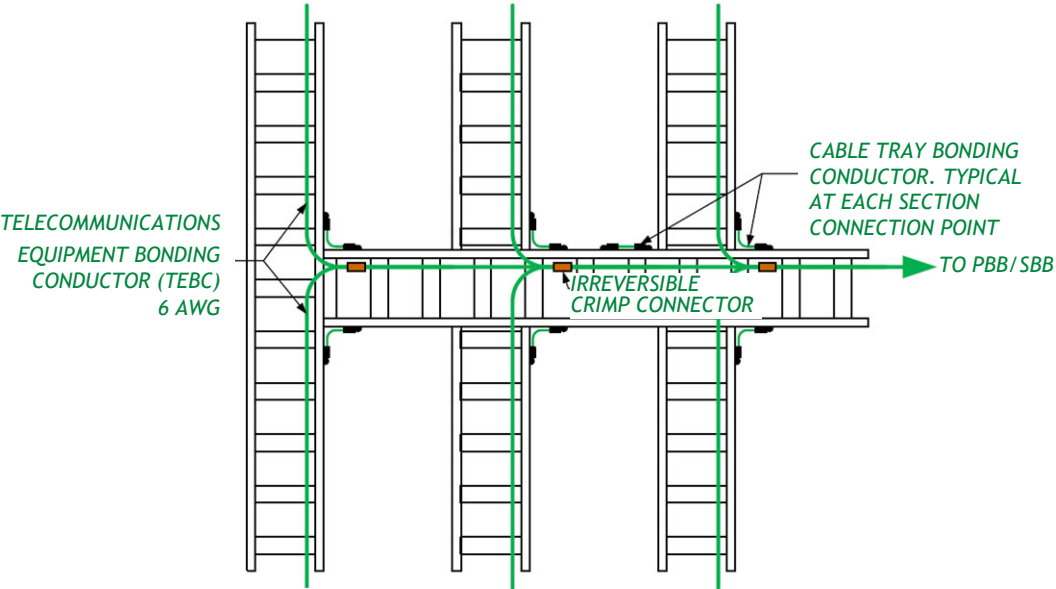
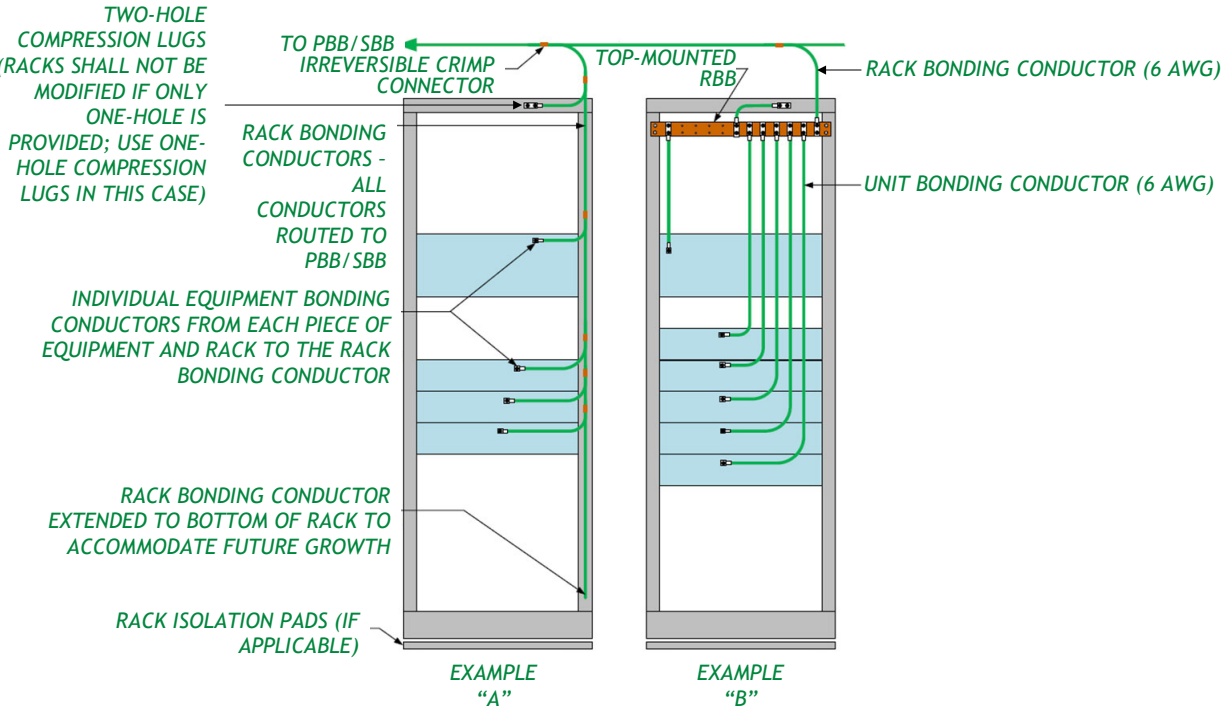
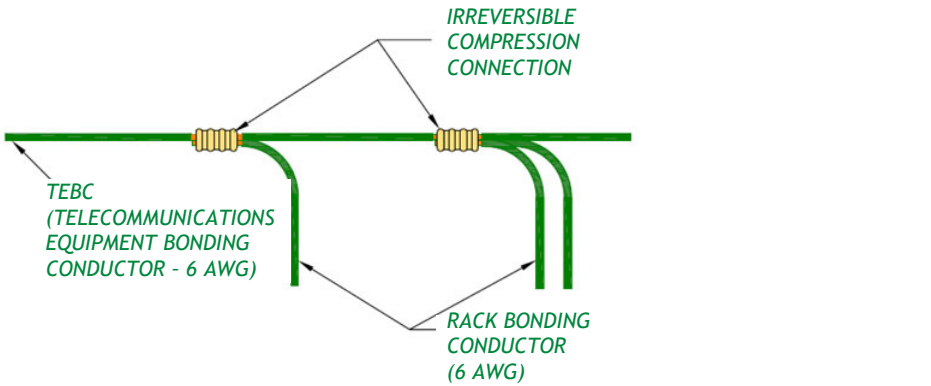
NETWORK EQUIPMENT MUST BE GROUNDED. NEVER DEFEAT THE GROUND CONDUCTOR OR OPERATE THE EQUIPMENT IN THE ABSENCE OF A SUITABLY INSTALLED GROUND CONDUCTOR. THESE DEVICES WILL BE BONDED TO THE RACK BONDING BUSBAR (RBB) IN THE SAME RACK. EACH DEVICE WILL HAVE A DEDICATED UNIT BONDING CONDUCTOR (UBC). THE UBC WILL NOT BE SHARED.

CISCO 6509

CISCO 9606



EARTHING EXAMPLE (CISCO 6509/9606)
Ref: CISCO PRODUCT INSTALLATION GUIDE



PATHWAY, RACK, BONDING CONDUCTOR JOIN REQUIREMENTS



PROJECT:

OIT DESIGN GUIDE TEMPLATES

PROJECT No:

N/A

MARK	DATE	DESCRIPTION
------	------	-------------

ISSUE:

DRAWING No:

FILE: OIT_DESIGN_GUIDE_TEMPLATE
S-04162021-V3.1.VSDX

DESIGN BY: KEVIN GRZELKA, CTDC

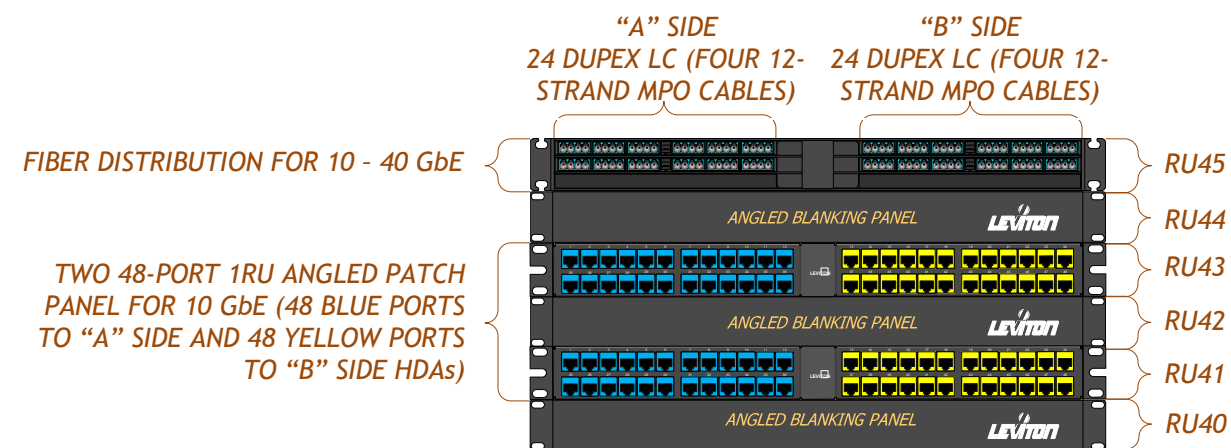
VERIFIED BY: MICHAEL JULIAN, RCDD
KELLY BATES, DCIE

DOC VERSION No: 3.1

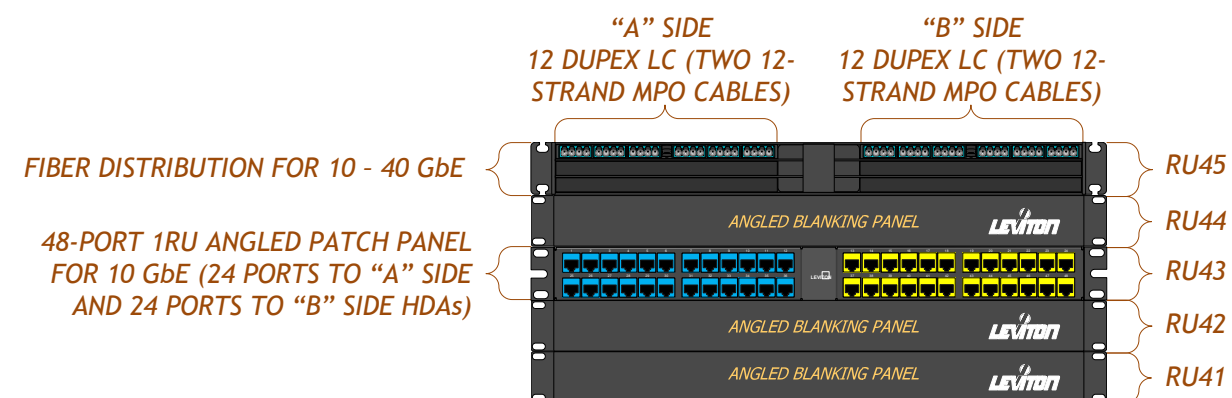
ISSUE DATE: Apr 16, 2021

SHEET TITLE

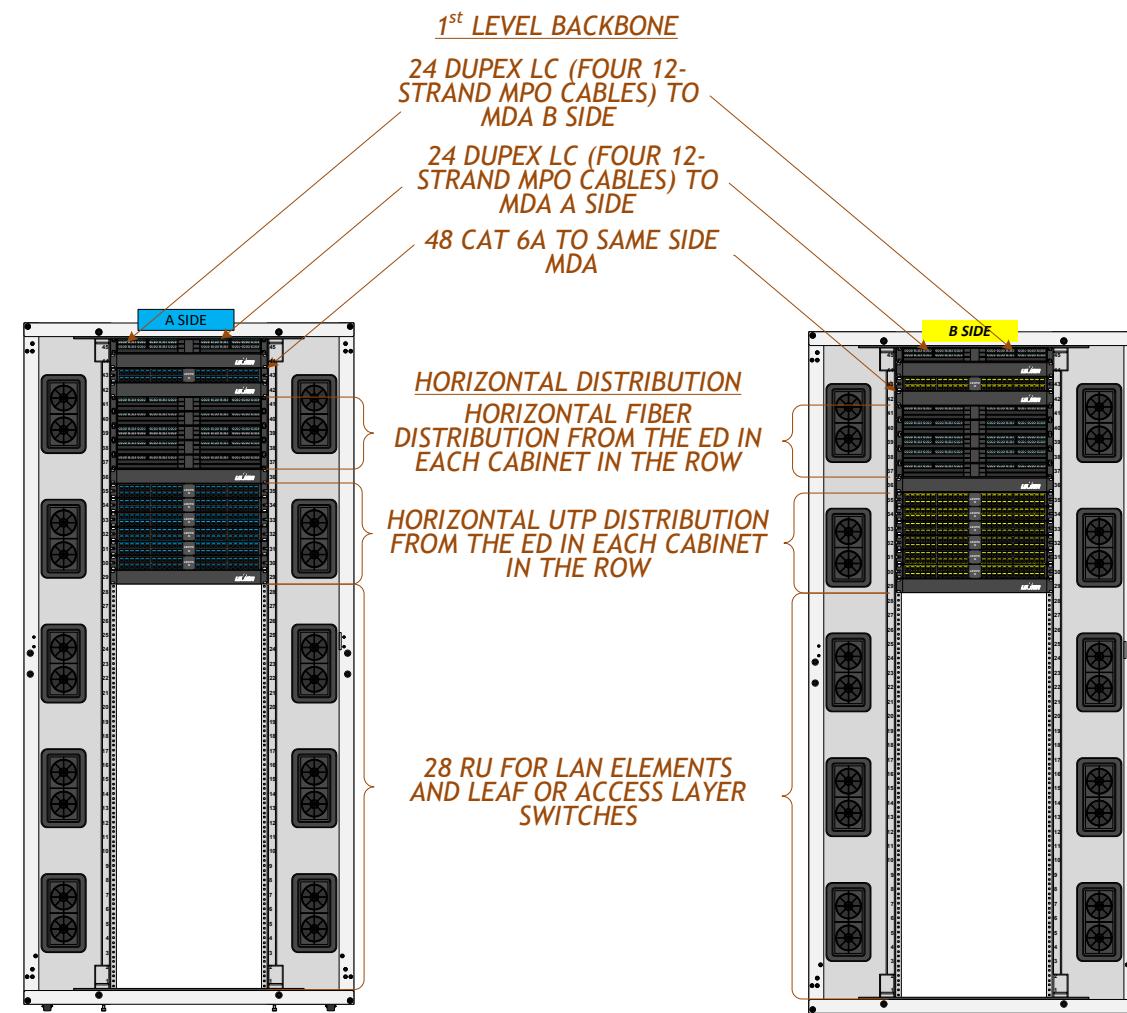
BONDING



3 HIGH DENSITY EQUIPMENT DISTRIBUTOR ELEVATION FOR CABINETS



2 STANDARD DENSITY EQUIPMENT DISTRIBUTOR ELEVATION FOR CABINETS



1 TYPICAL END OF ROW HORIZONTAL DISTRIBUTION AREA



**IT OPERATIONS AND SERVICES
SOLUTION DELIVERY**



EDCT

ENTERPRISE DATA CENTER AND
INFRASTRUCTURE ENGINEERING

PROJECT:

OIT DESIGN GUIDE TEMPLATES

PROJECT No:

N/A

MARK	DATE	DESCRIPTION

ISSUE:

DRAWING No:

FILE: OIT_DESIGN GUIDE TEMPLATE
S-04162021-V3.1.VSDX

DESIGN BY: KEVIN GRZELKA, CTDC

VERIFIED BY: MICHAEL JULIAN, RCDD
KELLY BATES, DCIE

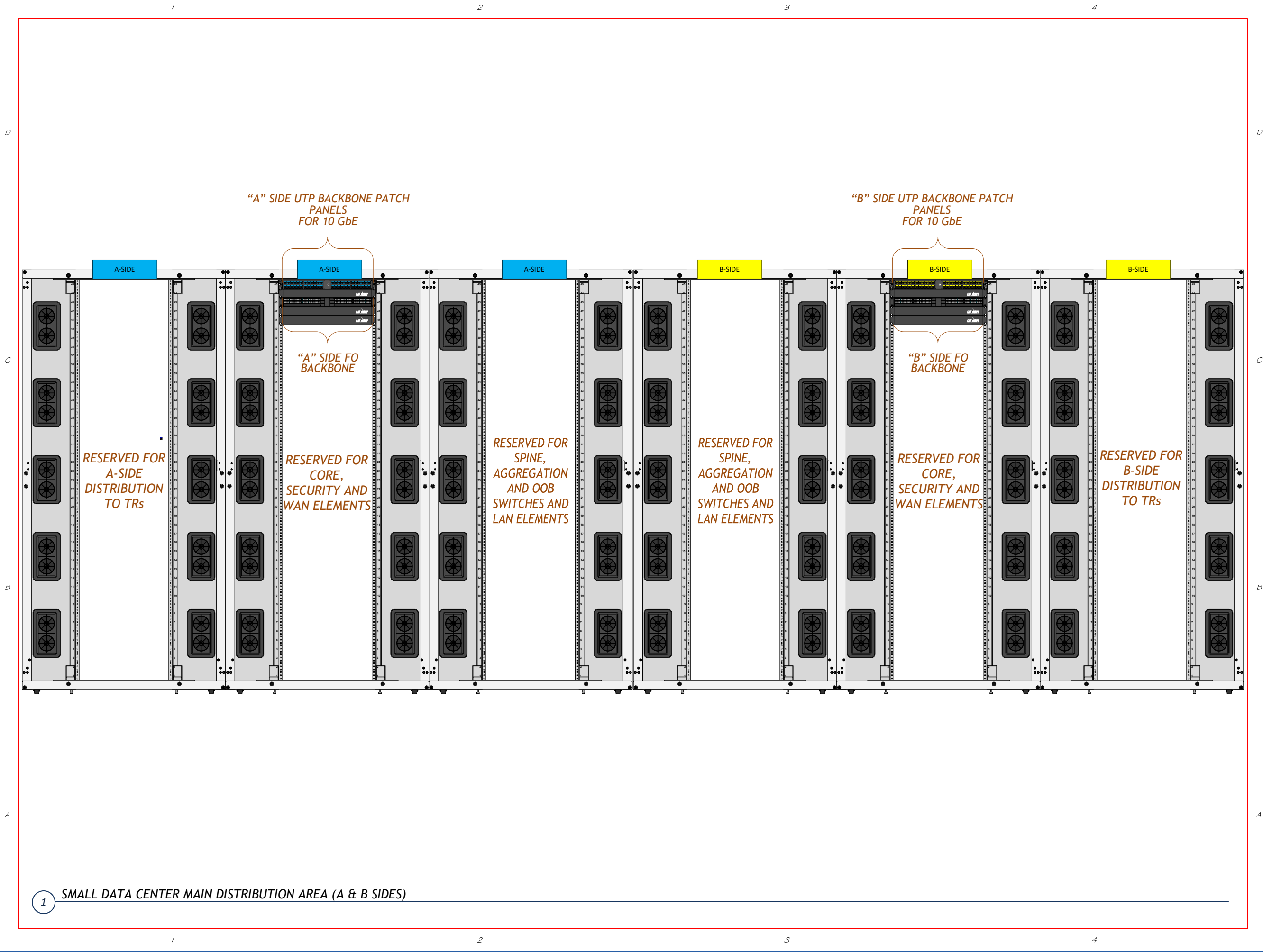
DOC VERSION No:	3.1
-----------------	-----

ISSUE DATE: Apr 16, 2021

SHEET TITLE

ED & HD (L, M, S)

SHEET: 25 OF 37



MARK	DATE	DESCRIPTION
------	------	-------------

ISSUE:

DRAWING No:

FILE: OIT_DESIGN_GUIDE_TEMPLATE
S-04162021-V3.1.VSDX

DESIGN BY: KEVIN GRZELKA, CTDC

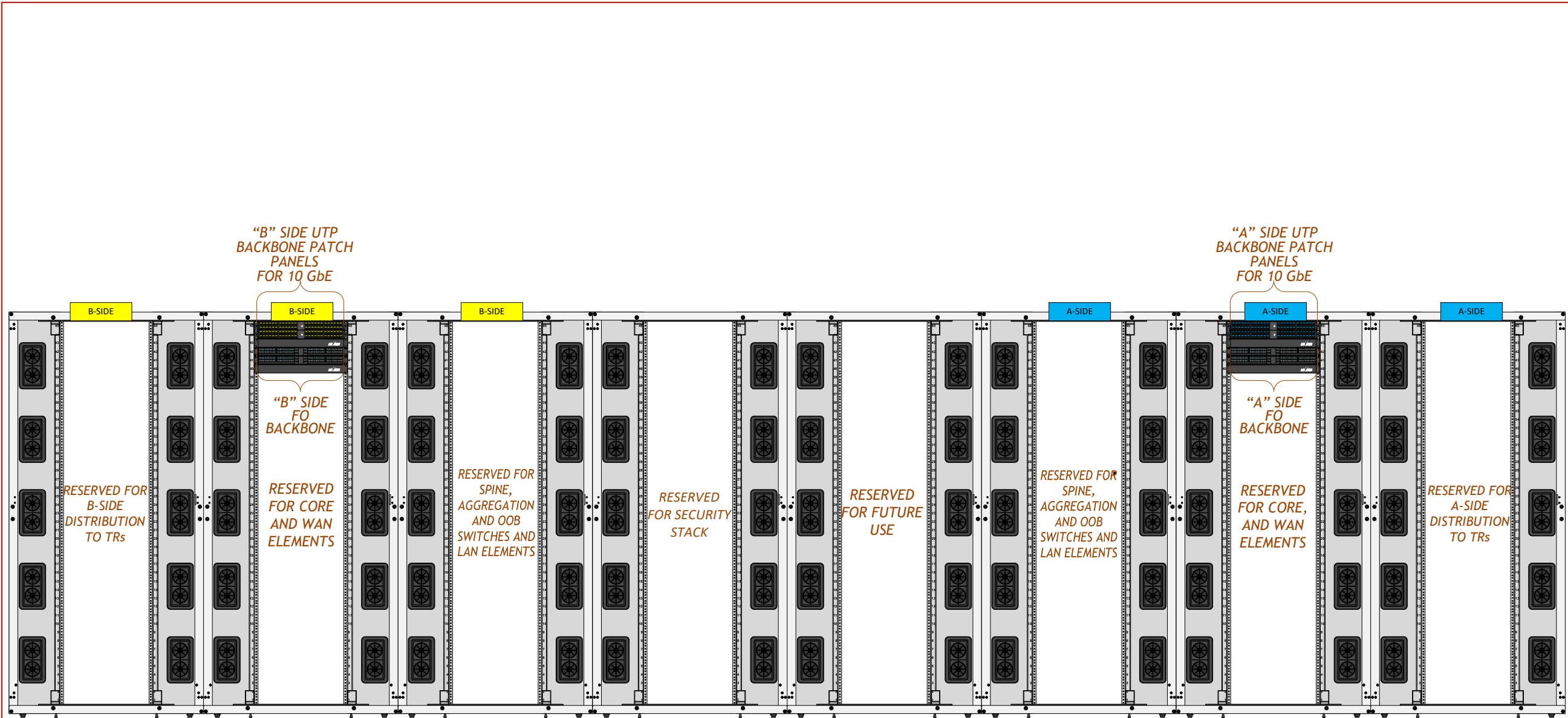
VERIFIED BY: MICHAEL JULIAN, RCDD
KELLY BATES, DCIE

DOC VERSION No: 3.1

ISSUE DATE: Apr 16, 2021

SHEET TITLE

MDA (S)



1 MEDIUM DATA CENTER MAIN DISTRIBUTION AREA (A & B SIDES)



PROJECT:

OIT DESIGN GUIDE TEMPLATES

PROJECT No:
N/A

MARK	DATE	DESCRIPTION

ISSUE:

DRAWING No:

FILE: OIT_DESIGN_GUIDE_TEMPLATE
S-04162021-V3.1.VSDX

DESIGN BY: KEVIN GRZELKA, CTDC

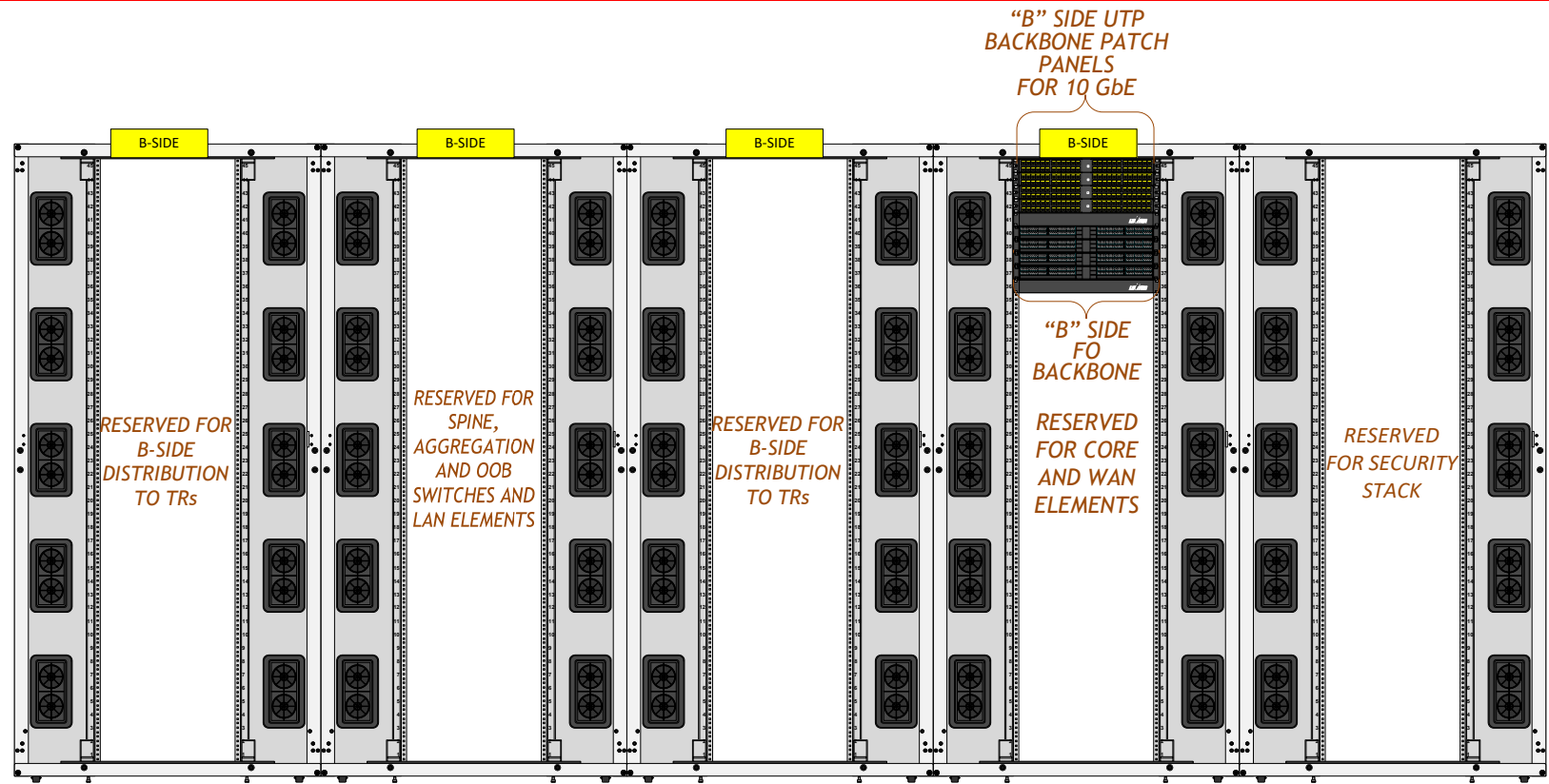
VERIFIED BY: MICHAEL JULIAN, RCDD
KELLY BATES, DCIE

DOC VERSION No: 3.1

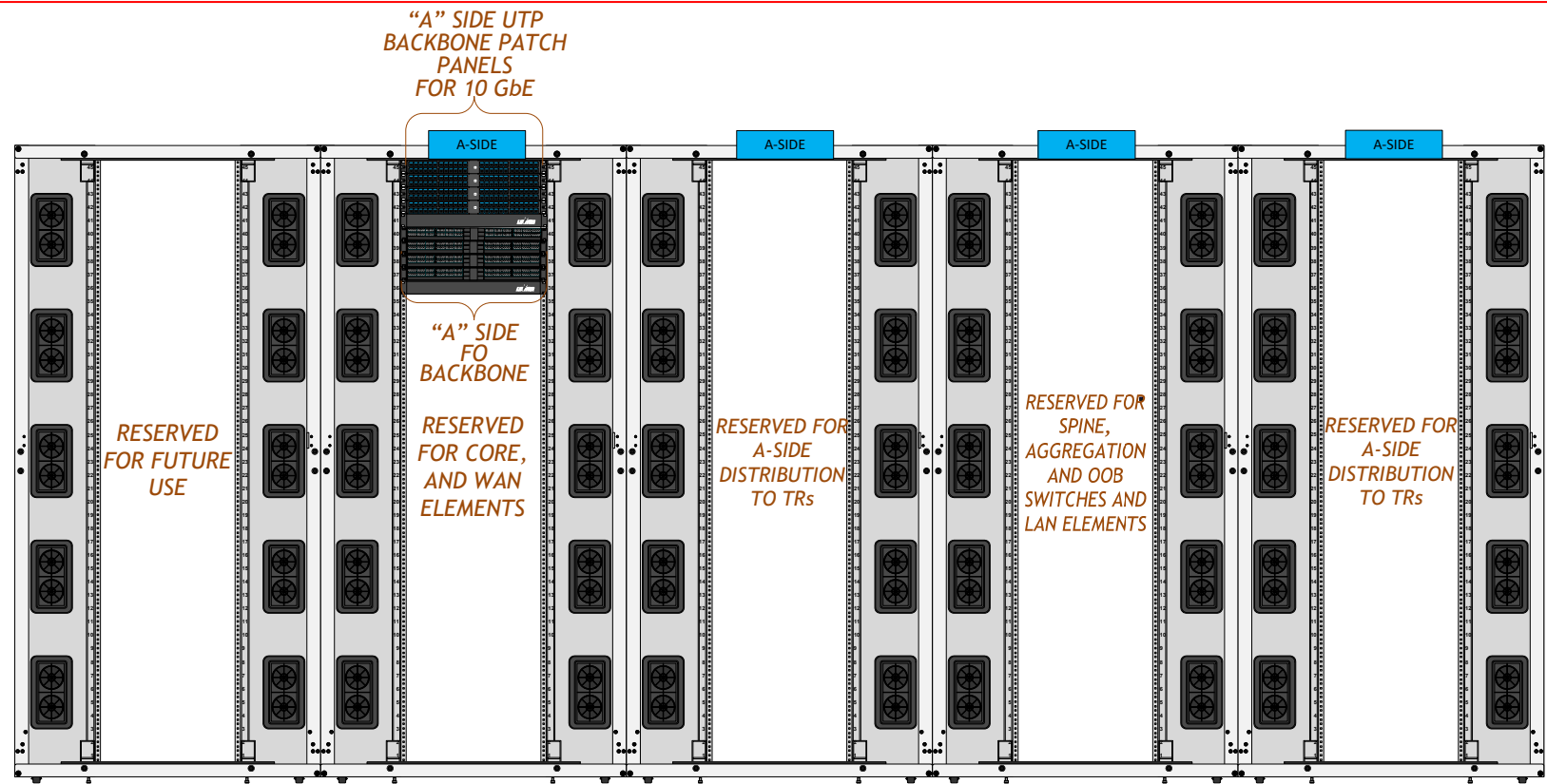
ISSUE DATE: Apr 16, 2021

SHEET TITLE

MDA (M)



2 LARGE DATA CENTER MAIN DISTRIBUTION AREA (B SIDE)



1 LARGE DATA CENTER MAIN DISTRIBUTION AREA (A SIDE)



PROJECT:

OIT DESIGN GUIDE TEMPLATES

PROJECT No:

N/A

MARK	DATE	DESCRIPTION
------	------	-------------

ISSUE:

DRAWING No:

FILE: OIT_DESIGN_GUIDE_TEMPLATE
S-04162021-V3.1.VSDX

DESIGN BY: KEVIN GRZELKA, CTDC

VERIFIED BY: MICHAEL JULIAN, RCDD
KELLY BATES, DCIE

DOC VERSION No: 3.1

ISSUE DATE: Apr 16, 2021

SHEET TITLE

MDA (L)



RECOMMENDED UTP MEDIA: ATLAS-X1 COPPER TRUNKS
CAT 6A UTP CMP CABLE ASSEMBLY
BUNDLE OF 6 BLUE CABLES
BERK-TEK - LANMARK-10G2 (UTP) BRANDED CABLE
TIA/EIA WIRING T568B
TESTED TO TIA 568-C.2
TERMINATION: ATLAS-X1 E2XHD CONNECTION

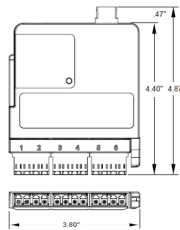
RECOMMENDED A-SIDE JACK
COLOR: BLUE



RECOMMENDED B-SIDE JACK COLOR:
YELLOW

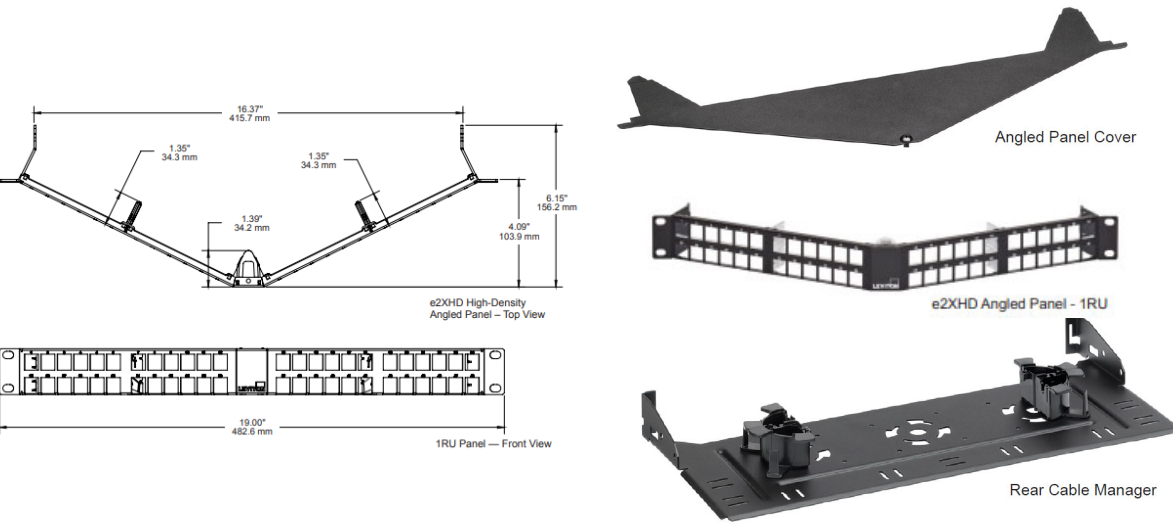


3 UTP MEDIA



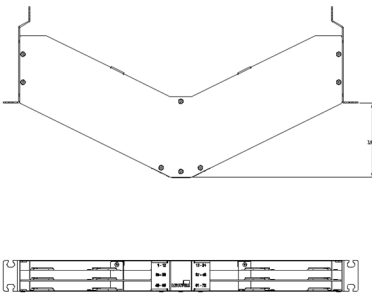
RECOMMENDED CASSETTE: LEVITON HDX ENTERPRISE MTP
FIBER TYPE: OM4/OS1
FIBER COUNT: 12-FIBER
CONNECTOR (FRONT): LC
CONNECTOR (REAR): MPO
POLARITY: TYPE A

4 FIBER CASSETTE



RECOMMENDED UTP PATCH PANEL:
LEVITON e2XHD-BLK HIGH-DENSITY ANGLED
48 POSITION
ONE RACK UNIT PER PANEL
WITH REAR CABLE MANAGER
AND ANGLED PANEL COVERS

2 UTP PATCH PANEL



RECOMMENDED FIBER DISTRIBUTION CABINET:
LEVITON ANGLED OPT-X UHDX
144 FIBER
ONE RACK UNIT

1 FIBER DISTRIBUTION CABINET

DEPARTMENT OF VETERANS
AFFAIRS



IT OPERATIONS AND SERVICES
SOLUTION DELIVERY



EDCT
ENTERPRISE DATA CENTER AND
INFRASTRUCTURE ENGINEERING

PROJECT:

OIT DESIGN GUIDE TEMPLATES

PROJECT No:

N/A

MARK	DATE	DESCRIPTION
------	------	-------------

ISSUE:

DRAWING No:

FILE: OIT_DESIGN_GUIDE_TEMPLATE
S-04162021-V3.1.VSDX

DESIGN BY: KEVIN GRZELKA, CTDC

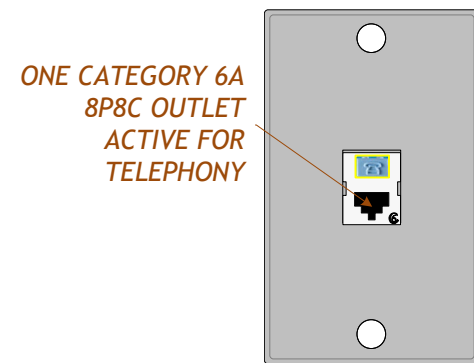
VERIFIED BY: MICHAEL JULIAN, RCDD
KELLY BATES, DCIE

DOC VERSION No: 3.1

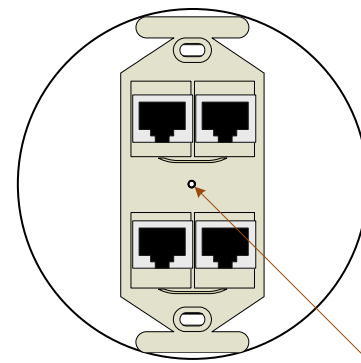
ISSUE DATE: Apr 16, 2021

SHEET TITLE

MEDIA DETAILS



3 TYPICAL WALL MOUNT PHONE VOIP OUTLET



4 TYPICAL WORK FLOOR MOUNT OUTLET

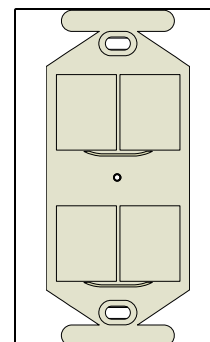
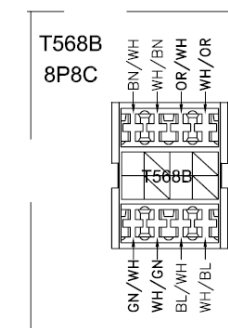
NOTES:

TYPICAL WORK AREA OUTLET FACEPLATE WILL BE INSTALLED WITH CATEGORY 6A COMPONENT-COMPLIANT 8P8C MEDIA INTERFACE CONNECTORS (RJ45). EACH CONNECTOR WILL BE TERMINATED TO HIGH QUALITY CATEGORY 6A HORIZONTAL CABLING WHICH WILL TERMINATE IN THE TELECOMMUNICATIONS ROOM AS SPECIFIED ELSEWHERE IN THIS DESIGN PACKAGE. ALL HORIZONTAL UTP SHALL BE CATEGORY 6A AND TERMINATED TO T568B.

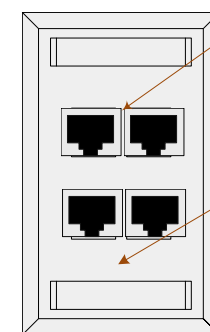
TYPICAL FACEPLATE WILL BE INSTALLED WITH TWO RJ45s. HIGH DENSITY FACEPLATES WILL BE INSTALLED WITH FOUR RJ45s.

FACEPLATE MATERIAL AND COLOR DETERMINED BY OTHERS.

THIS SHEET DETAILS A GENERIC FACEPLATE SPECIFICATION.



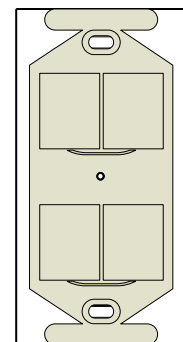
**SINGLE GANG WORKBOX
WITH TWO CONNECTOR
CHASSIS.**



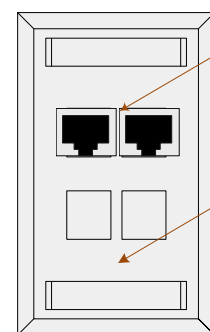
✓ FOUR CATEGORY 6A 8P8C
OUTLETS ACTIVE FOR
TELEPHONY/DATA

FOUR POSITION FACEPLATE
MOUNTED ON SINGLE GANG
WORKBOX. FACEPLATE COLOR
SPECIFIED BY OTHERS.

② TYPICAL HIGH DENSITY WALL MOUNTED WORK AREA
OUTLET CONFIGURATION



**SINGLE GANG
WORKBOX WITH ONE
CONNECTOR
CHASSIS.**



TWO CATEGORY 6A
8P8C OUTLETS ACTIVE
FOR TELEPHONY/DATA

FOUR POSITION FACEPLATE
MOUNTED ON SINGLE GANG
WORKBOX. FACEPLATE COLOR
SPECIFIED BY OTHERS.

1 STANDARD DENSITY WALL MOUNTED WORK AREA OUTLET CONFIGURATION

ALL LABELING SHALL BE ANSI/TIA/EIA/606C COMPLIANT. BLACK LETTERING ON WHITE FIELD. MACHINE PRINTED. FURTHER GUIDANCE ON ADMINISTRATION MAY BE SPECIFIED IN OTHER SECTIONS OF THIS DESIGN PACKAGE.

**IT OPERATIONS AND SERVICES
SOLUTION DELIVERY**



EDUCT
ENTERPRISE DATA CENTER AND
INFRASTRUCTURE ENGINEERING

PROJECT:

OIT DESIGN GUIDE TEMPLATES

PROJECT No:

N/A

MARK	DATE	DESCRIPTION

ISSUE:

DRAWING No:

FILE: OIT_DESIGN GUIDE TEMPLATE
S-0416202T-V3.1.VSDX

DESIGN BY: KEVIN GRZELKA, CTDC

VERIFIED BY: MICHAEL JULIAN, RCDD
KELLY BATES, DCIE

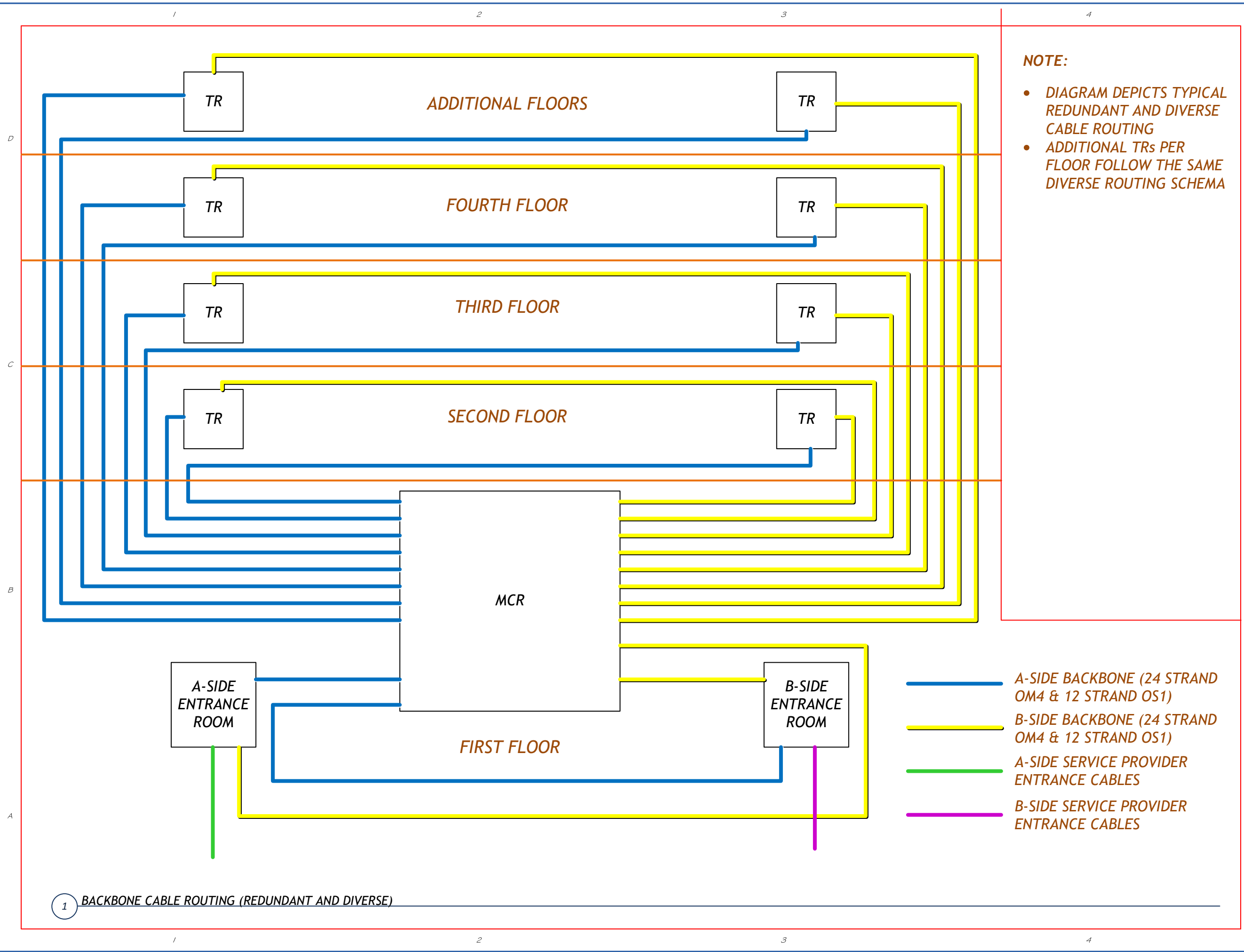
DOC VERSION No: 3.1

ISSUE DATE: Apr 16, 2021

SHEET TITLE

WORK AREA OUTLETS

SHEET: 30 OF 37



NOTE:

- DIAGRAM DEPICTS TYPICAL REDUNDANT AND DIVERSE CABLE ROUTING
- ADDITIONAL TRs PER FLOOR FOLLOW THE SAME DIVERSE ROUTING SCHEMA

- A-SIDE BACKBONE (24 STRAND OM4 & 12 STRAND OS1)
- B-SIDE BACKBONE (24 STRAND OM4 & 12 STRAND OS1)
- A-SIDE SERVICE PROVIDER ENTRANCE CABLES
- B-SIDE SERVICE PROVIDER ENTRANCE CABLES

1 BACKBONE CABLE ROUTING (REDUNDANT AND DIVERSE)

DEPARTMENT OF VETERANS AFFAIRS



IT OPERATIONS AND SERVICES
SOLUTION DELIVERY



EDCT
ENTERPRISE DATA CENTER AND
INFRASTRUCTURE ENGINEERING

PROJECT:

OIT DESIGN GUIDE TEMPLATES

PROJECT No: N/A

MARK	DATE	DESCRIPTION

ISSUE:

DRAWING No:

FILE: OIT_DESIGN_GUIDE_TEMPLATE
S-04162021-V3.1.VSDX

DESIGN BY: KEVIN GRZELKA, CTDC

VERIFIED BY: MICHAEL JULIAN, RCDD
KELLY BATES, DCIE

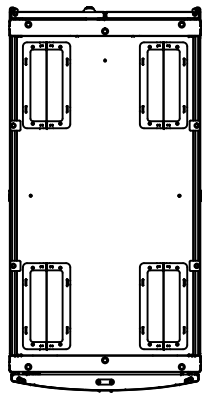
DOC VERSION No: 3.1

ISSUE DATE: Apr 16, 2021

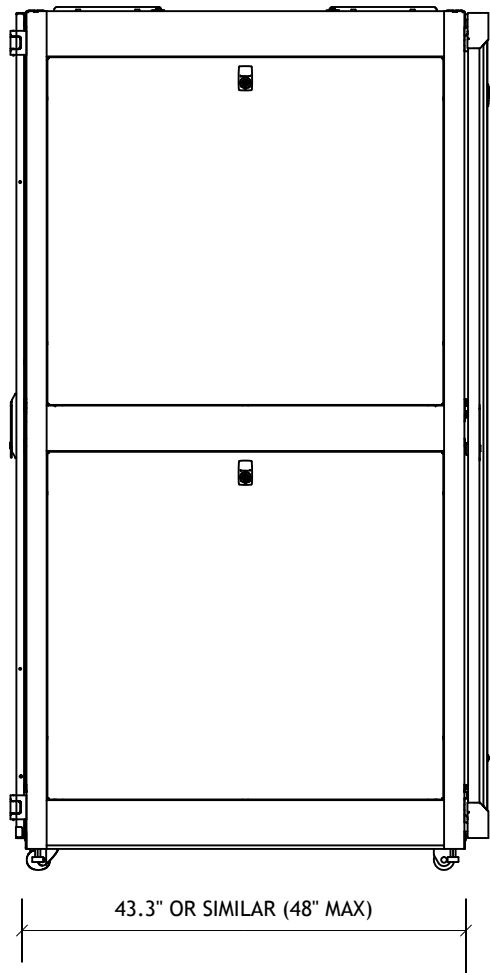
SHEET TITLE

BUILDING TELECOM CABLE ROUTING

SHEET: 31 OF 37

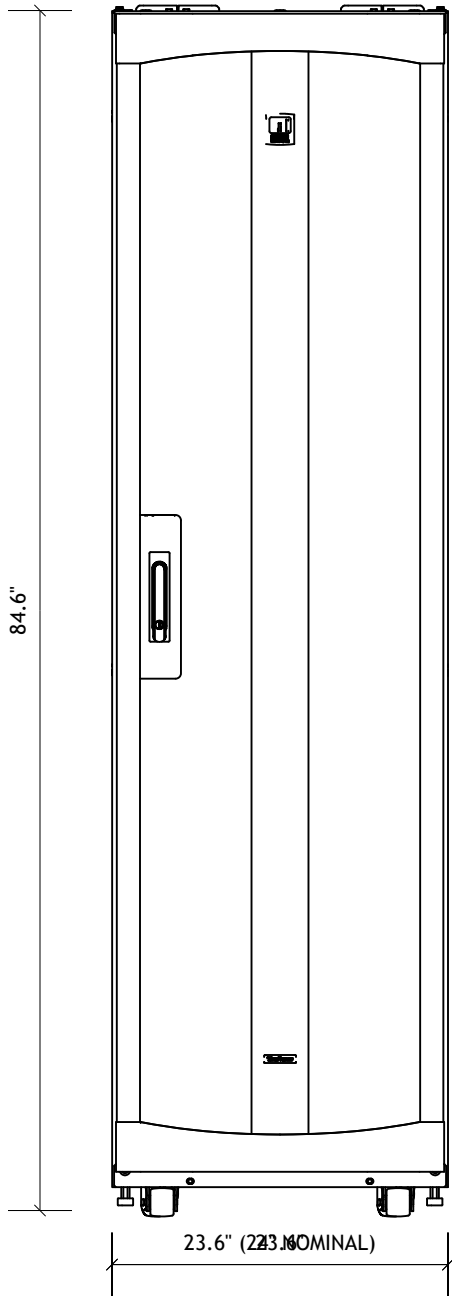


4 TYPICAL SERVER CABINET TOP VIEW WITH FOUR (4) BRUSHED GROMMET CABLE OPENINGS



3 TYPICAL SERVER CABINET SIDE VIEW

NOTE: SERVER CABINETS MAY ONLY HOUSE SERVER AND STORAGE EQUIPMENT AND SUPPORTING PATCH PANELS. SERVER CABINETS MAY NOT HOUSE TELECOMMUNICATIONS (NETWORK) EQUIPMENT



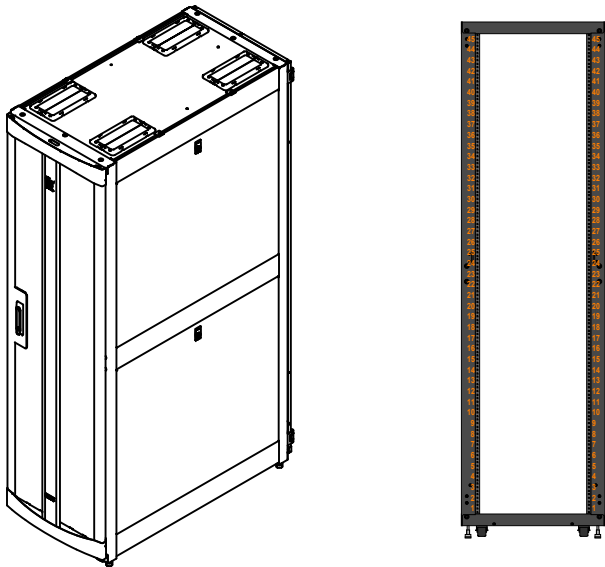
2 TYPICAL SERVER CABINET FRONT

F-Series TeraFrame Gen 3 Cabinet System with accessories installed (Or equivalent)
Configuration includes the following components:

Item Number	Description	(Qty)
FF1N-113C-E42-B	F-Series TeraFrame Gen 3 Cabinet System, 45 RMU, 84.6 in. (2149mm) H x 23.6" (600mm) W x 43.30" (1100mm) D Square-Punched Rails, 2-Pair, Single, Perforated Metal Front Door, Two Point, Keyed Swing Handle Lock, Double Perforated Metal Rear Door, Two Point, Keyed Swing Handle Latch, Server Top Panel, 2 Solid Side Panel, Glacier White, 6-Slide Frame	
024-739003-E63	Frame, 6 Slide, 23.6" (600 mm) W x 43.3" (1100 mm) D x 45 RMU, Glacier White	(1)
024-739060-E03	Door, TeraFrame, Front, Assembly, 23.6" (600 mm) W x 45 RMU, Glacier White	(1)
024-739064-703	Lock Kit, 2-Point, Keyed, Front Door, 45 RMU	(1)
024-739050-E03	Double Door Assembly, 23.6" (600 mm) W x 45 RMU, Glacier White	(1)
024-739054-703	Lock Kit, 2-Point Latching, Keyed, Perforated Door, 45 RMU	(1)
024-739140-E12	Top Panel, 2-Piece, Brush, 23.6" (600 mm) W x 43.3" (1100 mm) D, Glacier White	
024-739046-E63	Side Panel, 6 SLIDE, 43.3" (1100 mm) D x 45 RMU, Glacier White	(2)
024-739200-002	Kit, Common Parts Parts, Teraframe, UL, White	(1)
024-739110-701	PDU Bracket, Assembly, Standard, 0.7" (17 mm) W x 3.9" (100 mm) D x 2.4" (60 mm), Black	(1)
39110-C01	PDU Bracket, Assembly, Standard, 0.7" (17 mm) W x 3.9" (100 mm) D x 2.4" (60 mm), Black	(1)
024-739240-003	Kit, Packaging, 600 W x 1075-1200D x 42U - 48U	(1)
024-739029-001	Caster Kit, Two Swivel, Two Fixed, 1.6" (41 mm) W x 2" (51 mm) D x 2" (41 mm) H	(1)
39085-E03	Air Dam, 6 SLIDE, 23.6" (600 mm) W x 45 RMU, Glacier White	(1)
39137-703	Baying Seal Kit, 45U	(2)
39132-E00	Bottom Panel, With Brush, 23.6" (600 mm) W x 39.4" - 47.2" (1000 - 1200 mm) D, Glacier White	(1)
31920-E03	Finger Cable Manager no Cover	(2)

UL2416 This Cabinet and any included accessories are UL Listed under the NWIN category per the UL2416 Standard. UL file number E227626. Note, the UL Listing only applies to cabinets manufactured in the United States.

(VENDOR TO PROVIDE 45 RU OF BLANKING PANEL PER ENCLOSURE)



1 TYPICAL SERVER CABINET SPECIFICATION

DEPARTMENT OF VETERANS AFFAIRS



IT OPERATIONS AND SERVICES SOLUTION DELIVERY



EDCT
ENTERPRISE DATA CENTER AND
INFRASTRUCTURE ENGINEERING

PROJECT:

OIT DESIGN GUIDE TEMPLATES

PROJECT No:

N/A

MARK	DATE	DESCRIPTION
------	------	-------------

ISSUE:

DRAWING No:

FILE: OIT DESIGN GUIDE TEMPLATE
S-04162021-V3.1.VSDX

DESIGN BY: KEVIN GRZELKA, CTDC

VERIFIED BY: MICHAEL JULIAN, RCDD
KELLY BATES, DCIE

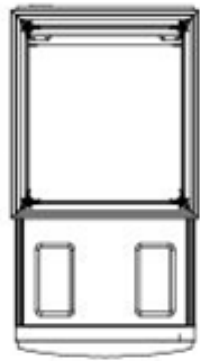
DOC VERSION No: 3.1

ISSUE DATE: Apr 16, 2021

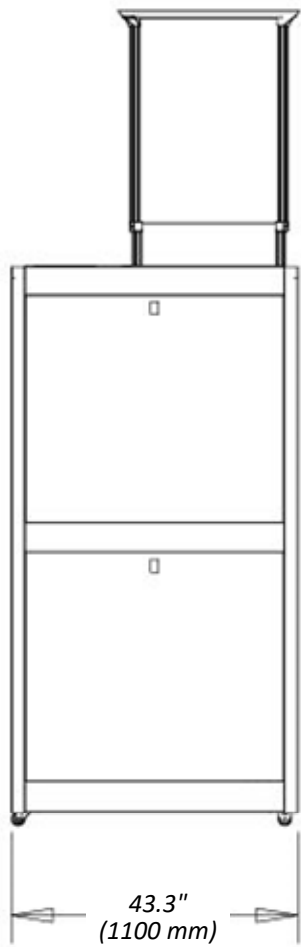
SHEET TITLE

TYPICAL SERVER CABINET

SHEET: 32 OF 37

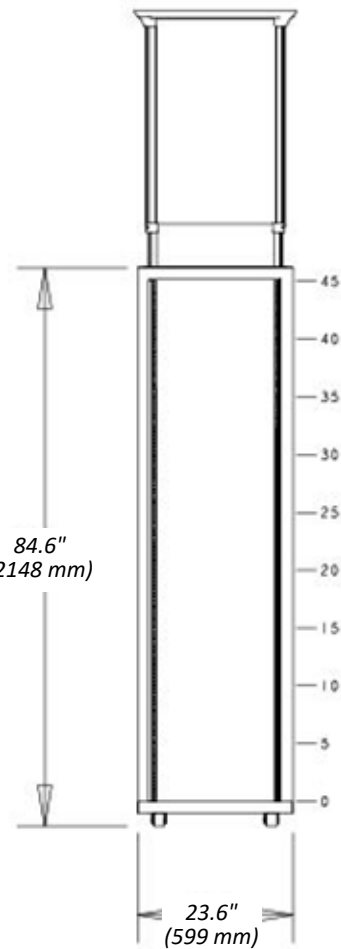


4 TYPICAL SERVER CABINET TOP VIEW WITH VERTICAL EXHAUST DUCT (VED) WITH BRUSHED GROMMET CABLE OPENINGS



3 TYPICAL SERVER CABINET SIDE VIEW WITH VERTICAL EXHAUST DUCT (VED)

NOTE: SERVER CABINETS MAY ONLY HOUSE SERVER AND STORAGE EQUIPMENT AND SUPPORTING PATCH PANELS. SERVER CABINETS MAY NOT HOUSE TELECOMMUNICATIONS (NETWORK) EQUIPMENT



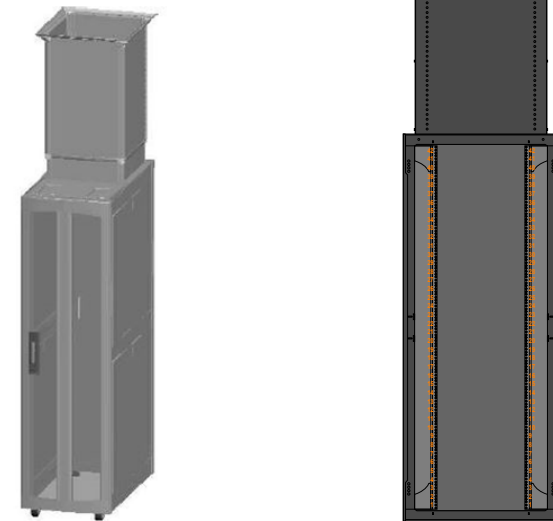
2 TYPICAL SERVER CABINET FRONT WITH VERTICAL EXHAUST DUCT (VED)

F-Series TeraFrame Gen 3 Cabinet System with accessories installed
Configuration includes the following components:

Item Number	Description	(Qty)
FF1N-111C-E52-B F-Series TeraFrame Gen 3 Cabinet System, 45 RMU, 84.6 in. (2149mm) H x 23.6" (600mm) W x 43.30" (1100mm), D Square-Punched Rails, 2-Pair, Single, Perforated Metal Front Door, Two Point, Keyed Swing Handle Lock, Single Solid Metal Rear Door, Two Point, Keyed Swing Handle Latch, Vertical Exhaust Duct System 14-20" (356-508mm), 2 Solid Side Panel, Glacier White, 6-Slide Frame (1)		
024-739003-E63	Frame, 6 Slide, 23.6" (600 mm) W x 43.3" (1100 mm) D x 45 RMU, Glacier White (1)	
024-739060-E03	Door, TeraFrame, Front, Assembly, 23.6" (600 mm) W x 45 RMU, Glacier White (1)	
024-739064-703	Lock Kit, 2-Point, Keyed, Front Door, 45 RMU (1)	
024-739070-E03	Door, Solid, Rear, Assembly, 23.6" (600 mm) W x 45 RMU, Glacier White (1)	
024-739074-703	Lock Kit, 2-Point, Keyed, Solid Door, 45 RMU (1)	
024-739191-E12	Vertical Exhaust Duct Kit, 20"H-34"H (508 mm-863 mm), TeraFrame, Cable Port, 23.6" (600 mm) W x 43.3" (1100 mm) Glacier White (1)	
024-739046-E63	Side Panel, 6 SLIDE, 43.3" (1100 mm) D x 45 RMU, Glacier White (2)	
024-739200-002	Kit, Common Parts Parts, Teraframe, UL, White (1)	
024-739110-701	PDU Bracket, Assembly, Standard, 0.7" (17 mm) W x 3.9" (100 mm) D x 2.4" (60 mm), Black (1)	
39110-C01	PDU Bracket, Assembly, Standard, 0.7" (17 mm) W x 3.9" (100 mm) D x 2.4" (60 mm), Black (1)	
024-739107-E00	Air Director, Glacier White (1)	
024-739240-003	Kit, Packaging, 600 W x 1075-1200D x 42U - 48U (1)	
024-739029-001	Caster Kit, Two Swivel, Two Fixed, 1.6" (41 mm) W x 2" (51 mm) D x 2" (41 mm) H	
024-735077-002	F-Series TeraFrame Cabinet with Caster Packaging, 600-700mm X 1025-1200mm D X 36-45U H (1)	
39085-E03	Air Dam, 6 SLIDE, 23.6" (600 mm) W x 45 RMU, Glacier White (1)	
39137-703	Baying Seal Kit, 45U (2)	
39132-E00	Bottom Panel with Brushes (1)	
31920-E03	Finger Cable Manager no Cover (2)	

UL2416 This Cabinet and any included accessories are UL Listed under the NWIN category per the UL2416 Standard. UL file number E227626. Note, the UL Listing only applies to cabinets manufactured in the United States.

(VENDOR TO PROVIDE 45 RU OF PLENAFILL BLANKING PANEL PER ENCLOSURE)



1 TYPICAL SERVER CABINET WITH VERTICAL EXHAUST DUCT (VED) SPECIFICATION

DEPARTMENT OF VETERANS AFFAIRS



IT OPERATIONS AND SERVICES
SOLUTION DELIVERY



EDCT
ENTERPRISE DATA CENTER AND
INFRASTRUCTURE ENGINEERING

PROJECT:

OIT DESIGN GUIDE TEMPLATES

PROJECT No:
N/A

MARK	DATE	DESCRIPTION

ISSUE:

DRAWING No:

FILE: OIT_DESIGN_GUIDE_TEMPLATE
S-04162021-V3.1.VSDX

DESIGN BY: KEVIN GRZELKA, CTDC

VERIFIED BY: MICHAEL JULIAN, RCDD
KELLY BATES, DCIE

DOC VERSION No: 3.1

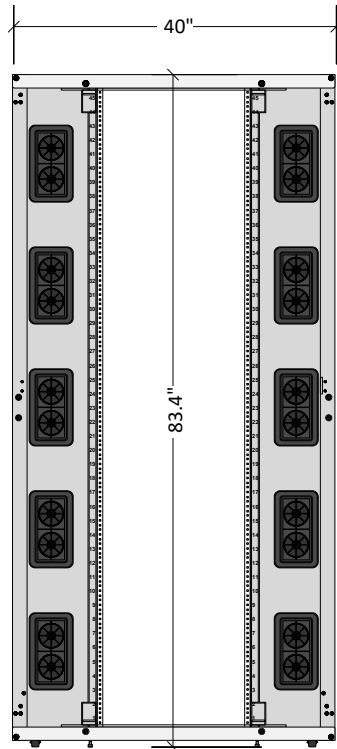
ISSUE DATE: Apr 16, 2021

SHEET TITLE

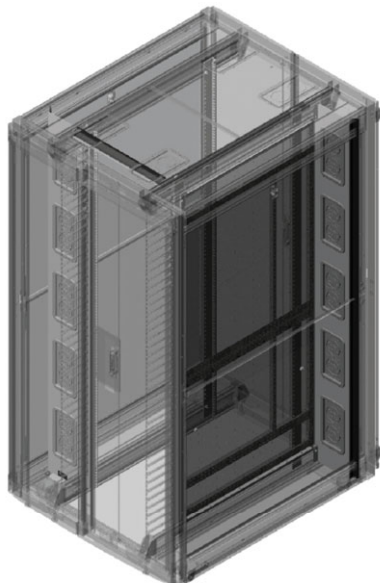
TYPICAL SERVER CABINET WITH
VERTICAL EXHAUST DUCT

SHEET: 33 OF 37

NOTE: NETWORK CABINETS ARE PURPOSE BUILT TO PROVIDE AMPLE ROOM FOR CABLING NEEDS OF THE HORIZONTAL DISTRIBUTION AREA AND THE MAIN DISTRIBUTION AREA AND MAY BE USED IN LIEU OF TELECOMMUNICATION CHANNEL RACKS FOR THIS PURPOSE. TELECOMMUNICATION CABINETS MAY NOT BE USED IN TLECOMMUNICATION ROOMS.



3 TYPICAL NETWORK CABINET FRONT VIEW DIMENSIONS



2 TYPICAL NETWORK CABINET ISOMETRIC VIEW WITH DOUBLE DOORS



4 SIDE PANELS (CHATSWORTH 39043-E63)

SIDE PANEL WITH BRUSH SEALED CABLE OPENINGS (CHATSWORTH 39043-E63)
SIDE PANEL WITH BRUSH SEALED CABLE OPENINGS FEATURE MULTIPLE BRUSH SEALED CABLE OPENINGS THAT ALLOW CABLES TO ENTER THE SIDE OF THE CABINET WHILE CONTAINING EXHAUST AIR WITHIN EACH CABINET

- TWO-PIECE DESIGN WITH TOP AND BOTTOM HALVES FOR EASIER HANDLING
- EIGHT CABLE OPENINGS PER SIDE, FOUR PER HALF-HEIGHT PANEL, ONE PER CORNER, ARE 4.5"W x 9"D WITHOUT COVER
- BRUSH COVER OVER CABLE OPENINGS PROTECTS CABLES AS THEY PASS THROUGH THE SIDE PANEL AND PROVIDES A 3.9"W X8.8"D PASS THROUGH AREA FOR CABLES
- TOOL-LESS REMOVAL, DROP IN DESIGN WITH INTEGRATED LOCKING SPRING LATCH SECURES THE SIDE PANEL TO THE CABINET FRAME
- INTEGRATED GROUNDING TABS TO ELECTRICALLY BOND THE SIDE PANEL TO THE CABINET FRAME

1. Height	
U	In mm
42	78.1 1984 C
43	79.9 2029 D
44	81.6 2073 E
45	83.4 2117 F
46	85.1 2162 G
47	86.9 2206 H
48	88.6 2251 J
49	90.4 2295 K
50	92.1 2340 L
51	93.9 2384 M
52	95.6 2429 N

Height includes transport casters.

2. Width	
In	mm
31.5	800
31.5	800
31.5	800
40.0	1016
40.0	1016
40.0	1016

3. Depth	
In	mm
39.4	1000
40.4	1025
41.3	1050
42.3	1075
43.3	1100
44.3	1125
45.3	1150
46.3	1175
47.2	1200

4. Equipment Mounting Rail Style	
Square-Punched, 2-Pair	1
Tapped, #12-24, 2-pairs	2

5. Front Door	
None	0
Perforated Metal, matches F-Series TeraFrame Gen 3 Cabinet	1
Perforated Metal, matches GF-Series GlobalFrame Gen 2 Cabinet	2

6. Rear Door	
None	0
Double, Perforated Metal	3
Solid Metal with Perforated Insert	4
Solid Metal	5

7. Latch/Locks Kits	
Two-Point, Keyed Lock	C
Two-Point, Keyed/Combination Lock	E
None	N

8. Color	
Black	C
Glacier White	E

9. Top Panel Style	
Standard Top	4
Vertical Exhaust Duct, 20"-34"H (508-863 mm)	5
Vertical Exhaust Duct, 34"-60"H (863-1523 mm)	6
Vertical Exhaust Duct, 14"-20"H (356-508 mm)	7

10. Side Panel Style	
None	0
Solid, 1 each	1
Solid, 2 each	2

11. Cable Entry Cover	
Plastic Grommet	0
Brush Sealed	1

40.0"W (1016 mm) cabinets are only available 44.3"D (1100 mm) or deeper.

Frame Depth does not include doors. Front door adds 2.4" (61 mm). Rear door adds 1.5" (38 mm). See Dimensional Drawing.

Single door on the 31.5"W (800 mm) cabinet. Double door on the 40.0"W (1016 mm) cabinet.

Options 5, 6, & 7 are for cabinets that are 41.3"D (1050 mm) or deeper. Use Solid Metal Rear Door or Solid Metal with Perforated Insert Rear Door with Vertical Exhaust Duct. Vertical Exhaust Duct includes a Bottom Panel.

Use Double Perforated Metal door with Standard Top panel. Use Solid Metal With Perforated Insert with Vertical Exhaust Duct and Side Intake Duct. Use Solid Metal door with Vertical Exhaust Duct and Air Dams. Solid Metal with Perforated Insert and Solid Metal doors are single doors on the 31.5"W (800 mm) cabinet and double doors on the 40.0"W (1016 mm) cabinet.

There is a single-Point latch on front door that matches GF-Series GlobalFrame Gen 2 Cabinet

Include side panels on all cabinets with Front Air Dams or Side Intake Duct Air Managers.

SIDE PANELS WITH BRUSH SEALED CABLE OPENINGS ARE MANDATORY AND MUST BE ORDERED SEPARATELY

1 TYPICAL NETWORK CABINET CONFIGURATION - CHATSWORTH NF8N-113N-C42-1



IT OPERATIONS AND SERVICES
SOLUTION DELIVERY



EDCT
ENTERPRISE DATA CENTER AND
INFRASTRUCTURE ENGINEERING

PROJECT:

OIT DESIGN GUIDE TEMPLATES

PROJECT No:

N/A

MARK DATE DESCRIPTION

ISSUE:

DRAWING No:

FILE: OIT DESIGN GUIDE TEMPLATE
S-04162021-V3.1.VSDX

DESIGN BY: KEVIN GRZELKA, CTDC

VERIFIED BY: MICHAEL JULIAN, RCDD
KELLY BATES, DCIE

DOC VERSION No: 3.1

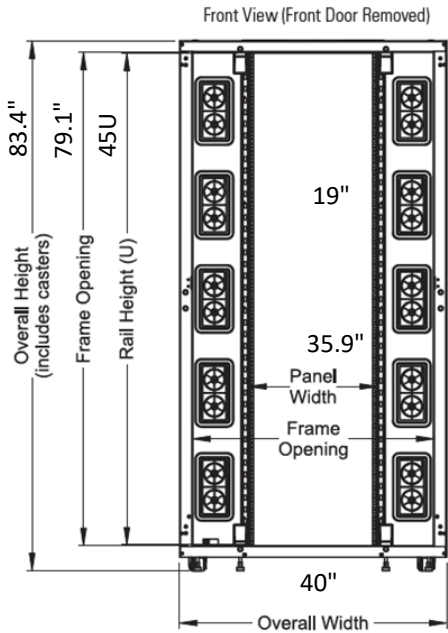
ISSUE DATE: Apr 16, 2021

SHEET TITLE

TYPICAL NETWORK CABINET

SHEET: 34 OF 37

NOTE: NETWORK CABINETS ARE PURPOSE BUILT TO PROVIDE AMPLE ROOM FOR CABLING NEEDS OF THE HORIZONTAL DISTRIBUTION AREA AND THE MAIN DISTRIBUTION AREA AND MAY BE USED IN LIEU OF TELECOMMUNICATION CHANNEL RACKS FOR THIS PURPOSE. TELECOMMUNICATION CABINETS MAY NOT BE USED IN TELECOMMUNICATION ROOMS.



3 FRONT VIEW DIMENSIONS



2 ISOMETRIC VIEW WITH DOORS AND VERTICAL EXHAUST DUCT (VED)



4 SIDE PANELS (CHATSWORTH 39043-E63)

SIDE PANEL WITH BRUSH SEALED CABLE OPENINGS (CHATSWORTH 39043-E63)
SIDE PANEL WITH BRUSH SEALED CABLE OPENINGS FEATURE MULTIPLE BRUSH SEALED CABLE OPENINGS THAT ALLOW CABLES TO ENTER THE SIDE OF THE CABINET WHILE CONTAINING EXHAUST AIR WITHIN EACH CABINET

- TWO-PIECE DESIGN WITH TOP AND BOTTOM HALVES FOR EASIER HANDLING
- EIGHT CABLE OPENINGS PER SIDE, FOUR PER HALF-HEIGHT PANEL, ONE PER CORNER, ARE 4.5"W x 9"D WITHOUT COVER
- BRUSH COVER OVER CABLE OPENINGS PROTECTS CABLES AS THEY PASS THROUGH THE SIDE PANEL AND PROVIDES A 3.9"W X 8.8"D PASS THROUGH AREA FOR CABLES
- TOOL-LESS REMOVAL, DROP IN DESIGN WITH INTEGRATED LOCKING SPRING LATCH SECURES THE SIDE PANEL TO THE CABINET FRAME
- INTEGRATED GROUNDING TABS TO ELECTRICALLY BOND THE SIDE PANEL TO THE CABINET FRAME

N				H		W		D		-		R		F		R		L		-		C		T		S		-		E	
1. Height																															
U	in	mm	C																												
42	78.1	1984	C																												
43	79.9	2029	D																												
44	81.6	2073	E																												
45	83.4	2117	F																												
46	85.1	2162	G																												
47	86.9	2206	H																												
48	88.6	2251	J																												
49	90.4	2295	K																												
50	92.1	2340	L																												
51	93.9	2384	M																												
52	95.6	2429	N																												
Height includes transport casters.																															
2. Width		Included Air Manager																													
in	mm																														
31.5	800	No Air Manager	0																												
31.5	800	Front Air Dams	7																												
31.5	800	Side Intake Duct	1																												
40.0	1016	No Air Manager	6																												
40.0	1016	Front Air Dams	8																												
40.0	1016	Side Intake Duct	5																												
40.0"W (1016 mm) cabinets are only available 44.3"D (1100 mm) or deeper.																															
3. Depth		With Doors																													
in	mm	in	mm																												
39.4	1000	43.3	1100	J																											
40.4	1025	44.3	1125	K																											
41.3	1050	45.3	1150	L																											
42.3	1075	46.3	1175	M																											
43.3	1100	47.3	1200	N																											
44.3	1125	48.2	1225	P																											
45.3	1150	49.2	1250	R																											
46.3	1175	50.2	1275	T																											
47.2	1200	51.2	1300	U																											
Frame Depth does not include doors. Front door adds 2.4" (61 mm). Rear door adds 1.5" (38 mm). See Dimensional Drawing.																															
4. Equipment Mounting Rail Style																															
Square-Punched, 2-Pair				1																											
Tapped, #12-24, 2-pairs				2																											
5. Front Door																															
None				0																											
Perforated Metal, matches F-Series TeraFrame Gen 3 Cabinet				1																											
Perforated Metal, matches GF-Series GlobalFrame Gen 2 Cabinet				2																											
Single door on the 31.5"W (800 mm) cabinet. Double door on the 40.0"W (1016 mm) cabinet.																															
10. Side Panel Style																															
None				0																											
Solid, 1 each				1																											
Solid, 2 each				2																											
Include side panels on all cabinets with Front Air Dams or Side Intake Duct Air Managers.																															
9. Top Panel Style																															
Standard Top				4																											
Vertical Exhaust Duct, 20"-34"H (508-863 mm)				5																											
Vertical Exhaust Duct, 34"-60"H (863-1523 mm)				6																											
Vertical Exhaust Duct, 14"-20"H (356-508 mm)				7																											
Options 5, 6, & 7 are for cabinets that are 41.3"D (1050 mm) or deeper. Use Solid Metal Rear Door or Solid Metal with Perforated Insert Rear Door with Vertical Exhaust Duct. Vertical Exhaust Duct includes a Bottom Panel.																															
8. Color																															
Black				C																											
Glacier White				E																											
7. Latch/Locks Kits																															
Two-Point, Keyed Lock				C																											
Two-Point, Keyed/Combination Lock				E																											
None				N																											
There is a single-Point latch on front door that matches GF-Series GlobalFrame Gen 2 Cabinet																															
6. Rear Door																															
None				0																											
Double, Perforated Metal				3																											
Solid Metal with Perforated Insert				4																											
Solid Metal				5																											
Use Double Perforated Metal door with Standard Top panel. Use Solid Metal With Perforated Insert with Vertical Exhaust Duct and Side Intake Duct. Use Solid Metal door with Vertical Exhaust Duct and Air Dams. Solid Metal with Perforated Insert and Solid Metal doors are single doors on the 31.5"W (800 mm) cabinet and double doors on the 40.0"W (1016 mm) cabinet.																															
11. Cable Entry Cover																															
Plastic Grommet				0																											
Brush Sealed				1																											

SIDE PANELS WITH BRUSH SEALED CABLE OPENINGS ARE MANDATORY AND MUST BE ORDERED SEPARATELY

SIZED PER REQUIREMENT

1 TYPICAL RACK CONFIGURATION - CHATSWORTH NF8N-115C-E52-1

DEPARTMENT OF VETERANS AFFAIRS



IT OPERATIONS AND SERVICES
SOLUTION DELIVERY



EDCT
ENTERPRISE DATA CENTER AND
INFRASTRUCTURE ENGINEERING

PROJECT:

OIT DESIGN GUIDE TEMPLATES

PROJECT No:

N/A

MARK DATE DESCRIPTION

ISSUE:

DRAWING No:

FILE: OIT_DESIGN_GUIDE_TEMPLATE
S-04162021-V3.1.VSDX

DESIGN BY: KEVIN GRZELKA, CTDC

VERIFIED BY: MICHAEL JULIAN, RCDD
KELLY BATES, DCIE

DOC VERSION No: 3.1

ISSUE DATE: Apr 16, 2021

SHEET TITLE

TYPICAL NETWORK CABINET
WITH VERTICAL EXHAUST DUCT

SHEET: 35 OF 37

Cable Lashing Manager

Cable Lashing Manager creates a simple vertical pathway for multiple cable bundles and has multiple attachment points for securing cables using CPI Saf-T-Grip® Straps or zip ties. Use to support network cables along the sides of the cabinet.

- Attaches to the frame, adjusts in depth independent of the equipment mounting rails
- Two sizes: Slim and Wide
- Slim manager is 5"W (130 mm) and has three vertical columns of attachment points for 1.5" (38 mm) diameter cable bundles, 5 in² (3200 mm²) cable fill area
- Wide manager is 9.8"W (250 mm) and has five vertical columns of attachment points for 1.5" (38 mm) diameter cable bundles, 9.8 in² (6320 mm²) cable fill area
- Includes: Bracket and installation hardware, order Saf-T-Grips or cable ties separately
- Material: Steel; Select part number to match usable height and color of the cabinet

Part Number		Height (U)	Shipping Weight lb (kg)
Slim	Wide		
5.1"W (130 mm)	9.8"W (250 mm)		
39380-X00	39381-X00	42	19 (8.9)
39380-X01	39381-X01	43	20 (9.1)
39380-X02	39381-X02	44	20 (9.1)
39380-X03	39381-X03	45	21 (9.5)

X=Color; C=Black, E=Glacier White. Shipping Weights are for Wide managers.



4 TYPICAL CABLE LASHING MANAGER

Integrated Cable Manager

Integrated Cable Manager are plastic, T-shaped cable guides that help organize patch cords and jumper cables as they enter and exit equipment at the front of the cabinet.

Key Features:

- Attaches to and adjusts with mounting rails
- Requires a minimum rail setback of 7.6" (193 mm) from the front of the cabinet frame
- Smooth, rounded surfaces on T-shaped cable guides to protect cables
- Openings in guides align with each rack-mount space (U) on the equipment mounting rails
- Openings are 6.6 in² (4260 mm²), large enough for 24 each .30" (7.6 mm) diameter cables

Specifications:

- Includes: One pair of full height Cable Managers
- Material: Plastic
- Finish: Black

Integrated Cable Manager			
Part Number - Cabinet Width		Height (U)	Shipping Weight lb (kg)
31.5"W (800 mm)	40.0"W (1016 mm)		
39375-000	39375-000	42	5 (2.3)
39375-001	39375-001	43	6 (2.7)
39375-002	39375-002	44	6 (2.7)
39375-003	39375-003	45	6 (2.7)

Color is Black.

3 TYPICAL INTEGRATED CABLE MANAGER



Part Number - Cabinet Width		Frame Depth in (mm)	Shipping Weight lb (kg)
31.5" (800 mm)	40.0" (1016 mm)		
39327-X01	39327-X01	Any	1 (0.5)

X=Color; C=Black, E=Glacier White.

6 TYPICAL RADIUS DROP



Part Number	Description	Shipping Weight lb (kg)
32697-001	Tool-less Bend Radius Kit for Fiber Patch Cords, Pack of 10	1 (0.5)
32697-002	Tool-less Bend Radius Kit for Fiber Patch Cords, Pack of 50	3 (1.4)

Notes: Color is Black.

5 TYPICAL TOOL-LESS BEND RADIUS KIT

Radius Drop

Radius Drop attaches to the top of the cabinet and provides a smooth curved support where cables exit or enter the cable management space along the sides of the cabinet.

- Attaches to the top of the cabinet to support cables as they exit/enter the cabinet
- Radius drop is 11"L (279 mm) with a 1.3" (33 mm) diameter radius
- Includes: radius drop, installation hardware
- Material: Aluminum; select part number to match the color of the cabinet

Tool-less Bend Radius Kit for Fiber Patch Cords

Tool-less Bend Radius Kit for Fiber Patch Cords ensures proper bend radius support for fiber optic patch cords when transitioning through T-shaped cable guides. It snaps onto the finger manager without any tools, and can be used for new or retrofit applications.

- Provides proper bend radius support for fiber optic patch cords
- Tool-less installation
- May be installed in upward or downward orientations on either side of a T-shaped cable guide
- Available in packs of 10 or 50



Telescoping Cable Manager

Telescoping Cable Manager attaches to the sides of the front and rear equipment mounting rail providing a front-to-rear or vertical pathway for cables along the side of equipment.

- Attaches to and adjusts in depth with mounting rails, fully compatible with Front Air Dams and Side Intake Duct
- Five smooth, plastic L-shaped spools support cables
- Spools on 31.5"W (800 mm) model are 2.5"H x 2.6"D (64 mm x 66 mm)
- Spools on 40.0"W (1016 mm) model are 3.3"H x 6.8"D (84 mm x 173 mm)
- Spools rotate and lock in 90-degree increments and can be adjusted to manage horizontal or vertical cable runs
- Includes: cable manager, five cable spools, hardware
- Material: Steel manager, plastic spools
- Select part number to match the color of the cabinet

Part Number - Cabinet Width		Frame Depth in (mm)	Shipping Weight lb (kg)
31.5" (800 mm)	40.0" (1016 mm)		
39377-X01	39377-X02	Any	5 (2.3)

X=Color; C=Black, E=Glacier White. Shipping Weights are for 1016 mm kits.

7 TYPICAL TELESCOPING CABLE MANAGER

Full Height Dual PDU Bracket

Full Height Dual PDU Bracket supports two vertical PDUs side-by-side and has multiple slots for CPI Saf-T-Grip® Straps or tie wraps to secure cords to the bracket.

- Attaches to the frame, adjusts in depth independent of the mounting rails
- Available in two widths: Dual and Wide
- Dual bracket is 4.8"W (121 mm) and supports two PDUs up to 2.2"W (56 mm) side-by-side
- Dual bracket supports most eConnect PDUs and power strips side-by-side
- Wide bracket is 5.8"W (147 mm) and supports two PDUs up to 2.7"W (69 mm) side-by-side
- Wide bracket is required to mount two 6-breaker eConnect PDUs (series P/N PX-3XXXX) side-by-side
- Tool-less mounting on 28" (711 mm), 61.25" (1556 mm) or 64.75" (1645 mm) centers
- Includes: brackets, installation hardware, order Saf-T-Grips or cable ties separately
- Material: Steel; Select part number to match the usable height and color of the cabinet

Part Number		Height (U)	Shipping Weight lb (kg)
Dual	Wide		
4.8"W (121mm)	5.8"W (147 mm)		
39086-X00	39221-X00	42	10 (4.5)
39086-X01	39221-X01	43	10 (4.5)
39086-X02	39221-X02	44	11 (5.0)
39086-X03	39221-X03	45	11 (5.0)

X=Color; C=Black, E=Glacier White.

1 TYPICAL PDU BRACKET

Ring Cable Manager

Ring Cable Manager creates a simple vertical pathway for patch and jumper cords and has C-shaped rings to organize and secure cables. Use to support patch cords and jumper cords near the front of the cabinet.

- Attaches to the frame, adjusts in depth independent of the equipment mounting rails
- Rings have an opening on the front for easy addition and removal of cables
- Includes: bracket, rings and installation hardware
- Material: Steel bracket, plastic rings
- Finish: Black or Glacier White; rings are black
- Select part number to match the width, usable height and color of the cabinet

Part Number - Cabinet Width		Height (U)	Shipping Weight lb (kg)
31.5" (800 mm)	40.0" (1016 mm)		
39128-X00	39129-X00	42	14 (6.4)
39128-X01	39129-X01	43	14 (6.4)
39128-X02	39129-X02	44	14 (6.4)
39128-X03	39129-X03	45	14 (6.4)

X=Color; C=Black, E=Glacier White. Shipping Weights are for 40.0"W (1016 mm) managers.

2 TYPICAL RING CABLE MANAGER

DEPARTMENT OF VETERANS AFFAIRS



IT OPERATIONS AND SERVICES SOLUTION DELIVERY



EDCT ENTERPRISE DATA CENTER AND INFRASTRUCTURE ENGINEERING

PROJECT:

OIT DESIGN GUIDE TEMPLATES

PROJECT No:

N/A

MARK	DATE	DESCRIPTION

ISSUE:

DRAWING No:

FILE: OIT_DESIGN_GUIDE_TEMPLATE S-04162021-V3.1.VSDX

DESIGN BY: KEVIN GRZELKA, CTDC

VERIFIED BY: MICHAEL JULIAN, RCDD KELLY BATES, DCIE

DOC VERSION No: 3.1

ISSUE DATE: Apr 16, 2021

SHEET TITLE

TYPICAL NETWORK CABINET ACCESSORIES

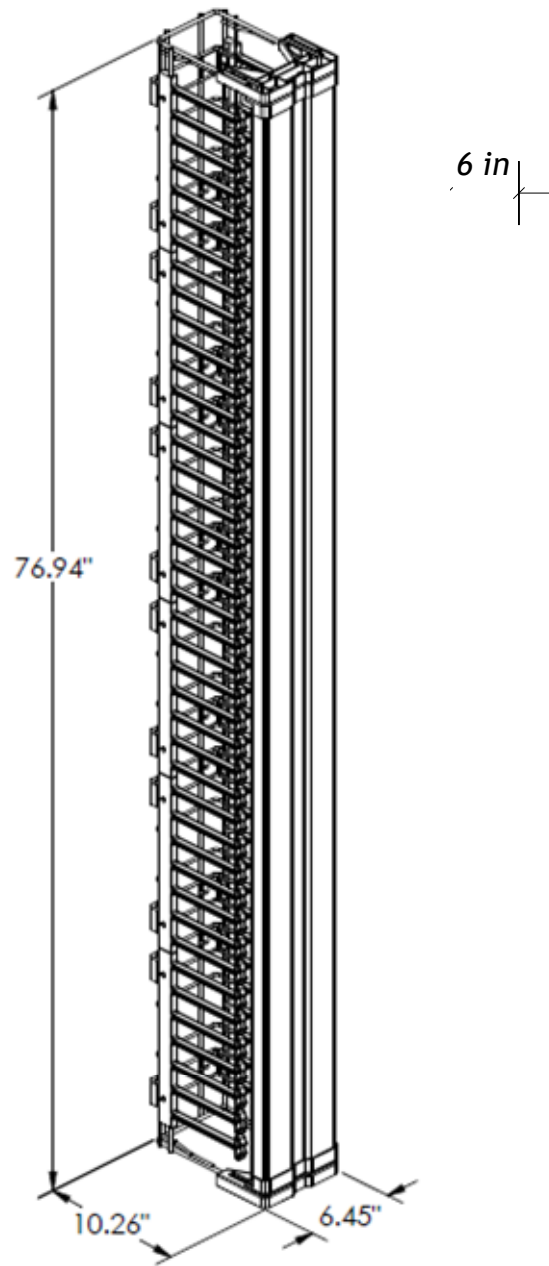
SHEET: 36 OF 37

D

C

B

A



P/N: OR-MM20VMD706 (OR EQUIVALENT)

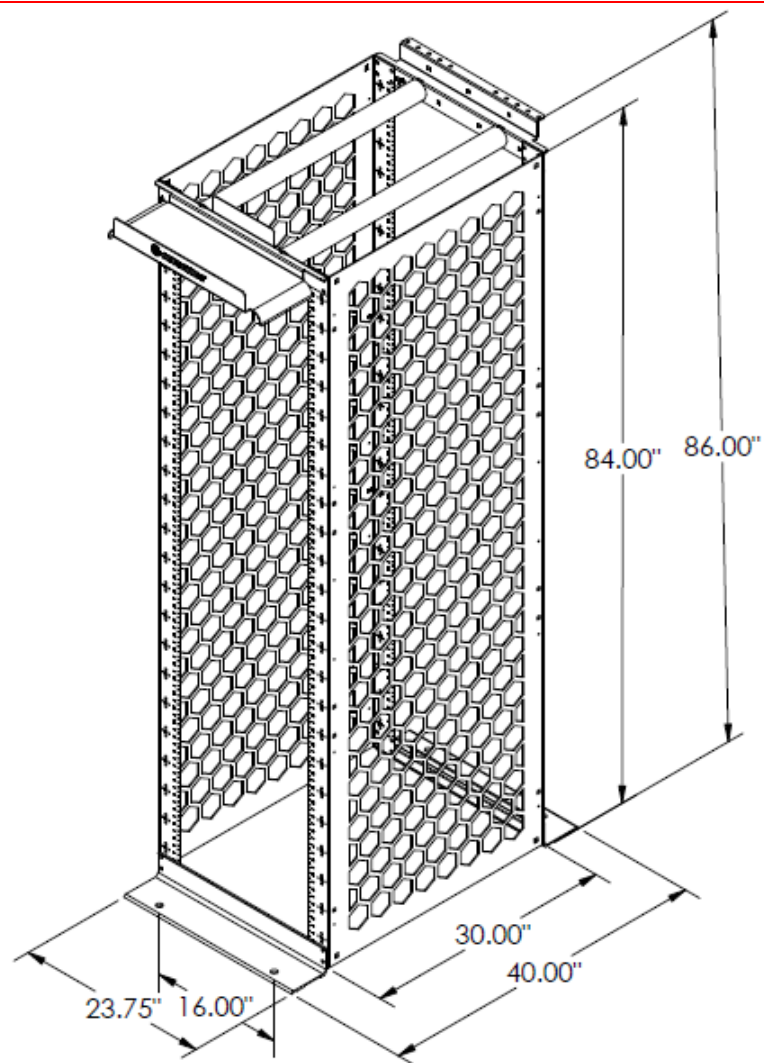
MM20 VERTICAL CABLE MANAGEMENT "CAGE",
WITH DOOR, 6.45" X 10.26" X 76.94" WHITE

CAPACITY: 350 CATEGORY 6 PATCH CORDS OR
245 CATEGORY 6A PATCH CORDS

LARGER VCM AUTHORIZED IN ACCORDANCE WITH
CABLING REQUIREMENT

3 6" WIDE VERTICAL CABLE MANAGEMENT

NOTE: TELECOMMUNICATION CHANNEL RACKS ARE
REQUIRED IN ALL TELECOMMUNICATION ROOMS.
TELECOMMUNICATION CHANNEL RACKS OR
TELECOMMUNICATION CABINETS MAY BE USED IN
THE HORIZONTAL AND MAIN DISTRIBUTION AREAS.



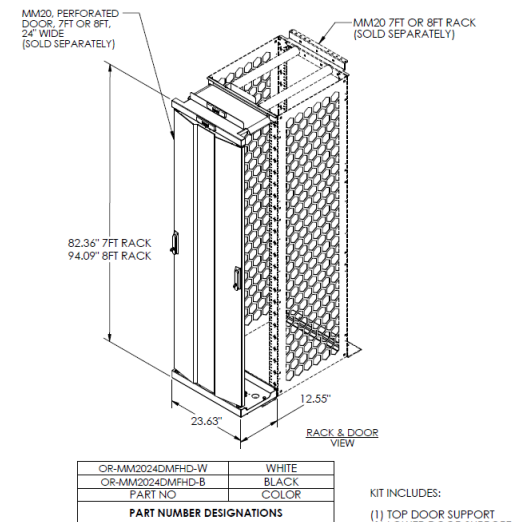
P/N: OR-MM2073038-W (OR EQUIVALENT)

MIGHTY MO 20 CABLE MANAGEMENT RACK
30" CHANNEL DEPTH, 7' HIGH, WHITE FINISH OR
EQUIVALENT

DIMENSIONS:
USABLE HEIGHT: 45RU
DEPTH (US): 30"
HEIGHT (US): 7'
WIDTH (US): 23.75"
STATIC LOAD CAPACITY: 2000 LBS
SQUARE HOLE

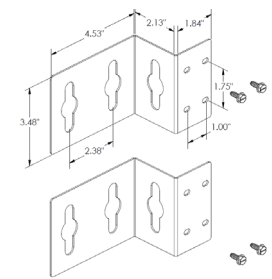
2 TYPICAL RACK DIMENSIONS AND SPECIFICATIONS

3



4 SECURITY DOOR FRONT
MOUNTING HARDWARE (OR-
MM2024DMFHD-W)

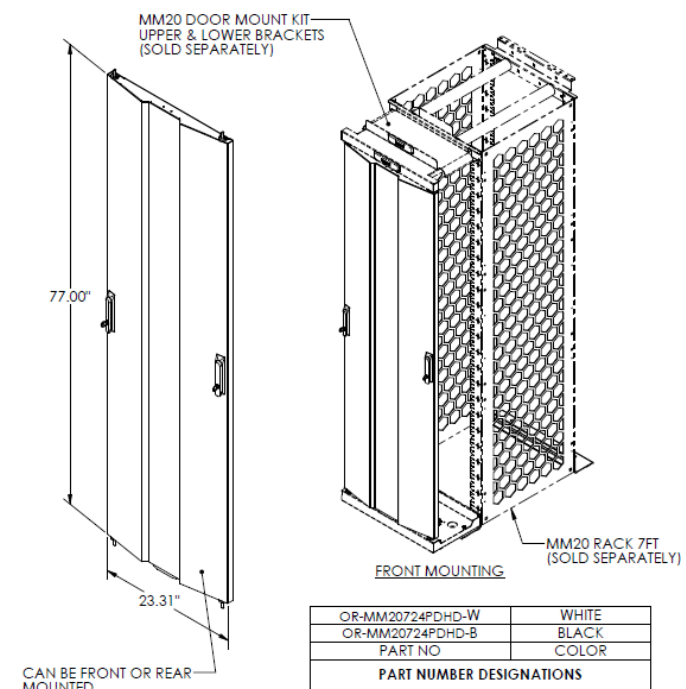
4



PDU MOUNTING BRACKET KIT
(TWO KITS PER RACK)
P/N: OR-MM20PDUMB1D2W-W



5 SECURITY DOOR REAR MOUNTING
HARDWARE (OR-MM2024DMRHD-
W)

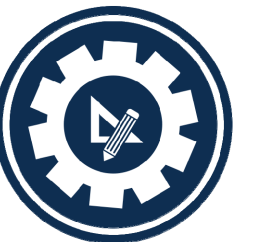


1 FRONT AND REAR HEAVY DUTY PERFORATED SECURITY DOOR (OR-
MM20724PDHD-W) (OPTIONAL)

DEPARTMENT OF VETERANS
AFFAIRS



IT OPERATIONS AND SERVICES
SOLUTION DELIVERY



EDCT
ENTERPRISE DATA CENTER AND
INFRASTRUCTURE ENGINEERING

PROJECT:

OIT DESIGN GUIDE TEMPLATES

PROJECT No:

N/A

MARK	DATE	DESCRIPTION
------	------	-------------

ISSUE:

DRAWING No:

FILE: OIT DESIGN GUIDE TEMPLATE
S-04162021-V3.1.VSDX

DESIGN BY: KEVIN GRZELKA, CTDC

VERIFIED BY: MICHAEL JULIAN, RCDD
KELLY BATES, DCIE

DOC VERSION No: 3.1

ISSUE DATE: Apr 16, 2021

SHEET TITLE

TYPICAL NETWORK RACK

SHEET: 37 OF 37